Smooth anisotropic Calderón uniqueness on the ball

Expanded proof manuscript for the frozen Lean target v002

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Abstract

We expand a proof route for the following precise target: two smooth positive-definite matrix metrics on $\mathbb{R}^3$, both Euclidean on the entire complement of the open unit ball, with equal minimum Dirichlet energies for every globally smooth exterior datum, are related by a smooth global diffeomorphism fixing that complement pointwise. The variational definition, its relation to boundary data, the passage to a common exterior source region, the harmonic-jet argument, the local continuation machinery, and the final global extension are kept explicit. The local analytic core of the supplied manuscript is retained, with additional derivations and a dependency ledger. Standard analytic interfaces are stated separately from the new continuation estimates. This is a mathematical working manuscript; its authoring checks are not Lean proof acceptance or an independent semantic certification. The frozen theorem and its definitions are unchanged.

How to read this version. Sections 1–4 contain the specialization and its detailed bridges. Sections 5–10 retain the local proof, with expansion notes after each major part. Section 12 treats the interior matching map, and Section 14 proves the exterior extension required by Lean. The final ledger distinguishes proved reductions, analytic interfaces still requiring exact source verification, and any unresolved mathematical obligations. Book theorem numbers are deliberately not invented where the chosen edition has not been supplied. No external mathematical axiom has been installed.

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1 The exact target and its conventions

Set $B = \{x \in \mathbb{R}^3 : |x| < 1\}$ and $N = \{x : |x| < 2\}$; a compact manifold denoted $\overline{N}$ below includes its boundary. Indices range over $1, 2, 3$; this is the relabeling of Lean's `Fin 3`. For a smooth real function u, write $\partial_i u(x) = Du(x)e_i$. For a smooth map Φ, the Jacobian matrix has entries $D\Phi(x)_{ij} = D\Phi(x)e_j|_i$. Matrices act on column vectors. The metric is covariant, so its inverse appears in the energy. The orthonormal coordinate identification of Lean's `EuclideanSpace` with $\mathbb{R}^3$ preserves its Euclidean norm and canonical volume; consequently the set integral in the frozen energy is exactly the Lebesgue integral written here. Entrywise smoothness of a finite matrix field is equivalent to smoothness as a finite-dimensional map.

Definition 1.1 (The frozen data)**.** An admissible metric is a smooth map $g : \mathbb{R}^3 \to \mathrm{Sym}_3^+$ such that $g(x) = I$ for $|x| \geq 1$. Put

$$\rho_g = \sqrt{\det g}, \qquad A_g = \rho_g g^{-1}, \qquad E_g(u) = \int_B (\nabla u)^T A_g \nabla u \, dx.$$

For $f \in C^\infty(\mathbb{R}^3; \mathbb{R})$, define

$$\mathcal{C}_f = \{u \in C^\infty(\mathbb{R}^3; \mathbb{R}) : u(x) = f(x) \text{ whenever } |x| \geq 1\}, \qquad m_g(f) = \inf\{E_g(u) : u \in \mathcal{C}_f\}.$$

The infimum is a real infimum, as in the frozen definition `minimalEnergy`, not an extended-real convention.

Theorem 1.2 (Selected target)**.** *Let g_1, g_2 be admissible metrics. Suppose $m_{g_1}(f) = m_{g_2}(f)$ for every real $f \in C^\infty(\mathbb{R}^3)$. Then there is a bijection $\Phi : \mathbb{R}^3 \to \mathbb{R}^3$ such that Φ and Φ^{-1} are smooth, $\Phi(x) = x$ for $|x| \geq 1$, and*

$$g_2(x) = D\Phi(x)^T g_1(\Phi(x)) D\Phi(x) \qquad (x \in \mathbb{R}^3). \tag{1.1}$$

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No real analyticity, smallness, curvature sign, or Euclidean collar inside B is assumed. Smoothness and the exterior identity do imply that every derivative of $g - I$ vanishes on ∂B, by continuity from the exterior; they do not imply that $g = I$ on an interior collar. The proof below never uses such a collar. The status of the proof of Theorem 1.2 is the status of the full dependency chain in the final ledger.

Lemma 1.3 (Coefficient and integrability facts)**.** *For each admissible g, ρ_g and A_g are smooth, $\rho_g > 0$, and A_g is symmetric positive definite. On every compact set K there are $0 < a_K \leq b_K < \infty$ with*

$$a_K|\xi|^2 \leq \xi^T A_g(x)\xi \leq b_K|\xi|^2 \quad (x \in K,\ \xi \in \mathbb{R}^3).$$

For every globally smooth u, $E_g(u)$ is a finite nonnegative real number. The set of competitor energies is nonempty and bounded below.

Proof. A real positive-definite matrix is symmetric with positive eigenvalues, so its determinant is positive and its inverse is positive definite. The cofactor formula for the inverse and smoothness of square root on $(0, \infty)$ prove smoothness. The claim for an empty compact set is vacuous. For nonempty K, the continuous positive function $(x, \xi) \mapsto \xi^T A_g(x)\xi$ on $K \times S^2$ has a positive minimum and a finite maximum. Homogeneity gives the displayed bounds. The integrand for smooth u is

continuous on $\overline{B}$ and bounded there; B has finite Lebesgue volume. It is pointwise nonnegative. Finally $f \in \mathcal{C}_f$, so the competitor set is nonempty, and zero is a lower bound. These facts justify ordinary real infimum rules in the later Lean bridge; no convention for an empty or unbounded set is being used. □

The positive Laplace operator convention is

$$L_g = -\Delta_g = -\rho_g^{-1}\partial_i(A_g^{ij}\partial_j), \qquad dV_g = \rho_g\,dx. \tag{1.2}$$

Thus the weak form of $L_g u = F$ is $\int A_g \nabla u \cdot \nabla v = \int F v\,dV_g$. All Green kernels below use this sign and metric volume. In the common Euclidean annulus, metric volume and ordinary volume coincide.

2 Precisely delimited standard analytic inputs

The following interfaces describe the classical analysis used by this proof. They are requirements for library reuse or a later explicitly authorized literature-relative theorem; they are not additional hypotheses silently inserted into Theorem 1.2. McLean [1], Gilbarg–Trudinger [2], and Hörmander [3, 4] are the preferred monograph sources. Exact edition-specific theorem passages remain to be checked before opaque formal premises are adopted. We state the needed specializations so this check has a concrete mathematical object.

S1: Sobolev spaces, traces, density, and variational calculus. On a bounded smooth domain D, the trace $H^1(D) \to H^{1/2}(\partial D)$ is continuous and onto, with kernel $H_0^1(D)$, the closure of $C_c^\infty(D)$ in H^1. On $H_0^1(D)$, $\|v\|_{L^2} \le C_D\|\nabla v\|_{L^2}$. Matching traces on either side of a smooth interface give an H^1 function on the union; zero trace on the outer boundary gives H_0^1 there. We use the Hilbert-space representation theorem for a bounded coercive symmetric bilinear form, interior compact Sobolev embedding, and finite-order interpolation. McLean, Chapters 2–3 and Appendices A–B, is the proposed primary source. The particular scaling interpolation used later is derived explicitly.

S2: Smooth Dirichlet solutions and estimates. For a smooth symmetric uniformly positive matrix A on $\overline{D}$, the weak problem $-\partial_i(A^{ij}\partial_j u) = F$ with a prescribed trace has the regularity implied by smooth coefficients, smooth data, and smooth boundary. In particular smooth data give $u \in C^\infty(\overline{D})$. For nested compact sets $K \Subset K' \Subset D$ and every integer m, homogeneous solutions obey $\|u\|_{C^m(K)} \le C_m\|u\|_{L^2(K')}$. Corresponding inhomogeneous and boundary estimates are used to differentiate fixed-domain inverse operators. Constants depend on a positive ellipticity lower bound, specified finitely many coefficient and boundary-chart norms, the order, and the separation of the compact sets. Uniformity must be verified against these inputs, not inferred from the word "smooth." Preferred sources: Gilbarg–Trudinger, Chapters 6, 8–9; McLean, Chapter 4. The weak and strong maximum principles for L_g are included here.

S3: The inverse and its distribution kernel. The zero-Dirichlet inverse $R_g = L_{g,D}^{-1}$ is symmetric with respect to dV_g. It has a real symmetric Green kernel $G_g(x,y)$, smooth off the diagonal and with zero boundary values, satisfying

$$L_{g,x}G_g(x,y) = \delta_y^g(x), \qquad R_gF(x) = \int_D G_g(x,y)F(y)\,dV_g(y).$$

For a distribution T supported on finitely many interior points, transposition defines R_gT on test functions by $\langle R_gT, \phi\rangle = \langle T, R_g\phi\rangle$. It satisfies $L_gR_gT = T$, and is smooth and harmonic off the

support of T. This last assertion uses local distributional elliptic regularity. The fundamental-solution singularity and finitely many off-diagonal derivative bounds are used under spatial dilation. The exact uniform rescaling statement needed in the local core is recorded there. Preferred sources: McLean, Chapter 6, and Hörmander III, elliptic operators and elliptic boundary problems. A kernel with another normalization must be converted before use; symmetry alone does not identify it.

S4: Unique continuation and strict weights. A smooth solution of a real scalar smooth uniformly elliptic second-order equation on a connected open set, with the lower-order coefficients used here smooth and bounded locally, which vanishes on a nonempty open subset vanishes everywhere. The quantitative input is the strict pseudoconvex L^2 Carleman estimate in Equation (5.6), for compactly supported test functions in a sufficiently small coordinate neighborhood. The strictness margin, coefficient bounds and support margins determine its constants. Hörmander IV, Chapter XXVIII, is preferred. Koch–Tataru [7, Definition 8.3 and Theorem 9(a)] is a published precise reference for the estimate; the local core also spells out the scalar commutator derivation. This interface gives neither the mixed-kernel extension nor metric uniqueness by itself.

S5: Compact spectral theory and boundary operator calculus. We use the spectral theorem for compact positive operators on Hilbert spaces and the finite-rank characterization of singular values. For a smooth uniformly invertible family of scalar elliptic Dirichlet problems on the unit ball, its normal trace operator is a classical sphere pseudodifferential operator of order one, with the principal symbol specified by the normal root. Composition, adjoints, Sobolev mapping bounds, smooth parameter differentiation and the smoothing remainder are required in the topology of finitely many symbol and kernel seminorms. The fixed-domain inverse formula $\partial R = -R(\partial L)R$ and regularity supply the bounds on the remainder. Preferred sources: Hörmander III and McLean, Chapter 2. The particular moving-sphere Taylor expansion is a derivation in Lemma 8.1, not a named black-box uniqueness theorem. The published scalar specialization of the boundary calculus in [8] is useful corroboration.

Inputs eliminated or reduced by the specialization. We do not need a boundary-jet recovery theorem, an analytic Calderón uniqueness theorem, quantitative Runge with an unspecified geometric constant, or a curvature-sign assumption. Source separation and second jets are proved below from S3–S4. Their established published context is [6, Appendix A]; A.7 alone is not a statement about all second jets. The final removal of a conformal factor is a direct calculation, so Baird–Wood [5] is background rather than a necessary opaque premise for this route. The new weighted raising, perturbative coercivity, uniform density and bootstrap remain project proof obligations.

3 From the frozen infimum to common exterior data

3.1 The weak energy problem

For $v, w \in H^1(B)$, put $a_g(v,w) = \int_B \nabla v^T A_g \nabla w \, dx$. This is bounded by $b\|\nabla v\|_2\|\nabla w\|_2$ and coercive on $H^1_0(B)$ by Lemma 1.3 and Poincaré's inequality. For smooth f, the representation theorem gives a unique $w \in H^1_0(B)$ with

$$a_g(w,z) = -a_g(f,z) \quad (z \in H^1_0(B)).$$

Write $H_g f = f + w$. This is the unique weak harmonic function with trace $f|_{\partial B}$. If $z \in H^1_0(B)$, expanding the square gives

$$a_g(H_g f + z, H_g f + z) = a_g(H_g f, H_g f) + a_g(z,z). \tag{3.1}$$

It is therefore the unique energy minimizer in the affine Sobolev class. Smooth regularity gives smoothness up to ∂B, but says nothing about matching all exterior normal derivatives of f.

Lemma 3.1 (The exact infimum bridge)**.** *For each admissible g and globally smooth f,*

$$m_g(f) = a_g(H_g f, H_g f).$$

In particular $m_g(f)$ depends only on $f|_{\partial B}$.

Proof. If $u \in \mathcal{C}_f$, continuity gives equality of boundary traces, so $u|_B - H_g f \in H_0^1(B)$. Equation (3.1) implies $E_g(u) \geq a_g(H_g f, H_g f)$. Thus the right side is a lower bound for all competitor energies.

Conversely choose $w_k \in C_c^\infty(B)$ converging to $w = H_g f - f$ in $H^1(B)$, by the definition of $H_0^1(B)$. Extend w_k by zero to $\mathbb{R}^3$ and set $u_k = f + w_k$. Each extension is globally smooth because its support is compactly contained in B, and $u_k = f$ on the full exterior. Moreover $u_k|_B \to H_g f$ in H^1. The estimate

$$|a_g(v,v) - a_g(u,u)| \leq b\|\nabla(v-u)\|_2(\|\nabla v\|_2 + \|\nabla u\|_2)$$

shows $E_g(u_k) \to a_g(H_g f, H_g f)$. Since the real infimum is no larger than any $E_g(u_k)$, it is no larger than their limit. Together with the lower bound this proves equality. No minimizer in $\mathcal{C}_f$ has been asserted. □

3.2 Polarization and all smooth boundary data

Every $h \in C^\infty(S^2)$ is the trace of a globally smooth function: choose $\chi \in C_c^\infty((1/2, 3/2))$ equal to one near 1 and put $f(x) = \chi(|x|)h(x/|x|)$ for $x \neq 0$, with $f = 0$ near 0. This formula is smooth, including the origin. For boundary data h, k, let $H_g h, H_g k$ denote their harmonic extensions and define

$$Q_g(h) = a_g(H_g h, H_g h), \qquad \mathcal{B}_g(h,k) = a_g(H_g h, H_g k).$$

Linearity and uniqueness of the weak problem give $H_g(h+k) = H_g h + H_g k$. Consequently

$$\mathcal{B}_g(h,k) = \tfrac{1}{4}\{Q_g(h+k) - Q_g(h-k)\}. \tag{3.2}$$

The hypothesis of Theorem 1.2 and Lemma 3.1 imply equality of Q and then of $\mathcal{B}$ for all smooth h, k. Define the density-valued weak Dirichlet-to-Neumann map by $\langle \Lambda_g h, k\rangle = \mathcal{B}_g(h,k)$. Green's identity identifies it for smooth data with the outward conormal flux $(A_g \nabla H_g h) \cdot \nu \, dS$. Since $g = I$ on ∂B, this is the usual outward normal derivative times the same Euclidean surface density for both metrics. Thus their full smooth DN maps agree. The following argument in fact needs only the equality of Q.

3.3 A fixed exterior annulus

Put $E = \{x : 1 < |x| < 2\} \subset N$. No metric extension is needed: the original globally smooth metrics restrict to $\overline{N}$ and coincide with I on E. Choose, once and for all, $U_0 = B((3/2, 0, 0), 1/8)$, whose closure lies in E.

Proposition 3.2 (Common source responses and Green data)**.** *Let $F \in C_c^\infty(E)$, and let $w_j \in H_0^1(N)$ solve $L_{g_j} w_j = F$. Then $w_1 = w_2$ on E. The actual zero-Dirichlet Green kernels on N satisfy $G_1(x,y) = G_2(x,y)$ for $x, y \in E$, $x \neq y$.*

Proof. The unique weak solution minimizes the strictly convex functional

$$J_j(v) = \tfrac{1}{2}\int_N \nabla v^T A_{g_j} \nabla v \, dx - \int_E Fv \, dx \quad (v \in H_0^1(N)).$$

Existence and uniqueness follow from the same coercivity argument as above. Since F vanishes near $\overline{B}$, $w_j|_B$ is harmonic; smooth regularity on $\overline{N}$ gives a smooth trace $h_j = w_j|_{\partial B}$. Form $\widetilde{w}_1$ by using w_1 on E and the g_2-harmonic extension of h_1 on B. Matching traces show $\widetilde{w}_1 \in H^1(N)$ and the outer trace is zero, so it belongs to $H_0^1(N)$. Its exterior energy and source pairing are unchanged. Its interior energy is $Q_{g_2}(h_1) = Q_{g_1}(h_1)$, which is exactly the original interior energy of w_1. Hence $J_2(\widetilde{w}_1) = J_1(w_1)$ and $\min J_2 \le \min J_1$. Swapping the indices proves equality. Uniqueness of the minimizer then gives $\widetilde{w}_1 = w_2$, and in particular $w_1 = w_2$ on E.

Normalize the Green kernels by

$$L_{g_j,x} G_j(x,y) = \delta_y^{g_j}(x), \qquad G_j(\cdot,y)|_{\partial N} = 0, \qquad R_j F(x) = \int_N G_j(x,y) F(y) \, dV_{g_j}(y). \tag{3.3}$$

The two source measures are ordinary volume on E. Equality of the responses for every test source implies equality of the kernel distributions restricted to $E \times E$. One can see the off-diagonal claim directly: fix $x \in E$, choose an open neighborhood of $y \in E \setminus \{x\}$ avoiding x, and test against sources there. The smooth function $G_1(x,\cdot) - G_2(x,\cdot)$ has zero integral against every such source, and must vanish. This proves the claim. □

4 Source evaluations, separation, and all harmonic two-jets

For $f \in C_c^\infty(U_0)$, write $u_f^j = R_j f$ and define

$$I_j(p)(f) = u_f^j(p). \tag{4.1}$$

The common-annulus proposition gives $I_1 = I_2$ on E. The functions u_f^j are harmonic on $N \setminus \overline{U_0}$. Source support is fixed throughout; we do not change it with the shrinking scales.

Lemma 4.1 (Source separation and the jet range)**.** *For one metric g on $\overline{N}$, let $R = L_{g,D}^{-1}$ and $I(p)(f) = Rf(p)$ for $f \in C_c^\infty(U_0)$. Then $I(p)$ is nonzero for interior p, is zero at boundary points, and separates distinct interior points. For fixed f it is continuous in $p \in \overline{N}$. If $p \notin \overline{U_0}$, the second jets of the functions Rf at p fill exactly the kernel of the linear functional $j_p^2 u \mapsto L_g u(p)$.*

Proof. Regularity gives continuity and the zero boundary values. For a nonnegative nonzero $f \in C_c^\infty(U_0)$ the weak and strong maximum principles give $Rf > 0$ in N: the boundary value is zero, $-\Delta_g Rf = f \ge 0$, and the solution cannot be identically zero. Thus every interior evaluation is nonzero.

Let V be the finite-dimensional real space of second jets at p, and let $S \subset V$ be the linear image of the source map. Clearly $S \subset \ker \ell$, where $\ell(j_p^2 u) = L_g u(p)$, provided $p \notin \overline{U_0}$. Suppose a linear functional on V annihilates S. Represent it by a distribution T supported at p of order at most two, with pairing against functions understood with the fixed metric density. Define $w = RT$ by transposition (S3). For every test source f in U_0, $\langle w, f\rangle = \langle T, Rf\rangle = 0$. Thus $w = 0$ on U_0. On $N \setminus \{p\}$ it is a smooth harmonic function. That punctured domain is connected: any path through p can be rerouted in a sufficiently small punctured three-dimensional coordinate ball. Unique continuation gives $\operatorname{supp} w \subset \{p\}$.

A point-supported distribution has the form $w = \sum_{|\alpha| \le k} c_\alpha \partial^\alpha \delta_p$. If $w \neq 0$ and k is its largest derivative order, the leading polynomial of $L_g w$ is the product of its nonzero degree-k coefficient

polynomial and the elliptic quadratic principal symbol at p. The polynomial ring over $\mathbb{R}$ has no zero divisors, so this product is nonzero and the order is exactly $k+2$. Because $L_g w = T$ has order at most two, $k = 0$, and $w = c\delta_p$. Self-adjointness with the metric density then gives $\langle T, u\rangle = cL_g u(p)$; equivalently the annihilator of S is exactly the span of ℓ. In finite dimensions the double annihilator is the subspace itself, so $S = \ker \ell$. The zero case $w = 0$ gives $T = 0$ and is included.

If $I(p) = I(q)$ for distinct interior points, repeat the argument with $T = \delta_p - \delta_q$. On the connected domain $N \setminus \{p, q\}$, unique continuation gives support only at p, q. This is valid even if a point lies in U_0, because the initial vanishing was distributional on the entire open source set. If w has a nonzero point-supported component, its elliptic image has order at least two at that point, whereas T has order zero. Different support points cannot cancel each other's distributions. Consequently $w = 0$, which contradicts $L_g w = T \neq 0$. This proves separation. □

Obtaining a finite coordinate and Hessian family. The jet equation can be written $a^{ij}H_{ij} + b^i\xi_i = 0$, with a positive definite. Every prescribed first derivative ξ can be completed by a Hessian: for example choose $H = cI$ with $c = -b^i\xi_i/\operatorname{tr} a$. The jet lemma therefore gives three source solutions with linearly independent differentials at p. Their tuple is a coordinate chart by the inverse function theorem; each coordinate is harmonic. In that chart

$$\Delta_g v = g^{ij}\partial_{ij} v,$$

because applying the coordinate expression of Δ_g to each coordinate function shows its first-order coefficient is zero. Consequently the ordinary coordinate Hessians of harmonic functions lie in the five-dimensional hyperplane $\{H \in \mathrm{Sym}_3 : g^{ij}H_{ij} = 0\}$. The full jet range realizes a basis of this hyperplane, with zero first derivatives if desired. After choosing five such sources, one nonzero 5×5 Gram minor remains bounded away from zero on a sufficiently small neighborhood. These sources and this neighborhood are fixed before any dilation or shrinking parameter.

Remark 4.2. The pairs (u_f^1, u_f^2) agree on the common annulus. The constant pair $(1, 1)$ is adjoined separately when needed; it is not a zero-Dirichlet source solution. The source-jet proof above is a derivation from standard distributional elliptic theory and unique continuation, rather than an assumption of Runge on a domain with unchecked connected complement. It retains the source label `data:jet-family` for correspondence with the supplied manuscript.

5 Local continuation and product solutions

5.1 The local metric extension theorem

The common cap gives an initial isometry. To enlarge it, the global argument will use source-generated harmonic coordinates to identify neighborhoods of two interior points. In these coordinates we must continue equality of the metrics across a point where equality is already known on one side. The following theorem is the local input.

Theorem 5.1 (Local metric extension)**.** *Let $n \geq 3$. Let (N_1, g_1) and (N_2, g_2) be smooth compact connected Riemannian manifolds with nonempty smooth boundary, with Dirichlet Green functions G_1, G_2. Identify two interior harmonic coordinate neighborhoods with one neighborhood $\Omega \subset \mathbb{R}^n$ of 0. Suppose an open $W \subset \Omega$ contains, near 0, one side of a smooth hypersurface through 0, and suppose*

$$g_1 = g_2 \text{ on } W, \qquad G_1(x, y) = G_2(x, y) \quad (x, y \in W,\ x \neq y).$$

Suppose there are finitely many smooth pairs u_j^a harmonic for g_j on Ω, agreeing on W, whose first-side coordinate Hessians at 0 span

$$\{H \in \mathrm{Sym}^2(\mathbb{R}^n) : g_1^{ij}(0)H_{ij} = 0\}.$$

Then the two coordinate metrics agree on a neighborhood of 0.

We prove this theorem in Section 10. Its proof first constructs a mixed Green kernel that is harmonic for the first metric in x and for the second metric in y: initially it is $G_1(x, y)$ when $y \in W$ and $G_2(x, y)$ when $x \in W$. For each prescribed compact region at positive separation from $x = y$, a sufficiently small spatial dilation allows the continuation constructed in this section to reach that region. Section 6 then continues it to a family of small spheres centered at y, under a temporary bound that controls a normalized difference of the inverse metrics, together with finitely many derivatives, by the current radius. Its flux transfers first-metric harmonic functions to second-metric harmonic functions. In particular, the resulting functional sends u_1^a to u_2^a and reproduces the constant and coordinate functions.

The remaining task is uniform control as the spheres shrink. Section 8 estimates the density representing the flux functional. Under the same temporary metric bound, this estimate and exact reproduction of affine functions give an L^2 bound for each paired harmonic difference divided by the squared radius. Section 10 combines it with the spanning harmonic Hessians to improve the temporary bound strictly. This improvement allows the shrinking parameter to tend to zero with the spatial dilation fixed, forcing the two metrics to be conformal near the contact point. The common harmonic coordinates remove the remaining conformal factor. The geometric continuation in this section provides existence away from the diagonal; its estimates need not remain bounded as the separation tends to zero.

5.2 Product solutions and extension from a cross

We begin with a statement for general scalar elliptic operators. Let L_i be a smooth scalar second-order elliptic operator in a coordinate domain, with real positive definite principal matrix g_i^{-1}. Lower-order terms are allowed. A *product solution* is a function $F(x, y)$ satisfying

$$L_1^x F = 0, \qquad L_2^y F = 0.$$

Such a function has ordinary unique continuation, since $L_1^x + L_2^y$ is elliptic on the product. We shall construct extensions explicitly; unique continuation will identify extensions already constructed.

The starting data consist of two compatible product sets, with one variable restricted to a smaller observed set in each piece. A common basis for a solution space and its restriction to the smaller set will construct the extension. Common-basis methods for separately harmonic extension have a classical antecedent in Zériahi [10]; singular systems also enter quantitative Runge approximation [9]. We give the variable-coefficient argument, including its quantitative dependence.

Lemma 5.2 (Extension from a cross)**.** *Let D_i be bounded connected coordinate domains and $S_i \Subset D_i$ nonempty open sets. Suppose a product solution F is given consistently on*

$$(D_1 \times S_2) \cup (S_1 \times D_2),$$

with the sum of its two L^2 norms at most M. Let $O_i \Subset D_i$ be open sets such that every L_i-solution w on D_i satisfies

$$\|w\|_{L^2(O_i)} \le C\|w\|_{L^2(S_i)}^{\gamma_i}\|w\|_{L^2(D_i)}^{1-\gamma_i}, \qquad 0 < \gamma_i < 1, \qquad \gamma_1 + \gamma_2 > 1. \tag{5.1}$$

Then F extends as a product solution to $O_1 \times O_2$, agreeing with both given pieces on their intersections. On every compact subset of this product its C^m norm is at most $C_m M$. The constants are uniform when the geometric margins, ellipticity, coefficient bounds, interpolation constants and positive exponent margin are uniform. Each specified output order uses only finitely many coefficient bounds.

Proof. Let $\mathcal{H}_1(D_1)$ be the closed subspace of $L^2(D_1)$ consisting of distributional L_1-solutions. Restriction $R : \mathcal{H}_1(D_1) \to L^2(S_1)$ is compact by interior elliptic estimates and injective by unique continuation. The spectral theorem for R^*R gives an orthonormal basis (w_j) of $\mathcal{H}_1(D_1)$ such that

$$(w_i, w_j)_{L^2(S_1)} = \mu_j^2 \delta_{ij}, \qquad 0 < \mu_j \leq 1. \tag{5.2}$$

For every N,

$$\mu_j \leq C_N (1 + j)^{-N}. \tag{5.3}$$

Indeed, multiply a solution by a cutoff equal to one near $\overline{S_1}$ and compactly supported in D_1. Its H^k norm is bounded by its $L^2(D_1)$ norm. Fourier truncation in a containing cube gives a rank $O(A^n)$ approximation with operator error $O(A^{-k})$. The minimax characterization of singular values yields $\mu_j \leq C_k j^{-k/n}$. All these constants have the stated uniformity.

For $y \in S_2$ expand $F(\cdot, y)$ on D_1:

$$F(x, y) = \sum_j w_j(x) b_j(y), \qquad \|b_j\|_{L^2(S_2)} \leq M.$$

Equation (5.2) continues these coefficients to D_2 by the formula

$$b_j(y) = \mu_j^{-2} \int_{S_1} F(x, y) \overline{w_j(x)} \, \mathrm{d}x, \qquad \|b_j\|_{L^2(D_2)} \leq M \mu_j^{-1}. \tag{5.4}$$

The integral solves $L_2 b_j = 0$ and agrees with the original coefficient on S_2. Applying (5.1) gives

$$\|w_j\|_{L^2(O_1)} \leq C \mu_j^{\gamma_1}, \qquad \|b_j\|_{L^2(O_2)} \leq C M \mu_j^{\gamma_2 - 1}.$$

Consequently the series converges absolutely in $L^2(O_1 \times O_2)$: its summands are bounded by $CM\mu_j^{\gamma_1 + \gamma_2 - 1}$. Choose N in (5.3) so that $N(\gamma_1 + \gamma_2 - 1) > 1$. The limit solves both equations in distributions. Interior estimates for their elliptic sum give the asserted smoothness and bounds on smaller sets.

We verify agreement with the second given piece; no assertion that an arbitrary local solution extends to D_1 is needed. Let z be orthogonal to the closure of $R\mathcal{H}_1(D_1)$ in $L^2(S_1)$. The function

$$y \longmapsto \int_{S_1} F(x, y) \overline{z(x)} \, \mathrm{d}x$$

solves the second equation and vanishes on S_2, so it vanishes on connected D_2. Thus each second-piece slice belongs to that closed range. Its expansion in the orthonormal basis w_j/μ_j is complete, and its coefficients are $\mu_j b_j(y)$ by (5.4). It follows that the series is F on $S_1 \times O_2$; the series converges there also by the positive exponent γ_2. On $O_1 \times S_2$ it is the original expansion, with positive exponent γ_1. This proves both agreements. Only upper bounds for the singular values were used, so uniformity does not require continuous choices of singular bases or positive lower bounds for their singular values. □

Figure 1 summarizes the sets and the two estimates in Lemma 5.2.

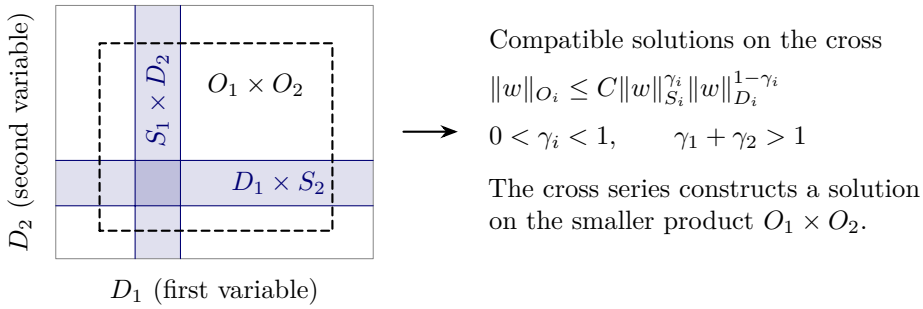


Figure 1: Compatible product solutions are given on the shaded cross. The L^2 interpolation estimates for single-variable solutions, with $\gamma_1 + \gamma_2 > 1$, make the singular-function series converge on $O_1 \times O_2$. The drawing is schematic: each factor is a multidimensional domain, and containment of S_i in O_i is not required.

5.3 A geometric criterion for local extension

We next build the interpolation estimates needed in Lemma 5.2. For a metric g, call a smooth real function θ a *strict weight* at a point where $\mathrm{d}\theta \neq 0$ if

$$\mathrm{Hess}_g\,\theta(e,e) + \mathrm{Hess}_g\,\theta(v,v) > 0 \quad \text{for all } v \perp e,\ |v|_g = 1, \qquad e = \frac{\nabla^g\theta}{|\nabla^g\theta|_g}. \tag{5.5}$$

The strict scalar Carleman estimate gives, after shrinking the coordinate neighborhood,

$$\tau^3\|e^{\tau\theta}w\|_{L^2}^2 + \tau\|\nabla(e^{\tau\theta}w)\|_{L^2}^2 \le C\|e^{\tau\theta}Lw\|_{L^2}^2, \qquad \tau \ge \tau_0, \tag{5.6}$$

for compactly supported w. This is the elliptic strict-weight estimate of the classical Carleman theory; see Hörmander [4, Chapter XXVIII] and Koch and Tataru [7, Definition 8.3 and Theorem 9(a)]. Here the hypotheses can also be checked directly. For $L = -\Delta_g$, write $v = e^{\tau\theta}w$ and

$$e^{\tau\theta}Le^{-\tau\theta} = A_\tau + B_\tau, \qquad A_\tau = -\Delta_g - \tau^2|\mathrm{d}\theta|_g^2, \quad B_\tau = \tau(2\nabla^g\theta\cdot\nabla + \Delta_g\theta).$$

Here A_τ is self-adjoint and B_τ is skew-adjoint for metric volume. Integration by parts gives

$$\|(A_\tau + B_\tau)v\|^2 = \|A_\tau v\|^2 + \|B_\tau v\|^2 + ([A_\tau, B_\tau]v, v),$$
$$([A_\tau, B_\tau]v, v) = 4\tau\int \mathrm{Hess}_g\,\theta(\nabla v, \nabla\overline{v}) + 4\tau^3\int \mathrm{Hess}_g\,\theta(\nabla\theta,\nabla\theta)|v|^2 - \tau\int(\Delta_g^2\theta)|v|^2.$$

Choose a real constant m strictly between the negative of the least unit tangential Hessian value and the unit normal Hessian value. After shrinking the patch, these inequalities have a positive margin. Add $4m\tau(A_\tau v, v)$ to the commutator form. Its gradient coefficient becomes $4\tau(\mathrm{Hess}_g\,\theta + mg)$, and its leading zero-order coefficient is

$$4\tau^3|\mathrm{d}\theta|_g^2(\,\mathrm{Hess}_g\,\theta(e,e) - m) > c\tau^3.$$

The tangential gradient form is positive. Its mixed and possibly negative normal terms cost at most $C\tau\|\nabla_e v\|^2$, and

$$\|\nabla_e v\|^2 \le C\tau^{-2}\|B_\tau v\|^2 + C\|v\|^2.$$

Finally $|4m\tau(A_\tau v, v)| \le \frac{1}{2}\|A_\tau v\|^2 + C\tau^2\|v\|^2$. For large τ, the squared parts absorb the normal derivative cost and the positive zero-order term absorbs the remaining errors. This yields (5.6).

Smooth first- and zero-order terms in L cost only $C(\|\nabla v\|^2 + \tau^2\|v\|^2)$ and are absorbed in the same way. The proof gives uniform constants on fixed compact sets of strict weights and bounded coefficients.

The existence criterion is a condition on a hypersurface in the product. Hessians in the following statement use the product connection of g_1 and g_2.

Proposition 5.3 (Product extension criterion)**.** *Let ρ be smooth near $z_0 = (x_0, y_0)$, with $\rho(z_0) = 0$ and $\mathrm{d}_x\rho, \mathrm{d}_y\rho \neq 0$. Put*

$$a_i = \frac{\nabla_i \rho}{|\nabla_i \rho|^2}, \qquad H = \operatorname{Hess} \rho.$$

Suppose that at z_0

$$H((a_1, -a_2), (a_1, -a_2)) + H((v_1, v_2), (v_1, v_2)) > 0 \quad (v_i \perp a_i,\ |v_i| = |a_i|)\,. \tag{5.7}$$

Every product solution given on $\{\rho > 0\}$ near z_0 extends to a neighborhood of z_0, agreeing on a smaller part of that side. In fact, the construction uses only two compact product sets in $\{\rho > \epsilon\}$ for some $\epsilon > 0$. The neighborhoods and estimates on smaller neighborhoods are uniform under small smooth perturbations preserving the strict hypotheses, provided the data on those compact sets are bounded.

Proof. We first split the product Hessian condition into a strict weight in each factor. Their sum will lie below a defining function for the given side, with a quadratic gap. That gap places two intersecting product sets in the given side; the weights then yield interpolation exponents greater than one half, allowing Lemma 5.2 to apply.

Splitting the weight between the factors. We seek symmetric forms P_i satisfying

$$P_i(a_i, a_i) + P_i(v_i, v_i) > 0, \qquad H + K\,\mathrm{d}\rho^{\otimes 2} > P_1 \oplus P_2$$

for some $K > 0$, with the vectors v_i as in (5.7). After division by $|a_i|^2$, the first inequality is the strict-weight condition for a function with gradient $\nabla_i \rho$ and Hessian P_i. The second inequality will give the quadratic gap. To obtain these forms by convex separation, the dual tests are positive semidefinite forms D whose diagonal blocks are

$$d_i a_i^{\otimes 2} + S_i, \qquad d_i \geq 0, \quad S_i \geq 0 \text{ on } a_i^{\perp}, \quad \operatorname{tr} S_i = d_i |a_i|^2.$$

This follows by separating the open convex cone consisting of positive definite forms plus the allowed $P_1 \oplus P_2$; the individual dual cones are generated by $a_i^{\otimes 2} + v_i^{\otimes 2}$. Normalize $\operatorname{tr} D = 1$. To obtain a suitable K, it suffices by compactness to prove $H : D > 0$ on tests with $D\,\mathrm{d}\rho = 0$.

Represent such a D as a covariance matrix. The normal components measured by the two partial covectors are opposite. The diagonal-block condition makes each normal component uncorrelated with its own tangential component, hence with both tangential components. Their variances coincide, giving $d_1 = d_2 = d > 0$ and

$$D = d(a_1, -a_2)^{\otimes 2} + D_{\perp}, \qquad \operatorname{tr}(D_{\perp})_{ii} = d|a_i|^2.$$

The case $d = 0$ would force $D = 0$. Among nonnegative tangential forms with these two fixed traces, every extreme point has rank one: on an image of rank $r \geq 2$, a nonzero symmetric perturbation preserves the two traces because $r(r+1)/2 > 2$, and small perturbations of both signs preserve

positivity. The rank-one tests are precisely those in (5.7), multiplied by d. The asserted positivity follows. Compactness of the normalized tests then supplies K, as required.

Prescribe gradients $\mathrm{d}_i\rho$ and Hessians P_i for smooth functions θ_i vanishing at the respective points. In small coordinates centered there, Taylor expansion gives

$$\theta_1(x) + \theta_2(y) \le \widetilde{\rho}(x,y) - c(|x|^2 + |y|^2), \qquad \widetilde{\rho} = \rho + K\rho^2/2, \tag{5.8}$$

with $c > 0$. Each θ_i satisfies (5.5). The positive side of $\widetilde{\rho}$ is the positive side of ρ in this neighborhood.

Placing the cross and obtaining interpolation. Use coordinates (s_i, z_i) with $s_i = \theta_i$. Fix a small $\kappa > 0$ so that $\phi(s_i) = s_i - \kappa s_i^2$ remains a strict weight. Choose successively a small lateral radius R, a height $l \ll cR^2$, and $e \ll \kappa l^2$. Take

$$D_i = \{-l < s_i < l + 4e,\ |z_i| < R\}.$$

Use a cutoff equal to one on the inner lateral half and for $-l + 3e < s_i < l + 2e$, supported in the cylinder and in $-l + 2e < s_i < l + 3e$. Let $S_i \Subset D_i$ include neighborhoods of its top and lateral derivative supports, where respectively

$$s_i > l + e/2 \quad \text{or} \quad |z_i| > R/3,$$

and an inner top slice near $s_i = l + 5e/4$. The two closed product sets $\overline{D_1 \times S_2}$ and $\overline{S_1 \times D_2}$ lie strictly in the given side. For a top slice, the sum of the two s coordinates is positive. For a lateral slice, it is greater than $-2l$, which is dominated by the quadratic gap in (5.8). Making the sets slightly smaller if necessary gives a common $\epsilon > 0$ with $\rho > \epsilon$ on their closures.

For an L_i-solution normalized by $\|w\|_{L^2(D_i)} = 1$, apply (5.6) to this cutoff times w. Interior estimates bound the bottom derivative terms using the large norm and the weight $\phi(-l + 3e)$. The remaining derivative terms use $\|w\|_{L^2(S_i)}$ and a weight at most $\phi(l + 3e)$. On a smaller inner cylinder beginning at $s_i = -e$, the weight is at least $\phi(-e)$. Balancing the two exponentials yields (5.1) with

$$\gamma_i = \frac{\phi(-e) - \phi(-l + 3e)}{\phi(l + 3e) - \phi(-l + 3e)} > \frac{1}{2}. \tag{5.9}$$

Indeed at $e = 0$ the quotient is $1/2 + \kappa l/2$. Values of the balancing parameter below τ_0 are covered by increasing the constant. The output cylinder may be chosen to include a connected tube from the origin to the indicated top slice.

Constructing and identifying the extension. Lemma 5.2 now gives a product solution on a neighborhood of z_0. It agrees with the old solution there on the positive side: from a sufficiently near positive-side point, increase s_2 into the top slice. Since $\partial_{s_2}\rho > 0$ locally, the segment stays on that side and in the larger constructed cylinder. Agreement on the slice and unique continuation along a neighborhood of the segment prove agreement at the original point. All choices have strict margins. Keeping their finitely many compact sets and shrinking their output neighborhoods once proves the stated perturbation uniformity by (5.6), Lemma 5.2 and interior elliptic estimates. □

5.4 Gluing along a compact family of surfaces

Proposition 5.3 provides a germ across one surface point. The following formulation records the geometry needed to combine these germs. In particular, agreement is proved on specified overlap components, rather than inferred from the existence of a cover.

Lemma 5.4 (Propagation through a compact cylinder)**.** *Suppose a region in a product patch has smooth coordinates (z, a) near $B \times [a_-, a_+]$, where B is a compact base, possibly with boundary or corners. A product solution is given on a neighborhood of the top and lateral boundary and on an open set $\mathcal{S}$. Assume that $\mathcal{S}$ is upward closed in a at fixed z and contains these boundary neighborhoods. If every level $a =$ constant satisfies Proposition 5.3 outside $\mathcal{S}$, with the positive side pointing toward increasing a, the solution extends to a neighborhood of the cylinder, agreeing on $\mathcal{S}$. For fixed coordinates, compact sets and strict reference inequalities, the extension has uniform compact-subset bounds under small smooth perturbations, in terms of bounds on finitely many compact subsets of the initially given region.*

Proof. Starting from the top, let a_* be the infimum of levels down to which an extension agreeing with the seed has been obtained. In the uniform version include uniform bounds in this property. At a non-seed point of $B \times \{a_*\}$, Proposition 5.3 uses compact data strictly above that level. Take a finite cover of the level by such output neighborhoods and seed neighborhoods. Their data are contained above $a_* + \eta$ for one $\eta > 0$, so an already reached level supplies them with the needed bounds.

Shrink the outputs in (z, a) coordinates to boxes $B_i \times (a_* - d_i, a_* + d_i)$. Every connected component of an overlap contains vertical segments to a common height greater than a_*. Both germs equal the preceding extension there. Unique continuation for $L_1^x + L_2^y$ makes them identical on that overlap component. The same argument gives agreement with $\mathcal{S}$, because a vertical segment moving upward from a seed point stays in the seed. Boundary neighborhoods use the given solution. A finite subcover has a positive minimum output width and therefore passes the putative last level, or includes the endpoint a_-. This proves continuation on the whole cylinder. The same finite constructions retain all strict margins under small perturbations. Their Carleman, series and interior estimates prove the uniform assertion. Data bounded up to a limiting surface are never required: each local construction uses compact sets strictly above it. □

5.5 Initialization at a fixed distance from the diagonal

We apply the criterion to a mixed Green kernel. Let g_1, g_2 be smooth metrics in coordinate patches centered at 0, obtained from interior patches of compact Dirichlet manifolds. Suppose they agree on an open set W containing the lower side of a smooth hypersurface through 0. After a linear coordinate change, assume

$$g_1(0) = g_2(0) = I, \qquad \{x_n < q(x')\} \subset W \text{ near } 0, \qquad q(0) = 0, \quad \mathrm{d}q(0) = 0,$$

for a smooth function q. No regularity of the rest of ∂W is assumed. Let G_j be the Dirichlet Green kernels, normalized with metric volume, and suppose $G_1 = G_2$ on $W \times W$. For a small dilation parameter $\lambda > 0$, put

$$g_{j,\lambda}(z) = g_j(\lambda z), \qquad G_{j,\lambda}(x, y) = \lambda^{n-2} G_j(\lambda x, \lambda y), \qquad W_\lambda = \lambda^{-1} W. \tag{5.10}$$

On each fixed bounded patch the metrics tend smoothly to the Euclidean metric, and W_λ contains every fixed compact subset of $\{z_n < 0\}$ for sufficiently small λ. Initially define, off the diagonal,

$$\mathcal{G}_\lambda(x, y) = \begin{cases} G_{1,\lambda}(x, y), & y \in W_\lambda, \\ G_{2,\lambda}(x, y), & x \in W_\lambda. \end{cases} \tag{5.11}$$

The pieces fit and solve the first equation in x and the second in y.

Proposition 5.5 (Fixed-separation initialization)**.** *In this setting, let $n \geq 3$ and fix $B > 0$, $d > 0$ and $\eta > 0$. For all sufficiently small λ, the data (5.11) have a product-solution continuation to a neighborhood of*

$$\{(x, y) : |x|, |y| \leq B,\ |x - y| \geq d\}.$$

Every fixed C^m norm on a smaller neighborhood is bounded uniformly in λ. The continuation agrees with $G_{1,\lambda}$ where $y_n < -\eta$ and with $G_{2,\lambda}$ where $x_n < -\eta$, on the corresponding overlaps. The smallness threshold for λ and the constants may depend on B, d, η, m.

Proof. We first cross the limiting plane, then deform variable-radius tubes to reach the prescribed separations. All the sets and positive separation margins used in the two steps are fixed before choosing λ.

Crossing the plane and obtaining a common seed. We first extend slightly across the plane in one variable while keeping the other at a comparable positive separation. The first interpolation exponent can be made close to one; this compensates for the fixed positive exponent obtained by propagation in the separated second variable.

Fix a point x_0 of height zero and a separation scale $s > 0$. In the second factor take

$$D_2 = \{s/2 < |y - x_0| < 2s\}$$

and an observed ball S_2 about $x_0 - se_n$. In the first take a ball of radius $3\tau_0 s$ centered at $x_0 - \tau_0 s e_n$, with $\tau_0 > 0$ fixed small enough that the two large domains are separated. Its observed ball has radius $(1 - \alpha)\tau_0 s$, with $\alpha > 0$ to be chosen. These observed balls lie strictly in the lower half-space.

There is a two-constants estimate from S_2 to a slightly enlarged closed annulus $4s/5 \leq |y - x_0| \leq 6s/5$, with an exponent $\gamma_2 > 0$ independent of α. Here is a direct way to obtain it with room for perturbations. On a Euclidean annulus the weight $|z - z_*|^{-1}$ is strict: its radial Hessian is $2|z - z_*|^{-3}$ and each tangential Hessian value is $-|z - z_*|^{-3}$. Radial cutoffs in (5.6), followed by exponential balancing, propagate smallness from an inner ball to an intermediate ball using the norm on an outer ball. A finite chain of overlapping balls with enlarged balls inside D_2 connects S_2 to the target annulus. The annulus is connected since $n \geq 3$. Multiplying the finitely many positive exponents gives the asserted γ_2. The same weights and margins work for sufficiently small perturbations of the Euclidean metric.

In the first factor use this radial argument centered at $x_0 - \tau_0 s e_n$. Keep the outer cutoff a fixed distance beyond radius $\tau_0 s$, and put the inner cutoff just inside radius $(1 - \alpha)\tau_0 s$. Take a concentric target ball of radius $(\tau_0 + c)s$, with $c > 0$ small; it contains the observed ball and a ball of radius cs about x_0. As α and c decrease, the target weight approaches the inner weight. The interpolation exponent therefore tends to one. Choose these constants so that $\gamma_1 + \gamma_2 > 1$. Lemma 5.2 extends the kernel to a small ball about x_0 times the target annulus. On these fixed separated sets the rescaled Green kernels and their derivatives are uniformly bounded: the local Green singularity is $O(|x - y|^{2-n})$, and interior elliptic estimates give derivative bounds away from the pole.

The extensions agree with both lower pieces. Agreement for $y \in S_2$ comes from the first expansion. For fixed x below a sufficiently small negative height, continue in y through the connected target annulus to compare with $G_{2,\lambda}$. For a second point below that height, compare with $G_{1,\lambda}$ by continuation in the first variable from its observed ball. All comparisons remain separated.

The local extensions must next be compared on overlaps. Retain much smaller first balls and narrower annuli for coverage, keeping the larger products as comparison domains. The unused margins allow both variables to move into the known lower region. Overlapping retained products have scales comparable to $|x - y|$. For two products of the same orientation, lower their near-plane

variable into the known region; the path stays inside both larger first balls. For products of opposite orientations, both variables lie in retained near-plane balls. On the fixed compact range of locations and positive separation scales, choose the retained products with strict ball and annular margins, then take $\eta_0 > 0$ much smaller than the smallest retained width. Lower both variables by

$$h = \max(x_n, y_n, 0) + 2\eta_0.$$

By making the retained balls a sufficiently small fraction of the larger balls, the path stays in both larger products: its displacement is smaller than their ball and annular margins. Its endpoint has both heights below $-\eta_0$, where both germs are the original kernel. Unique continuation along a neighborhood of the path proves agreement. Choose $\eta_0 \le \eta$ and then λ small enough for this truncated lower half-space to lie in W_λ on all sets used.

Interchanging the two variables and covering gives, for some fixed $c_0 > 0$, a single-valued continuation in

$$\min(x_n, y_n) < c_0|x - y|, \tag{5.12}$$

on every fixed compact range of bounded locations and positive separations. The estimates are uniform in λ on that range. For coverage near the plane, take $x_0 = (x', 0)$ and $s = |y - x_0|$ when the first variable has the smaller nonnegative height; then x is in the retained first ball and y in the retained annulus if c_0 is small. Points already below the plane are covered by the same construction near it or by the given cross farther below it. The preceding overlap argument makes these choices compatible.

Sweeping tubes inward from the common seed. The region (5.12) is now the seed for the second stage. At fixed center y and direction ω, we move x along the ray from y: large radii lie in the seed, and decreasing the radius will reach the target pairs. Choose $M > B + 3$ and introduce

$$t = M + y_n + e_0|y'|^2, \qquad r_a(y) = at^{1+\sigma}, \qquad x = y + r_a(y)\omega, \quad |\omega| = 1, \tag{5.13}$$

where $\sigma, e_0 > 0$ will be small. The base is

$$y_n \ge -B - 1, \qquad t \le T_0, \qquad \omega \in \mathbb{S}^{n-1},$$

and a decreases through a fixed interval $[a_{\min}, a_{\max}]$. The term $e_0|y'|^2$ makes this base compact. The parameter a decreases while (y, ω) remains fixed, as required by the compact-cylinder propagation lemma. Take $a_{\min}$ small enough that the target separations exceed $r_{a_{\min}}(y)$, and $a_{\max}$ large enough that its whole level satisfies (5.12). The bottom base boundary has $y_n < 0$. Take T_0 so large that $c_0 a_{\min} T_0^\sigma > 1$; its top boundary also satisfies (5.12). Here the superlinear power $1 + \sigma$ ensures that $r_a/t = at^\sigma$ is large on this top face. Thus the initial a-level and both base boundary faces are supplied by the seed. This known region is upward closed in a: since $\min(x_n, y_n) = y_n + r\min(\omega_n, 0)$, its defining inequality is $y_n < (c_0 - \min(\omega_n, 0))r$, whose radius coefficient is positive.

It remains to check the strict extension criterion outside that region. At the Euclidean metrics use

$$\rho = \tfrac{1}{2}(|x - y|^2 - r_a(y)^2), \qquad b = \nabla r_a.$$

Away from $\omega + b = 0$, multiply the vectors in (5.7) by the common positive factor r_a and write their real and imaginary parts as

$$\begin{aligned} U_1 &= \omega + iv, && |v| = 1, \quad v \perp \omega, \\ U_2 &= \frac{\omega + b}{|\omega + b|^2} + iw, && w \perp (\omega + b), \quad |w| = |\omega + b|^{-1}. \end{aligned} \tag{5.14}$$

The identity $\omega \cdot (U_1 - U_2) = b \cdot U_2$ cancels the normal part of the distance Hessian. Consequently the test is

$$\left|\pi_{\omega^\perp}(U_1 - U_2)\right|^2 - (r_a \partial^2 r_a)(U_2, \overline{U_2}). \tag{5.15}$$

For bounded b, outside fixed small neighborhoods of $b = -\omega$ and $b = -2\omega$, the first term is at least $c|b|^2$. To see this, a zero forces the real part of U_2 to be parallel to ω, and equality of the projected imaginary lengths gives $|\omega + b| = 1$. Thus $b = 0$ or $b = -2\omega$. Near $b = 0$, the real projected difference controls $|\pi_{\omega^\perp} b|$ to first order; also

$$|\pi_{\omega^\perp} w| = 1 - \omega \cdot b + O(|b|^2),$$

so the imaginary projected difference controls $|\omega \cdot b|$ to first order. This proves the quadratic lower bound near zero; compactness proves it on the remaining range.

Outside (5.12), $y_n \geq c_0 r_a$, so $r_a/t \leq 1/c_0$ and

$$b = (1 + \sigma)(r_a/t)\nabla t$$

stays bounded. When ∇t is close to e_n, the two excluded neighborhoods already lie in (5.12). Indeed,

$$\frac{x_n}{r_a} \leq \frac{(1+\sigma)|\nabla t|}{|b|} + \omega_n,$$

which is arbitrarily small near $b = -\omega$ and negative near $b = -2\omega$ as σ and the gradient error tend to zero. On the complement $|U_2|$ is bounded and differentiation of (5.13) gives

$$r_a |\partial^2 r_a| \leq C(\sigma + T_0 e_0)|b|^2. \tag{5.16}$$

Choose the excluded-neighborhood sizes first in terms of c_0. Use the positive lower bound on the remaining bounded b-range to choose σ small, then choose the a-interval and T_0 as above, and finally $e_0 > 0$ so small that (5.16) is absorbed. Choosing the transverse compactification last keeps its contribution to the Hessian test small. These choices also make the gradient error small, since $|\nabla t - e_n| \leq 2\sqrt{e_0 T_0}$ on the base. Thus (5.15) is strictly positive wherever continuation is needed. Both partial gradients are nonzero there.

All these sets are bounded and stay at separation at least $a_{\min}(M - B - 1)^{1+\sigma} > 0$. Smooth convergence of the metrics therefore preserves the strict tests for sufficiently small λ. Lemma 5.4 continues the kernel through the cylinder, with uniform estimates and agreement on the seed. A target pair has $a = |x - y|/t^{1+\sigma} \geq a_{\min}$; if $a > a_{\max}$ it is already in the coarse region. Hence every target pair is covered.

We also identify the extension with both pieces of the full cross (5.11). Perform the construction with a larger bound $B' > B$ and a smaller separation $0 < d' < d$, chosen so that for every $|y| \leq B$ the domain

$$D_y = B(0, B') \setminus \overline{B(y, d')}$$

is connected and contains a fixed lower open ball with $x_n < -\eta$. Choose B' large enough that every excluded closed ball lies strictly inside $B(0, B')$ and is disjoint from that lower ball. For fixed $y \in W_\lambda$ in the target range, the continuation and $G_{1,\lambda}$ solve the same first-variable equation on D_y. On the lower ball the continuation equals $G_{2,\lambda}$, which equals $G_{1,\lambda}$ because both variables lie in W_λ. Unique continuation on D_y proves agreement on the required overlap. Interchanging the variables proves agreement with the other piece. Keeping slightly larger target bounds and a slightly smaller separation before restricting to the prescribed range gives the claimed neighborhood and its compact-subset estimates. □

The constants in Proposition 5.5 may deteriorate as $d \downarrow 0$. Its role is to initialize a construction at finitely many fixed positive scales. Lemma 5.4 will also provide qualitative existence at each positive radius in a shrinking family; the uniform estimate for that family is established below.

6 Continuation to shrinking spheres

We now apply the product continuation criterion near a point where an isometry is already known on one side. The purpose of this section is to construct a family of functionals on harmonic functions, represented on small spheres. Their existence will follow from the geometry of the spheres. A uniform bound for the functionals, needed when the spheres shrink, will follow from the estimate in the next section.

6.1 The local data and their normalization

Throughout Sections 6–10, fix the manifolds (N_j, g_j), actual Dirichlet Green functions G_j, common harmonic coordinate neighborhood Ω, matching set W, and finite harmonic pairs (u_1^a, u_2^a) of Theorem 5.1. Thus the coordinate metrics and paired functions agree on W, and the Green kernels agree on $W \times W$ off the diagonal. The first-side coordinate Hessians span the hyperplane annihilated by $g_1^{-1}(0)$. An interior ball in W tangent at 0 supplies the one-sided hypersurface required by that theorem. We construct the sphere functionals for these data.

A linear change of harmonic coordinates makes $g_1(0) = g_2(0) = I$ and writes the separating hypersurface as a graph tangent to $x_n = 0$, with the known side below it. For a spatial dilation $\lambda > 0$, write

$$g_{j,\lambda}(x) = g_j(\lambda x), \qquad G_{j,\lambda}(x, y) = \lambda^{n-2} G_j(\lambda x, \lambda y). \tag{6.1}$$

Dilate the harmonic pairs at the same time:

$$u_{j,\lambda}^a(x) = u_j^a(\lambda x).$$

They remain harmonic for $g_{j,\lambda}$ and agree on the dilated known set. The metrics and Green kernels in (6.1) have the corresponding normalization on the dilated neighborhoods. On each fixed bounded coordinate set the metrics converge smoothly to I as $\lambda \downarrow 0$. The dilated known set contains every fixed compact set lying strictly below $x_n = 0$, once λ is sufficiently small.

Use x for the first variable and y for the second. The mixed kernel $\mathcal{G}$ is initially $G_{1,\lambda}$ when y is in the known set, and $G_{2,\lambda}$ when x is in that set. These prescriptions agree on their overlap and solve the first equation in x and the second in y, away from the diagonal. Proposition 5.5 continues this kernel to any specified compact region of fixed positive separation, with smooth bounds uniform for small λ. We shall use that result only after all the positive separation ranges in question have been fixed.

The geometry of a moving sphere dictates the next choices. In the strict extension test, the leading terms involving the gradient of its radius cancel; strict concavity of the logarithmic radius supplies the remaining positive term (see (6.12)). We therefore choose a conformal reference metric for which a depth function is strictly concave. Separately, a scalar normalization of the second inverse metric will give the two inverse matrices the same trace relative to the reference metric. The harmonic Hessians will determine their remaining difference.

Fix the coordinate cylinder

$$Y = \{y = (y', y_n) : |y'| \le 1,\ -1 \le y_n \le 1\}, \qquad t_0(y) = y_n + |y'|^2.$$

Take a sufficiently large fixed number L and put

$$\gamma = e^{-2Lt_0} g_{1,\lambda}, \qquad Q = \frac{n\, g_{2,\lambda}^{-1}}{\operatorname{tr}(\gamma g_{2,\lambda}^{-1})}, \qquad h = Q - \gamma^{-1}. \tag{6.2}$$

Thus Q is the principal matrix of a positive multiple of the second equation and $\operatorname{tr}(\gamma h) = 0$. The first equation can likewise be multiplied by a positive function to give principal metric γ. Such multiplication does not change its solutions. The metric γ will also identify the two tangent spaces at a common coordinate point. All these coefficients have uniform smooth bounds on a fixed neighborhood of Y for sufficiently small λ, and $h \to 0$ in every fixed smooth norm as $\lambda \downarrow 0$.

Define the depth function

$$t(y) = \int_{1/8}^{t_0(y)} e^{-2Ls}\, ds. \tag{6.3}$$

It is negative at 0 and has nonvanishing differential on Y. The conformal connection formula gives

$$\operatorname{Hess}_\gamma t = e^{-2Lt_0} (\operatorname{Hess}_{g_{1,\lambda}} t_0 - L|dt_0|^2_{g_{1,\lambda}} g_{1,\lambda}) \le -c\gamma. \tag{6.4}$$

Indeed $|dt_0|$ is bounded below, whereas the first Hessian on the right is uniformly bounded. Choose L first and then an upper bound for λ so that the displayed strict inequality holds with fixed $c > 0$.

6.2 Radii and cutoff regions

For a fixed integer $p \ge 2$, a positive amplitude R_*, and $0 < \delta \le 1$, define

$$\ell_\delta(t) = \delta \log(1 + e^{t/\delta}), \qquad r_\delta(y) = R_* \ell_\delta(t(y))^p, \qquad f_\delta = \log r_\delta, \quad \beta_\delta = |df_\delta|_\gamma. \tag{6.5}$$

The balls will be the γ-geodesic balls with center y and radius $r_\delta(y)$, lying in the first coordinate neighborhood. As the lemma below shows, they collapse where $t \le 0$, including a neighborhood of the contact point, but their radii stay bounded below where t is bounded away from zero on the positive side. This distinction will keep cutoff derivatives away from uncontrolled shrinking spheres.

Lemma 6.1 (Profile bounds)**.** *On Y, after the preceding choices of geometry, the following bounds hold uniformly for small λ, $0 < \delta \le 1$, and $p \ge 2$:*

$$\beta_\delta \ge cp, \qquad \operatorname{Hess}_\gamma f_\delta \le -c\beta_\delta \gamma, \qquad |\partial^j f_\delta| \le C_j \beta_\delta^j \quad (j \ge 1). \tag{6.6}$$

The constants c, C_j do not depend on p. The functions $r_\delta^{1/p}$ have a Lipschitz bound independent of δ and λ for fixed p, R_*. The radii increase with δ, tend to zero on $\{t \le 0\}$ as $\delta \downarrow 0$, and satisfy

$$\sup_{\delta, y} r_\delta \beta_\delta^2 \longrightarrow 0 \quad \text{as } R_* \downarrow 0 \tag{6.7}$$

for each fixed p.

Proof. Write $a_\delta = (\log \ell_\delta)'$. In terms of $s = t/\delta$,

$$a_\delta(t) = \delta^{-1} A(s), \qquad A(s) = \frac{e^s}{(1 + e^s)\log(1 + e^s)}.$$

The function A is positive and decreasing. At the negative end it tends to one, and at the positive end it is comparable to $(1 + s)^{-1}$. Differentiating the formula, or its convergent expressions at the

two ends, gives $|A^{(j)}(s)| \le C_j A(s)^{j+1}$. On the remaining compact interval the same inequalities follow by positivity. On the fixed t interval this gives a positive lower bound for a_δ and

$$a'_\delta \le 0, \qquad |\partial_t^j \log \ell_\delta| \le C_j a_\delta^j, \qquad a_\delta \le \ell_\delta^{-1}.$$

Since $|dt|_\gamma$ is bounded above and below, $\beta_\delta = p a_\delta |dt|_\gamma$. Equation (6.4) and

$$\operatorname{Hess}_\gamma f_\delta = p a_\delta \operatorname{Hess}_\gamma t + p a'_\delta \, dt \otimes dt$$

prove the Hessian bound. The chain rule proves the other derivative bounds, with constants independent of $p \ge 2$ because $p a_\delta^j \le p^j a_\delta^j$.

The bound $0 < \ell'_\delta < 1$ gives the asserted Lipschitz estimate for $r_\delta^{1/p} = R_*^{1/p} \ell_\delta \circ t$. Moreover,

$$\partial_\delta \ell_\delta = \log(1 + e^s) - \frac{s e^s}{1 + e^s} > 0.$$

The function on the right increases up to $s = 0$ and decreases afterward, with limit zero at both ends. The pointwise limit of ℓ_δ is $\max(t, 0)$. Finally,

$$r_\delta \beta_\delta^2 \le C R_* p^2 \ell_\delta^{p-2},$$

and ℓ_δ is uniformly bounded on the fixed interval. This proves (6.7). □

Choose $t_b > 0$ so small that $3t_b < t(1/4)$, where on the right t denotes the increasing function of t_0 in (6.3), and set

$$K = \{y \in Y : y_n \ge -1/2, \ t(y) \le 2t_b\}. \tag{6.8}$$

Then $K \Subset Y^\circ$. We fix a smooth cutoff $\chi \in C_c^\infty(Y^\circ)$ equal to one on a neighborhood of K, with

$$\operatorname{supp}(d\chi) \subset \{y_n < -1/4\} \cup \{t > t_b\}. \tag{6.9}$$

For example, multiply a cutoff that changes from zero to one between $y_n = -3/4$ and $y_n = -1/2$ by a cutoff of t that changes from one to zero between $2t_b$ and $3t_b$, moving the endpoints slightly to make it one on a neighborhood of K. The inequality $3t_b < t(1/4)$ keeps its support away from the lateral and upper faces of Y. This gives fixed room around K and places every cutoff derivative either in the known lower region or where

$$r_\delta \ge R_* t_b^p > 0.$$

Thus the later estimates for the sphere functionals can use the original Green kernel or fixed-separation continuation on every cutoff derivative support.

6.3 The strict test at a moving sphere

For coordinate derivatives of the tensor h, use the pointwise notation

$$H_8(y) = \sum_{|\alpha| \le 8} |\partial_y^\alpha h(y)|.$$

We shall temporarily impose

$$H_8 \le \varepsilon r_\delta, \qquad r_{\delta'} \beta_{\delta'}^2 \le \varepsilon \quad (\delta \le \delta' \le 1) \quad \text{on } Y. \tag{6.10}$$

The first inequality measures how closely the two principal metrics agree compared with the current radius. It will be improved in Section 10.

Proposition 6.2 (Continuation to the sphere family)**.** *Choose p sufficiently large and then $\varepsilon > 0$ sufficiently small in terms of the fixed background bounds. Choose $R_* > 0$ small enough that all the balls lie in normal neighborhoods and (6.7) is at most ε. For all sufficiently small spatial dilations λ, and every $0 < \delta \le 1$ for which (6.10) holds, the mixed kernel extends to a neighborhood of*

$$\{(x, y) : y \in Y,\ d_\gamma(x, y) = r_\delta(y)\}.$$

The extension solves the two homogeneous equations and agrees with the prescribed lower cross and the fixed-separation continuation. It is smooth for each positive δ; no uniform norm as $\delta \downarrow 0$ is asserted here.

Proof. We check Proposition 5.3 at each intermediate radius $r = r_{\delta'}$. Put $f = \log r$, $\beta = |df|_\gamma$, and $b = \nabla^\gamma r$, so $|b| = r\beta$. Use the defining function

$$\rho(x, y) = \tfrac{1}{2}(d_\gamma(x, y)^2 - r(y)^2).$$

The positive side is the outside of the sphere. Parallel translation along its radial geodesic identifies the two tangent spaces for γ; write ω for the unit vector from y to x.

In the normalization of the strict test, the normal and tangential vectors can be combined into complex vectors

$$U_1 = \omega + iv, \quad v \perp \omega, \quad |v| = 1, \qquad U_2 = q + iw,$$

where the conditions for the second principal metric Q^{-1} give

$$(\omega + b) \cdot U_2 = 1, \qquad q = \frac{\omega + b}{|\omega + b|^2} + O(|h|), \qquad |w| = |\omega + b|^{-1} + O(|h|). \tag{6.11}$$

Here and below lengths and dot products in these formulas use γ and its parallel identification. The first equation includes $(\omega + b) \cdot w = 0$. Since $|b|$ and $|h|$ are small, both partial gradients of ρ are nonzero and $|U_2|$ is bounded above and below.

The Hessian of half the squared distance, with the product γ-connection, evaluated on these endpoint vectors is

$$|U_1 - U_2|^2 + O(r^2)(|U_1|^2 + |U_2|^2).$$

To see the error size, parametrize the radial geodesic on $[0, 1]$. Its Jacobi field with prescribed endpoint values differs, in a parallel frame, from the affine interpolant by $O(r^2)$ in C^1, by the Jacobi equation and bounded curvature. The second variation formula then gives the displayed bound. Replacing the second connection by that of Q^{-1} costs $O(r^2)$ as well: the connection difference is $O(|h| + |\partial h|) = O(\varepsilon r)$ and $|d\rho| = O(r)$.

Now $\omega \cdot (U_1 - U_2) = b \cdot U_2$ and $\operatorname{Hess}(r^2/2) = 2\, dr \otimes dr + r^2 \operatorname{Hess} f$. Consequently the strict Hessian expression is

$$|\pi_{\omega^\perp}(U_1 - U_2)|^2 - |b \cdot U_2|^2 - r^2 \operatorname{Hess}_\gamma f(U_2, \overline{U}_2) + O(r^2). \tag{6.12}$$

For a real Hessian, its value on a complex vector and its conjugate is the sum of its values on the real and imaginary parts, as required by the criterion.

The real part of the projected difference is $-\pi_{\omega^\perp} b + O(|b|^2 + |h|)$. For its imaginary part, (6.11) gives $|\pi_{\omega^\perp} w| = 1 - \omega \cdot b + O(|b|^2 + |h|)$; compare this with $|v| = 1$. Squaring these two bounds yields

$$|\pi_{\omega^\perp}(U_1 - U_2)|^2 \ge |b|^2 - C(|b|^3 + |b||h| + |h|^2).$$

The real and imaginary directions of U_2 are almost orthonormal, so

$$|b \cdot U_2|^2 \le (1 + C(|b| + |h|))|b|^2.$$

By Lemma 6.1, the expression (6.12) is therefore bounded below by

$$cr^2\beta - C\big(r^2 + r^3\beta^3 + \varepsilon r^2\beta + \varepsilon^2 r^2\big).$$

Choose p so that $\inf \beta$ absorbs the Cr^2 term. Then choose ε small. The inequality $r\beta^2 \le \varepsilon$ absorbs $r^3\beta^3$, and the expression is strictly positive.

It remains to specify the starting and boundary data for this local test. Deform δ' from 1 to δ, using (y, ω) as base variables. The inequality $H_8 \le \varepsilon r_\delta$ implies its counterpart for every $\delta' \ge \delta$, because the radii increase with δ'. At $\delta' = 1$ the required separations have a fixed positive minimum. The same is true wherever $t > t_b$, since $r_{\delta'} \ge R_* t_b^p$. Proposition 5.5 supplies uniformly bounded data on these fixed separation ranges, with open room. Where $y_n < -1/4$, use the original $G_{1,\lambda}$; for small λ this is part of the known lower cross. These regions cover the lateral and end boundaries of the cylinder that are not supplied by the initial parameter, and their prescriptions agree on overlaps.

The base sphere bundle over the closed cylinder is compact. On each fixed interval $\delta \le \delta' \le 1$, the radii are positive, all surfaces lie in normal neighborhoods, and the strict tests just proved apply away from the prescribed regions. Lemma 5.4 therefore supplies compatible extension germs on the whole family. Its finite-cover gluing also retains agreement with the prescribed regions. This is an existence argument for each finite interval; it does not assert uniform continuation bounds when the lower endpoint tends to zero. □

6.4 Functionals represented on the spheres

The mixed kernel now defines a map from first-metric harmonic functions to second-metric harmonic functions. Its representation on a small sphere will allow us to estimate this map without estimating the kernel through every continuation step. Fix $0 < \delta \le 1$ and write $r = r_\delta$ in this subsection.

For a function u harmonic for $g_{1,\lambda}$ near $\overline{B_\gamma(y, r(y))}$, define

$$T_y u = \int_{\partial B_\gamma(y, r(y))} \big(\mathcal{G}(x,y)\partial_{\nu_1} u(x) - u(x)\partial_{\nu_1}\mathcal{G}(x,y)\big)\, dS_1(x). \tag{6.13}$$

Here ν_1 is the outer unit normal and dS_1 the hypersurface measure for the actual first metric $g_{1,\lambda}$, not for γ. Let S_y be the unit sphere of $(T_y\mathbb{R}^n, \gamma_y)$, with its rotation-invariant probability measure $d\sigma_y$. Parametrize the integration sphere by $x = \exp_y^\gamma(r(y)\omega)$.

Proposition 6.3 (The transfer density)**.** *Under the assumptions of Proposition 6.2, there is a smooth function $\psi(y, \omega)$ on the unit sphere bundle over Y° such that*

$$T_y u = \int_{S_y} \psi(y,\omega) u\big(\exp_y^\gamma(r(y)\omega)\big)\, d\sigma_y(\omega). \tag{6.14}$$

For every fixed center y_0, if u is first-metric harmonic on a neighborhood of $\overline{B_\gamma(y_0, r(y_0))}$, then $y \mapsto T_y u$ is defined and second-metric harmonic on a sufficiently small neighborhood of y_0. This neighborhood may depend on u. In particular, for functions harmonic on the fixed coordinate neighborhood required by the sphere family,

$$\int_{S_y} \psi\, d\sigma_y = 1, \qquad T_y x^i = y^i, \qquad T_y u^a_{1,\lambda} = u^a_{2,\lambda}(y). \tag{6.15}$$

On the support of derivatives of χ, every fixed angular smooth norm of ψ and its first base derivatives is bounded independently of δ and small λ, after p, R_* and the fixed geometry have been chosen.

Proof. For each ball, its smooth Dirichlet solution operator determines the normal derivative of a harmonic function from its boundary trace. Move this boundary operator in the first term of (6.13) onto the smooth boundary value of $\mathcal{G}$ by adjunction, and then pull back to S_y. The resulting smooth density is ψ. Smooth dependence of the Dirichlet problem and of the sphere parametrization gives its smooth dependence on y for each positive radius. This construction initially applies to smooth boundary values and hence to every harmonic function used in (6.14).

Fix a center y_0 and a function u harmonic on a neighborhood of its closed ball. The continued kernel is defined on an open neighborhood of the compact sphere over y_0. Choose a fixed slightly larger surrounding surface inside this kernel neighborhood and the domain of u. After restricting the center neighborhood, it surrounds every moving sphere and the whole intervening collar remains in that common domain. Green's identity on the collar replaces the moving integral in (6.13) by the integral over this fixed surface, since both first-variable equations are homogeneous there. Applying the second-variable operator under the fixed integral proves the stated local harmonicity. This argument uses arbitrary local harmonic functions, not only the source-generated pairs.

On the lower known region $\mathcal{G} = G_{1,\lambda}$, so Green's formula gives $T_y u = u(y)$. Unique continuation on the connected set Y° now proves all identities in (6.15), first for the constant and coordinate pairs and then for each given harmonic pair.

For the uniform bounds on the cutoff region, distinguish the two sets in (6.9). On the known lower set, T_y is ordinary evaluation. Its density is the Poisson density at the center of the ball, expressed in normalized coordinates. After pulling the equation back by $z \mapsto \exp_y^\gamma(rz)$ and multiplying by the Laplacian scaling, the coefficients form a smooth uniformly elliptic family down to $r = 0$, whose limiting equation is the Euclidean Laplace equation. The interior evaluation kernel for the Dirichlet problem consequently has uniform angular smooth bounds and smooth center–radius dependence. This follows directly by differentiating the Dirichlet equation and using its uniform elliptic estimates; the center stays a fixed positive distance from the unit sphere. The chain rule for a base derivative costs $|dr| = r\beta$, which is uniformly bounded by (6.10).

On the other cutoff set, $t > t_b$. There the radii have a fixed positive lower bound and all their required derivatives are bounded uniformly in δ. The fixed-separation bounds of Proposition 5.5, followed by the smooth Dirichlet construction of ψ, give the same conclusion. Compactness of the cutoff derivative supports completes the proof. □

Figure 2 records the transfer and the two parameter roles. The next section establishes a bound for its density that remains uniform as the spheres shrink.

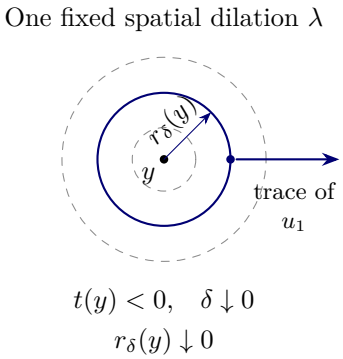


Figure 2: A normal-coordinate section of γ-geodesic spheres in the first variable, where γ is the conformal reference metric. The center y satisfies $t(y) < 0$. The spatial dilation λ is fixed while δ decreases. The Green functional transfers paired harmonic functions and reproduces affine harmonic coordinates exactly. The unit-sphere measure $d\sigma_y$ has total mass one; the density ψ_δ need not be positive.

7 Expanded checks for the cross and shrinking-sphere construction

This section supplements the retained original Sections 3–4. It records authoring calculations, not an independent review or a formal proof certificate. Source labels below refer to the retained manuscript. Standard analytic interfaces S2–S5 remain subject to their separate source verification and eventual formal proof. None is a new authorized axiom. The local metric extension theorem remains a claimed result until the later angular estimate and bootstrap have been established.

7.1 The common basis and the exponent margin

For `cross:series`, distributional solution space $\mathcal{H}(D) = \{w \in L^2(D) : Lw = 0\}$ is closed: testing an L^2 limit against $L^*\varphi$, $\varphi \in C_c^\infty(D)$, preserves the equation. Interior regularity and compact Sobolev embedding make restriction to $S \Subset D$ compact. Unique continuation makes it injective. The spectral theorem therefore gives the asserted basis with positive singular values; there is no omitted zero-eigenspace. A finite-dimensional solution space, if it occurs, simply gives a finite sum.

To quantify decay, fix a cutoff supported inside D, equal to one near $\overline{S}$, and a cube containing its support. Interior estimates bound the cutoff solution in H^k by its $L^2(D)$ norm. Projection onto Fourier modes $|\xi| \le A$ has rank at most CA^n and error at most $C_k A^{-k}$ in L^2. Restriction to S preserves this bound. The minimax principle gives $\mu_j \le C_k j^{-k/n}$. This uses upper bounds only.

For a slice on $D_1 \times S_2$, Parseval gives $b_j(y) = (F(\cdot, y), w_j)_{D_1}$. Restricting its expansion to S_1 and using $(w_i, w_j)_{S_1} = \mu_j^2 \delta_{ij}$ proves the coefficient formula in `cross:coefficient-extension`. Cauchy–Schwarz gives $\|b_j\|_{D_2} \le M/\mu_j$. Fubini and testing under the integral prove $L_2 b_j = 0$ distributionally. Consequently, with $\alpha = \gamma_1 + \gamma_2 - 1 > 0$,

$$\|w_j b_j\|_{L^2(O_1 \times O_2)} \le CM\mu_j^\alpha, \qquad \sum_j \mu_j^\alpha < \infty.$$

One may choose any integer k with $k\alpha/n > 1$. This makes explicit why a strictly positive exponent margin, rather than its limiting value zero, is indispensable. Distributional convergence preserves both equations, and interior estimates for the elliptic sum give the claimed smooth bounds.

The range-completeness check in the source is essential. If $z \perp \overline{R\mathcal{H}(D_1)}$, its pairing with the second-piece slice solves L_2 and vanishes on S_2, hence on connected D_2. The second-piece slices

therefore belong to the closure of that restriction range, as proved in Section S29.2. Their complete orthonormal expansion uses w_j/μ_j and coefficients $\mu_j b_j$. Absolute convergence on $S_1 \times O_2$ follows from $\sum_j \mu_j^{\gamma_2} < \infty$; on $O_1 \times S_2$ it follows from $\sum_j \mu_j^{\gamma_1} < \infty$. These observations prove agreement on the actual intersections even if the observed sets are not in the output sets.

For `cross:exponent`, put $\phi(s) = s - \kappa s^2$. Direct subtraction gives the exact identity

$$\gamma - \frac{1}{2} = \frac{\kappa l^2 - 4e + 8\kappa e^2}{2l(1 - 6\kappa e)}. \tag{7.1}$$

Choose the weight and lateral radius first, then l sufficiently small compared with the lateral quadratic gap, and finally $0 < e \le \kappa l^2/8$ with $6\kappa e < 1/2$. The numerator is then at least $\kappa l^2/2$, so both exponents exceed one half by a fixed positive amount after these choices are frozen. The balancing step has the form $\|w\|_O \le C(e^{-a\tau} + se^{b\tau})$ for normalized $\|w\|_D = 1$ and $s = \|w\|_S$, with $a = \phi(-e) - \phi(-l + 3e) > 0$ and $b = \phi(l + 3e) - \phi(-e) > 0$. Taking $\tau = \log(1/s)/(a + b)$ gives $s^{a/(a+b)}$ when $\tau \ge \tau_0$. Otherwise $s \ge e^{-(a+b)\tau_0}$ and the trivial $\|w\|_O \le 1$ is absorbed by a larger constant. The case $s = 0$ follows from unique continuation. Polynomial Carleman prefactors can be discarded for $\tau \ge \tau_0 \ge 1$ in this upper bound.

7.2 The finite-dimensional separation step

Write $\ell = d\rho$ and normalize dual positive semidefinite tests by $\operatorname{tr} D = 1$. The dual cone of the allowed ith weight Hessians is generated by $a_i \otimes a_i + v_i \otimes v_i$ with $v_i \perp a_i$, $|v_i| = |a_i|$. Its elements have diagonal blocks $d_i a_i \otimes a_i + S_i$ with $S_i \ge 0$ tangential and $\operatorname{tr} S_i = d_i |a_i|^2$. This characterization follows by diagonalizing S_i; its eigenvalues express it as a nonnegative sum of the generating tangential rank-one forms.

Regard D as the covariance of a centered finite-dimensional random vector. This is shorthand for a Gram factorization $D = CC^T$; no probabilistic result is needed for the following matrix calculation. If $D\ell = 0$, the two scalar normal components measured by $d_x\rho, d_y\rho$ sum to zero almost surely. Their variances are equal, so $d_1 = d_2 = d$. The diagonal-block condition says each normal component is uncorrelated with its own tangent component; the opposite-normal relation makes it uncorrelated with both. Hence

$$D = d(a_1, -a_2) \otimes (a_1, -a_2) + D_\perp, \qquad \operatorname{tr}(D_\perp)_{ii} = d|a_i|^2.$$

If $d = 0$, positivity and the trace conditions force $D = 0$, a contradiction. In fact normalization gives $2d(|a_1|^2 + |a_2|^2) = 1$. In the compact convex set of admissible $D_\perp$, a matrix of rank $r \ge 2$ is not extreme: the space of symmetric forms on its image has dimension $r(r + 1)/2 > 2$, leaving a nonzero perturbation preserving both block traces. Small perturbations of both signs preserve positivity. Every extreme point thus has rank one and equals $d(v_1, v_2) \otimes (v_1, v_2)$ with the required lengths. Finite-dimensional convexity reduces positivity of $H : D$ exactly to `cross:hessian-test`.

The set of normalized dual tests is compact. On its subset $D\ell = 0$, $H : D$ has a positive minimum. It remains positive on a neighborhood of that subset. On the complement, $D : (\ell \otimes \ell)$ has a positive minimum; choosing K large there makes $(H + K\ell \otimes \ell) : D > 0$ on every normalized test. Separation from the open cone of positive definite forms plus allowed block Hessians yields the desired P_1, P_2. Prescribe their covariant Hessians and the gradients $d_i\rho$ by quadratic coordinate polynomials. Taylor expansion of $\rho + K\rho^2/2 - \theta_1 - \theta_2$, whose value and first derivative vanish and whose Hessian is positive definite, supplies the quadratic gap. This proves the linear-algebra step rather than assuming a splitting theorem.

7.3 The tube geometry and its parameter order

For `cross:tubes`, write $r = at^{1+\sigma}$ and $t = M + y_n + e_0|y'|^2$. Differentiation gives

$$b = (1+\sigma)\frac{r}{t}\,dt, \qquad r\partial^2 r = \frac{\sigma}{1+\sigma} b \otimes b + (1+\sigma)\frac{r^2}{t}\partial^2 t.$$

Since $|dt| \geq 1$, $\|\partial^2 t\| = 2e_0$, and $t \leq T_0$, the second term is bounded by $CT_0e_0|b|^2$. Outside the seed, $y_n \geq c_0 r$ gives $r/t \leq 1/c_0$, hence a bounded b-range. The zero cases for the projected square are $b = 0, -2\omega$; $b = -\omega$ is separately excluded because the second partial gradient vanishes. Near $b = 0$, the real projection controls b perpendicular to ω, and the change in imaginary projected length controls its parallel component. Thus the lower bound is $c|b|^2$ there as well.

The parameter order avoids circular dependence. Fix the coarse-seed constant c_0, $M > B + 3$, and the two excluded-neighborhood sizes first. Choose $\sigma > 0$ small. Restrict eventually to $e_0 \leq 1/(1 + B^2)$, so target values of t have a fixed upper bound. Choose $a_{\min} > 0$ small enough to lie below all target separations, and $a_{\max}$ large enough for its level to lie in the seed. Next choose T_0 with $c_0 a_{\min} T_0^{\sigma} > 1$. Finally choose e_0 so small that T_0e_0 and $\sqrt{T_0e_0}$ satisfy the required error margins. The transverse base may become very large, but is still compact; only after fixing it is the dilation λ chosen. Near $b = -\omega$ or $b = -2\omega$, the inequality $x_n/r \leq (1+\sigma)|dt|/|b| + \omega_n$ places the pair in the seed for these choices. On the remaining range the displayed Hessian error is absorbed by $c|b|^2$. Its strict margin is positive on the compact non-seed cylinder. This argument gives no uniform estimate as the prescribed separation tends to zero.

7.4 Profile bounds and the moving-sphere cancellation

For `balls:profiles`, put $z = e^{t/\delta}$. Logarithmic differentiation of $A = z/((1+z)\log(1+z))$ gives $A'/A = (1 - z/\log(1+z))/(1+z) < 0$. At the negative end $A = 1 + O(e^s)$; at the positive end $A = s^{-1}(1 + O(e^{-s}(1+s)))$. Differentiating these convergent expressions gives $|A^{(j)}| \leq C_j A^{j+1}$, with the remaining compact interval covered by positivity. Thus the chain rule and the fixed upper and lower bounds on $|dt|_\gamma$ give the stated derivative bounds for $f = p\log\ell + \log R_*$. The factor p is absorbed into p^j for $j \geq 1$. Also $a = \ell'/\ell \leq 1/\ell$, so $r\beta^2 \leq CR_*p^2\ell^{p-2}$; this is uniform in δ for fixed $p \geq 2$. The derivative in δ of ℓ is positive, since its derivative as a function of s is $-se^s/(1+e^s)^2$ and its two endpoint limits are zero. These calculations justify monotonicity used in propagation.

For `balls:test-expression`, all complex quadratic expressions mean the real form evaluated on a vector and its conjugate. Since $\omega \cdot (U_1 - U_2) = b \cdot U_2$ and $\mathrm{Hess}(r^2/2) = 2dr \otimes dr + r^2 \,\mathrm{Hess}\, f$, the positive normal square in the distance Hessian cancels one of the two copies of $|b \cdot U_2|^2$. The surviving expression is exactly the projected square minus $|b \cdot U_2|^2$ minus the Hessian term. The real projection is $-b_\perp + O(|b|^2 + |h|)$, while the reverse triangle inequality for imaginary projections gives a square at least $(\omega \cdot b)^2 - C(|b|^3 + |b||h| + |h|^2)$. The two real directions of U_2 are orthogonal up to $O(|h|)$, with lengths $1 + O(|b| + |h|)$, so Bessel's inequality with this perturbation bounds $|b \cdot U_2|^2$ by $(1 + C(|b| + |h|))|b|^2$. Finally $|h| + |\partial h| \leq C\varepsilon r$ makes the connection error $O(\varepsilon r^2)$, and the Jacobi-field error is $O(r^2)$. The resulting lower bound is $cr^2\beta - C(r^2 + r^3\beta^3 + \varepsilon r^2\beta + \varepsilon^2 r^2)$. Choose p first so that the lower bound for β absorbs Cr^2, then ε, then R_* so that $r\beta^2 \leq \varepsilon$. The spatial dilation is chosen last and remains fixed while δ decreases. The temporary H_8 inequality is a hypothesis at this stage.

7.5 Flux density and the exact remaining obligations

Let k_y be the boundary value of the mixed kernel and let Λ_y be the Dirichlet-to-Neumann map on the first-metric ball. Green's identity makes Λ_y self-adjoint with respect to dS_1. If the sphere

parametrization pulls dS_1 back to $J_y\,d\sigma_y$, the density is explicitly $\psi = J_y(\Lambda_y k_y - \partial_{\nu_1}\mathcal{G})$. The sign gives $T_y u = u(y)$ when the kernel is the positively normalized inverse of $-\Delta_{g_1}$, because Green's second identity then has integrand $G\partial_\nu u - u\partial_\nu G$. For a fixed center, compactness of its sphere gives a common open collar in the kernel domain. A slightly larger fixed surface in this collar surrounds all sufficiently nearby moving spheres. Green's identity in the intervening annulus fixes the integration surface; applying L_2 under that integral proves local harmonicity of $T_y u$. Unique continuation from the known lower region then gives reproduction for every fixed paired harmonic function, including constants and coordinates.

The cutoff bounds require parameter-dependent smooth Dirichlet regularity on a fixed unit ball after rescaling, including the center evaluation kernel and its angular derivatives. On the lower region the limiting principal metric is a scalar multiple of the Euclidean one, because γ is conformal to g_1; the limiting harmonic equation is therefore Euclidean. Center differentiation introduces dr, bounded by $r\beta$. On $t > t_b$ the fixed positive lower radius permits fixed-separation Green bounds instead. These uses belong to S2, S3 and S5 and require precise interfaces with uniform constants. S4 supplies unique continuation and the strict-weight Carleman estimate. The common-basis spectral theorem belongs to S5. Exact published locators remain pending separately from the derivations above; monographs are preferred and no lecture notes have been adopted. Sections 3–4 give only conditional existence of the sphere transfer, not its uniform norm at shrinking radius. The latter and closure of $H_8 \le \varepsilon r_\delta$ remain the substantive downstream obligations. No contradiction in the calculations checked here was identified; that observation is not a proof of the frozen theorem.

8 An estimate on the shrinking spheres

The functional in Proposition 6.3 is defined at every positive radius. We now estimate its representing density uniformly as the radius shrinks. The equation for this density has second-order angular and base terms. Comparing a degree-raising operator with a degree-lowering operator will control both kinds of derivatives; a second comparison will retain the principal terms caused by the changing metric.

8.1 The equation for the density

We use the reference metric γ, the normalized second principal matrix Q, and $h = Q - \gamma^{-1}$ from (6.2). In this section the subscript δ on the positive radius r is suppressed. Put

$$f = \log r, \qquad \beta = |df|_\gamma.$$

The required profile bounds are

$$\beta \ge \beta_0, \qquad \operatorname{Hess}_\gamma f \le -c_0\beta\gamma, \qquad |\nabla^j f| \le C_j\beta^j \quad (1 \le j \le 8), \tag{8.1}$$

where $c_0 > 0$ and C_j are fixed. Lemma 6.1 provides these bounds with β_0 arbitrarily large by choosing p sufficiently large. All background metric bounds below are on a fixed relatively compact coordinate region; their constants are uniform in the spatial dilation.

Let $S_\gamma Y$ be the bundle of γ-unit tangent spheres over the base region Y. Each sphere has probability measure $d\sigma_y$. Horizontal differentiation ∇ uses parallel transport for γ, including the covariant tensor indices of an operator or section. This connection preserves the sphere measure. Unless otherwise specified, all norms and full adjoints use $dV_\gamma(y)\,d\sigma_y(\omega)$.

On each tangent unit ball, extend a sphere function harmonically for the Euclidean metric $\gamma(y)$. Denote the boundary radial derivative of this extension by B, and its ambient derivatives in an

orthonormal frame by A_i. Thus $B = k$ on the space $\mathcal{H}_k$ of spherical harmonics of degree k, and $A_i : \mathcal{H}_k \to \mathcal{H}_{k-1}$. The star in A_i^* denotes the sphere adjoint. These operators are parallel, with the indicated tensor indices, and

$$[B, A_i^*] = A_i^*, \qquad \sum_i A_i^* A_i^* = 0. \tag{8.2}$$

Indeed differentiation lowers homogeneous degree by one, and the second identity is the adjoint of the vanishing Laplacian of a harmonic extension. For $s \in \mathbb{R}$, use the fiber Sobolev norm

$$\|v(y,\cdot)\|_s = \|(1+B)^s v(y,\cdot)\|_{L^2(S_y)}.$$

It is equivalent to the usual Sobolev norm on the unit sphere. Tensor-valued versions use the sum over an orthonormal frame.

We write Ψ^m for the classical pseudodifferential operators of order m on the sphere. For a positive function $a(y)$, the notation $a\Psi^m$ means that, after division by a, the finitely many symbol and operator seminorms used below are bounded. A statement with j base derivatives permits the indicated additional factor β^j.

Lemma 8.1 (Moving trace operators)**.** *Suppose that the normal balls and the functional T_y of Proposition 6.3 are defined, and that*

$$\sum_{|\alpha|\le 8} |\partial_y^\alpha h| \le \varepsilon r, \qquad r\beta^2 \le \varepsilon. \tag{8.3}$$

If u is first-metric harmonic near one such ball, set

$$U(y,\omega) = u\big(\exp_y^\gamma(r(y)\omega)\big).$$

There are sphere operators $\mathcal{C}_i$ such that $\nabla_i U = \mathcal{C}_i U$. Their sphere adjoints satisfy

$$\mathcal{C}_i^* = r^{-1}A_i^* + f_i B + E_i + F_i, \qquad E_i \in r\Psi^1, \quad F_i \in \Psi^0, \qquad f_i = \nabla_i f. \tag{8.4}$$

The normalized remainders $r^{-1}E_i$ and F_i have uniform symbol bounds after up to six base derivatives, with the allowance $C_j\beta^j$.

Write the normalized second equation, in the reference connection, as $Q^{ij}\nabla_i\nabla_j + b_2^i\nabla_i$. Its drift b_2 is uniformly controlled. If $\psi(y,\omega)$ represents T_y against $d\sigma_y$, then

$$P\psi = 0, \qquad P = r\left[Q^{ij}(\nabla_i + \mathcal{C}_i^*)(\nabla_j + \mathcal{C}_j^*) + b_2^i(\nabla_i + \mathcal{C}_i^*)\right]. \tag{8.5}$$

The products in this formula are covariant products. In particular, Q is not differentiated by taking a base adjoint.

Proof. Normalize the principal part of the first equation to γ^{-1} and pull it back by $z \mapsto \exp_y^\gamma(rz)$ to the unit ball. After multiplication by r^2, its principal coefficients are $\delta^{ij} + O(r^2)$ and its first-order coefficients are $O(r)$. The resulting smooth family remains uniformly elliptic and has an invertible Dirichlet problem, including at $r = 0$. Its radial derivative operator therefore has the expansion

$$B + rB_1(y) + r^2 B_2(y,r), \qquad B_1 \in \Psi^0, \quad B_2 \in \Psi^1. \tag{8.6}$$

Here is the parameter justification. The classical boundary-symbol construction is described in [8, Theorem 1.1(i) and Proposition 3.1]; we keep the center and radius dependence explicitly. In

boundary coordinates the decaying normal root of the principal symbol constructs the Dirichlet parametrix. Successive lower-order transport equations construct its symbols smoothly in (y, r). The first normal derivative trace is an operator of order one, whose principal symbol depends only on the principal coefficients and the differentiating vector. Those coefficients have zero first r-derivative at $r = 0$, so that derivative of the trace operator has order zero. Taylor's formula in this symbol class gives (8.6) to smoothing error. The true solution differs from the parametrix by the Dirichlet inverse of a smooth error; differentiated Dirichlet estimates control this remainder in every fixed seminorm. More explicitly, on a compact center set the rescaled Dirichlet operator is smoothly invertible between each fixed pair of Sobolev spaces, uniformly for $0 \le r \le r_0$. Differentiating that inverse controls every fixed number of parameter derivatives of the smoothing correction; increasing the Sobolev orders controls its smooth kernel seminorms. Thus the Taylor expansion holds in the complete classical symbol topology, including the smoothing remainder, and not merely in one Sobolev operator norm. This proves the stated expansion and its parameter bounds.

Differentiate the exponential map in its center, transporting ω parallel to the center velocity. The Jacobi field has initial derivative zero. After inverting the differential in the z variable, the velocity acting on the unit-ball solution is

$$r^{-1}(e_i + O(r^2)) + f_i\omega.$$

The coefficient error is smooth in (y, r, ω). Tangential derivatives act directly on the boundary value; the radial derivative is (8.6). At $r = 0$ the first term is $r^{-1}A_i$. The order-one errors are $O(r)$, and the order-zero errors are bounded: in particular rf_iB_1 is bounded because $r\beta \le \varepsilon/\beta$. Taking sphere adjoints gives (8.4). Differentiating the radius costs at most one factor β per derivative, by (8.1); the asserted parameter estimates follow.

For fixed u, differentiate $T_y u = \langle \psi(y, \cdot), U(y, \cdot)\rangle_{S_y}$. Metric compatibility and $\nabla_i U = \mathcal{C}_i U$ put the differentiated operator on ψ as $\nabla_i + \mathcal{C}_i^*$. The local second-metric harmonicity in Proposition 6.3 proves that (8.5) pairs to zero with the trace of every function harmonic near the fixed closed ball. No global extension or source-family approximation is used in this step. At a fixed sphere these traces are dense among smooth boundary data: solve the smooth Dirichlet problem on slightly enlarged balls, using boundary values transported from the limiting sphere, and use smooth dependence of the solution on the radius. Thus the paired equation is the stated equation for ψ. □

8.2 The model operator and its principal perturbations

We separate a degree-raising model operator from the remaining principal and lower-order terms of the moving trace equation. We continue under the hypotheses of Lemma 8.1, including (8.3).

Define

$$D_i^+ = \nabla_i + f_iB, \qquad D_i^- = -\nabla_i + f_iB, \qquad T = \sum_i A_i^* D_i^+, \quad T^\flat = \sum_i A_i D_i^-. \tag{8.7}$$

The symbol T here denotes a differential operator on sphere functions; the transfer functional continues to be denoted T_y. Expand (8.5) by (8.4). The identities (8.2) give

$$D_i^+(r^{-1}A_j^*) = r^{-1}A_j^* D_i^+, \qquad \sum_i A_i^* A_i^* = 0.$$

Thus, in ordinary base coordinates and an orthonormal sphere trivialization,

$$P = 2T + U + J, \tag{8.8}$$

where J is a uniformly bounded family in Ψ^1, with no base derivatives, and

$$U = \sum_{|\alpha|\le 2} U_\alpha(y)\partial_y^\alpha, \qquad U_\alpha \in \varepsilon\beta^{-|\alpha|}\Psi^{2-|\alpha|}. \tag{8.9}$$

Up to five derivatives of these coefficients have the additional allowance $C_j\beta^j$. The second-base-derivative term is precisely

$$rQ^{ij}\partial_{y_i}\partial_{y_j}; \tag{8.10}$$

its coefficients are real scalars independent of the sphere variable.

We detail these bounds. The pure angular second derivative contracted with γ^{-1} vanishes. The remaining contraction is $(h^{ij}/r)A_i^*A_j^*$, and belongs to the zero-base-derivative term of (8.9). The mixed correction is $2h^{ij}A_i^*D_j^+$ and has the stated small bounds. The D^+D^+ term has coefficient r; its base orders two, one and zero have coefficients $O(r)$, $O(r\beta)$ and $O(r\beta^2)$, respectively. The condition $r\beta^2 \le \varepsilon$ gives (8.9) for all three. Terms containing E_i obey the same bounds. Terms containing F_i also do so, except for their cross terms with $r^{-1}A_j^*$, which are bounded angular order-one terms and belong to J. The drift term with A_i^* belongs to J as well. Differentiating any of these expressions gives the stated coefficient bounds by (8.1) and (8.3). A horizontal connection written in this trivialization adds a bounded angular first derivative, which preserves the same estimates. The bounds for J use the fixed background and remain uniform as β_0 is increased.

The decomposition identifies the model operator $2T$ and the second-order perturbation U. To carry the model estimate over to P we shall need more than an ordinary first-derivative estimate: the error forms also contain mixed base and angular derivatives. For a sphere function φ, put

$$\mathcal{N}(\varphi)^2 = \int_Y \left(\|\nabla\varphi\|_0^2 + \beta^2\|\varphi\|_1^2 + \beta^{-1}\|\nabla\varphi\|_{1/2}^2 + \beta\|\varphi\|_{3/2}^2\right) dV_\gamma. \tag{8.11}$$

The last two terms of $\mathcal{N}$ place half an angular derivative on $\nabla\varphi$ and three halves on φ. They will bound the remaining error forms left by the comparison with P.

8.3 A comparison of raising and lowering operators

The next estimate supplies these two derivative strengths for the model operator. Spherical harmonics are trace-free symmetric tensors, and comparing their raising and lowering operators is familiar from energy identities in tensor tomography; see, for example, [11, 12]. Here the negative Hessian of the spatial weight supplies the positive term that absorbs the bounded curvature. We give the calculation with its degree factors.

Lemma 8.2 (Weighted raising estimate)**.** *Let $n \ge 3$, and suppose (8.1) holds on a fixed base region with uniformly bounded reference geometry. There are β_0 and $c > 0$, depending only on that geometry and the constants in (8.1), such that every smooth, compactly base-supported function with zero fiber mean satisfies*

$$\|T\varphi\|^2 - \|T^\flat\varphi\|^2 \ge c\mathcal{N}(\varphi)^2. \tag{8.12}$$

Proof. Both operators change angular degree by exactly one, so it suffices to prove the estimate at one input degree $k \ge 1$. Identify that component with a harmonic homogeneous polynomial $v(z)$ of degree k. Use the Fischer inner product, in which the monomials z^α are orthogonal with squared norm $\alpha!$. Polynomial differentiation $a_i = \partial_{z_i}$ is adjoint to multiplication $M_i = z_i$. On harmonic polynomials of degree k the Fischer squared norm is the normalized squared sphere norm multiplied by

$$N_k = n(n+2)\cdots(n+2k-2), \qquad N_0 = 1.$$

For completeness, integration against a standard real Gaussian identifies the Fischer norm on trace-free homogeneous polynomials with their Gaussian L^2 norm; integrating radially then gives the factor N_k. These classical identities are also recorded in [13, Equation (2.1) and Theorem 2.1].

Set $d = k + n/2 - 1$ and use $D_i^{\pm} = \pm\nabla_i + k f_i$ at this fixed degree. For harmonic polynomials q_i of degree k, the Laplacian of $\sum_i M_i q_i$ is $2\sum_i a_i q_i$. Since

$$\Delta_z(|z|^2 s) = 4d\, s \qquad (s \in \mathcal{H}_{k-1}),$$

its harmonic projection is $\sum_i M_i q_i - |z|^2 \sum_i a_i q_i/(2d)$. Orthogonality of this projection and its complement shows that its squared norm is

$$\sum_i \|q_i\|^2 + \sum_{i,j} (a_j q_i, a_i q_j) - d^{-1} \Big\| \sum_i a_i q_i \Big\|^2.$$

Changing from Fischer adjoints to sphere adjoints gives

$$A_i^*|_{\mathcal{H}_k} = \frac{N_{k+1}}{N_k} \Pi_{k+1} M_i,$$

where Π_{k+1} is harmonic projection in homogeneous degree $k+1$. Consequently N_k times the left side of (8.12) is

$$(2d+2) \left[\|D^+ v\|^2 + \sum_{i,j} (a_j D_i^+ v, a_i D_j^+ v) - \frac{1}{d} \Big\| \sum_i a_i D_i^+ v \Big\|^2 \right] - 2d \Big\| \sum_i a_i D_i^- v \Big\|^2. \tag{8.13}$$

All norms in this calculation include integration over the base and use the polynomial inner product in the fibers. The factors are $N_{k+1}/N_k = 2d+2$ and $N_k/N_{k-1} = 2d$.

Write $C_{ij} = 2k(\mathrm{Hess}_\gamma f)_{ij}$ and

$$C[av] = \int_Y \sum_{i,j} C_{ij} \langle a_i v, a_j v \rangle \, dV_\gamma.$$

Covariant integrations by parts give

$$\sum_{i,j} (a_j D_i^+ v, a_i D_j^+ v) = \Big\| \sum_i a_i D_i^- v \Big\|^2 - C[av] + O(k^2)\|v\|^2, \tag{8.14}$$

$$\Big\| \sum_i a_i D_i^+ v \Big\|^2 \le k \|D^- v\|^2 - C[av] + O(k^2)\|v\|^2, \tag{8.15}$$

$$\|D^- v\|^2 = \|D^+ v\|^2 + \int_Y \mathrm{tr}\, C\, |v|^2 \, dV_\gamma. \tag{8.16}$$

Here and below a form denoted $O(k^2)\|v\|^2$ has absolute value at most a fixed constant times that quantity. To verify the first two formulas, move a D_j^+ across the pairing and commute $D_j^- D_i^+$ to $D_i^+ D_j^- - C_{ij}$. The reordered term in (8.15) is at most $k\|D^- v\|^2$, because $\sum_i |a_i w|^2 = k|w|^2$ for $w \in \mathcal{H}_k$. The extra commutator is curvature: it acts as a sum of rotations of norm $O(k)$ at degree k, and the two contractions supply one more factor k. This proves the claimed error bounds. Expanding the two squares proves (8.16). For complex functions all the displayed forms are understood through their real parts.

Substituting (8.14) into (8.13) rewrites that expression as

$$\begin{aligned}(2+2/d)\left[d\|D^+v\|^2-\Big\|\sum_i a_iD_i^+v\Big\|^2-dC[av]\right]\\+2\Big\|\sum_i a_iD_i^-v\Big\|^2+O(k^3)\|v\|^2.\end{aligned}\tag{8.17}$$

We must control the remaining negative divergence square while retaining both an ordinary derivative estimate and an additional half-order angular derivative estimate. The latter requires a coefficient of size k/β in front of $|D^+v|^2$, including when k is much larger than β. Choose a small constant $\vartheta>0$. On the fraction $1-\vartheta$ of its negative divergence square use (8.15)–(8.16). On the remaining fraction use the pointwise convex combination

$$\begin{aligned}\Big|\sum_i a_iD_i^+v\Big|^2\le(1-\zeta)(k+n-1)|D^+v|^2\\+\zeta\left(2\Big|\sum_i a_iD_i^-v\Big|^2+8k^3\beta^2|v|^2\right),\qquad\zeta=c_1/\beta.\end{aligned}\tag{8.18}$$

The fraction $1-\vartheta$ in the integrated estimates is constant; the y-dependent ζ is used only in this pointwise inequality. The first bound follows from the adjoint multiplication operators, since the contraction $(q_i)\mapsto\sum_i a_iq_i$ has squared norm at most $k+n-1$ on polynomials of degree k. The second follows from $D_i^++D_i^-=2kf_i$ and $|\sum_i f_ia_iv|^2\le k\beta^2|v|^2$.

Combining these bounds in (8.17), the coefficient retained for $|D^+v|^2$ is

$$(2+2/d)\big[d-k-\vartheta(n-1)+\vartheta\zeta(k+n-1)\big].\tag{8.19}$$

Since $d-k=(n-2)/2$, a sufficiently small fixed ϑ makes this at least a fixed positive constant plus a positive multiple of c_1k/β. The coefficient of $|\sum_i a_iD_i^-v|^2$ is nonnegative when β_0 is sufficiently large. The Hessian contribution is exactly

$$-(2+2/d)\left[(d-1+\vartheta)C[av]+(1-\vartheta)k\int_Y\operatorname{tr}C\,|v|^2\,dV_\gamma\right].\tag{8.20}$$

For $k\ge1$ and $n\ge3$, both coefficients inside this expression are positive. The Hessian hypothesis bounds it below by $ck^3\int\beta|v|^2$. This absorbs the curvature error $O(k^3)\|v\|^2$ by increasing β_0, and absorbs the last term of (8.18) by choosing c_1 sufficiently small. We have therefore retained positive multiples of

$$\|D^+v\|^2,\qquad\int_Y\frac{k}{\beta}|D^+v|^2\,dV_\gamma,\qquad k^3\int_Y\beta|v|^2\,dV_\gamma.$$

The first controls both $\|\nabla v\|^2$ and $k^2\int\beta^2|v|^2$, since its cross term is $-k\int\Delta_\gamma f\,|v|^2\ge0$. For the weighted derivative use the pointwise inequality

$$\frac{k}{\beta}|\nabla v|^2\le2\frac{k}{\beta}|D^+v|^2+2k^3\beta|v|^2.$$

No differentiation of β^{-1} is required. Dividing by N_k, and then summing the orthogonal degree components, proves (8.12); the weights $(1+k)^s$ are comparable to k^s because the zero degree is excluded. □

8.4 Keeping the principal perturbations

The raising estimate controls the model operator. The actual equation (8.5) also contains second base derivatives and a small second-order angular perturbation. We retain these terms in a comparison of squared norms.

Proposition 8.3 (Coercivity for the moving trace equation)**.** *Fix a base region with bounded reference geometry and the constants in* (8.1)*. Suppose P is the operator* (8.5)*, with the trace operators of Lemma 8.1. There exist β_0 sufficiently large, $\varepsilon_0 > 0$ and $C < \infty$, depending only on these fixed bounds, such that* (8.3) *with $\varepsilon \le \varepsilon_0$ implies*

$$\mathcal{N}(\varphi) \le C\|P\varphi\| \tag{8.21}$$

for every smooth, compactly base-supported function φ with zero mean on each sphere. These constants are independent of the shrinking parameter. Only the eight indicated derivatives of the small tensor h are required.

Proof. Let a dagger denote the full adjoint for $dV_\gamma d\sigma$. Covariant integration and the degree shift give

$$Z := T^\flat - T^\dagger = \sum_i f_i A_i. \tag{8.22}$$

A direct expansion yields

$$\begin{aligned} &\|(2T+U)\varphi\|^2 - \|(2T^\flat + U^\dagger)\varphi\|^2 \\ &\qquad = 4\big(\|T\varphi\|^2 - \|T^\flat\varphi\|^2\big) + \langle \mathcal{E}\varphi, \varphi\rangle, \end{aligned} \tag{8.23}$$

where

$$\mathcal{E} = 2[T^\dagger, U] + 2[U^\dagger, T] + [U^\dagger, U] - 2\big(UZ + (UZ)^\dagger\big). \tag{8.24}$$

We will prove

$$|\langle \mathcal{E}\varphi, \varphi\rangle| \le C\varepsilon \mathcal{N}(\varphi)^2. \tag{8.25}$$

Here are the complete order and weight counts needed for this claim. An operator has count (m, s) if its coefficient at ∂_y^α is angular order $m - |\alpha|$, of size $O(\beta^{s-|\alpha|})$, with the derivative allowances already specified. Adjoints preserve the count, and products add the counts. When a base derivative hits a coefficient it consumes one differential order and adds a factor β, so it preserves the weight index. A commutator of scalar angular operators loses one angular order: the products of their principal symbols cancel. Hence a scalar commutator loses one in total order while preserving the sum of the weight indices. The counts of $T, T^\dagger, U, Z$ are, respectively,

$$(2,1), \quad (2,1), \quad (2,0), \quad (1,1),$$

and every occurrence of U carries the factor ε. Thus (8.24) has count at most $(3,1)$ and a factor $O(\varepsilon)$; the quadratic U term is smaller.

There is a further cancellation that the total count alone does not express: no third base derivative survives. To check it explicitly, use an orthonormal sphere trivialization and write the base measure as $\mu(y)\,dy$. The sphere probability measure is then fixed. Write

$$U = a^{ij}\partial_i\partial_j + \mathcal{B}^i\partial_i + \mathcal{C}, \qquad a^{ij} = rQ^{ij} = a^{ji} \in \mathbb{R}.$$

The coefficients $\mathcal{B}^i$ and $\mathcal{C}$ are angular operators. Replacing a horizontal derivative by a coordinate derivative plus its angular connection term changes only base orders at most one, so these are all the second-base coefficients. If a star denotes the sphere adjoint, direct integration by parts gives

$$\begin{aligned} U^\dagger - U = &\left(2\mu^{-1}\partial_j(\mu a^{ij}) - \mathcal{B}^i - (\mathcal{B}^i)^*\right)\partial_i \\ &+ \mu^{-1}\partial_i\partial_j(\mu a^{ij}) - \mu^{-1}\partial_i(\mu(\mathcal{B}^i)^*) + \mathcal{C}^* - \mathcal{C}. \end{aligned}$$

In particular it has base order at most one. In $[T^\dagger, U]$ and its partner, the possible third-base-derivative coefficient is a commutator of an angular coefficient with rQ^{ij}, which is zero by (8.10). For $[U^\dagger, U]$, first write it as $[U^\dagger - U, U]$. The operator $U^\dagger - U$ has no second base derivative, since the coefficient in (8.10) is real. Taking adjoints against the smooth base density only produces lower base orders. Its remaining possible third-base-derivative coefficients commute for the same reason. Finally, UZ already has at most two base derivatives. It follows that $\mathcal{E}$ is a sum of terms of precisely the following three permitted types:

$$\varepsilon\beta^{-1}\Psi^1\partial_y^2, \qquad \varepsilon\Psi^2\partial_y, \qquad \varepsilon\beta\Psi^3. \tag{8.26}$$

This argument also covers lower orders, since $\beta \geq 1$.

For the first type in (8.26), integrate once in the base. The principal form is bounded by $C\varepsilon \int \beta^{-1}\|\partial_y\varphi\|_{1/2}^2$. A derivative of its coefficient costs one factor β and leaves an $\varepsilon\Psi^1\partial_y$ term. This includes differentiation of the weight itself: $|d\beta| \leq C\beta^2$ implies $|d(\beta^{-1})| \leq C$. The second type has the form bound

$$C\varepsilon \int_Y \|\partial_y\varphi\|_{1/2}\|\varphi\|_{3/2}\, dV_\gamma \leq C\varepsilon \int_Y \left(\beta^{-1}\|\partial_y\varphi\|_{1/2}^2 + \beta\|\varphi\|_{3/2}^2\right) dV_\gamma.$$

The third type is bounded by $C\varepsilon \int \beta\|\varphi\|_{3/2}^2$. The lower-order term from differentiating a coefficient satisfies the same bound. Replacing coordinate derivatives by horizontal derivatives adds a bounded angular first derivative, again controlled by these norms. A fixed finite partition of the base region has the same harmless lower-order errors. This proves (8.25).

The constants in the preceding estimates are uniform for all $\beta \geq \beta_0$ once a fixed lower threshold is met. Choose ε_0 small relative to the constant in Lemma 8.2. Dropping the nonnegative second squared norm in (8.23) gives $\mathcal{N}(\varphi) \leq C\|(2T + U)\varphi\|$. Finally,

$$\|J\varphi\| \leq C\|\varphi\|_{L^2(Y;H^1(S_y))} \leq C\beta_0^{-1}\mathcal{N}(\varphi),$$

so a sufficiently large β_0 absorbs J and proves (8.21).

All these operations use angular symbol estimates pointwise in the base, and ordinary base differential operators of degree at most two. Formal adjoints, the commutator expansion and the one integration by parts use at most the five base coefficient derivatives specified in (8.9). Construction of these coefficients differentiates the moving trace operators once and the second metric to only finitely many orders below eight. Higher angular symbol seminorms depend on the fixed smooth first-metric family; differentiating in the angular variable does not differentiate the center-dependent tensor h. Thus the eight smallness derivatives in (8.3) suffice in every fixed dimension. No smallness assumption on all derivatives of h is used. □

8.5 The constant mode and the uniform density bound

We now apply the coercive estimate to the transfer density constructed in Section 6, restoring the notation $r = r_\delta$. Its normalization is $\int_{S_y} \psi\, d\sigma_y = 1$, so $\chi(\psi - 1)$ has zero fiber mean for the fixed cutoff (6.9). This is the mean-zero condition required by Proposition 8.3.

The decomposition (8.8) gives a uniform bound for $P1$: its raising part annihilates 1, only U's bounded zero-base-derivative coefficient acts on 1, and J is uniformly bounded in Ψ^1. There is no factor growing with β in this assertion.

For any fixed smooth base cutoff, the same decomposition gives

$$\|[P,\chi]w\| \le C_\chi \left(\|w\|_{L^2(\operatorname{supp} d\chi; H^1(S_y))} + \|r\nabla w\|_{L^2(\operatorname{supp} d\chi \times S_y)} \right), \tag{8.27}$$

where the support can be replaced by a fixed small neighborhood of the supports of all derivatives of χ. Indeed $[2T,\chi] = 2\sum_i A_i^*(\nabla_i\chi)$, the second base derivative of U contributes $2rQ^{ij}(\partial_i\chi)\partial_j$ and a bounded zeroth-base-derivative term, and its first base derivative contributes a bounded angular order-one term. The operator J commutes with χ.

Lemma 8.4 (Uniform transfer density)**.** *Fix the structural parameters so that Propositions 6.2 and 8.3 apply. There is a constant C independent of δ and of sufficiently small spatial dilations λ such that*

$$\|\chi\psi\|_{L^2(S_\gamma Y)} \le C \tag{8.28}$$

whenever the coefficient condition (6.10) holds.

Proof. Set $\varphi = \chi(\psi - 1)$, which is compactly supported in the base and has zero fiber mean. For the actual moving-sphere operator P, Equation (8.5) gives

$$P\varphi = P\big(\chi(\psi-1)\big) = -\chi P1 + [P,\chi](\psi - 1). \tag{8.29}$$

The cutoff estimate (8.27) involves only angular derivatives through order one and $r\nabla\psi$ on the supports of derivatives of χ. Proposition 6.3 bounds these there uniformly: on the lower region the density is that of ordinary evaluation, and on the other region the radii stay at a fixed positive separation. Together with the bound for $P1$, this gives a uniformly bounded L^2 norm for the right side of (8.29). Proposition 8.3 bounds φ; adding the fixed function χ proves (8.28). □

The uniform bound now applies to the density representing the actual Green functional at every radius allowed by (6.10). Exact affine moments will turn it into a two-power bound for the paired harmonic differences.

9 Expanded angular estimate on the moving spheres

This section expands the argument labelled `sec:angular` in the source. It is a candidate proof calculation for the selected three-dimensional problem. The algebra below is given for $n \ge 3$, since the positive degree factor explains exactly where the dimension restriction enters. The construction of the transfer functional and its local harmonicity are inputs from `balls:functional`; they are not consequences of this section. The standard elliptic and pseudodifferential facts needed to make the calculation analytic are listed in the final subsection as explicit proof and reference obligations. In particular, the coercive estimate proved below is not being adopted as an external premise.

9.1 Spaces, measures, and parameter conventions

Fix a relatively compact base region containing the support of the cutoff χ in `balls:cutoff-region`. Write $d\mu = dV_\gamma d\sigma_y$, where $d\sigma_y$ has total mass one on the γ_y-unit sphere S_y. The metric connection on this sphere bundle differentiates both tensor indices and the sphere argument by parallel transport. It preserves $d\sigma_y$. All integrations by parts below are for sections with compact base support; there

is no boundary integral at the boundary of the cylinder. For complex sections use the Hermitian pairing and take real parts of quadratic expressions.

The fiberwise Euclidean harmonic extension defines its radial derivative B and its ambient derivatives A_i. If $\mathcal{H}_k(y)$ is the space of harmonic homogeneous polynomials of degree k, restricted to S_y, then $B = k$ and $A_i : \mathcal{H}_k \to \mathcal{H}_{k-1}$. In a parallel orthonormal frame these operators are parallel with their tensor index. Sphere adjoints are denoted by a star. Full adjoints for $d\mu$ will be denoted by a dagger. In particular,

$$[B, A_i^*] = A_i^*, \qquad \sum_i A_i^* A_i^* = 0. \tag{9.1}$$

The second identity follows by taking the sphere adjoint of $\sum_i A_i A_i = 0$ on every homogeneous component, including components where a derivative is zero.

Define

$$\|v\|_s^2 = \sum_{k \geq 0} (1+k)^{2s} \|v_k\|^2_{L^2(S_y)}.$$

Since the positive spherical Laplacian has eigenvalues $k(k+n-2)$, this is equivalent to the usual $H^s(S_y)$ norm, with constants depending only on n, s. The condition of zero fiber mean is exactly $v_0 = 0$. The initial test class throughout is $C_c^\infty(S_\gamma Y)$ with this condition. No assertion concerning an unspecified closed realization of P is needed: for each fixed $\delta > 0$ the cutoff transfer density belongs to this test class. If a Hilbert-space formulation is desired, complete this class in the graph norm $\|v\| + \|Pv\|$ for fixed δ; the final inequality extends to that completion by definition.

Put $r = r_\delta > 0$, $f = \log r$, and $\beta = |df|_\gamma$. The assumptions used in the calculations are

$$\beta \geq \beta_0 \geq 1, \qquad \mathrm{Hess}_\gamma f \leq -c_0 \beta \gamma, \qquad |\nabla^j f| \leq C_j \beta^j \quad (1 \leq j \leq 8), \tag{9.2}$$

and

$$\sum_{|\alpha| \leq 8} |\partial^\alpha h| \leq \varepsilon r, \qquad r\beta^2 \leq \varepsilon, \qquad Q = \gamma^{-1} + h. \tag{9.3}$$

Coordinate and covariant seminorms are equivalent here with fixed background constants. Repeated differentiation gives, in every order used below,

$$|\partial^j r| \leq C_j r \beta^j, \qquad |\partial^j (r^{-1})| \leq C_j r^{-1} \beta^j, \qquad |\partial^j \beta| \leq C_j \beta^{j+1} \quad (0 \leq j \leq 7).$$

Consequently $|\partial^j (h/r)| \leq C_j \varepsilon \beta^j$ for $j \leq 8$, and $|\partial(\beta^{-1})| \leq C$. These are pointwise estimates. They do not assert a uniform upper bound on β as δ tends to zero.

For clarity, a statement $K \in a\Psi^m$ means a bound on a fixed finite collection of classical symbol seminorms, together with smoothing-kernel seminorms sufficient for the Sobolev mappings used here, after division by $a(y)$. Angular variables lie on the compact sphere. No base pseudodifferential calculus or frequency cutoff in the base is used.

9.2 The moving trace equation

Normalize the first principal part to γ^{-1}. In γ-normal coordinates based at y, write $x = \exp_y^\gamma(rz)$, $|z| \leq 1$, and multiply its equation by r^2. Its coefficients have the form

$$L_{y,r} = \Delta_z + r c^i(y,z,r)\partial_{z_i} + r^2 c^{ij}(y,z,r)\partial_{z_i}\partial_{z_j},$$

with smooth coefficients down to $r = 0$, uniform on a compact set of centers. This uses the vanishing of the first derivatives of the metric in normal coordinates; the normalized first equation may have a drift, producing the $r\partial_z$ term. Shrink the fixed upper bound for r so that the Dirichlet problem

remains uniformly elliptic. Constants solve the homogeneous equation, and the maximum principle gives uniqueness.

Let $\Lambda(y,r)$ be the radial derivative trace of its Dirichlet solution. The required boundary-calculus conclusion is smooth dependence in the complete Ψ^1 topology, including smoothing kernels. Its principal symbol depends only on the principal part and the differentiating vector. The first r derivative of those principal coefficients is zero at 0. Thus $\partial_r\Lambda(y,0)\in\Psi^0$, and Taylor's formula reads

$$\Lambda(y,r)=B+rB_1(y)+r^2B_2(y,r),\qquad B_1\in\Psi^0,\quad B_2\in\Psi^1, \tag{9.4}$$

where, explicitly, $B_2(y,r)=\int_0^1(1-s)\partial_r^2\Lambda(y,sr)\,ds$. This is the precise analytic expansion needed; an $H^s\to H^{s-1}$ estimate alone would not justify subsequent commutator order reductions.

For a fixed first-metric harmonic function u put $U(y,\omega)=u(\exp_y^\gamma(r(y)\omega))$. The Jacobi field obtained by moving y and parallel transporting ω has initial values $J(0)=e_i$, $\nabla_tJ(0)=0$. On a bounded-curvature normal neighborhood its endpoint, expressed in the endpoint parallel frame, is $e_i+O(r^2)$. The angular differential of the exponential map is $r(I+O(r^2))$. Their quotient therefore gives the vector acting on the unit-ball solution as

$$r^{-1}(e_i+O(r^2))+f_i\omega.$$

Tangential derivatives of U are direct sphere derivatives. Radial derivatives use (9.4). Splitting the resulting operators by angular order, and then taking the sphere adjoint, gives

$$\nabla_iU=\mathcal{C}_iU,\qquad \mathcal{C}_i^*=r^{-1}A_i^*+f_iB+E_i+F_i,\qquad E_i\in r\Psi^1,\quad F_i\in\Psi^0. \tag{9.5}$$

For example, the term f_irB_1 is included in F_i, since $r\beta\le\varepsilon/\beta$; $f_ir^2B_2$ is included in E_i. After up to six base derivatives, E_i/r and F_i have the additional allowance $C_j\beta^j$. This chain-rule count requires derivatives of f through order seven, since the un-differentiated expression contains f_i. It requires no derivative of the tensor h, which belongs to the second equation.

Write that second equation as $L_2=Q^{ij}\nabla_i\nabla_j+b_2^i\nabla_i$ in the reference connection. For a locally harmonic transfer functional,

$$\nabla_i(T_yu)=\int_{S_y}(\nabla_i+\mathcal{C}_i^*)\psi\,U\,d\sigma_y.$$

Here and in the iterated derivative, covariant tensor indices are retained. Applying L_2 gives

$$P\psi=0,\qquad P=r\{Q^{ij}(\nabla_i+\mathcal{C}_i^*)(\nabla_j+\mathcal{C}_j^*)+b_2^i(\nabla_i+\mathcal{C}_i^*)\}. \tag{9.6}$$

There is no base adjoint in this derivation and no derivative of Q. To justify equality as a smooth function, fix the center and test against all traces of harmonic functions defined near the closed ball. For any $g\in C^\infty(S_y)$, solve Dirichlet problems on radius $r+\eta$, with g transported radially to that boundary. In fixed-ball coordinates the coefficients and boundary data converge smoothly as $\eta\downarrow0$. Smooth parameter dependence and elliptic boundary regularity imply that the solutions converge in each C^m on the closed limiting ball. Their restrictions to its boundary converge to g. Thus the test traces are dense in $C^\infty(S_y)$, not merely in the finite harmonic family.

9.3 Decomposition with the exact second base coefficient

Set

$$D_i^+=\nabla_i+f_iB,\qquad D_i^-=-\nabla_i+f_iB,\qquad T=\sum_i A_i^*D_i^+,\quad T^\flat=\sum_i A_iD_i^-. \tag{9.7}$$

The identity $[B, A_j^*] = A_j^*$ cancels the derivative of r^{-1}: $D_i^+(r^{-1}A_j^*) = r^{-1}A_j^*D_i^+$, as operator products. The contraction of $r^{-2}A_i^*A_j^*$ with γ^{ij} vanishes. In a fixed orthonormal sphere trivialization,

$$P = 2T + U + J, \tag{9.8}$$

where $J \in \Psi^1$ has no base derivative and uniform bounded seminorms, and

$$U = \sum_{|\alpha|\le 2} U_\alpha \partial_y^\alpha, \qquad \partial_y^j U_\alpha \in C_j \varepsilon \beta^{j-|\alpha|}\Psi^{2-|\alpha|} \quad (0 \le j \le 5). \tag{9.9}$$

Crucially, its second-base part is exactly

$$U_2 = rQ^{ij}\partial_i\partial_j. \tag{9.10}$$

Its coefficient is a real scalar on every sphere. The reference connection contributes an angular first derivative to a base derivative, so replacing horizontal derivatives affects only base orders zero and one.

Here is an explicit allocation of terms. The pure angular perturbation $(h^{ij}/r)A_i^*A_j^*$ is $\varepsilon\Psi^2$. The mixed perturbation $2h^{ij}A_i^*D_j^+$ has first-base coefficient $O(\varepsilon r)\Psi^1$ and zero-base coefficient $O(\varepsilon r\beta)\Psi^2$; these obey the stated weaker bounds. The coefficients of $rQ^{ij}D_i^+D_j^+$ at base orders two, one, and zero are respectively $O(r)$, $O(r\beta)\Psi^1$, and $O(r\beta^2)\Psi^2$. This includes the derivative $rQ^{ij}f_{j;i}B$. Every product with E_i is bounded by the same classes. A product between $r^{-1}A_i^*$ and F_j, after multiplication by the outside r, is a uniformly bounded Ψ^1 term and is put in J. All other F terms, including $r\nabla_i F_j$, have the small bounds in (9.9). The term $b_2^iA_i^*$ is put in J; remaining drift terms have the small bounds. The five-derivative coefficient allowance is inclusive of full adjoints, commutators, and the final integration by parts; these are not extra derivatives added to five. The final error-form calculation needs at most four derivatives of each U coefficient. Constructing these coefficients costs at most one additional trace-coefficient derivative, so six trace derivatives and seven derivatives of f suffice. The stated eight-derivative allowance for the small tensor is sufficient at the coefficient-count level. Angular symbol seminorms require additional fixed smooth bounds for the first metric, but no additional small derivatives of the center-dependent tensor h.

Define the energy

$$\mathcal{N}(v)^2 = \int_Y \left(\|\nabla v\|_0^2 + \beta^2\|v\|_1^2 + \beta^{-1}\|\nabla v\|_{1/2}^2 + \beta\|v\|_{3/2}^2\right) dV_\gamma. \tag{9.11}$$

9.4 The raising estimate, degree by degree

We prove, for large enough β_0 depending only on the background geometry and profile constants,

$$\|Tv\|^2 - \|T^\flat v\|^2 \ge c\mathcal{N}(v)^2 \quad \text{when } v_0 = 0. \tag{9.12}$$

First let v have one degree $k \ge 1$. Write $a_i = \partial_{z_i}$ and $M_i = z_i$ in the Fischer space, where monomials have squared norm $\alpha!$. Then a_i and M_i are adjoints. The Fischer norm and the probability sphere norm on harmonic degree k satisfy

$$\|v\|_F^2 = N_k\|v\|_{S_y}^2, \qquad N_k = \prod_{j=0}^{k-1}(n+2j), \quad N_0 = 1.$$

One elementary verification contracts the coefficients of trace-free symmetric tensors against Gaussian moments. Pairings within a tensor vanish by trace-freeness; the surviving cross-pairings give its

Fischer norm. Gaussian radial integration gives $\mathbb{E}|X|^{2k} = N_k$ for a standard real Gaussian X. Also $\sum_i \|a_i v\|_F^2 = k\|v\|_F^2$.

Let $d = k + n/2 - 1$, and let q_i be harmonic of degree k. For $s = \sum_i a_i q_i$, the polynomial s is harmonic of degree $k-1$, and

$$\Delta_z \left(\sum_i M_i q_i \right) = 2s, \qquad \Delta_z(|z|^2 s) = 4ds.$$

Hence harmonic projection of $\sum_i M_i q_i$ is $\sum_i M_i q_i - |z|^2 s/(2d)$. Its two summands in the orthogonal harmonic decomposition have squared norms

$$\left\| \Pi_{k+1} \sum_i M_i q_i \right\|_F^2 = \sum_i \|q_i\|_F^2 + \sum_{ij} (a_j q_i, a_i q_j)_F - d^{-1}\|s\|_F^2.$$

Indeed the unprojected squared norm is the first two terms, by $a_j M_i = M_i a_j + \delta_{ij}$; the complement has squared norm $(2d)^{-2}(s, \Delta_z(|z|^2 s))_F = d^{-1}\|s\|_F^2$. Taking account of the sphere normalization gives $A_i^*|_{\mathcal{H}_k} = (N_{k+1}/N_k)\Pi_{k+1} M_i$.

At this degree $D_i^{\pm} = \pm\nabla_i + k f_i$. All norms in the next calculation use the Fischer inner product and base integration. Put $X = \|D^+ v\|_F^2$, $F = \|\sum_i a_i D_i^+ v\|_F^2$, $G = \|\sum_i a_i D_i^- v\|_F^2$. Then N_k times the left side of (9.12) is exactly

$$(2d+2)\left[X + \sum_{ij} (a_j D_i^+ v, a_i D_j^+ v)_F - d^{-1} F \right] - 2dG. \tag{9.13}$$

This includes the factors $N_{k+1}/N_k = 2d+2$ for raising and $N_k/N_{k-1} = 2d$ for lowering.

Set $C_{ij} = 2k f_{;ij}$ and $C[av] = \int \sum_{ij} C_{ij}(a_i v, a_j v)_F \, dV_\gamma$. For compactly supported tensor sections, the covariant commutator is

$$[D_j^-, D_i^+] = -C_{ij} + \mathcal{R}_{ji}.$$

Here $\mathcal{R}$ is a curvature rotation acting on degree k tensors. In fixed dimension its norm is at most Ck: each of the k tensor slots contributes a bounded curvature endomorphism. The derivative contractions contribute another factor k, so every curvature form in the next two lines has absolute value at most $Ck^2\|v\|_F^2$. Moving one covariant derivative across each pairing, and using the commutator displayed above, yields

$$\sum_{ij} (a_j D_i^+ v, a_i D_j^+ v)_F = G - C[av] + R_1, \quad |R_1| \le Ck^2\|v\|_F^2, \tag{9.14}$$

$$F \le k\|D^- v\|_F^2 - C[av] + R_2, \quad |R_2| \le Ck^2\|v\|_F^2, \tag{9.15}$$

$$\|D^- v\|_F^2 = X + \int \operatorname{tr} C \, |v|_F^2 dV_\gamma. \tag{9.16}$$

For the inequality, after reordering, the remaining mixed sum is bounded by $\sum_{ij} \|a_j D_i^- v\|_F^2 = k\|D^- v\|_F^2$ using $2\operatorname{Re}(a,b) \le |a|^2 + |b|^2$ and interchanging i, j. The last identity follows by expanding both squares; its difference is $2k \int \Delta_\gamma f \, |v|_F^2$. These steps use the metric connection on all contraction indices; differentiating only the scalar coefficients in a nonparallel frame would omit curvature terms.

Substitution in (9.13) gives

$$(2 + 2/d)(dX - F - dC[av]) + 2G + R_3, \qquad |R_3| \le Ck^3\|v\|_F^2. \tag{9.17}$$

We now retain a positive coefficient of k/β in front of $|D^+v|_F^2$ without differentiating β^{-1}. The contraction map $(q_i) \mapsto \sum_i a_i q_i$ on all homogeneous degree k polynomials has norm at most $\sqrt{k+n-1}$: the composition with its adjoint is $\sum_i a_i M_i = k+n-1$ on degree $k-1$. On harmonic subspaces its norm is no larger. Moreover, $D_i^+ + D_i^- = 2kf_i$ and $|\sum_i f_i a_i v|_F^2 \le k\beta^2 |v|_F^2$. Thus, for $0 \le \zeta \le 1$, pointwise

$$\left|\sum_i a_i D_i^+ v\right|_F^2 \le (1-\zeta)(k+n-1)|D^+v|_F^2 + \zeta\left(2\left|\sum_i a_i D_i^- v\right|_F^2 + 8k^3\beta^2|v|_F^2\right). \tag{9.18}$$

Fix $0 < \vartheta < (n-2)/(4(n-1))$. Apply (9.15)–(9.16) to the fraction $1-\vartheta$ of F in (9.17), and use the pointwise inequality on the remaining fraction with $\zeta = c_1/\beta$. The resulting coefficient of $|D^+v|_F^2$ is

$$(2+2/d)\left[d - k - \vartheta(n-1) + \vartheta c_1(k+n-1)/\beta\right]. \tag{9.19}$$

Because $d-k = (n-2)/2$, its constant part is positive; in dimension three one may take $\vartheta = 1/16$. The coefficient of the lowering square is $2 - 2(2+2/d)\vartheta c_1/\beta$, which is nonnegative after the fixed lower threshold for β is increased. The complete remaining Hessian term is

$$-(2+2/d)\left[(d-1+\vartheta)C[av] + (1-\vartheta)k\int \operatorname{tr} C\,|v|_F^2 dV_\gamma\right]. \tag{9.20}$$

The first coefficient obeys $d-1+\vartheta \ge c_n k$ for every $k \ge 1$. Also $-C[av] \ge 2c_0k^2\int\beta|v|_F^2$, and $-\operatorname{tr} C \ge 2nc_0k\beta$. Therefore this Hessian term is at least $c_H k^3 \int \beta|v|_F^2$, with $c_H > 0$ independent of k, δ, and the spatial dilation. The last term of (9.18) costs at most $C\vartheta c_1 k^3 \int \beta|v|_F^2$. Choose c_1 so small that it uses at most $c_H/4$ of the positive coefficient. Then choose β_0 so large that the curvature errors use at most another $c_H/4$. The result is

$$N_k(\|Tv\|^2 - \|T^\flat v\|^2) \ge c\left(X + \int (k/\beta)|D^+v|_F^2 + k^3\int \beta|v|_F^2\right).$$

The constants now fixed do not depend on k.

Finally

$$X = \|\nabla v\|_F^2 + k^2\int\beta^2|v|_F^2 - k\int \Delta_\gamma f\,|v|_F^2 \ge \|\nabla v\|_F^2 + k^2\int\beta^2|v|_F^2,$$

and $\frac{k}{\beta}|\nabla v|_F^2 \le 2\frac{k}{\beta}|D^+v|_F^2 + 2k^3\beta|v|_F^2$. Divide by N_k and sum the degree components. Both raising and lowering send distinct input degrees to distinct output degrees, so their norms sum separately without cross terms. For smooth sections first use finite harmonic sums and then their convergence in all the finite Sobolev norms occurring here. Since $1+k$ and k are comparable for $k \ge 1$, this proves (9.12).

9.5 Coercivity and cancellation of third base derivatives

The full adjoint of T lowers degree. On input degree k, applying A_i first changes the multiplication by B to $k-1$. Hence

$$Z := T^\flat - T^\dagger = \sum_i f_i A_i. \tag{9.21}$$

Expanding the two squared norms gives the exact identity

$$\|(2T+U)v\|^2 - \|(2T^\flat + U^\dagger)v\|^2 = 4(\|Tv\|^2 - \|T^\flat v\|^2) + (\mathcal{E}v, v), \tag{9.22}$$

with

$$\mathcal{E} = 2[T^\dagger, U] + 2[U^\dagger, T] + [U^\dagger, U] - 2(UZ + (UZ)^\dagger). \tag{9.23}$$

We claim the form estimate

$$|(\mathcal{E}v, v)| \le C\varepsilon \mathcal{N}(v)^2. \tag{9.24}$$

The following explicit coefficient calculation is necessary: merely calling U a small operator of order two would not prove this claim.

For coefficient bookkeeping assign count (m, s) to a base differential operator whose coefficient at ∂_y^α is $\beta^{s-|\alpha|}\Psi^{m-|\alpha|}$, with one additional factor of β per coefficient derivative. Products add the counts. To see this, a term in the Leibniz formula with j derivatives landing on a coefficient has base degree lowered by j and coefficient weight raised by j. The angular order stays fixed, so it also meets the asserted upper bound. For a commutator, the terms with no coefficient derivative have scalar angular principal symbols and lose one angular order. Every other term already loses at least one base derivative. A scalar commutator therefore has count at most $(m_1 + m_2 - 1, s_1 + s_2)$. The counts here are

$$T, T^\dagger : (2, 1), \qquad U, U^\dagger : (2, 0), \qquad Z : (1, 1).$$

Every term containing U carries ε; the quadratic commutator has ε^2. Thus $\mathcal{E}$ has count at most $(3, 1)$ with coefficient factor $C\varepsilon$ when $\varepsilon \le 1$.

This count alone would permit a scalar coefficient of a third base derivative, which is not controlled by $\mathcal{N}$. Here is why none survives. In an orthonormal sphere trivialization, let $\mu(y)dy$ be the base volume and write

$$U = a^{ij}\partial_i\partial_j + \mathcal{B}^i\partial_i + \mathcal{C}, \qquad a^{ij} = rQ^{ij} = a^{ji} \in \mathbb{R}.$$

The sphere measure is fixed in this trivialization. Integrating twice in the base, keeping the base density, gives

$$\begin{aligned} U^\dagger - U &= V^i\partial_i + W, \\ V^i &= 2\mu^{-1}\partial_j(\mu a^{ij}) - \mathcal{B}^i - (\mathcal{B}^i)^*, \\ W &= \mu^{-1}\partial_i\partial_j(\mu a^{ij}) - \mu^{-1}\partial_i(\mu(\mathcal{B}^i)^*) + \mathcal{C}^* - \mathcal{C}. \end{aligned}$$

For any $T^\dagger = t^\ell\partial_\ell + t^0$, the coefficient of $\partial_\ell\partial_i\partial_j$ in $[T^\dagger, U]$ is $[t^\ell, a^{ij}] = 0$, since a^{ij} is scalar in the angular variable. The same calculation applies to $[U^\dagger, T]$. For the quadratic commutator use $[U^\dagger, U] = [U^\dagger - U, U]$; its possible cubic coefficient is now $[V^\ell, a^{ij}] = 0$. All coefficients where a base derivative lands on a, V, or the density have base degree at most two. Finally UZ has at most two base derivatives, as does its full adjoint. This checks each possible source of a cubic derivative, rather than relying on its principal symbol in the combined base and angular cotangent variables.

Together the count and cancellation leave precisely the classes

$$\varepsilon\beta^{-1}\Psi^1\partial_y^2, \qquad \varepsilon\Psi^2\partial_y, \qquad \varepsilon\beta\Psi^3, \tag{9.25}$$

and terms of lower angular order. For the first type, integrate one base derivative. Its principal form is bounded by $C\varepsilon\int \beta^{-1}\|\partial_y v\|_{1/2}^2$, because a Ψ^1 operator maps $H^{1/2}$ to $H^{-1/2}$. A derivative landing on its coefficient, on β^{-1}, or on the smooth measure leaves $\varepsilon\Psi^1\partial_y$. For the second type, the mapping $\Psi^2 : H^{1/2} \to H^{-3/2}$ gives

$$C\varepsilon\int \|\partial_y v\|_{1/2}\|v\|_{3/2} \le \frac{C\varepsilon}{2}\int \left(\beta^{-1}\|\partial_y v\|_{1/2}^2 + \beta\|v\|_{3/2}^2\right).$$

For the third type use $\Psi^3 : H^{3/2} \to H^{-3/2}$. The extra $\Psi^1\partial_y$ term satisfies the same weaker bound. An ordinary base derivative differs from the horizontal derivative by a bounded angular first derivative. Thus $\|\partial_y v\|_{1/2} \le C(\|\nabla v\|_{1/2} + \|v\|_{3/2})$; the new weighted term is absorbed since $\beta^{-1} \le \beta$. A finite fixed atlas, with a subordinate partition in these form calculations, has the same controlled coefficient derivatives. This proves (9.24).

Let c_R be the constant in (9.12). Choose ε_0 so that $C\varepsilon_0 \le 2c_R$. Dropping the nonnegative second norm in (9.22) yields $\mathcal{N}(v) \le C_R\|(2T + U)v\|$. Since $J \in \Psi^1$, $\|Jv\| \le C_J\|v\|_{L^2(H^1)} \le C_J\beta_0^{-1}\mathcal{N}(v)$. Increase β_0 so that $C_R C_J/\beta_0 \le 1/2$; the profile constants and the bound for J are independent of that lower threshold. It follows that

$$\mathcal{N}(v) \le 2C_R\|Pv\| \qquad (v_0 = 0). \tag{9.26}$$

All constants were fixed independently of δ and of sufficiently small spatial dilations. This proof used finite angular symbol seminorms pointwise in the base, and a single integration by parts in the final error forms. It did not use uniform bounds for higher base derivatives of the unknown density.

9.6 Cutoff terms and the density bound

Let D be a fixed compact neighborhood of the supports of all nonzero derivatives of χ. Choose it within the two regions in `balls:cutoff-region`. The given construction of the transfer density satisfies $\int \psi\, d\sigma_y = 1$. In particular $v = \chi(\psi - 1)$ has zero fiber mean exactly, even though P need not preserve the mean-zero subspace.

The model operator annihilates 1 because $B1 = 0$ and $\nabla 1 = 0$. In (9.8), only the zero-base coefficient of U acts on 1. Its bounded $\varepsilon\Psi^2$ seminorms give $\|U1\|_{L^2(S_y)} \le C\varepsilon$, and $\|J1\| \le C$. Hence $\|\chi P1\| \le C$, without any positive power of β.

There is also a cutoff estimate

$$\|[P, \chi]w\| \le C_\chi\big(\|w\|_{L^2(D;H^1)} + \|r\nabla w\|_{L^2(D)}\big). \tag{9.27}$$

Indeed $[2T, \chi] = 2A_i^*(\nabla_i\chi)$. The second-base part gives $[a^{ij}\partial_i\partial_j, \chi] = 2a^{ij}(\partial_i\chi)\partial_j + a^{ij}(\partial_i\partial_j\chi)$. The first-base coefficient gives $\mathcal{B}^i\partial_i\chi$, a bounded Ψ^1 family because its size is at most $C\varepsilon\beta^{-1}$. Both J and zero-base coefficients commute with multiplication by χ. When the base derivative is rewritten horizontally its additional angular first derivative has coefficient $O(r)$ and is included in the first norm.

For the actual smooth density at a fixed $\delta > 0$,

$$P(\chi(\psi - 1)) = -\chi P1 + [P, \chi](\psi - 1). \tag{9.28}$$

The cutoff right side is uniformly bounded as follows. On the lower known region the transfer is evaluation, so ψ is the Poisson density at the center of the rescaled unit ball. The center remains at unit distance from the rescaled boundary. Its angular smooth norms and its derivatives with respect to (y, r) are uniformly bounded down to $r = 0$ by the interior evaluation-kernel estimates for the uniformly elliptic Dirichlet family. The chain rule introduces dr, whose norm is $r\beta \le \varepsilon$. On the other part of D, $t > t_b$ and $r \ge R_* t_b^p > 0$; after p, R_*, t_b are fixed, all needed radius derivatives are bounded, uniformly in δ. The fixed-separation kernel bounds and the same Dirichlet construction give uniform angular H^1 bounds and first base derivative bounds there. Compactness of D therefore makes the right side of (9.28) uniformly bounded in L^2.

Applying (9.26), and using $\beta \ge 1$ to control the L^2 norm, proves the source conclusion

$$\|\chi\psi\|_{L^2(S_\gamma Y)} \le \|\chi(\psi - 1)\|_{L^2} + \|\chi\|_{L^2} \le C. \tag{9.29}$$

The argument is an a priori bound for an already constructed smooth density for each positive δ. It does not claim convergence of ψ_δ, positivity of ψ_δ, or an inverse for P.

9.7 Precisely retained analytic obligations

The derivation above fills in the algebraic and weighted form estimates of `angular:raising` and `angular:coercivity`, including the potential third-base-derivative terms. It does not supply an accepted formalization of the following standard analytic machinery. Each item must receive an exact published reference interface and applicability check, or a direct proof, before this section is marked dependency-closed.

1. The rescaled smooth Dirichlet family has uniformly bounded inverse between the required Sobolev spaces; boundary and interior regularity persist after a finite number of parameter derivatives. In particular, the radial derivative trace is a smooth family in the complete classical Ψ^1 topology, its first parameter derivative at $r = 0$ has order zero, and its smoothing remainder satisfies the same parameter estimates. The explicit interface is (9.4) with six center derivatives after composition with $r(y)$, not just pointwise existence of a Dirichlet-to-Neumann operator. The source cites Joshi–Lionheart, *Asymptotic Analysis* 41 (2005), 93–106, Theorem 1.1(i) and Proposition 3.1. This is published boundary-symbol background; its parameter consequence must be derived, not silently attributed verbatim to either numbered result.

2. The scalar classical sphere calculus has continuous composition and adjoint maps, an order drop by one in scalar commutators, and Sobolev mapping bounds controlled by finitely many symbol and smoothing seminorms. The selected monograph source is Hörmander, *The Analysis of Linear Partial Differential Operators III*, Chapter XVIII, Section 18.1. The exact edition and theorem texts must still be fixed before this interface is accepted; the book's general scope does not replace that check.

3. The uniform cutoff-region bounds and local harmonicity of the transfer density depend on the actual product continuation and functional construction in `balls:continuation` and `balls:functional`. The angular calculation supplies no substitute for these inputs. Failure of either input prevents (9.29) from being applied.

For the elementary Fischer normalization, the matching published calculation is Render, *Archiv der Mathematik* 91 (2008), 136–144, Equation (2.1) and Theorem 2.1; the author-repository text was checked. The calculation was given directly above, so no novelty is outsourced to that reference. The tensor-tomography papers cited in the source are context for raising and lowering methods, not replacements for the weight-dependent estimate (9.12).

No new hypothesis on the selected metrics, no conditional replacement for the frozen uniqueness theorem, and no trusted analytic axiom is introduced by these authoring notes. Until the listed dependencies and the complete proof route are checked, this remains an expanded candidate argument.

10 From harmonic differences to local metric extension

Lemma 8.4 bounds the transfer density uniformly on the shrinking spheres. Exact reproduction of affine functions now turns that bound into a two-power estimate for the difference of paired harmonic functions. The harmonic Hessians convert derivatives of these differences into metric derivatives. Finite-order interpolation will then improve the temporary metric bound and allow the radii to shrink with the spatial dilation held fixed.

10.1 From the functional to harmonic jets

Return to the finite harmonic family in Theorem 5.1. Its values and first derivatives at 0 agree between the two metrics, because the functions agree on the one-sided open set and are smooth. After dilation define

$$v^a_{j,\lambda}(y) = \lambda^{-2}\big(u^a_j(\lambda y) - u^a_1(0) - \lambda\, du^a_1(0)\cdot y\big). \tag{10.1}$$

The subtracted functions are harmonic because the coordinates are harmonic. These new pairs still agree on the known side and are transferred by T_y. For each fixed integer m they have uniform C^m bounds on every required fixed bounded coordinate neighborhood. For derivative orders $j \geq 2$, this follows from the factor λ^{j-2}; orders zero and one follow from the second-order Taylor remainder. Moreover,

$$\partial^2 v^a_{1,\lambda}(y) \longrightarrow \partial^2 u^a_1(0)$$

smoothly on fixed sets. Thus the Hessian spanning condition has a uniform rank bound after sufficiently small dilation. Write $V^a = v^a_{2,\lambda} - v^a_{1,\lambda}$.

Lemma 10.1 (Two-power moment bound)**.** *Under the coefficient condition* (6.10)*, the paired differences satisfy*

$$\left\| \frac{\chi V^a}{r_\delta^2} \right\|_{L^2(Y)} \leq C \tag{10.2}$$

with a constant independent of δ and sufficiently small λ.

Proof. Fix a center y and subtract from $v^a_{1,\lambda}$ its affine coordinate Taylor polynomial at y. Exact constant and coordinate reproduction in (6.15) cancels this polynomial in $T_y v^a_{1,\lambda} - v^a_{1,\lambda}(y) = V^a(y)$. The remainder on $\partial B_\gamma(y,r)$ is bounded by Cr^2, since coordinate distance and γ-distance are uniformly comparable and the functions have uniform second derivatives. Hence

$$|V^a(y)| \leq C r_\delta(y)^2 \|\psi(y,\cdot)\|_{L^2(S_y)}.$$

Lemma 8.4 proves (10.2). The coordinate volume measure and dV_γ are uniformly comparable. No derivative in y of the continued density is used here. □

We next express metric derivatives in terms of derivatives of the paired differences. Recovery of the conformal metric class from the hyperplane of harmonic Hessians is also used in [6, Proposition 4.1 and Remark 4.2]. All Hessians in this step are ordinary coordinate Hessians, not covariant Hessians. Because the coordinates are harmonic, the equations for $v^a_{j,\lambda}$ have no first-order term in these coordinates:

$$(g^{-1}_{k,\lambda})^{ij}\partial_{ij} v^a_{k,\lambda} = 0 \qquad (k = 1, 2).$$

Multiplication by the scalar factors in (6.2) therefore gives

$$h^{ij}\partial_{ij} v^a_{1,\lambda} = -Q^{ij}\partial_{ij} V^a, \qquad \operatorname{tr}(\gamma h) = 0. \tag{10.3}$$

Harmonicity puts each $\partial^2 v^a_{1,\lambda}(y)$ in the hyperplane annihilated by $\gamma^{-1}(y)$. The uniform rank bound established above shows that these matrices span this hyperplane for small λ. The extra equation $\operatorname{tr}(\gamma h) = 0$ completes an injective system on symmetric contravariant matrices: an element annihilating all those Hessians is a multiple of γ^{-1}, and its γ-trace is n times that multiple. The associated finite matrix has a smooth left inverse with uniform derivatives on Y. For example, in fixed coordinate bases use $(A^*A)^{-1}A^*$, where A is this injective matrix. Its smallest singular

value stays bounded below by the spanning condition and smooth convergence under dilation. Differentiating (10.3) at most eight times gives

$$H_8(y) \le C \sum_a \sum_{|\alpha|\le 10} |\partial^\alpha V^a(y)|. \tag{10.4}$$

The derivatives of Q, γ, and of the finite harmonic family are uniformly bounded. In particular, this inversion introduces no factor involving r_δ^{-1}.

The metric estimate requires derivatives of the paired differences through order ten, whereas Lemma 10.1 controls only their weighted L^2 norm. The following finite-order interpolation estimate bridges these two statements using the already fixed smooth bounds.

Lemma 10.2 (Finite-order interpolation)**.** *Let V be smooth on a neighborhood of $\overline{B(y,s)} \subset \mathbb{R}^n$, with $0 < s \le 1$ and $\|V\|_{C^m} \le A_m$ there. For every multi-index α of length $j < m$,*

$$|\partial^\alpha V(y)| \le C_{m,n}\big(s^{-j-n/2}\|V\|_{L^2(B(y,s))} + A_m s^{m-j}\big). \tag{10.5}$$

Proof. Expand $V(y+sz)$ at $z=0$ through degree $m-1$ on the unit ball. The remainder has absolute value at most $C_{m,n}A_m s^m$. The L^2 norm of the Taylor polynomial is therefore at most $s^{-n/2}\|V\|_{L^2(B(y,s))} + C_{m,n}A_m s^m$. On the finite-dimensional space of polynomials of degree at most $m-1$, every coefficient is bounded by a constant times the L^2 norm on the unit ball. Its z^α coefficient is $s^j\partial^\alpha V(y)/\alpha!$, giving the claim. □

10.2 A strict improvement of the metric bound

Set

$$D = 10 + n/2, \qquad \eta = \frac{1}{2D}, \qquad p > 2D, \qquad m \ge \lceil 10 + 3D \rceil. \tag{10.6}$$

We also take the integer p large enough for Proposition 8.3. Thus $\eta > 1/p$, and m is one fixed finite smoothness order. The constants may depend on these choices, which precede the final spatial dilation and all shrinkage.

Lemma 10.3 (Superlinear improvement)**.** *After the choices in (10.6), there are $C > 0$ and $\rho_0 > 0$, independent of δ and sufficiently small λ, such that the coefficient condition (6.10) implies*

$$H_8(y) \le C r_\delta(y)^{3/2} \quad \text{for } y \in K \text{ with } r_\delta(y) < \rho_0. \tag{10.7}$$

Proof. Put $r = r_\delta(y)$ and $s = r^\eta$. By Lemma 6.1, $r_\delta^{1/p}$ has a fixed Lipschitz constant. Since

$$\frac{s}{r^{1/p}} = r^{\eta - 1/p} \longrightarrow 0,$$

a sufficiently small threshold ρ_0 ensures

$$\tfrac{1}{2} r^{1/p} \le r_\delta(z)^{1/p} \le \tfrac{3}{2} r^{1/p} \quad (z \in B(y,s)).$$

The same fixed threshold ensures that this ball lies where $\chi = 1$, because K has a fixed buffer inside that set. The upper comparison and Lemma 10.1 imply

$$\|V^a\|_{L^2(B(y,s))} \le C r^2.$$

Apply Lemma 10.2 and the fixed uniform C^m bounds of the renormalized functions. For $j \le 10$ it gives

$$|\partial^\alpha V^a(y)| \le C_m\big(r^{2-\eta(j+n/2)} + r^{\eta(m-j)}\big) \le C_m r^{3/2}.$$

Both exponents are at least $3/2$ by (10.6). Equation (10.4) completes the proof. □

The gain in (10.7) is stronger than the temporary bound $H_8 \le \varepsilon r_\delta$. Its usefulness is that a single choice of the spatial dilation makes the improvement strict at every radius, including those above the small threshold.

10.3 Local extension of the isometry

Proof of Theorem 5.1. Make the linear normalization and spatial dilation of Section 6. All harmonic functions and their finite spanning family are chosen before the dilation. We now specify the remaining choices and hold each one fixed before making the next.

First choose the fixed conformal geometry, cylinder, depth function, and cutoff, keeping uniform background smooth bounds for $0 < \lambda \le \lambda_0$. Choose p, η, m as in (10.6), with p large enough for the geometric and angular estimates. Choose ε for those estimates and then R_* so that the balls lie in normal neighborhoods and $r_\delta \beta_\delta^2 \le \varepsilon$ for every δ. These choices fix every separation range needed to start the continuation and to bound the cutoff terms. Restrict λ_0 further so that Proposition 5.5 applies to these fixed ranges, with bounds uniform for $0 < \lambda \le \lambda_0$.

The constants C, ρ_0 of Lemma 10.3 are now fixed. Choose

$$0 < \rho < \rho_0, \qquad C\rho^{1/2} \le \varepsilon/2. \tag{10.8}$$

At points with $r_\delta < \rho$, that lemma improves the coefficient bound to $H_8 \le \varepsilon r_\delta/2$ whenever the temporary bound holds. At points with $r_\delta \ge \rho$, the same improvement will hold if

$$\sup_Y H_8 \le \varepsilon\rho/2. \tag{10.9}$$

Since $h \to 0$ smoothly under dilation, choose one $\lambda \le \lambda_0$ satisfying (10.9) and the additional initial bounds described next. This λ is kept fixed for the rest of the proof.

There is a uniform reason the coefficient bound holds off K for every δ. Outside K, either $y_n < -1/2$, where $h = 0$ after the fixed dilation is sufficiently small, or $t > 2t_b$, where

$$r_\delta \ge R_*(2t_b)^p > 0.$$

Smooth smallness of h therefore gives $H_8 \le \varepsilon r_\delta/2$ off K for every δ by reducing the same dilation once. At $\delta = 1$, the radius has a positive minimum on all of Y, so we can also ensure the initial coefficient bound there. These conditions involve only fixed positive constants, and are compatible with (10.9).

For this fixed metric pair on the dilated neighborhood, set

$$\mathcal{A} = \{\delta \in (0,1] : H_8(y) \le \varepsilon r_\delta(y) \text{ for every } y \in K\}.$$

It contains 1. At any $\delta \in \mathcal{A}$, the bound already established off K gives the full hypothesis (6.10). Proposition 6.2 and the uniform density estimate of Lemma 8.4 then apply. The moment bound, metric system, and interpolation argument give Lemma 10.3. The small-radius improvement (10.8) and the large-radius bound (10.9) give

$$H_8 \le \tfrac{1}{2}\varepsilon r_\delta \quad \text{on } K.$$

For every fixed positive δ, r_δ has a positive minimum on the compact set K. Thus $\delta \mapsto \max_K(H_8/r_\delta)$ is continuous on $(0,1]$. The set $\mathcal{A}$ is relatively closed, and the strict improvement makes it relatively open. Connectedness implies $\mathcal{A} = (0,1]$.

Finally let $\delta \downarrow 0$, with the spatial dilation, harmonic family, cutoff and all structural parameters fixed. On the fixed open neighborhood of 0 where $t < 0$, the radii tend to zero. Hence $h = 0$ there.

By (6.2), $g_{2,\lambda}$ is conformal to $g_{1,\lambda}$ on that neighborhood. Write $g_{2,\lambda} = e^{2\omega} g_{1,\lambda}$. The conformal change formula for the Laplace–Beltrami operator is

$$\Delta_{e^{2\omega} g} u = e^{-2\omega} \big(\Delta_g u + (n-2) \langle d\omega, du \rangle_g \big).$$

Apply it to every common harmonic coordinate. Since their differentials form a basis and $n-2 \neq 0$, it forces $d\omega = 0$. Since $g_{1,\lambda}(0) = g_{2,\lambda}(0) = I$, this constant is zero. This is the local harmonic-morphism rigidity mechanism of [14, Section 8], in the restricted-family form used in [6, Proposition 4.1]. The metrics therefore agree on a connected neighborhood of 0; undoing the dilation proves the theorem. □

For clarity, the two limits in this proof have different roles: spatial dilation is used once to obtain a fixed small metric difference, and the subsequent shrinking family improves the bound for that same difference. The parameter order is

$$\begin{aligned} \text{fixed geometry and harmonic family} \;&\longrightarrow\; (p, \eta, m) \;\longrightarrow\; (\varepsilon, R_*) \\ &\longrightarrow\; \rho \;\longrightarrow\; \lambda \text{ fixed} \;\longrightarrow\; \delta \downarrow 0. \end{aligned} \tag{10.10}$$

All estimates use finitely many derivatives selected before either limiting step. Theorem 5.1 is the local input that will prevent a maximal matched region from having an interior frontier.

11 Conditional completion of the shrinking-sphere bootstrap

Scope and dependency status. This section expands the implications in the original `06-bootstrap.tex`; it does not supply a new assumption to the selected unconditional theorem. Its analytic prerequisite is the following assertion about the *actual* continued Green functional. For the geometry, radius family, cutoff, and small coefficient condition defined below, the construction in the original Sections 3–5 must give a transfer density with exact affine and paired-function reproduction, and a uniform weighted-base L^2 bound. Those construction and estimate claims remain obligations until their proofs are checked. In particular, an arbitrarily prescribed bounded functional is not a substitute for the functional obtained from the two actual Dirichlet Green kernels. The deductions below establish the local metric extension conclusion *conditional on those precise earlier obligations*; they do not certify the local extension theorem by assuming that theorem itself.

11.1 Fixed local data and the exact analytic input

Use common harmonic coordinates near a contact point, taken to be 0, for two smooth three-dimensional metrics. Their coordinate matrices agree on an open set containing one side of a smooth hypersurface through 0. The actual Dirichlet Green kernels agree on the square of that open set away from the diagonal. Fix finitely many harmonic pairs u_j^a, $j = 1, 2$, agreeing on that set, whose first-side ordinary Hessians at 0 span

$$\{ H \in \mathrm{Sym}^2(\mathbb{R}^3) : (g_1^{-1}(0))^{ij} H_{ij} = 0 \}.$$

By continuity of each derivative, agreement on the open known side implies agreement of all metric and paired-function jets at 0. Make a fixed linear change of coordinates so that $g_1(0) = g_2(0) = I$. A further orthogonal change makes the tangent hyperplane horizontal, with the known side below it. Affine changes preserve harmonic coordinates and the Hessian spanning property.

On a fixed coordinate neighborhood large enough for every ball used below, set

$$g_{j,\lambda}(y) = g_j(\lambda y), \qquad G_{j,\lambda}(x, y) = \lambda G_j(\lambda x, \lambda y), \qquad 0 < \lambda \leq \lambda_0 \leq 1.$$

The Green normalization is the one for dimension three. The functions $u_j^a(\lambda y)$ and coordinate functions remain harmonic for $g_{j,\lambda}$. On each fixed compact coordinate set the two metric families tend to I smoothly as $\lambda \downarrow 0$. Every fixed compact subset of $\{y_3 < 0\}$ lies in the dilated known set for all sufficiently small λ.

The structural geometry is that of the original Section 4:

$$Y = \{(y', y_3) : |y'| \le 1,\ -1 \le y_3 \le 1\}, \qquad t_0(y) = y_3 + |y'|^2,$$
$$\gamma_\lambda = e^{-2Lt_0} g_{1,\lambda}, \qquad t(y) = \int_{1/8}^{t_0(y)} e^{-2Ls}\, ds,$$
$$Q_\lambda = \frac{3g_{2,\lambda}^{-1}}{\operatorname{tr}(\gamma_\lambda g_{2,\lambda}^{-1})}, \qquad h_\lambda = Q_\lambda - \gamma_\lambda^{-1}.$$

Here L is fixed first. The preceding geometric argument is required to give uniform normal-coordinate bounds and strict concavity of t for all sufficiently small λ. The identities

$$\operatorname{tr}(\gamma_\lambda h_\lambda) = 0, \qquad h_\lambda \longrightarrow 0 \quad \text{in } C^k(Y) \text{ for each fixed } k$$

follow directly from these formulas and smooth convergence of the metrics. If $g_{1,\lambda} = g_{2,\lambda}$ on an open set, then $h_\lambda = 0$ there, including every coordinate derivative of h_λ.

Choose $t_b > 0$ as in the source and let

$$K = \{y \in Y : y_3 \ge -1/2,\ t(y) \le 2t_b\} \Subset Y^\circ.$$

Fix a smooth compactly supported cutoff χ on Y° equal to one on a neighborhood of K, and put $b = \operatorname{dist}(K, \{\chi \ne 1\} \cup \partial Y) > 0$, with the distance to an empty set ignored. We may decrease b to ensure every Euclidean b-ball about K lies in the common coordinate neighborhood. All these sets and b are independent of δ and λ. Define

$$\ell_\delta(t) = \delta \log(1 + e^{t/\delta}), \qquad r_\delta(y) = R_* \ell_\delta(t(y))^p,$$
$$\beta_\delta(y) = |d \log r_\delta(y)|_{\gamma_\lambda}, \qquad H_8(y) = \sum_{|\alpha| \le 8} |\partial^\alpha h_\lambda(y)|.$$

The profile estimates needed here are

1. $r_\delta > 0$ for $\delta > 0$, and r_δ is continuous in (y, δ) on $Y \times (0, 1]$;
2. r_δ increases with δ;
3. $r_\delta(y) \to 0$ if $t(y) \le 0$;
4. $r_\delta(y) \ge R_* a^p$ whenever $t(y) \ge a > 0$;
5. for fixed p, R_*, $r_\delta^{1/p}$ has a Lipschitz constant L_r independent of δ and small λ;
6. R_* can be chosen so that $r_\delta \beta_\delta^2 \le \varepsilon$ for every $y \in Y$ and $0 < \delta \le 1$.

For example, the Lipschitz assertion follows from $r_\delta^{1/p} = R_*^{1/p} \ell_\delta \circ t$ and $0 < \ell_\delta' < 1$. Monotonicity and the smallness assertion allow the prior continuation argument to use all intermediate radii from 1 down to a fixed positive δ.

Transfer obligation used below. After the fixed choices of p, ε, R_* and a sufficiently small upper bound λ_0, the following must hold with one constant C_T for every $0 < \lambda \le \lambda_0$ and every $0 < \delta \le 1$ satisfying $H_8 \le \varepsilon r_\delta$ on all of Y. On the unit sphere S_y for $\gamma_\lambda(y)$, with probability measure $d\sigma_y$, the actual Green transfer functional has a smooth density $\psi_{\lambda,\delta}(y,\omega)$ such that

$$T_y u = \int_{S_y} \psi_{\lambda,\delta}(y,\omega) u(\exp_y^{\gamma_\lambda}(r_\delta(y)\omega))\, d\sigma_y, \tag{11.1}$$

$$T_y 1 = 1, \qquad T_y x^i = y^i, \qquad T_y(u_1^a(\lambda\,\cdot)) = u_2^a(\lambda y), \tag{11.2}$$

$$\int_Y \chi(y)^2 \int_{S_y} |\psi_{\lambda,\delta}|^2\, d\sigma_y\, dV_{\gamma_\lambda}(y) \le C_T^2. \tag{11.3}$$

Equations (11.1)–(11.3) are the original transfer, reproduction, and density claims. Their source labels are `balls:functional`, `balls:reproduction`, and `local:density`. They depend on `balls:continuation` and `angular:coercivity`. Smoothness for each positive radius alone does not imply (11.3). The bound is a base-and-fiber L^2 bound, not an assumed pointwise bound in y.

11.2 Renormalized pairs and two exact powers of radius

For every fixed pair define

$$v_{j,\lambda}^a(y) = \lambda^{-2}(u_j^a(\lambda y) - u_1^a(0) - \lambda\, du_1^a(0)\cdot y), \qquad V^a = v_{2,\lambda}^a - v_{1,\lambda}^a.$$

The subtracted affine function is harmonic for both metrics because these are harmonic coordinates. Its constant and linear coefficients are the correct Taylor coefficients for both pairs because their first jets at 0 agree. Taylor's formula with integral remainder bounds $v_{j,\lambda}^a$ and its first derivatives uniformly on the fixed coordinate neighborhood. For $k \ge 2$,

$$D^k v_{j,\lambda}^a(y) = \lambda^{k-2}(D^k u_j^a)(\lambda y).$$

Consequently every fixed C^m norm of every $v_{j,\lambda}^a$ has a bound independent of small λ. In particular, $\|V^a\|_{C^m} \le A_m$ for a common finite A_m after increasing it over the finite family. Also $D^2 v_{1,\lambda}^a \to D^2 u_1^a(0)$ smoothly on fixed compact sets. Linearity and (11.2) show that $T_y v_{1,\lambda}^a = v_{2,\lambda}^a(y)$.

For a fixed center y, subtract the first-order coordinate Taylor polynomial

$$P_y(x) = v_{1,\lambda}^a(y) + \partial_i v_{1,\lambda}^a(y)(x^i - y^i).$$

Exact reproduction gives $T_y P_y = P_y(y) = v_{1,\lambda}^a(y)$. The ordinary Taylor bound and uniform comparison of coordinate and γ_λ distances give

$$|v_{1,\lambda}^a(x) - P_y(x)| \le C r_\delta(y)^2 \quad \text{if } d_{\gamma_\lambda}(x,y) = r_\delta(y).$$

Since $d\sigma_y$ has mass one, Cauchy–Schwarz yields

$$|V^a(y)| = |T_y(v_{1,\lambda}^a - P_y)| \le C r_\delta(y)^2 \left(\int_{S_y} |\psi_{\lambda,\delta}|^2\, d\sigma_y\right)^{1/2}.$$

The coordinate volume density and dV_{γ_λ} have uniform positive upper and lower bounds. Squaring and integrating therefore proves

$$\left\| \frac{\chi V^a}{r_\delta^2} \right\|_{L^2(Y,dy)} \le C_M, \tag{11.4}$$

where C_M is independent of δ and λ in the range of the transfer obligation. No derivative of the density in the base variable has been taken.

11.3 Recovery of the normalized metric and its first eight jets

Harmonic coordinates remove the first-order part of the scalar Laplacian. Thus

$$(g_{j,\lambda}^{-1})^{ik}\partial_{ik}v_{j,\lambda}^{a}=0.$$

Multiplying by the positive scalar defining Q_λ or γ_λ^{-1} and subtracting gives

$$h_\lambda^{ik}\partial_{ik}v_{1,\lambda}^{a}=-Q_\lambda^{ik}\partial_{ik}V^{a},\qquad \operatorname{tr}(\gamma_\lambda h_\lambda)=0. \tag{11.5}$$

At every point the matrices $M_a=\partial^2 v_{1,\lambda}^a$ annihilate γ_λ^{-1} under contraction. Their limiting matrices span the five-dimensional trace-free hyperplane. Compactness of Y and smooth convergence give a uniform lower rank bound for all sufficiently small λ.

More explicitly, on the six-dimensional space of symmetric contravariant matrices set

$$\mathcal{A}_y(H)=((H:M_a(y))_a,\ \gamma_\lambda(y):H).$$

If $\mathcal{A}_y(H)=0$, the spanning condition implies $H=c\gamma_\lambda^{-1}$, and its last component is $3c$. Hence $H=0$. Choose fixed Euclidean bases for these finite-dimensional spaces. Then

$$\mathcal{L}_y=(\mathcal{A}_y^{*}\mathcal{A}_y)^{-1}\mathcal{A}_y^{*}$$

is a left inverse. Its derivatives through order eight are uniformly bounded: the smallest singular value is uniformly positive, the coefficients of $\mathcal{A}$ have uniform derivatives, and each derivative of a matrix inverse is obtained by differentiating $B^{-1}B=I$. No radius occurs in this inversion.

Applying this left inverse to (11.5), and then differentiating at most eight times, gives

$$H_8(y)\le C_H\sum_a\sum_{|\alpha|\le 10}|\partial^\alpha V^a(y)|. \tag{11.6}$$

Indeed each Leibniz term contains a derivative through order eight of $\mathcal{L}$ or Q_λ, multiplying a derivative of V^a of order between two and ten. The larger sum in (11.6) is convenient and valid. Uniform bounds for derivatives of Q_λ are background smooth bounds; they do not require solving the bootstrap inequality. This avoids a circular estimate for derivatives of h_λ on the right-hand side.

11.4 Finite-order interpolation with explicit three-dimensional exponents

For $0<s\le 1$, expand a smooth function $V(y+sz)$ to degree $m-1$ on $|z|\le 1$. If $\|V\|_{C^m(B(y,s))}\le A_m$, its Taylor remainder is bounded by $C_mA_ms^m$. The polynomial P in this expansion therefore satisfies

$$\|P\|_{L^2(B(0,1))}\le s^{-3/2}\|V\|_{L^2(B(y,s))}+C_mA_ms^m.$$

Each coefficient functional on polynomials of degree at most $m-1$ is bounded in the finite-dimensional $L^2(B(0,1))$ norm. The z^α coefficient is $s^{|\alpha|}\partial^\alpha V(y)/\alpha!$. For $j=|\alpha|<m$ this proves

$$|\partial^\alpha V(y)|\le C_m\left(s^{-j-3/2}\|V\|_{L^2(B(y,s))}+A_ms^{m-j}\right). \tag{11.7}$$

Fix once and for all

$$\eta=\frac{1}{23},\qquad p\in\mathbb{N},\quad p>23,\quad p\text{ also meets the earlier coercivity threshold},\qquad m=45. \tag{11.8}$$

In particular $\eta - 1/p > 0$. Let $y \in K$ and write $r = r_\delta(y)$ and $s = r^{1/23}$. Choose $\rho_0 \in (0,1)$ so small that

$$L_r \rho_0^{1/23-1/p} \le \tfrac{1}{2}, \qquad \rho_0^{1/23} < b.$$

If $L_r = 0$ only the second condition is needed. For $r < \rho_0$ and $z \in B(y,s)$ the Lipschitz bound gives

$$\tfrac{1}{2} r^{1/p} \le r_\delta(z)^{1/p} \le \tfrac{3}{2} r^{1/p}.$$

Thus $B(y,s) \subset \{\chi = 1\}$ and $r_\delta(z) \le (3/2)^p r$. Applying (11.4) on this ball shows

$$\|V^a\|_{L^2(B(y,s))} \le (3/2)^{2p} r^2 C_M.$$

No factor $s^{3/2}$ is asserted: the information available is the global weighted L^2 bound. Formula (11.7) now gives, for $0 \le j \le 10$,

$$\begin{aligned} |\partial^\alpha V^a(y)| &\le C\left(r^{2-(j+3/2)/23} + r^{(45-j)/23}\right) \\ &\le C' r^{3/2}. \end{aligned}$$

The smallest first exponent occurs at $j = 10$ and equals $2 - (23/2)/23 = 3/2$. The smallest second exponent is $(45-10)/23 = 35/23 > 3/2$. Since $0 < r < 1$, both terms are bounded by a constant times $r^{3/2}$. Combining with (11.6) proves

$$H_8(y) \le C_* r_\delta(y)^{3/2} \quad (y \in K,\ r_\delta(y) < \rho_0), \tag{11.9}$$

whenever the full coefficient condition holds. Constants may depend on the fixed p and the chosen finite family, but are independent of δ and every λ in the already chosen sufficiently small range.

11.5 A single dilation, an open-and-closed argument, and conformal rigidity

The order of choices is part of the assertion. First fix the original harmonic family and local charts; then fix the geometry and cutoff; then choose p, η, m; then choose ε and R_*. The fixed positive separation ranges at $\delta = 1$ and on the cutoff region are now determined. The earlier positive-separation continuation must give one upper bound $\lambda_0 > 0$ and uniform constants throughout $0 < \lambda \le \lambda_0$ for those ranges. Only after this uniform input may C_* and ρ_0 above be fixed. Choose

$$0 < \rho < \rho_0, \qquad C_* \rho^{1/2} \le \varepsilon/2.$$

Now select a *single* $\lambda \le \lambda_0$ with

$$\sup_Y H_8 \le \frac{\varepsilon}{2} \min\left\{\rho,\ R_*(2t_b)^p,\ \min_Y r_1\right\}, \tag{11.10}$$

and so small that $\{y \in Y : y_3 \le -1/2\}$ lies in the dilated known side. All numbers inside the minimum are fixed and strictly positive; smooth convergence of h_λ makes the choice possible. The fixed λ is not changed again.

For every $\delta > 0$, the coefficient bound already holds with factor $1/2$ off K. Indeed outside K either $y_3 < -1/2$, where $H_8 = 0$, or $t > 2t_b$, where $r_\delta \ge R_*(2t_b)^p$ and (11.10) applies. At $\delta = 1$ it also holds on K. Define

$$F(\delta) = \max_{y \in K} \frac{H_8(y)}{r_\delta(y)}, \qquad \mathcal{A} = \{\delta \in (0,1] : F(\delta) \le \varepsilon\}.$$

The numerator now denotes one fixed metric difference. For each $\delta_0 > 0$, r_δ has a positive lower bound on $K \times [\delta_0/2, \min(1, 3\delta_0/2)]$. Uniform continuity on this compact set shows that the ratio varies continuously in the uniform norm, so F is continuous on $(0, 1]$.

We have $1 \in \mathcal{A}$. If $\delta \in \mathcal{A}$, the bound off K provides the full hypothesis of the transfer obligation. At points of K with $r_\delta < \rho$, (11.9) gives $H_8 \le \varepsilon r_\delta/2$. At the other points (11.10) gives the same inequality. Thus $F(\delta) \le \varepsilon/2$ for every $\delta \in \mathcal{A}$. Continuity makes $\mathcal{A}$ relatively closed. The strict margin and continuity make it relatively open: near such a δ, $F < 3\varepsilon/4 < \varepsilon$. Connectedness of $(0, 1]$ therefore gives $\mathcal{A} = (0, 1]$.

Because $t(0) < 0$ and 0 lies in the interior of K, choose a connected open neighborhood O of 0 with $O \subset K \cap \{t < 0\}$. For each fixed $y \in O$, send $\delta \downarrow 0$ in $H_8(y) \le \varepsilon r_\delta(y)$. The left side is fixed and nonnegative, and the right side tends to zero; hence $h_\lambda = 0$ throughout O. The trace normalization then implies $g_{2,\lambda} = c(y) g_{1,\lambda}$ for a smooth positive c. Write $c = e^{2\omega}$. In dimension three the direct conformal Laplacian calculation is

$$\Delta_{e^{2\omega} g} u = e^{-2\omega} \left(\Delta_g u + \langle d\omega, du \rangle_g \right).$$

Apply this to the three common harmonic coordinates. It yields $(g_{1,\lambda}^{-1})^{ik} \partial_i \omega = 0$ for every k, so $d\omega = 0$. Connectedness of O makes ω constant, and $g_{1,\lambda}(0) = g_{2,\lambda}(0) = I$ makes the constant zero. Thus the coordinate metrics agree near 0. Undoing the one fixed dilation and the affine normalization gives the asserted local metric extension, subject exactly to the prior transfer obligations.

12 The interior global isometry

We now apply Theorem 5.1 to the data constructed in Section 3. The source evaluations identify which points should correspond. Their separation makes this correspondence single-valued, and their behavior at the boundary prevents its escape from the interior. The local theorem then removes every interior frontier of the isometric correspondence. The use of a maximal matching set and limits of source evaluations follows the source-embedding architecture in [6, Section 6.1]; Theorem 5.1 supplies the smooth continuation step needed here.

For this specialization put $M_1^e = M_2^e = \overline{N}$. Retain their common annulus E, the open source set $U_0 \Subset E$, and the functionals I_j of Equation (4.1). For this section an *admissible match* is a smooth local isometry

$$\Phi : W \longrightarrow (M_1^e)^\circ, \qquad W \subset (M_2^e)^\circ,$$

where W is connected, open, and contains E, such that

$$\Phi|_E = \mathrm{Id}, \qquad I_2(p) = I_1(\Phi(p)) \quad (p \in W). \tag{12.1}$$

Thus $\Phi^* g_1 = g_2$ on W. The identity on E is an admissible match by Proposition 3.2.

Proposition 12.1 (No interior frontier)**.** *There is an isometry*

$$\Phi : (M_2^e)^\circ \longrightarrow (M_1^e)^\circ$$

onto the first interior satisfying Equation (12.1)*.*

Proof. The maximal match and its Green kernel. Two admissible matches agree wherever their domains overlap, since I_1 separates interior points by Lemma 4.1. Their union therefore defines a single smooth local isometry on the union W of all admissible domains. This union is connected because every domain contains the connected set E. The union map is injective: equal images give

equal I_2 evaluations, hence equal original points. It is itself an admissible match and is maximal by construction.

We first record the Green equality on this larger region:

$$G_2(p,q) = G_1(\Phi(p),\Phi(q)), \qquad p,q \in W, \quad p \neq q. \tag{12.2}$$

For fixed $p \in W$, equality of its evaluations and the common source density give this equality for the other variable in U_0, initially in the distributional sense. Both Green functions are smooth away from their pole. After pullback by the local isometry, their difference is harmonic on $W \setminus \{p\}$. This set is connected, and equality holds on a nonempty open part of it. Unique continuation proves Equation (12.2).

A frontier partner and common coordinates. Suppose W is not the entire second interior. Connectedness gives an interior frontier point p. If $p_k \in W$ tends to p, compactness of M_1^e gives a subsequential limit q of $\Phi(p_k)$. For every fixed source f, continuity up to the boundary gives

$$I_2(p)(f) = I_1(q)(f).$$

The point q is interior: all boundary evaluations are zero, whereas $I_2(p)$ is nonzero. Any two such limits have the same evaluations and so coincide. It follows that every matched sequence tending to p has partners tending to this unique q. In particular, nearby matched points have partners in any prescribed neighborhood of q.

Both p and q are outside $\overline{U_0}$. The first statement holds because $\overline{U_0} \subset E \subset W$. If q belonged to $\overline{U_0}$, the common-cap identification would give $I_1(q) = I_2(q)$; separation on the second manifold would force $p = q \in E$, a contradiction.

Choose sources $f_1, \dots, f_n$ so that

$$y = (u^2_{f_1}, \dots, u^2_{f_n})$$

has independent differentials at p, and put $x = (u^1_{f_1}, \dots, u^1_{f_n})$ near q. The second tuple is also a coordinate system. Indeed the isometries at the points p_k give equality of the gradient Gram matrices

$$\left(\langle du^2_{f_a}, du^2_{f_b}\rangle_{g_2}(p_k)\right)_{a,b} = \left(\langle du^1_{f_a}, du^1_{f_b}\rangle_{g_1}(\Phi(p_k))\right)_{a,b}.$$

The first limit is positive definite, and so is the second. The coordinate values at p, q coincide. Restricting the charts gives a common coordinate ball V about this value, with closures away from $\overline{U_0}$. The preceding convergence observation ensures that every already matched point sufficiently near p has its partner in the first chart. In these coordinates its identification is $x = y$.

Let $A \subset V$ be the coordinate image of the already matched second-side points. It is open and has the center value in its relative frontier. The two coordinate metrics and every paired source function agree on A. By Lemma 4.1, choose finitely many first-side source functions whose ordinary Hessians span the hyperplane annihilated by g_1^{-1} at the center. After shrinking V, their spanning rank has a positive lower bound throughout V. Their second-side partners are harmonic there and agree with them on A.

A touching ball removes the frontier. We can now find a smooth ball touching the unknown part, without assuming any regularity of the frontier of A. Choose $c \in A$ sufficiently close to the center that

$$r = \operatorname{dist}(c, V \setminus A) > 0, \qquad \overline{B(c,r)} \Subset V.$$

For example, if the center has distance R from ∂V, take c within $R/3$ of it. Then $r < R/3$, and a minimizing sequence stays in a compact subset of V. Thus the distance is attained at some

$$z_* \in \partial B(c,r) \cap (V \setminus A).$$

On $B(c,r)$ the metrics, the finite harmonic pairs, and the Green kernels in both variables agree. Smoothness gives equality of the metric and harmonic-function jets at z_*, and the Hessian spanning condition holds there by our choice of V. All the inputs of Theorem 5.1 are therefore satisfied at z_*. It gives equality of the coordinate metrics on a neighborhood of z_*. We may take a small connected ball there meeting $B(c,r)$. Each paired source difference is now harmonic for the common metric and vanishes on that intersection. Unique continuation makes every pair agree on the same ball. The resulting local isometry thus matches all evaluations.

This isometry agrees with the existing one on every overlap, by separation of evaluations. Its union with W is an admissible match whose domain contains z_*, contradicting maximality. Consequently $W = (M_2^e)^\circ$.

Surjectivity. The image is open because Φ is a local diffeomorphism. It is also closed in the first interior. Indeed, if $\Phi(p_k)$ tends to an interior point q, a subsequence of p_k converges in the compact second manifold to p. Evaluation continuity again gives $I_2(p) = I_1(q)$, so p cannot lie on the boundary. Then p is in the domain and $\Phi(p) = q$. Connectedness of the first interior makes the image all of it. An injective surjective local diffeomorphism has a smooth inverse, proving the proposition. □

13 Globalization on the ball of radius two

This section specializes the original `07-globalization.tex` to the selected metrics on $\mathbb{R}^3$. Put $M = \overline{B(0,2)}$ and $E = \{x : 1 < |x| < 2\}$. The original metrics restrict to smooth metrics on M and coincide with the Euclidean metric on E. Fix a nonempty open $U_0 \Subset E$. For $f \in C_c^\infty(U_0)$ let

$$-\Delta_{g_j} u_f^j = f \quad \text{in } B(0,2), \qquad u_f^j|_{\partial B(0,2)} = 0, \qquad I_j(p)(f) = u_f^j(p).$$

The sign convention only fixes the Green normalization; both sides use the same convention. The energy-to-source-data bridge is a separate obligation. Its required conclusion here is

$$I_1(p) = I_2(p) \quad (p \in E). \tag{13.1}$$

The source-family arguments must also supply the following facts:

1. for fixed f, u_f^j is smooth up to the boundary;
2. $I_j(p) = 0$ on the boundary and $I_j(p) \neq 0$ in the interior;
3. I_j separates distinct interior points;
4. away from $\overline{U_0}$, the source solutions have arbitrary compatible second jets, and hence supply harmonic coordinates and a finite spanning family of ordinary harmonic Hessians;
5. the actual Dirichlet Green representation and interior unique continuation hold for these smooth elliptic metrics.

These are the precise uses of the original `data:jet-family` and standard elliptic prerequisites. None is being adopted as an extra trusted axiom. Together with the local extension implication proved above once its obligations are discharged, they suffice for the following global construction.

13.1 The maximal source-matching local isometry

An admissible match is a smooth local isometry $\Phi_W : W \to B(0,2)$, where $W \subset B(0,2)$ is connected and open and contains E, with

$$\Phi_W|_E = \mathrm{Id}, \qquad I_2(p) = I_1(\Phi_W(p)) \quad (p \in W).$$

The identity on E is admissible by (13.1). Any two admissible maps agree on every point of their overlapping domains: their image points have identical I_1 functionals, so separation on the first ball identifies them. Their union is therefore a well-defined smooth local isometry on the union W of all admissible domains. This domain is open and connected, since every constituent domain is connected and contains E. The union still matches every evaluation and fixes E; it is itself admissible and maximal. It is also injective: equal images imply equal I_2 evaluations, and separation on the second ball identifies the original points. No application of Zorn's lemma is needed.

For this maximal map one has

$$G_2(p,q) = G_1(\Phi_W(p), \Phi_W(q)) \quad (p,q \in W,\ p \neq q). \tag{13.2}$$

To prove it, fix $p \in W$. The evaluation identity and Green representation give equality of the two kernels in the source variable as distributions on U_0. The integration density there is the same Euclidean density for both metrics, and Φ_W is the identity there. Thus the asserted equality holds pointwise on $U_0 \setminus \{p\}$, using smoothness away from the pole. Its difference, viewed as a function of q, is harmonic on $W \setminus \{p\}$: pullback by a local isometry preserves harmonicity, and injectivity ensures there is no second preimage of the pole. This punctured open set is connected in dimension three. Indeed a path through the removed point can be modified inside a small coordinate ball to avoid it, using connectedness of the punctured ball. Interior unique continuation proves (13.2) throughout $W \setminus \{p\}$.

13.2 A unique partner for an arbitrary interior frontier point

Suppose $W \neq B(0,2)$. Since W is nonempty and open and the ball is connected, there exists $p \in \overline{W} \setminus W$ in the interior. Choose $p_k \in W$ with $p_k \to p$. Compactness of M gives a subsequence with $\Phi_W(p_k) \to q \in M$. For every fixed source f, continuity and matching imply

$$I_2(p)(f) = \lim_k I_2(p_k)(f) = \lim_k I_1(\Phi_W(p_k))(f) = I_1(q)(f).$$

The same subsequence proves this for each f; no compactness of a space of source functions and no diagonal extraction are required. The point q is interior, since boundary evaluations vanish and $I_2(p) \neq 0$. Every possible subsequential limit has the same functional $I_1(q)$ and therefore equals q. It follows that every matched sequence tending to p has partners tending to q. Equivalently, for each neighborhood V_q of q, all matched points sufficiently close to p have partners in V_q. Otherwise a contrary sequence and compactness would produce a different cluster behavior.

Both points lie outside $\overline{U_0}$. This is immediate for p because $\overline{U_0} \subset E \subset W$. If $q \in \overline{U_0}$, then $q \in E$ and $I_1(q) = I_2(q)$, whence $I_2(p) = I_2(q)$. Separation would give $p = q \in E$, a contradiction.

Choose sources f_1, f_2, f_3 such that $y = (u^2_{f_1}, u^2_{f_2}, u^2_{f_3})$ has independent differentials at p. Set $x = (u^1_{f_1}, u^1_{f_2}, u^1_{f_3})$ near q. These are also independent at q. To see this without assuming a derivative limit for Φ_W, use the isometry and evaluation equality at p_k to obtain equality of the Gram matrices

$$\left(\langle du^2_{f_a}, du^2_{f_b}\rangle_{g_2}(p_k)\right)_{a,b} = \left(\langle du^1_{f_a}, du^1_{f_b}\rangle_{g_1}(\Phi_W(p_k))\right)_{a,b}.$$

Passing to the limit makes the second matrix positive definite, because the first is positive definite at p. Both tuples are therefore harmonic coordinate systems. Evaluation equality gives $x(q) = y(p)$; subtract this common value.

After restricting both charts, choose a common coordinate ball $V = B(0,R)$ with compact closure in their images. Shrink it further so that the coordinate neighborhoods avoid $\overline{U_0}$ and every matched point in the second chart has its partner in the first chart. The latter is possible by the

partner convergence just proved. For such matched points the equality $x \circ \Phi_W = y$ makes the coordinate identification exactly the identity. Define

$$A = \{z \in V : y^{-1}(z) \in W\}.$$

Then A is open, $0 \in \overline{A} \setminus A$, the coordinate metrics agree on A, all coordinate source pairs agree on A, and (13.2) supplies equality of their Green kernels on $A \times A$ off the diagonal. Choose first-side sources whose ordinary Hessians span the five-dimensional harmonic Hessian hyperplane at 0. Shrink V once more so that the same finite family spans the corresponding hyperplane at every point of V, with a positive lower rank bound. This is possible by continuity; harmonicity ensures each Hessian stays in the correct hyperplane.

13.3 A touching ball; no regularity of the frontier is assumed

Choose $c \in A$ with $|c| < R/4$ and set $r = \operatorname{dist}(c, V \setminus A)$. Since A is open, $r > 0$. Since $0 \in V \setminus A$, $r \le |c| < R/4$. In particular $\overline{B(c,r)} \subset B(0, R/2) \Subset V$. A sequence in $V \setminus A$ with distances from c tending to r lies eventually in a compact ball strictly inside V. As $V \setminus A$ is relatively closed, a subsequence has a limit

$$z_* \in V \setminus A, \qquad |z_* - c| = r.$$

Every point of $B(c,r)$ belongs to A, by the definition of the distance. Thus the smooth sphere $\partial B(c,r)$ touches the unknown set at z_*, with its whole interior known. We have made no claim that ∂A is smooth or even a hypersurface.

Near z_* the ball contains one side of its smooth boundary sphere. On that side the metrics and the chosen harmonic pairs coincide, and the actual Green kernels coincide in both variables. Every derivative of a difference that vanishes on the ball also vanishes at z_* by continuity. The chosen Hessian rank condition holds at z_* because it holds throughout V. Translate the contact point to the origin and, if needed, make the affine normalization used above. These are exactly the hypotheses of the local extension argument, with genuine Green kernels on the compact ball M. Once its transfer obligations are proved, that argument yields equality of the two coordinate metrics on a neighborhood of z_*.

Take a small connected coordinate ball D about z_* in that neighborhood, with $D \Subset V$ and $D \cap B(c,r) \neq \varnothing$. For each source f, the difference of its coordinate representatives is harmonic for the now common metric on D and vanishes on the nonempty open set $D \cap B(c,r)$. Unique continuation gives equality on D for every f. The coordinate identity therefore gives a local isometry matching *all* source evaluations, not just the finite family used for the Hessian calculation.

On every overlap with W this local map agrees with Φ_W by separation of I_1. This reasoning works even if the overlap has more than one connected component. The union map is smooth, is a local isometry, fixes E, and matches all evaluations. Its domain is connected because the added ball meets W, and it contains $y^{-1}(z_*) \notin W$. This contradicts maximality. Consequently $W = B(0,2)$.

13.4 Surjectivity and smooth extension to the selected global target

Write Φ for the resulting injective local isometry of the second open ball into the first. Its image is open. It is also closed in the first open ball: if $\Phi(p_k) \to q \in B(0,2)$, choose a subsequence $p_k \to p \in M$. Evaluation continuity gives $I_2(p) = I_1(q)$. The right side is nonzero, so p is interior. The map is defined and continuous at p, hence $\Phi(p) = q$. Sequential closedness is closedness here, and connectedness of $B(0,2)$ implies surjectivity. A bijective local diffeomorphism has a smooth inverse, as follows by using its local inverse on each coordinate neighborhood.

14 Completing the exact global conclusion

For our specialization, set $M_1^e = M_2^e = \overline{N}$ in Proposition 12.1. Their underlying point sets are the same; their metrics can differ. The common connected cap in that proposition is the entire annulus E. The compactness used to take limits is ordinary compactness of $\overline{N}$, and the evaluation functions vanish on the outer sphere $|x| = 2$.

Lemma 14.1 (Smooth extension fixing the entire exterior)**.** *Suppose $\Psi : N \to N$ is a smooth diffeomorphism satisfying $\Psi^* g_1 = g_2$ and $\Psi|_E = \mathrm{Id}$. Then there is a smooth diffeomorphism $\Phi : \mathbb{R}^3 \to \mathbb{R}^3$ satisfying Equation (1.1) and $\Phi(x) = x$ for $|x| \geq 1$.*

Proof. The open sets N and $O = \{x : |x| > 1\}$ cover $\mathbb{R}^3$ and intersect in E. Define $\Phi = \Psi$ on N and $\Phi = \mathrm{Id}$ on O. These definitions agree on the intersection, so Φ is a well-defined smooth map. To check bijectivity without any boundary extension theorem, construct the inverse in the same way. The inverse Ψ^{-1} is the identity on E: for $y \in E$, $\Psi(y) = y$ and injectivity gives its unique preimage. Thus Ψ^{-1} on N and identity on O glue to a smooth map Θ on $\mathbb{R}^3$. For $x \in N$, $\Phi(x) = \Psi(x) \in N$, so $\Theta(\Phi(x)) = x$; for $x \in O$ the same identity follows from $\Phi(x) = x$ and $\Theta(x) = x$. The reverse composition is identical.

At a point x with $|x| = 1$, choose $x_k = (1 + 1/k)x \in E$ for $k \geq 2$. Continuity in N gives $\Phi(x) = \lim_k \Psi(x_k) = x$. This proves the identity for the closed exterior, including its boundary. The pullback equation holds on N by hypothesis and on O because both metrics and the map there are Euclidean. These open sets cover all points, including ∂B, so the equation holds everywhere. □

Assembly of the selected theorem. Lemma 1.3 makes the frozen real energies and infima legitimate. Lemma 3.1 and polarization give equality of the boundary quadratic forms. Proposition 3.2 then gives common source responses and Green data on E. Lemma 4.1 supplies separation, coordinates and a finite harmonic Hessian family. The local theorem 5.1, with the estimates and verification obligations recorded below, removes the interior frontier in Proposition 12.1. The resulting map Ψ meets Lemma 14.1. Its global extension is the bijection required by Theorem 1.2.

This assembly is also a precise logical reduction: all dependencies of the new local continuation theorem must be discharged to turn it into a proof of the selected theorem. The local theorem is not among the standard literature inputs and must not be replaced by an unproved axiom.

14.1 Why a common harmonic coordinate system removes scale

The last step of the local bootstrap can be checked directly. If $g_2 = e^{2\omega} g_1$, then

$$\sqrt{\det g_2} = e^{n\omega}\sqrt{\det g_1}, \qquad g_2^{-1} = e^{-2\omega} g_1^{-1}.$$

Substitution into the divergence formula, including the derivative of $e^{(n-2)\omega}$, gives for every smooth u

$$\Delta_{g_2} u = e^{-2\omega}(\Delta_{g_1} u + (n-2) g_1^{ij}(\partial_i \omega)(\partial_j u)). \tag{14.1}$$

In coordinates harmonic for both metrics take $u = x^a$ in this identity. Both Laplacians vanish, so $(n-2) g_1^{ia} \partial_i \omega = 0$ for every a. For $n = 3$, invertibility of g_1^{-1} gives $d\omega = 0$. It is constant on the connected coordinate neighborhood and zero on the known matching open set; hence it is zero throughout. This avoids importing a theorem about harmonic morphisms with a stronger hypothesis than the finite family actually available.

14.2 The matrix equation and the Lean structure fields

For tangent column vectors v, w at x, the isometry assertion says

$$v^T g_2(x) w = (D\Phi(x)v)^T g_1(\Phi(x))(D\Phi(x)w).$$

Taking $v = e_i, w = e_j$ for all basis vectors is precisely Equation (1.1). The construction gives the fields of `IsIsometryFixingExterior` in this order: the bijection and its inverse supply the Lean equivalence; smooth gluing supplies `smooth` and `smooth_inv`; the continuity argument at $|x| = 1$ supplies `fixes_exterior`; the bilinear calculation supplies `isometry`. There is no reversal of g_1, g_2 or of the Jacobian transpose convention. Each matrix equality concerns the globally given fields, rather than only equivalence classes of metrics.

15 Dependency, correspondence, and verification ledger

15.1 What was specialized and what was retained

The immutable source is the supplied 24 September 2026 archive. This expansion uses its local Sections 3–6 and the interior part of Section 7. Their original labels are retained even though the section numbers in this expanded version differ. New expansion labels use `expanded:`, or an authoring-fragment prefix; these labels are source correspondence aids, not Lean declaration names.

The original boundary-jet determination and cap attachment are replaced by the exact infimum bridge and the fixed radius-two ball. Its scalar conductivity consequence, arbitrary measured patches, nonorientable manifolds and multiple boundary components are outside the selected completion scope. The local lemmas remain stated for $n \geq 3$ where that shortens their algebra; they are used only at $n = 3$. The original general boundary-completion proof is replaced by smooth gluing on overlapping open subsets of $\mathbb{R}^3$.

Proof block	Source / expansion locator	Dependency and formal role
Frozen definitions	Definition 1.1; Lemma 1.3	Coordinate derivatives, determinant density, real infimum and exterior conditions; unchanged target meanings.
Infimum and DN bridge	Lemma 3.1; Equation (3.2)	S1 and ellipticity; new target-specific proof, no smooth-minimizer assumption.
Exterior source data	Proposition 3.2	S1–S3 and equal quadratic forms; replaces source boundary-jet recovery and cap geometry.
Separation and jets	Lemma 4.1	S2–S4, point-supported distributions and finite-dimensional duality.
Cross extension	Lemma 5.2 and subsequent geometric criterion	S2, S4, compact spectral theory; constructive series, exponent sum strictly greater than one.
Shrinking transfer	Section 6	Cross extension and actual Green kernels; conditional on the temporary $H_8 \leq \varepsilon r$ bound. Positive-radius existence alone gives no uniform limiting bound.

Proof block	Source / expansion locator	Dependency and formal role
Trace and raising	Lemmas 8.1, 8.2	S5 with checked parameter bounds; sphere/Fischer algebra and the negative Hessian weight. New moving-trace specialization is retained.
Perturbed coercivity	Proposition 8.3	Raising estimate, scalar second-base symbol, commutator and form bounds; not an external literature premise.
Uniform density	Lemma 8.4	Actual-transfer equation, coercivity, fixed cutoff regions and constant-mode normalization.
Bootstrap	Section 10	Uniform density, exact affine moments, finite Hessian system and finite-order interpolation; fix λ before $\delta \downarrow 0$.
Global matching	Proposition 12.1	Local extension, source separation and compactness; touching balls require no regular frontier hypothesis.
Exact global map	Lemma 14.1	Interior matching plus identity on the common annulus; gives all frozen fields.

15.2 Literature-relative staging without changing the target

For a later relative formalization one may introduce an explicitly named record of the exact standard interfaces S1–S5 and prove a separate theorem from that record. Its assumption list must be reported. Only after each field is proved from checked libraries or after an explicit change of authority approved by the user can that work count toward the appropriate completion claim. No such record or additional Lean axiom is created by this manuscript. The current unconditional target stays fixed.

The source-supported Runge alternatives in [6] require their precise support and connected-complement hypotheses. Our point-distribution argument avoids importing those alternatives as extra black boxes. Published quantitative Runge is background for the singular-system idea, not a bound uniform over the collapsing spheres. Similarly, the negative curvature theorems in the tensor-tomography references are not used: only the sphere identities discussed in the local calculation are relevant.

15.3 Checks still required before formal proof acceptance

1. **Exact analytic interfaces.** Pin the relevant theorem passages in the specified monograph editions and check signs, density conventions, parameter hypotheses, finite derivative orders and uniform constants. The proposed source families are not a certification that an unread book contains the complete specialized moving-family statement.

2. **Moving-sphere uniformity.** Verify the normal-trace symbol Taylor expansion including smoothing remainders, its base derivatives, and the complete eight-derivative budget in the actual operator. The expanded calculations identify the needed estimates; formal proof and an independent mathematical review remain outstanding. The uniform shrinking estimate cannot be inferred from off-diagonal Green regularity.

3. **Dependency closure.** The compact spectral, distribution, elliptic, sphere and Sobolev results must resolve to checked library theorems or explicitly approved relative interfaces with identical hypotheses. The original main theorem still has its unproved Lean body.

4. **Correspondence and checking.** New supporting declarations must be mapped to these locators, compiled with the frozen dependency baseline, independently reviewed against exact inputs, and checked through the existing trusted boundary. Authoring and PDF compilation do not perform those steps. Human semantic ratification remains pending.

The companion machine-readable correspondence and authoring record gives exact artifact hashes and records any specific unresolved mathematical finding from this expansion pass. It does not promote parallel authoring checks to independent semantic reviews. In particular, absence of a counterexample in this pass is not a certificate for the new local theorem.

S1 Frozen-source provenance and a practical reading order

The selected theorem file has SHA-256

`0d6df55af2a96f44872d5a4bfe52e3256bf8`
`c2051352c501b4aa6bf5bda00437`

and the preserved manuscript archive has SHA-256

`e8c81ef0a68cba233975ada7386a77456e15`
`ea03a38fd40f78d355d0ab2544e3.`

The line breaks above are typographical; each hash is a single 64-digit hexadecimal string. The freeze is version 002; this expansion has its own version 001 and does not replace the freeze.

For a rapid mathematical review, first read the exact target and the infimum bridge, then the error-form estimate in Proposition 8.3 together with its detailed expansion notes. Next check that Lemma 8.4 applies to the actual transfer operator, with constants independent of the shrinking parameter. Finally read the explicit bootstrap exponents and the touching-ball globalization. This order puts the most consequential analytic work before the straightforward assembly into the global Lean structure.

The most useful additional reference texts are McLean (Chapters 2–4, 6 and Appendices A–C), Hörmander III (elliptic operators, pseudodifferential calculus and boundary problems), and Hörmander IV (Chapter XXVIII). Gilbarg–Trudinger, Chapters 6, 8 and 9, supplies the preferred classical regularity statements. Baird–Wood is optional for the route written here because Equation (14.1) replaces its metric-recovery use by a direct calculation.

References

[1] W. McLean, *Strongly Elliptic Systems and Boundary Integral Equations*, Cambridge University Press, 2000. Chapters 2–4, 6 and Appendices A–C. Publisher front matter.

[2] D. Gilbarg and N. S. Trudinger, *Elliptic Partial Differential Equations of Second Order*, second edition, Classics in Mathematics reprint, Springer, 2001. Chapters 6, 8–9. Publisher record.

[3] L. Hörmander, *The Analysis of Linear Partial Differential Operators III: Pseudo-Differential Operators*, Classics in Mathematics, Springer, 2007. Chapters XVII–XX. Publisher record.

[4] L. Hörmander, *The Analysis of Linear Partial Differential Operators IV: Fourier Integral Operators*, Classics in Mathematics, Springer, 2009. Chapter XXVIII. Publisher record.

[5] P. Baird and J. C. Wood, *Harmonic Morphisms Between Riemannian Manifolds*, London Mathematical Society Monographs, New Series 29, Oxford University Press, 2003. Publisher record.

[6] M. Lassas, T. Liimatainen and M. Salo, The Poisson embedding approach to the Calderón problem, *Mathematische Annalen* 377 (2020), 19–67. Version of record.

[7] H. Koch and D. Tataru, Dispersive estimates for principally normal pseudodifferential operators, *Communications on Pure and Applied Mathematics* 58 (2005), 217–284. Published article.

[8] M. S. Joshi and W. R. B. Lionheart, An inverse boundary value problem for harmonic differential forms, *Asymptotic Analysis* 41 (2005), 93–106. Published article.

[9] A. Rüland and M. Salo, Quantitative Runge approximation and inverse problems, *International Mathematics Research Notices* 2019 (2019), 6216–6234. Published article.

[10] A. Zériahi, Bases communes dans certains espaces de fonctions harmoniques et fonctions séparément harmoniques sur certains ensembles de $\mathbb{C}^n$, *Annales de la Faculté des Sciences de Toulouse*, series 5, 4 (1982), 75–102. Journal archive.

[11] G. P. Paternain, M. Salo and G. Uhlmann, Invariant distributions, Beurling transforms and tensor tomography in higher dimensions, *Mathematische Annalen* 363 (2015), 305–362. Published article.

[12] C. Guillarmou, G. P. Paternain, M. Salo and G. Uhlmann, The X-ray transform for connections in negative curvature, *Communications in Mathematical Physics* 343 (2016), 83–127. Published article.

[13] H. Render, Reproducing kernels for polyharmonic polynomials, *Archiv der Mathematik* 91 (2008), 136–144. Published article.

[14] B. Fuglede, Harmonic morphisms between Riemannian manifolds, *Annales de l'Institut Fourier* 28 (1978), no. 2, 107–144. Journal archive.

S2 Finite-degree algebra retained for the first formal components

These lemmas formalize algebra used in the original angular estimate. They do not assert the existence of the moving trace density, the differential identities, or the pseudodifferential bounds. Those remain separate proof obligations. The constants below are convenient fixed choices, not a change to the selected metrics or theorem.

Lemma S2.1 (Positive degree coefficient)**.** *Let n, k, β, c be real numbers with $n \geq 3$, $k \geq 1$, $\beta > 0$ and $c \geq 0$. Put $d = k + n/2 - 1$. Then*

$$(2 + 2/d)\{d - k - (n-1)/8 + c(k+n-1)/(8\beta)\} \geq (n-2)/2 + ck/(4\beta).$$

Proof. Here $d > 0$. The expression in braces is at least $(n-2)/4 + ck/(8\beta)$: the difference between its constant terms is $(n-3)/8$, and the omitted fraction is $c(n-1)/(8\beta) \geq 0$. This lower bound is nonnegative. Multiplication by $2 + 2/d \geq 2$ gives the displayed result. This fixes $\vartheta = 1/8$ in (8.19); no upper bound on k/β is used. □

Lemma S2.2 (Squared-norm comparison absorption)**.** *Let $a, b, N, e, c, \varepsilon$ be real numbers with $N \geq 0$, $c > 0$, $\varepsilon \geq 0$, $\varepsilon \leq c/2$, and $b \geq 0$. Suppose*

$$a - b \geq cN + e, \qquad |e| \leq \varepsilon N.$$

Then $a \geq (c/2)N$. In particular $N \leq 2a/c$.

Proof. The absolute-value bound gives $e \geq -\varepsilon N$. Hence $a \geq b + (c - \varepsilon)N \geq (c/2)N$. Divide by the positive number $c/2$. For the angular application, a and b are squared norms and N is $\mathcal{N}(\varphi)^2$. The estimate on e must be proved from the actual commutator form; this lemma is not an analytic assumption supplying it. □

Lemma S2.3 (Bounded remainder absorption)**.** *Let n, p, j, C, a be nonnegative real numbers with $a \leq 1/2$, $n \leq C(p + j)$, and $Cj \leq an$. Then $n \leq 2Cp$.*

Proof. The two estimates give $(1 - a)n \leq Cp$, while $(1 - a)n \geq n/2$. Multiplication by two yields the claim. In the angular application $p = \|P\varphi\|$, $j = \|J\varphi\|$ and $n = \mathcal{N}(\varphi)$. The small remainder estimate supplying $Cj \leq an$ remains an analytic obligation. □

Lemma S2.4 (Exact adjoint comparison)**.** *Let E be a real inner-product space, not necessarily complete. Let $T, T^\dagger, U, U^\dagger, Z, Z^\dagger : E \to E$ satisfy the adjoint pairing identities for all pairs of elements of E. Thus, for example, $\langle Tx, y\rangle = \langle x, T^\dagger y\rangle$. No boundedness or everywhere-defined closed realization on the ambient L^2 completion is being asserted. For $v \in E$, set $T^\flat v = T^\dagger v + Zv$ and*

$$\mathcal{E}v = 2(T^\dagger Uv - UT^\dagger v) + 2(U^\dagger Tv - TU^\dagger v) + (U^\dagger Uv - UU^\dagger v) - 2(UZv + Z^\dagger U^\dagger v).$$

Then

$$\|2Tv + Uv\|^2 - \|2T^\flat v + U^\dagger v\|^2 = 4(\|Tv\|^2 - \|T^\flat v\|^2) + \langle \mathcal{E}v, v\rangle.$$

Proof. Expanding both squared norms leaves, after removing their fourfold model terms,

$$4\langle Tv, Uv\rangle - 4\langle T^\dagger v + Zv, U^\dagger v\rangle + \|Uv\|^2 - \|U^\dagger v\|^2.$$

The six adjoint pairing identities, the symmetry of a real inner product, and $\langle w, w\rangle = \|w\|^2$ identify this expression with $\langle \mathcal{E}v, v\rangle$. This is exactly the algebra in (8.23)–(8.24). For the original compactly supported smooth test class, the pairing identities still require covariant integration by parts. This component proves the algebra once those identities are supplied; it introduces no literature premise granting the operator estimate or its error bound. □

All-degree polynomial calculation for the original angular argument

This preparation proves a formal polynomial identity. It does not assert that formal partial derivatives have already been identified in Lean with analytic partial derivatives, that a Dirichlet solution depends smoothly on parameters, or that the actual moving-sphere Dirichlet-to-Neumann operator has a controlled remainder. Those analytic bridges remain separate proof obligations. The algebra here is one exact component of the original argument, and does not change the frozen metric-uniqueness target or add an analytic hypothesis to it.

Let I be any finite set of variables, let $n = |I|$, and work in $\mathbb{R}[X_i : i \in I]$. Set

$$R(X) = \sum_{i \in I} X_i^2, \qquad \Delta_{\mathrm{pol}} q = \sum_{i \in I} \partial_i \partial_i q.$$

The operators ∂_i are formal polynomial derivations. A polynomial is homogeneous of degree $m \in \mathbb{N}$ if every monomial with nonzero coefficient has total degree m. The zero polynomial is homogeneous in every degree. A polynomial is harmonic here precisely when $\Delta_{\mathrm{pol}} q = 0$.

Lemma S2.5 (Unrestricted radius product rule)**.** *For every polynomial q,*

$$\Delta_{\mathrm{pol}}(Rq) = R\Delta_{\mathrm{pol}} q + 4 \sum_{i \in I} X_i \partial_i q + 2nq.$$

Proof. Differentiate the finite sum defining R. For each i, the derivative of X_j is 1 when $j = i$ and 0 otherwise. Hence $\partial_i R = 2X_i$ and $\partial_i^2 R = 2$. The product rule gives

$$\begin{aligned} \partial_i(Rq) &= 2X_i q + R\partial_i q, \\ \partial_i^2(Rq) &= 2q + 2X_i \partial_i q + 2X_i \partial_i q + R\partial_i^2 q \\ &= 2q + 4X_i \partial_i q + R\partial_i^2 q. \end{aligned}$$

Sum over i. The constant summand contributes $2nq$, the cross term contributes $4\sum_i X_i \partial_i q$, and the final summand is $R\Delta_{\mathrm{pol}} q$. This proves the stated identity without any homogeneity or harmonicity assumption. □

Lemma S2.6 (Euler specialization)**.** *If q is homogeneous of degree m, then*

$$\Delta_{\mathrm{pol}}(Rq) = R\Delta_{\mathrm{pol}} q + (4m + 2n)q.$$

Proof. Euler's homogeneous identity is $\sum_i X_i \partial_i q = mq$. Substitution in Lemma S2.5 gives the claim. In Lean the existing theorem used is `MvPolynomial.IsHomogeneous.sum_X_mul_pderiv`; its natural number scalar multiplication is converted to multiplication by the corresponding constant polynomial. This step covers $m = 0$ and $q = 0$; there is no division by the degree. □

Proposition S2.7 (All-degree radial identity)**.** *If q is homogeneous of degree m and harmonic, then*

$$\boxed{\Delta_{\mathrm{pol}}((R-1)q) = (4m + 2n)q}.$$

Proof. Linearity of the formal Laplacian gives

$$\Delta_{\mathrm{pol}}((R-1)q) = \Delta_{\mathrm{pol}}(Rq) - \Delta_{\mathrm{pol}} q.$$

By Lemma S2.6, this is $R\Delta_{\mathrm{pol}} q + (4m + 2n)q - \Delta_{\mathrm{pol}} q$. Since $\Delta_{\mathrm{pol}} q = 0$, both terms containing $\Delta_{\mathrm{pol}} q$ vanish. No condition on m other than $m \in \mathbb{N}$ is needed. Even an empty variable type causes no problem for this identity: both finite sums and n are then zero. □

Proposition S2.8 (Explicit zero-boundary first variation)**.** *Set $\kappa = 4m + 2n$. If $\kappa \neq 0$ and q satisfies the hypotheses of Proposition S2.7, define*

$$w(X) = -\kappa^{-1}(R(X) - 1)q(X).$$

Then $\Delta_{\mathrm{pol}} w = -q$. Moreover for every real assignment $z : I \to \mathbb{R}$ with $\sum_i z_i^2 = 1$, polynomial evaluation gives $w(z) = 0$.

Proof. Formal differentiation is linear over real constants, so

$$\Delta_{\mathrm{pol}} w = -\kappa^{-1}\Delta_{\mathrm{pol}}((R - 1)q) = -\kappa^{-1}\kappa q = -q.$$

The nonzero condition is used only for canceling $\kappa^{-1}\kappa$. Evaluation is a ring homomorphism and $R(z) - 1 = 0$ on the stated sphere, so the factor $(R - 1)$ makes $w(z) = 0$. For the selected dimension three, $\kappa = 4m + 6 > 0$ automatically; the displayed nonzero condition is an algebraic condition for the general finite-variable corollary, not a new condition on the selected metrics. □

Exact use in the radial trace calculation. If v is harmonic homogeneous of degree $k \geq 1$ and $q = c^i \partial_i v$ for a constant real vector c, then q is homogeneous of degree $m = k - 1$ and harmonic: degree drops by one under formal differentiation, and partial derivatives commute with the Laplacian. The first fact is available as `MvPolynomial.IsHomogeneous.pderiv`; the derivative-commutation and analytic interpretation must be supplied if this corollary is used in the full formalization. Here

$$\kappa = 4(k - 1) + 2n = 2(2k + n - 2), \qquad w = -\frac{R - 1}{2(2k + n - 2)}\, c^i \partial_i v.$$

Thus the formula is exactly the polynomial solution of $\Delta w = -c \cdot \nabla v$ with zero boundary data on the unit sphere. At $k = 0$, $c \cdot \nabla v = 0$, and the first variation is zero without using the displayed denominator.

This supports the original source `05-angular-estimate.tex`, lines 107–133, equation (8.6), and the neutral expansion `expanded.tex`, lines 2725–2752, angular-expanded:radial-expansion. The same unshifted radius identity supports the harmonic projection calculation at original lines 285–295 and expansion lines 2891–2908: for harmonic s of degree $k - 1$, $\Delta_{\mathrm{pol}}(Rs) = [4(k - 1) + 2n]s = 4ds$ where $d = k + n/2 - 1$. To use this for harmonic projection, one must additionally show $s = \sum_i \partial_i q_i$ is harmonic and homogeneous of degree $k - 1$, and provide the Fischer-inner-product orthogonality argument. Those additional conclusions are not silently included in this algebra lemma.

Formal scope. The diagnostic Lean candidate quantifies over an arbitrary finite variable type and uses `MvPolynomial` over $\mathbb{R}$, not a finite list of degrees or numerical samples. Its main result is exactly Proposition S2.7. The auxiliary declarations prove the finite-sum product rule, Euler specialization, constant-multiplication linearity, first-variation forcing equation, and unit-sphere evaluation. No custom axiom or admission is permitted. Captured development compilation is separate from the trusted checking boundary and from registration in the working claim catalog.

S3 Scope, fixed quantities, and source authority

This is an abstract supporting component of the supplied original Section 6, not a metric uniqueness theorem. Its source is the exact neutral copy of `06-bootstrap.tex`, SHA256 `c1456afb32ed8d2e8097ccc4e02827a67e179330387ddf86d986a82db8b0aaa4`, especially lines 239–265, with parameter order at lines 284–296. This historical component used a neutral precursor expansion with SHA256 `3caecc6fc60cfef588d4f12ad6ee57d2046bc9aaf23332f5bea211e49276dae2`. Those line ranges refer to that historical precursor, not the current full manuscript; its corresponding passages were lines 3500–3527 and 3931–3957. The frozen full-boundary three-dimensional target is unchanged.

The structural choices, finite harmonic family, cutoff, exponents, thresholds, and spatial dilation λ have already been fixed. Only the shrinking parameter δ varies below. In particular the quantity $H(y)$ is a function of y, with no δ argument. The intended numerator is

$$H(y) = H_8(y) = \sum_{|\alpha|\le 8} |\partial^\alpha h_\lambda(y)|.$$

Its definition occurs at neutral expansion lines 1477–1478 and 3662. This component neither constructs h_λ nor proves its analytic estimates. All continuity, pointwise control, initialization, strict improvement, and radius limits below are theorem hypotheses; none is an axiom or a literature premise.

Definition S3.1. Let $I = \{\delta \in \mathbb{R} : 0 < \delta \le 1\}$ with the subspace topology inherited from $\mathbb{R}$. Let $\delta_1 \in I$ be the point with value 1. Relative openness and closedness mean openness and closedness as subsets of this space I. In Lean this is the subtype `PositiveUnitInterval := ↑(Set.Ioc (0 : ℝ) 1)`.

S4 Topological closure

Lemma S4.1. *For every subset $A \subset I$, if A is relatively closed and relatively open and A is nonempty, then $A = I$.*

Proof. The interval $(0, 1]$ is preconnected in $\mathbb{R}$, by the connectedness of real order intervals. Therefore its subtype I is a preconnected topological space. A nonempty clopen subset of a preconnected space is the whole space: otherwise it and its nonempty complement would give a separation by disjoint nonempty open sets. Apply this fact to A. The Lean proof obtains the subtype instance from `isPreconnected_Ioc` and `Subtype.preconnectedSpace`, then applies `IsClopen.eq_univ`. No compactness of $(0, 1]$ or membership of 0 is asserted or needed. □

Lemma S4.2. *Let X be any topological space, let $F : X \to \mathbb{R}$ be continuous, and let $\varepsilon \in \mathbb{R}$ satisfy $\varepsilon > 0$. Suppose*

$$\forall x \in X, \quad F(x) \le \varepsilon \implies F(x) \le \varepsilon/2. \tag{S}$$

Then $A = \{x \in X : F(x) \le \varepsilon\}$ is clopen in X.

Proof. Since F is continuous and $(-\infty, \varepsilon]$ is closed, A is closed. Since $\varepsilon > 0$, one has $\varepsilon/2 < \varepsilon$. If $x \in A$, hypothesis (S) gives $F(x) \le \varepsilon/2 < \varepsilon$. Conversely $F(x) < \varepsilon$ implies $F(x) \le \varepsilon$. Thus

$$A = \{x \in X : F(x) < \varepsilon\}.$$

The latter set is open by continuity. The Lean proof uses `isClosed_le`, `half_lt_self`, and `isOpen_lt` after proving this set equality. □

Lemma S4.3. *Let $F : I \to \mathbb{R}$ be continuous and let $\varepsilon > 0$. Suppose $F(\delta_1) \le \varepsilon$ and*

$$\forall \delta \in I, \quad F(\delta) \le \varepsilon \Longrightarrow F(\delta) \le \varepsilon/2. \tag{P}$$

Then $F(\delta) \le \varepsilon$ for every $\delta \in I$.

Proof. Define $A = \{\delta \in I : F(\delta) \le \varepsilon\}$. Lemma S4.2 shows A is clopen, while the initial bound gives $\delta_1 \in A$, hence $A \neq \varnothing$. Lemma S4.1 gives $A = I$. Evaluating membership at an arbitrary δ yields the asserted inequality. The initial point is exactly 1; no small-δ initialization is used. The proof does not change any of the fixed structural parameters as δ varies. □

S5 Collapse of a fixed nonnegative quantity

Lemma S5.1. *Let J be any type, let ℓ be a nontrivial filter on J, let $H, \varepsilon \in \mathbb{R}$, and let $r : J \to \mathbb{R}$. Suppose*

$$0 \le H, \qquad \forall i \in J,\ H \le \varepsilon r(i), \qquad r \longrightarrow 0 \text{ along } \ell.$$

Then $H = 0$.

Proof. The constant function $i \mapsto \varepsilon$ converges to ε, so multiplication of limits gives $\varepsilon r(i) \to \varepsilon \cdot 0 = 0$ along ℓ. The constant function $i \mapsto H$ converges to H. The inequalities are valid everywhere, hence eventually along ℓ. The order relation $\le$ is closed in $\mathbb{R}^2$; because ℓ is nontrivial, taking these limits yields $H \le 0$. Together with $0 \le H$, antisymmetry gives $H = 0$. This proof uses `tendsto_const_nhds.mul` and `le_of_tendsto_of_tendsto'`. Neither the sign of ε nor the sign of r is needed once the displayed hypotheses are given. Nontriviality is needed: at the bottom filter, every function has every limit, so $H = r = \varepsilon = 1$ would otherwise meet the limit and inequality hypotheses while $H \neq 0$. □

Lemma S5.2. *Let Y be any type, $K \subset Y$, $H : Y \to \mathbb{R}$, $r : \mathbb{R} \to Y \to \mathbb{R}$, and $\varepsilon \in \mathbb{R}$. Suppose:*

$$\forall y \in K, \quad 0 \le H(y), \tag{C1}$$

$$A = \{\delta \in I : \forall y \in K,\ H(y) \le \varepsilon r(\delta, y)\} \text{ is clopen in } I, \tag{C2}$$

$$\forall y \in K, \quad H(y) \le \varepsilon r(1, y), \tag{C3}$$

$$\forall y \in K, \quad r(\delta, y) \longrightarrow 0 \text{ as } \delta \to 0 \text{ within } (0, 1]. \tag{C4}$$

Then $H(y) = 0$ for all $y \in K$.

Proof. By (C3), $\delta_1 \in A$. Hence (C2) and Lemma S4.1 yield $A = I$. Thus for each real δ with $0 < \delta \le 1$ and each $y \in K$, $H(y) \le \varepsilon r(\delta, y)$.

Fix $y \in K$. Use the within-neighborhood filter

$$\ell = \mathcal{N}[(0, 1]](0) = \mathcal{N}(0) \wedge \text{principal}((0, 1]).$$

This filter is nontrivial because every neighborhood of 0 meets $(0, 1]$: it contains some $0 < \delta \le 1$. In Lean the exact fact is `left_nhdsWithin_Ioc_neBot zero_lt_one`. Membership in $(0, 1]$ holds eventually along ℓ by `self_mem_nhdsWithin`. Therefore the bounds obtained from $A = I$ hold eventually along ℓ. By (C4) and multiplication of limits, $\varepsilon r(\delta, y) \to 0$ along ℓ. The constant left side has limit $H(y)$, so `le_of_tendsto_of_tendsto` gives $H(y) \le 0$. Use (C1) and antisymmetry to conclude $H(y) = 0$. This proves the assertion for every $y \in K$. □

S6 Composition through a continuous ratio bound

Lemma S6.1. *Let Y be any type, $K \subset Y$, $H : Y \to \mathbb{R}$, $r : \mathbb{R} \to Y \to \mathbb{R}$, $F : I \to \mathbb{R}$, and $\varepsilon \in \mathbb{R}$. Assume all of the following:*

$$\forall y \in K, \quad 0 \le H(y), \tag{R1}$$
$$\forall \delta \in I \ \forall y \in K, \quad 0 < r(\delta, y), \tag{R2}$$
$$\forall \delta \in I \ \forall y \in K, \quad H(y) \le F(\delta) r(\delta, y), \tag{R3}$$
$$F \text{ is continuous on } I, \tag{R4}$$
$$0 < \varepsilon, \tag{R5}$$
$$F(\delta_1) \le \varepsilon, \tag{R6}$$
$$\forall \delta \in I, \quad F(\delta) \le \varepsilon \Longrightarrow F(\delta) \le \varepsilon/2, \tag{R7}$$
$$\forall y \in K, \quad r(\delta, y) \longrightarrow 0 \text{ as } \delta \to 0 \text{ within } (0, 1]. \tag{R8}$$

Then $H(y) = 0$ for every $y \in K$.

Proof. Apply Lemma S4.3 to (R4)–(R7) to obtain $F(\delta) \le \varepsilon$ for all $\delta \in I$. For arbitrary $\delta \in I$ and $y \in K$, multiply this inequality by the nonnegative number $r(\delta, y)$, using (R2), and combine with (R3). This gives

$$H(y) \le F(\delta) r(\delta, y) \le \varepsilon r(\delta, y).$$

Consequently the admissible set from Lemma S5.2 is all of I, hence clopen, and its initial bound at 1 holds. Apply that lemma using (R1) and (R8). The result is $H = 0$ on K. No topology, compactness, or smoothness structure on Y is needed in this abstract lemma; the only continuity assumption is (R4). □

S7 Exact original application and remaining obligations

For the original application, distinguish the bootstrap set K_{boot} from the collapse set K_{zero}. Take

$$F(\delta) = \max_{y \in K_{\text{boot}}} \frac{H_8(y)}{r_\delta(y)}, \qquad K_{\text{zero}} = O \subset K_{\text{boot}} \cap \{t < 0\}.$$

The maximum and positive radii give (R3) on K_{boot}, hence on O; (R4) concerns this same maximum on K_{boot}. The input set K in Lemma S6.1 is O. The original radius profile tends to zero on $\{t < 0\}$, so (R8) applies on O. The lemma does not claim collapse at points with $t > 0$, where the original radius need not tend to zero. One may alternatively use Lemma S4.3 first on the full maximum, then apply the scalar limit argument at each $y \in O$.

The component's completed conclusions are exactly relative clopen closure and collapse of a fixed nonnegative scalar quantity. To use them for the original local proof, future components must still prove: the fixed parameter order and choice of λ; existence and continuity of the maximum; radius positivity and its negative-side limit; the initial coefficient bound; and the uniform strict improvement assembled from the original continuation, density, moment, metric-system, and interpolation estimates. Hypothesis (R7) expresses that last project proof obligation and is not adopted as a standard analytic theorem. The later implication from $H_8 = 0$ to $h_\lambda = 0$, conformal normalization, removal of the conformal factor by common harmonic coordinates, undoing dilation, and globalization are outside this component.

The type of H encodes its independence from δ. Allowing an independent H_δ would change the conclusion: for instance $H_\delta = r_\delta = \delta$ tends to zero but is positive at every positive δ. The original fixed-dilation proof must supply one H, as it explicitly does at source lines 239 and 262–263.

The candidate Lean file proves all six lemmas by existing mathlib connectedness, clopen, order, and limit APIs. No custom axiom or `sorry` is used. This preparatory development compilation is separate from trusted source-correspondence acceptance of the main theorem.

S8 Clarifications and an elementary distributional derivation

These are authoring repairs to the exact expanded route. They introduce no extra hypothesis on the target metrics and do not assert independent validation. The original v001 audit inputs remain preserved byte for byte.

Inclusive coefficient derivative accounting. In (9.9), the allowance through five base derivatives includes the derivatives used in full adjoints, the commutator expansion, and the final integration by parts. They are not additional derivatives beyond that allowance. A conservative bound is four derivatives of each coefficient of base order zero, one, or two in the final error form. Construction of a U coefficient takes at most one derivative of a trace coefficient, so the stated five-derivative U allowance needs at most six trace derivatives and seven derivatives of f. Bounds for β are asserted only through the orders actually used: $|\partial^j \beta| \le C_j \beta^{j+1}$ for $j \le 7$ follows from f derivatives through eight. No eighth derivative of β is needed. The small tensor h is a center-dependent tensor, and angular symbol derivatives do not increase its required small base-derivative order.

Reference seed width before the final cylinder. The coarse retained relative width c_0 in the fixed-separation argument is selected in the scale-invariant Euclidean ball/annulus construction, using fixed reference shapes and interpolation exponents. This selection precedes the final transverse tube cylinder. Once that bounded cylinder, its positive separation range and derivative orders are fixed, choose the dilation threshold so that the rescaled smooth coefficients approximate those reference configurations on all compact sets used. The threshold may depend on that cylinder; the already selected relative reference width does not. This specifies the noncircular interpretation of the choice order in `cross:initial`.

Lemma S8.1 (Point-supported distributions)**.** *A distribution on a Euclidean open set whose support is one interior point p is a finite linear combination of derivatives of δ_p. A distribution supported on finitely many points is the sum of such components.*

Proof. Translate p to zero and take a compact neighborhood K of zero inside the open set. Continuity of the distribution on the test functions supported in K gives some integer m and constant C with

$$|T\phi| \le C \max_{|\alpha| \le m} \|\partial^\alpha \phi\|_\infty .$$

This is the usual test-function topology definition of a distribution; it is not an elliptic estimate. Suppose every derivative of ϕ at zero through order m vanishes. For a smooth compactly supported cutoff χ equal to one near zero, the support condition gives $T\phi = T(\chi(x/\rho)\phi)$ for every sufficiently small $\rho > 0$. Taylor's theorem gives

$$\sup_{|x| \le A\rho} |\partial^j \phi(x)| = o(\rho^{m-j}) \qquad (0 \le j \le m),$$

for the fixed radius A containing $\operatorname{supp} \chi$. Leibniz's rule therefore makes every derivative through order m of $\chi(x/\rho)\phi$ tend uniformly to zero: a cutoff derivative of order ℓ contributes $\rho^{-\ell} o(\rho^{m-j+\ell})$. The finite-order continuity bound forces $T\phi = 0$.

Thus T factors through the finite jet $(\partial^\alpha \phi(0))_{|\alpha| \le m}$. Every such jet is attained by a polynomial multiplied by a cutoff equal to one near zero. A linear functional on this finite-dimensional jet space is a linear combination of the coordinate evaluations; using $(\partial^\alpha \delta_0)(\phi) = (-1)^{|\alpha|} \partial^\alpha \phi(0)$ gives the assertion. For finitely many support points, use disjoint cutoffs equal to one near each point; their sum equals one near the support, so the corresponding point-supported components sum to T. □

The jet and separation arguments following `data:jet-family` may use this derivation. They still require the exact transposed Dirichlet inverse, distributional elliptic regularity, and scalar unique continuation. This elementary support argument does not supply any of those analytic inputs or the novel local continuation theorem.

S9 Derived standard boundary-family supplement

This supplement supplies a mathematical derivation for two standard analytic interfaces in the frozen expansion: its unit-ball radial trace family and its center Poisson kernel. This section gives the mathematical source derivation. Its procedural review record is kept separately; no Lean proof acceptance is asserted here. No theorem about the mixed continuation, the transfer equation, or its coercivity is a premise here. The full-boundary smooth three-dimensional target and its hypotheses are unchanged. All radius restrictions below concern auxiliary small balls in a fixed smooth background.

The exact source passages are listed in Section S17 and in `source-ledger.json`. Throughout, a numbered statement attributed to a monograph is a standard input. A statement about joint center–radius parameters, or about the particular family below, is our derived consequence of the stated construction and estimates; none is attributed verbatim to a monograph. The separate `bibliographic-erratum.json` corrects the earlier Vol. IV edition metadata without changing any immutable input.

S10 The family and its precise topology

Let $\Omega = \{z \in \mathbb{R}^n : |z| < 1\}$, $S = \partial\Omega$, $n \geq 2$, and let $\mathrm{d}\sigma = |S|^{-1}\mathrm{d}S$ be the Euclidean probability measure. Let Θ be a compact set in a finite-dimensional parameter space. All coefficients are smooth on an open neighborhood of $\Theta \times [0, r_0] \times \overline{\Omega}$; this also means smooth extension across parameter endpoints. Consider the real scalar family

$$L_{\theta,r} = a^{ij}_{\theta,r}(z)\partial_i\partial_j + rc^j(\theta, z, r)\partial_j, \qquad a^{ij}_{\theta,r} = \delta^{ij} + r^2C^{ij}(\theta, z, r), \qquad C^{ij} = C^{ji}. \tag{S10.1}$$

There is no zeroth-order term. Its small fixed upper radius is chosen so that all principal matrices are positive and the elementary coercivity bound in Section S12 holds. The same choice works for every regularity order, although the constants at higher orders depend on higher background coefficient bounds.

For $v \in C^\infty(S)$, define $K_{\theta,r}v$ to be the Dirichlet solution of $L_{\theta,r}u = 0$, $u|_S = v$. Define the *radial*, rather than conormal, trace

$$\Lambda(\theta, r)v = \partial_{\nu_0}K_{\theta,r}v|_S, \qquad \nu_0(z) = z \quad (z \in S). \tag{S10.2}$$

Thus the differentiating vector and the domain are fixed. Let B be the same operator for Δ_z; it acts by k on degree k spherical harmonics.

Here is the meaning of the complete classical topology used below. Fix a finite atlas, fixed cutoffs and quantizations on S. A localized operator of order m is represented by $p(z', \xi)$ and a smooth kernel, where

$$p \sim \sum_{j\geq 0} p_{m-j}, \qquad p - \sum_{j<N} \chi(\xi)p_{m-j} \in S^{m-N}.$$

Each p_{m-j} is homogeneous for $|\xi| \geq 1$. The seminorms are all derivatives of p_{m-j} on the compact unit cosphere, all weighted derivative seminorms of each indicated remainder, and all derivatives of the smooth residual kernels, including kernels for separated chart supports. Smooth dependence in this topology means that every parameter derivative has these properties and depends continuously in each such seminorm. A bound for a prescribed finite collection of seminorms also includes the smoothing-kernel seminorms needed for the prescribed Sobolev mappings. It is stronger than one $H^s \to H^{s-m}$ operator bound.

Theorem S10.1 (Derived Dirichlet-family interface)**.** *For* (S10.1) *with sufficiently small fixed* r_0*:*

1. *The Dirichlet problem is uniquely solvable. For every integer* $s \geq 0$*,*

$$\mathfrak{D}_{\theta,r} : u \longmapsto (L_{\theta,r}u, u|_S) : H^{s+2}(\Omega) \longrightarrow H^s(\Omega) \oplus H^{s+3/2}(S) \tag{S10.3}$$

is a bounded bijection. Its inverse is uniformly bounded and smoothly parameter dependent in these fixed Banach spaces, with the same property after every prescribed finite number of parameter derivatives.

2. $\Lambda(\theta, r)$ *is a smooth family in* $\Psi^1_{\mathrm{cl}}(S)$ *in the complete topology just specified. This assertion includes its actual smoothing correction, not only its symbol modulo smoothing operators. All prescribed finite seminorms and finite parameter derivatives are uniformly bounded on* $\Theta \times [0, r_0]$*.*

3. *There is an exact Taylor identity*

$$\Lambda(\theta, r) = B + rB_1(\theta) + r^2 B_2(\theta, r), \qquad B_1 \in C^\infty(\Theta; \Psi^0_{\mathrm{cl}}), \quad B_2 \in C^\infty(\Theta \times [0, r_0]; \Psi^1_{\mathrm{cl}}), \tag{S10.4}$$

where

$$B_1 = \partial_r \Lambda(\theta, 0), \qquad B_2(\theta, r) = \int_0^1 (1 - t)\partial_r^2 \Lambda(\theta, tr)\, \mathrm{d}t. \tag{S10.5}$$

4. *The center evaluation kernel* $p(\theta, r, \omega)$ *defined by*

$$(K_{\theta,r}v)(0) = \int_S p(\theta, r, \omega)v(\omega)\, \mathrm{d}\sigma(\omega) \tag{S10.6}$$

is jointly smooth down to $r = 0$*, with all prescribed finite parameter and angular derivatives uniformly bounded. It satisfies* $p(\theta, 0, \omega) = 1$ *and* $\int_S p\, \mathrm{d}\sigma = 1$*.*

The first part is proved by a weak inverse and boundary regularity. The second and fourth parts require more: a boundary parametrix with smooth errors and an interior evaluation kernel. These are constructed in Sections S13 and S16.

S11 Why this is the normalized first equation

Use the expansion's $g_{1,\lambda}(x) = g_1(\lambda x)$ on a fixed bounded coordinate neighborhood. The function $(\lambda, x) \mapsto g_1(\lambda x)$ is smooth including $\lambda = 0$; compactness gives uniform bounds at every fixed finite order. Fix L, $t_0(y) = y_n + |y'|^2$, and

$$\gamma_\lambda = e^{-2Lt_0} g_{1,\lambda}.$$

Take $\theta = (y, \lambda)$ with centers in a compact neighborhood of Y and $0 \leq \lambda \leq \lambda_0$. This has uniformly positive metric matrices and uniform smooth bounds. A smooth orthonormal frame is, for example, the positive matrix square root $E_{y,\lambda} = \gamma_\lambda(y)^{-1/2}$ in the original coordinate trivialization. The frame is merely a coordinate device.

Choose the positive sign for the harmonic equation, and multiply it by e^{2Lt_0}. Its principal part is then γ_λ^{-1}, and it can be written as $\Delta_{\gamma_\lambda} + Z_\lambda$. The scalar conformal formula gives

$$e^{2Lt_0}\Delta_{g_{1,\lambda}} = \Delta_{\gamma_\lambda} + (n - 2)L\nabla_{\gamma_\lambda} t_0 \cdot \nabla. \tag{S11.1}$$

Indeed, inserting $\gamma = e^{-2Lt_0} g$ into $\Delta_\gamma = |\gamma|^{-1/2}\partial_i(|\gamma|^{1/2}\gamma^{ij}\partial_j)$ yields $e^{2Lt_0}(\Delta_g - (n-2)Lg^{ij}(\partial_i t_0)\partial_j)$. This proves (S11.1) without requiring harmonic coordinates for the calculation. In dimension three $Z = L\nabla_\gamma t_0$.

Let w be γ_λ normal coordinates at y in frame $E_{y,\lambda}$. Uniform existence and smooth dependence of the geodesic ODE on a compact parameter set provide one normal radius and coefficients smooth jointly in (y, λ, w). On these coordinates the normalized operator is

$$a^{ij}(\theta, w)\partial_{w_i}\partial_{w_j} + d^j(\theta, w)\partial_{w_j}, \quad a(\theta, 0) = I, \quad \partial_w a(\theta, 0) = 0.$$

The Laplace–Beltrami first-order coefficients vanish at $w = 0$, so $d(\theta, 0)$ is precisely the frame components $b(\theta)$ of $Z(y)$. Set $w = rz$, pull back, and multiply by r^2. The result is $a(\theta, rz)\partial_z^2 + rd(\theta, rz)\partial_z$. Taylor's integral remainder in w factors $a(\theta, rz) - I$ as $r^2 C$:

$$C^{ij}(\theta, z, r) = \int_0^1 (1-t) z_k z_\ell \, \partial_{w_k}\partial_{w_\ell} a^{ij}(\theta, trz)\, \mathrm{d}t.$$

Consequently (S10.1) is an exact smooth extension of the rescaled operator to $r = 0$, and

$$c^j(\theta, z, 0) = b^j(\theta), \qquad \partial_r L_{\theta,0} = b(\theta) \cdot \nabla_z. \tag{S11.2}$$

Taking r_0 smaller lets these coefficients be defined on $|z| < 2$ as well. Thus the surrounding manifold needed for the parametrix can be fixed. No inverse of the degenerate map at $r = 0$ is used: its coefficients are defined by these normal-coordinate formulas. Only the fixed first background metric enters; smallness or higher derivatives of h are not used for this interface.

S12 A uniform ordinary inverse and its derivatives

Rewrite $-L$ in divergence form:

$$-L = -\partial_i(a^{ij}\partial_j) + d_r^j\partial_j, \qquad d_r^j = \partial_i a^{ij} - rc^j = O(r)$$

uniformly in the parameters. On $H_0^1(\Omega)$ its form is

$$\mathfrak{a}(u, v) = \int_\Omega a^{ij}\partial_j u \partial_i \bar{v} + \int_\Omega d_r^j \partial_j u \bar{v}.$$

Let C_P be the fixed unit-ball Poincare constant. For example, choose r_0 so that $a \geq \frac{1}{2}I$ and $C_P\|d_r\|_\infty \leq \frac{1}{4}$ for the entire parameter set. Then

$$\operatorname{Re}\mathfrak{a}(u, u) \geq \left(\tfrac{1}{2} - C_P\|d_r\|_\infty\right)\|\nabla u\|_2^2 \geq \tfrac{1}{4}\|\nabla u\|_2^2. \tag{S12.1}$$

McLean's Lemma 2.32, printed p. 43, gives the weak zero-Dirichlet inverse $H^{-1}(\Omega) \to H_0^1(\Omega)$ with a uniform bound. The sign switch from $-L$ to L changes no invertibility claim. A fixed trace lifting from McLean, Lemma 3.36 and Theorem 3.37, pp. 101–102, treats nonzero boundary values; it is independent of (θ, r).

McLean's Theorem 4.18(i), pp. 137–139, gives local boundary regularity and its estimate. Its proof explicitly includes the $s = 0$ step. Interior regularity is Theorem 4.16, pp. 135–137. A finite covering of the fixed ball and boundary yields, for each integer $s \geq 0$,

$$\|u\|_{H^{s+2}} \leq C_s(\|Lu\|_{H^s} + \|u|_S\|_{H^{s+3/2}} + \|u\|_{H^1}).$$

The proof uses coefficient derivatives at a finite order, an ellipticity lower bound, and fixed coordinate cutoffs. All are uniformly bounded on the compact parameter set. The weak estimate bounds the last term by the data in (S10.3); hence

$$\|\mathfrak{D}_{\theta,r}^{-1}(f, g)\|_{H^{s+2}} \leq C_s(\|f\|_{H^s} + \|g\|_{H^{s+3/2}}). \tag{S12.2}$$

This proves bijectivity, not just a Fredholm statement. One r_0 chosen for (S12.1) suffices at all orders. Fractional nonnegative orders follow by interpolation of the consistent solution operators if needed. No assertion of a boundary trace on an arbitrary negative bulk Sobolev space is required for the argument.

Put $G_{\theta,r} = \mathfrak{D}_{\theta,r}^{-1}$. Between each fixed pair of spaces in (S10.3), $\mathfrak{D}$ is smoothly dependent in operator norm. The resolvent identity and (S12.2) show that its inverse is continuous and differentiable, with

$$\partial_\eta G = -G(\partial_\eta \mathfrak{D})G. \tag{S12.3}$$

Here η is any center, dilation or radius coordinate. Higher derivatives are finite sums of ordered products

$$G(\partial^{\alpha_1}\mathfrak{D})G\cdots(\partial^{\alpha_k}\mathfrak{D})G, \qquad \alpha_j \neq 0, \quad \sum_j \alpha_j = \alpha,$$

with their usual integer coefficients and signs. Every factor is between the same fixed spaces, since $\partial\mathfrak{D}$ has zero boundary component and is a differential operator of order at most two. This gives the required finite parameter bounds at every fixed regularity order. It does not differentiate a moving domain.

S13 A complete boundary family, including its smooth correction

We now spell out why the classical boundary construction is parameter compatible. The source is Hörmander III, corrected second printing 1994, reprinted 2007, Chapter XVIII and Section 20.1. The parameter argument below is an application of its proofs, not a quoted parameter theorem.

In a fixed boundary chart with inward coordinate t and tangential coordinate s, the principal expression is

$$a^{tt}\partial_t^2 + 2a^{ta}\partial_t\partial_{s_a} + a^{ab}\partial_{s_a}\partial_{s_b}.$$

For tangential frequency $\xi \neq 0$, write $c = a^{ta}\xi_a$ and $q = a^{ab}\xi_a\xi_b$. The stable principal solution with boundary amplitude one is $e^{is\cdot\xi}e^{-\kappa t}$, where

$$\kappa = \frac{ic + \sqrt{a^{tt}q - c^2}}{a^{tt}}, \qquad \operatorname{Re}\kappa \geq c_*|\xi|. \tag{S13.1}$$

The positive square root is used. The Schur complement of a uniformly positive matrix proves this bound uniformly in charts and parameters. It also gives uniform separation of the two normal roots from the real axis after $|\xi| = 1$ normalization. There is one decaying mode, and Dirichlet evaluation on that mode is an isomorphism. This verifies the elliptic boundary condition of Definition 20.1.1, pp. 233–234, for this actual scalar family; lower-order drift has no effect on that check.

The ordinary parametrix T of the extended differential operator has order -2 and satisfies

$$TL = I + R_E, \qquad LT = I + R_F, \tag{S13.2}$$

with smooth kernels. This is (20.1.3), p. 234. Its homogeneous symbol coefficients are obtained recursively by inversion of the positive principal polynomial and finite differentiations and compositions. They are smooth in our parameters. Since L is differential, they have the rational normal-frequency form used in the boundary construction and satisfy the transmission condition. These properties persist under parameter differentiation.

There is no unproved parameter-dependent asymptotic summation here. In the proof of Proposition 18.1.3, pp. 66–67, choose the cutoff scale for its jth symbol term to make the tail bound 2^{-j} hold also for all parameter derivatives of order at most j, uniformly on the compact parameter

set. There are only finitely many conditions at each stage. The same cutoffs are used for every parameter. The resulting locally finite sum and each differentiated sum converge in every prescribed remainder seminorm once enough initial terms are removed. This gives a smooth family of full symbols, with smooth parameter-dependent smoothing errors in (S13.2). Fixed chart cutoffs and partitions of unity introduce no parameter-dependent domains. For a finite output order, only a finite symbol construction and a sufficiently regular error are needed; the smooth all-order family can alternatively be obtained with this diagonal choice of cutoffs. Its bounds refer to the fixed smooth background, rather than an eight-derivative bound on the metric difference h.

Let P_c be the boundary distribution operator obtained by applying L to a function extended by zero; see (20.1.4)–(20.1.5), p. 235. Write $\Gamma u = (u|_S, \partial_t u|_S)$; replacing Hörmander's D_t by ∂_t only changes fixed factors of i. The operator

$$Q = \Gamma T P_c$$

is the approximate Calderon projector (20.1.7). Its entries have orders $k - \ell$, $k, \ell \in \{0, 1\}$. Theorem 18.2.17, pp. 110–111, computes the full boundary symbols by the normal-frequency integrals (18.2.24), and (20.1.8), p. 235, computes their leading terms by residues. For a rational symbol the integral is a sum of residues, as explained after Lemma 18.2.16 on p. 109. On the unit cosphere choose contours enclosing the stable poles with a fixed positive distance from the other poles; compactness and (S13.1) permit finitely many uniform choices. Differentiate under each such contour integral. The estimates in the proof of Theorem 18.2.17 apply also to the differentiated remainders. Thus Q is a smooth complete classical family. The proof of Theorem 20.1.3, pp. 236–237, gives $Q^2 - Q$ with a smooth kernel by exactly the same Green-potential and smooth-error operations; the preceding parameter construction makes this kernel smooth jointly with the parameters.

The graded version of Proposition 20.1.5, pp. 237–239, provides a column $S_D = (S_0, S_1)^T$, with $S_k \in \Psi^k_{\mathrm{cl}}$, such that

$$\Gamma_0 S_D = I \pmod{\Psi^{-\infty}_{\mathrm{cl}}}, \qquad Q S_D = S_D \pmod{\Psi^{-\infty}_{\mathrm{cl}}}. \tag{S13.3}$$

Here Γ_0 selects the first Cauchy component. Its principal inverse exists by the one-dimensional stable-mode check above. The proof constructs the right inverse of $\Gamma_0 Q$ and then composes with Q; its recursion uses the smooth principal inverse and the composition formula. Applying the same parameter summation just described yields smooth S_k, with smooth-kernel remainders in (S13.3).

Consequently the approximate Poisson operator

$$K^{\mathrm{app}} = T P_c S_D \tag{S13.4}$$

has errors

$$F = L K^{\mathrm{app}} : \mathcal{D}'(S) \longrightarrow C^\infty(\overline{\Omega}), \qquad H = \Gamma_0 K^{\mathrm{app}} - I : \mathcal{D}'(S) \longrightarrow C^\infty(S). \tag{S13.5}$$

This is the homogeneous-interior-data case of (20.1.13)–(20.1.14), p. 239, and Theorem 20.1.7, p. 241. Their proof shows the smooth errors explicitly as $R_F P_c S_D$ and $\Gamma_0 Q S_D - I$, respectively. Thus all parameter derivatives have the same smoothing properties and uniform bounds in each $H^{-M}(S) \to H^N(\Omega)$ or $H^{-M}(S) \to H^N(S)$ norm needed below. Smoothness up to the interior boundary in the first error is part of the source statement, not merely off-diagonal regularity.

Correct both errors by the *already proved ordinary* inverse:

$$K = K^{\mathrm{app}} - G(F, H). \tag{S13.6}$$

For smooth boundary data this solves $LK = 0$, $\Gamma_0 K = I$, and uniqueness identifies it with the true Dirichlet solution. The correction is smoothing jointly in boundary and bulk variables and all parameters. To justify the kernel claim, fix desired angular derivative order q and take $M > q + (n-1)/2$. The map $\omega \mapsto \delta_\omega$ is C^q with values in $H^{-M}(S)$, by local Fourier estimates for the delta distribution and its derivatives. Apply (S13.5), (S12.2), and (S12.3), choosing the bulk Sobolev order larger than the desired bulk derivative order plus $n/2$. Sobolev embedding gives a joint C^q kernel on $\overline{\Omega} \times S$, with all prescribed parameter derivatives. Increasing M and the output order proves a jointly smooth kernel. This is why abstract inverse differentiation actually controls a smoothing *kernel* here.

Finally, $\partial_{\nu_0} K^{\mathrm{app}}|_S$ is a smooth classical operator of order one: it is the first-normal Cauchy row of QS_D, in fixed charts, with fixed tangential coordinate changes. The normal trace of the correction in (S13.6) is a smooth kernel. This proves the complete-topology assertion in Theorem S10.1. Hörmander's Theorems 18.1.7–18.1.8, pp. 70–72, and 18.1.13, p. 76, give the continuous adjoint and composition operations and the $H^s \to H^{s-1}$ mapping for every real s. Those Sobolev mappings are consequences of the established complete family; they do not replace its construction.

S14 Order-zero first variation and the exact Taylor remainder

For the fixed Euclidean outward radial trace in (S10.2), the principal boundary symbol in an inward boundary chart is κ from (S13.1). More generally the fixed radial vector has the corresponding fixed normal and tangential components; the formula is obtained by applying it to $e^{is\cdot\xi}e^{-\kappa t}$. On the unit sphere it is the Euclidean outward normal, so the collar may be chosen to make it exactly $-\partial_t$. All dependence of this order-one symbol on r comes from the principal coefficients. Their first derivative at $r = 0$ is zero by (S10.1). Therefore the homogeneous degree-one term of $\partial_r \Lambda(\theta, 0)$ vanishes. The complete symbol expansion then places this derivative in Ψ^0_{cl}, including its smoothing part. It depends smoothly on θ in that lower-order topology because all the remaining homogeneous terms and remainders do. The drift can and generally does contribute at order zero.

There is an additional direct check in the geometric family. Let $b = b(\theta)$ be as in (S11.2), let $\mathcal{H}$ denote Euclidean harmonic extension, and put $m_b(\omega) = b \cdot \omega$. Differentiating the exact Dirichlet problem at zero gives

$$\Delta v = -b \cdot \nabla \mathcal{H} f, \qquad v|_S = 0.$$

Since $\Delta((b \cdot z)\mathcal{H} f) = 2b \cdot \nabla \mathcal{H} f$, uniqueness gives

$$v = \tfrac{1}{2}\big(\mathcal{H}(m_b f) - (b \cdot z)\mathcal{H} f\big).$$

Taking its outward radial trace proves the exact formula

$$B_1 = \tfrac{1}{2}\big([B, M_{m_b}] - M_{m_b}\big). \tag{S14.1}$$

Here M_{m_b} is multiplication. The scalar composition formula gives $[B, M_{m_b}] \in \Psi^0_{\mathrm{cl}}$, with complete seminorm bounds, confirming the symbol argument and including the first-order normalized drift. Formula (S14.1) is asserted for (S11.2); in a general family (S10.1) with $c(\theta, z, 0)$ varying in z, the principal symbol argument still proves $B_1 \in \Psi^0_{\mathrm{cl}}$ but this explicit formula need not hold.

At $r = 0$ the full Dirichlet operator is Euclidean, so $\Lambda(\theta, 0) = B$ exactly, including its smoothing term. Taylor's integral identity in each of the compatible complete-symbol and smooth-kernel seminorms proves (S10.4)–(S10.5). The integral is a genuine smooth family in the complete Ψ^1_{cl} topology. There is no unspecified smoothing error left in the asserted expansion.

S15 Six center derivatives after substitution of the radius

Let $r = r(y) > 0$, $f = \log r$, and $\beta = |\mathrm{d}f|_\gamma \geq 1$. Assume the expansion's finite bounds $|\partial^j f| \leq C_j \beta^j$ and $r\beta^2 \leq \varepsilon$, with $r \leq r_0 \leq 1$. In a fixed coordinate trivialization repeated differentiation of e^f gives

$$|\partial^j r| \leq C_j r \beta^j. \tag{S15.1}$$

For any of the smooth operator families $A(y, \lambda, r)$ just constructed, the chain rule at order j is a finite sum of terms of the form

$$(\partial_y^\alpha \partial_r^k A)(y, \lambda, r(y)) \prod_{\ell=1}^{k} \partial_y^{\nu_\ell} r, \qquad |\alpha| + \sum_\ell |\nu_\ell| = j.$$

In every chosen finite classical and smoothing seminorm its size is at most $C_j r^k \beta^{\sum |\nu_\ell|} \leq C_j \beta^j$. Parameter derivative bounds for B_2 use the needed additional radius derivatives of Λ, which Section S13 supplies at every fixed finite order. Thus this applies in particular for $j \leq 6$, uniformly in λ and the shrinking parameter. Horizontal derivatives differ from coordinate derivatives by a fixed smooth connection acting on fiber coordinates and tensor indices; its angular vector fields are smooth on the compact sphere. Commuting these through the local symbols adds only finitely many angular derivatives and preserves the same bounds, as does taking fiber adjoints by Theorem 18.1.7.

For completeness the trace remainder allowance in the expansion can be read directly from this family. Under center motion parallel-transport ω. The center Jacobi field has $J(0) = e_i$, $J'(0) = 0$, hence $J(r) = \mathrm{PT}(e_i + O(r^2))$. The differential of the exponential map in z is $r\,\mathrm{PT}(I + O(r^2))$. These statements follow by integrating the Jacobi ODE twice; curvature and all required parameter derivatives are bounded on the fixed normal neighborhood. The smooth integral remainders also show that the $O(r^2)$ terms are smooth after division by r^2. The inverse endpoint differential applied to center velocity is therefore

$$r^{-1} e_i + r V_i(y, \lambda, r, \omega) + f_i \omega, \tag{S15.2}$$

where V_i is smooth with all prescribed parameter and angular bounds. The last term is exact: varying r by $\partial_i r = r f_i$ has endpoint velocity $r f_i\,\mathrm{PT}\,\omega$, and the radial z vector maps to $r\,\mathrm{PT}\,\omega$.

Split V_i into its tangent part and scalar radial part. On boundary data, e_i acts as its tangential derivative plus $\omega_i \Lambda$; at $r = 0$ this is A_i, the ambient derivative of Euclidean harmonic extension. Consequently (S15.2) and (S10.4) give

$$\mathcal{C}_i = r^{-1} A_i + f_i B + F_i + E_i, \tag{S15.3}$$

where a valid choice in this horizontal frame is

$$F_i = (\omega_i + r f_i) B_1, \tag{S15.4}$$

$$r^{-1} E_i = \omega_i B_2 + V_i^{\tan} \cdot \nabla_S + V_i^{\mathrm{rad}} \Lambda + r f_i B_2. \tag{S15.5}$$

Multiplication precedes each displayed operator. Coordinate frame changes give the covariant version of the same formula. Here $F_i \in \Psi^0_{\mathrm{cl}}$ and $r^{-1} E_i \in \Psi^1_{\mathrm{cl}}$. For $j \leq 6$,

$$|\partial^j (r f_i)| \leq C_j r \beta^{j+1} \leq C_j \varepsilon \beta^{j-1} \leq C_j \beta^j.$$

This uses derivatives of f only through order 7, within the eight orders supplied in the expansion. Combining it with the chain-rule bounds proves the precise allowance $C_j \beta^j$ for both normalized remainders, including their smoothing kernels. Fiber adjoints give $\mathcal{C}_i^* = r^{-1} A_i^* + f_i B + F_i^* + E_i^*$ with the same allowance. This checks the stated use of the radial family in angular-expanded: radial-expansion; it proves no coercive estimate for the subsequent transfer operator.

S16 The center Poisson density and its first center derivative

The correction term in (S13.6) already has a jointly smooth kernel on $\overline{\Omega} \times S$. For the approximate term, fix $|z| \le \frac{1}{2}$. Its kernel is smooth: the kernel of T and all its derivatives are smooth when the interior point and the source point on S are separated. Applying P_c differentiates that kernel finitely in the source variables, and applying the transpose of S_D on the boundary preserves a smooth boundary function. The same statements hold after all prescribed parameter and interior derivatives, uniformly because the separation is fixed. Hence K has a joint smooth kernel on $\{|z| \le \frac{1}{2}\} \times S$ down to $r = 0$. Converting its boundary measure from $\mathrm{d}S$ to $\mathrm{d}\sigma$ multiplies the kernel by the fixed constant $|S|$. Evaluation at $z = 0$ proves the assertion about p in Theorem S10.1.

Constants solve (S10.1), so uniqueness gives $K1 = 1$ and thus $\int_S p \, \mathrm{d}\sigma = 1$. At $r = 0$ Euclidean mean value gives $p = 1$ for all θ. Taylor's identity for this smooth kernel writes

$$p(\theta, r, \omega) = 1 + rq(\theta, r, \omega), \qquad q = \int_0^1 \partial_r p(\theta, tr, \omega) \, \mathrm{d}t. \tag{S16.1}$$

For every fixed angular order a and every fixed finite center derivative order, q has uniform corresponding $C^a(S)$ bounds. In particular, where the center derivative holds r fixed,

$$\|p\|_{C^a(S)} + \|\partial_r p\|_{C^a(S)} \le C_a, \qquad \|p - 1\|_{C^a(S)} + \|\partial_y p\|_{C^a(S)} \le C_a r. \tag{S16.2}$$

For the geometric constant drift one also obtains from the calculation in Section S14

$$\partial_r p(\theta, 0, \omega) = \tfrac{1}{2} b(\theta) \cdot \omega.$$

This optional formula makes explicit that the density is not generally constant at positive radius, even though its limit is constant.

Now view the density as a function on $S_\gamma Y$, using the orthonormal frame. A horizontal derivative equals a center coordinate derivative plus a fiber rotation vector field with uniformly bounded smooth coefficients, and an equivalent bounded change of base components. The rotation differentiates $p - 1$; it annihilates 1. For $p_\delta(y, \omega) = p(y, \lambda, r_\delta(y), \omega)$, the chain rule therefore gives, using one more angular derivative in (S16.2),

$$\|\nabla_y p_\delta(y, \cdot)\|_{C^a(S_y)} \le C_a(r_\delta + |\mathrm{d}r_\delta|_\gamma) = C_a(r_\delta + r_\delta \beta_\delta). \tag{S16.3}$$

The constants are uniform in λ and δ. The expansion assumes $r_\delta \beta_\delta^2 \le \varepsilon$ and $\beta_\delta \ge 1$, so $r_\delta \beta_\delta \le \varepsilon/\beta_\delta \le \varepsilon$. Together with (S16.2), this proves the required uniform angular H^1 bounds and first horizontal-center-derivative bounds as the balls shrink. In fact $r\beta \le \sqrt{\varepsilon r}$, giving the stronger vanishing estimate $C_a(r + \sqrt{\varepsilon r})$ when the radius tends to zero. No lower bound on the positive radius or upper bound on β is used for this lower-region statement.

The exact use for the lower known transfer region

The result just proved is a Dirichlet evaluation density. To apply it to the manuscript's ψ, use its actual flux definition on the lower known region, where the kernel is the actual positively normalized Green function of $-\Delta_{g_{1,\lambda}}$. For each smooth boundary datum v, its Dirichlet solution is smooth up to the ball boundary. Green representation gives

$$\int_{\partial B_\gamma(y,r)} (G_1 \partial_{\nu_1} u - u \partial_{\nu_1} G_1) \, \mathrm{d}S_1 = u(y).$$

The identity follows either by the distribution equation $-\Delta_{g_1} G_1(\cdot, y) = \delta_y$ with metric volume, or by Green's second identity on the punctured ball and its small-sphere limit. The smooth-boundary Green identity is McLean's Theorem 4.4(iii), p. 118; the actual Green normalization is a separate original manuscript input. Here $\Lambda_y^{g_1,\mathrm{con}}$ denotes the physical first-metric outward-unit-normal Dirichlet-to-Neumann map on the actual ball, self-adjoint against its physical area measure $\mathrm{d}S_1$. It is distinct from the normalized unit-ball radial trace $\Lambda(\theta, r)$. Even in the Euclidean case the physical map is $r^{-1}\Lambda$ after pullback. The Green flux below uses the physical map, physical normal derivative, and the retained Jacobian J_y.

Taking the physical boundary Dirichlet-to-Neumann adjoint in this identity shows that the density $J_y(\Lambda_y^{g_1,\mathrm{con}} k_y - \partial_{\nu_1} G_1)$, expressed against $\mathrm{d}\sigma_y$, represents $u(y)$ for *every* smooth boundary datum. Thus it equals $p(y, \lambda, r, \omega)$ by testing against all $v \in C^\infty(S)$. No density assertion about traces of functions extending harmonically outside the closed ball is needed. Scalar normalization of the equation does not change these Dirichlet solutions.

This identification is conditional on the stated lower-region equality of the actual kernel. It constructs no mixed continuation. On the upper cutoff region, where r has a fixed positive lower bound, the same ordinary-family construction supplies the Dirichlet part of smooth density bounds; a bound for the boundary mixed kernel and its first parameter derivative is still a distinct continuation input. The standard center kernel above supplies no such bound.

S17 Source locators and limits of the deliverable

The neutral frozen expansion used here is work/overnight-run/audit-inputs-v001/expanded.tex. Its exact relevant locators are: normalization, lines 1336–1348; profile estimates, lines 2693–2717; normalized unit-ball family and angular-expanded:radial-expansion, lines 2729–2750; six-center-derivative trace interface, lines 2062–2075 and 3220–3228; lower cutoff-region Poisson estimates, lines 3183–3190; flux and lower evaluation, lines 1620–1625, 1683–1703 and 1956–1967. These are correspondence locators, not assertions that any downstream novel theorem has been accepted. The source ledger records hashes and the exact extraction-line ranges read.

Monograph passage	**Standard content used**
McLean, Lemma 2.32, p. 43	Inverse for a positive bounded-below bounded operator from a Hilbert space to its dual. We verify its bound for the specific small-radius scalar form.
McLean, Lemma 3.36 and Theorem 3.37, pp. 101–102	Trace right inverse and trace continuity on smooth domains. Our lifting is fixed on the unit ball.
McLean, Theorem 4.18(i), pp. 137–139; Theorem 4.16, pp. 135–137	Boundary and interior regularity with estimates. Compact uniform coefficients and (S12.3) yield our parameter consequences.
McLean, Theorem 4.4(iii), p. 118	Green's second identity. Its use for the actual Green function presupposes that function's original normalization, not the novel continuation.
Hörmander III, Definition 18.1.1 and Proposition 18.1.3, pp. 65–67	Symbol seminorms and asymptotic summation. We add uniform finite parameter derivatives to the proof's diagonal cutoff conditions.

Monograph passage	Standard content used
Hörmander III, Theorems 18.1.7–18.1.8, pp. 70–72; Theorem 18.1.13, p. 76	Adjoint, composition and Sobolev mapping; scalar commutator order reduction follows from cancellation of the leading product.
Hörmander III, Theorem 18.2.15 and Lemma 18.2.16, pp. 108–109; Theorem 18.2.17 and (18.2.24), pp. 110–111	Transmission, the residue interpretation, and full tangential classical symbols of boundary Green-potential traces. We differentiate uniformly separated residue contours and the proof's remainder estimates.
Hörmander III, Definition 20.1.1, pp. 233–234	Elliptic boundary condition, checked directly by (S13.1).
Hörmander III, (20.1.3)–(20.1.8), pp. 234–235; Theorem 20.1.3, pp. 236–237	Extended parametrix, Green potential and approximate Calderon projector, including its smooth projection error.
Hörmander III, Proposition 20.1.5 and its graded extension, pp. 237–239	Dirichlet boundary inverse modulo smoothing. We retain the parameters in the symbol recursion.
Hörmander III, (20.1.13)–(20.1.14), p. 239; Proposition 20.1.6, pp. 239–241; Theorem 20.1.7, p. 241	Approximate Poisson solution, smooth interior and boundary errors, and its Sobolev mappings. Our ordinary inverse corrects both errors exactly.

The supplied III scan identifies itself as the 2007 reprint of the corrected second printing 1994, ISBN 978-3-540-49937-4. The supplied McLean scan is Cambridge University Press, first published 2000, ISBN 0-521-66332-6 / 0-521-66375-X. The supplied IV file identifies itself as Springer 1985, Grundlehren 275, ISBN 3-540-13829-3 / 0-387-13829-3; its filename's 2009 date is not the scan's edition. Vol. IV is used here only for the separate bibliographic erratum, not as a premise in the Dirichlet-family derivation.

The mathematical derivations of these two interfaces are complete at the classical analytic level, subject to fresh review of this authored argument and its source coverage. Their status is *derived for review*, not accepted. Remaining outside this work unit are the mixed kernel's existence and uniform continuation estimates, the upper cutoff-region mixed-kernel input, the transfer equation and coercive estimate, the bootstrap and global uniqueness, all formalization, correspondence registration and semantic ratification. No new axiom, opaque analytic premise, library pin or trusted checking code has been introduced.

References

[1] L. Hörmander, *The Analysis of Linear Partial Differential Operators III: Pseudo-Differential Operators*, corrected second printing 1994, Classics in Mathematics reprint, Springer, 2007.

[2] W. McLean, *Strongly Elliptic Systems and Boundary Integral Equations*, Cambridge University Press, 2000.

[3] L. Hörmander, *The Analysis of Linear Partial Differential Operators IV: Fourier Integral Operators*, Grundlehren der mathematischen Wissenschaften 275, Springer, 1985. Bibliographic erratum only in this supplement.

S18 Finite polynomial Fischer algebra

Historical source locator: the then-frozen neutral precursor, lines 2875–2907 of work/overnight-run/audit-inputs-v001/expanded.tex; original angular estimate manuscript `05-angular-estimate.tex`. This preparation proves only the algebraic steps, not sphere normalization or analytic estimates.

Let σ be any finite variable set. For a finitely supported exponent $\alpha : \sigma \to \mathbb{N}$, put $w(\alpha) = \prod_{i\in\sigma} \alpha_i!$. For real finite polynomials define

$$[p, q]_F = \sum_{\alpha\in\operatorname{supp} p} p_\alpha q_\alpha w(\alpha).$$

All sums are finite. Missing coefficients are zero. Distributing multiplication over the coefficient sums proves additivity in both variables, and multiplication by a constant multiplies this pairing by that constant. For monomials the pairing is $[aX^\alpha, bX^\beta]_F = \delta_{\alpha\beta} abw(\alpha)$. This is symmetric. Expanding both polynomials as finite sums of monomials proves symmetry generally, including zero coefficients. Every $w(\alpha)$ is positive, and $[p, p]_F$ is a finite sum of the nonnegative quantities $p_\alpha^2 w(\alpha)$. If that sum vanishes, each supported coefficient vanishes, so the support is empty and $p = 0$.

S18.1 Adjoint calculus

For each variable i let $a_i = \partial_i$ and $M_i p = X_i p$. The factorial recurrence gives $w(\alpha+e_i) = (\alpha_i+1)w(\alpha)$: the factors at all other indices are unchanged and the factor at i is $(\alpha_i + 1)! = (\alpha_i + 1)\alpha_i!$. The coefficient derivative rule is $(a_i q)_\alpha = (\alpha_i + 1)q_{\alpha+e_i}$. Consequently

$$[M_i(aX^\alpha), q]_F = aq_{\alpha+e_i} w(\alpha + e_i) = [aX^\alpha, a_i q]_F.$$

Additivity and finite monomial expansion imply $[M_i p, q]_F = [p, a_i q]_F$. Symmetry gives the required adjoint identity $[a_i p, q]_F = [p, M_i q]_F$.

S18.2 Homogeneous derivative norm

If p is homogeneous of degree k, the all-degree Euler identity is $\sum_i M_i a_i p = kp$. Thus

$$\sum_i [a_i p, a_i p]_F = \sum_i [p, M_i a_i p]_F = [p, \sum_i M_i a_i p]_F = k[p, p]_F.$$

These arguments do not assume any of the Fischer identities as hypotheses.

S19 Harmonic raising and the orthogonal complement

This extends the expansion before its Lean translation. Let q_i be harmonic homogeneous degree $k \geq 1$, put $m = k-1$, $n = |\sigma|$, and assume $n > 0$. Define $s = \sum_i a_i q_i$, $U = \sum_i X_i q_i$ and $c = 4m+2n$. Formal partial derivatives commute: their coefficient at α is $q_{\alpha+e_i+e_j}(\alpha_i + 1)(\alpha_j + 1)$ when $i \neq j$, and $q_{\alpha+2e_i}(\alpha_i + 2)(\alpha_i + 1)$ when $i = j$. Thus $\Delta a_i = a_i \Delta$. Euler homogeneity lowers the degree of each $a_i q_i$ to m, so s is harmonic and homogeneous degree m. The derivative product rule gives $\Delta(X_i q_i) = X_i \Delta q_i + 2a_i q_i$, hence $\Delta U = 2s$. The previously proved all-degree radius product rule gives $\Delta(|X|^2 s) = cs$. As $c > 0$, define

$$H = U - \frac{2}{c}|X|^2 s.$$

It follows directly that $\Delta H = 0$, and both terms have degree $k+1$. This is exactly the source's $U - |X|^2 s/(2d)$, because $d = k + n/2 - 1$ and $c = 4d$.

Twice applying the proven Fischer adjoint identity gives $[|X|^2 p, q]_F = [p, \Delta q]_F$. Thus $[H, |X|^2 t]_F = 0$ for every polynomial t, and the decomposition $U = H + (2/c)|X|^2 s$ is orthogonal. Moreover

$$[|X|^2 s, |X|^2 s]_F = c[s,s]_F, \qquad [H,H]_F = [U,U]_F - \frac{4}{c}[s,s]_F.$$

For arbitrary (even nonharmonic) q_i, adjointness and the product rule imply

$$\begin{aligned}[U,U]_F &= \sum_{ij} [q_i, a_i(X_j q_j)]_F \\ &= \sum_i [q_i, q_i]_F + \sum_{ij} [a_j q_i, a_i q_j]_F.\end{aligned}$$

The scalar term is the Kronecker delta from $a_i X_j = \delta_{ij}$; the remaining term moves X_j to the other side by the proven adjoint identity. Combining these identities gives exactly

$$[H,H]_F = \sum_i [q_i, q_i]_F + \sum_{ij} [a_j q_i, a_i q_j]_F - d^{-1}[s,s]_F.$$

Finite polynomial orthogonal projection is thereby characterized algebraically. Relating this projection to the Hilbert space of sphere harmonics, proving N_k normalization, base integration, and the covariant estimates are separate explicit bridges. They are not hypotheses used to prove these finite identities.

S20 The finite mixed contraction bound

Source: the mixed-sum explanation for `angular-expanded:ibp-two`, neutral expansion lines 2947–2949; original angular section lines 334–337. This is the algebraic bound after the covariant integrations by parts, not a proof of those integrations.

For any real finite polynomials p, q, positive definiteness and bilinearity give

$$0 \le [p-q, p-q]_F = [p,p]_F + [q,q]_F - 2[p,q]_F.$$

Thus $2[p,q]_F \le [p,p]_F + [q,q]_F$. Let now q_i be homogeneous of degree k (harmonicity is unnecessary), and set

$$S = \sum_{i,j} [a_j q_i, a_i q_j]_F, \qquad E = \sum_{i,j} [a_j q_i, a_j q_i]_F.$$

Apply the preceding inequality to $(a_j q_i, a_i q_j)$ for each ordered pair and sum the finite inequalities. This gives

$$2S \le \sum_{i,j} [a_j q_i, a_j q_i]_F + \sum_{i,j} [a_i q_j, a_i q_j]_F = 2E,$$

where the second sum equals E by exchanging the two finite indices. For every fixed i, the already proved homogeneous derivative identity gives $\sum_j [a_j q_i, a_j q_i]_F = k[q_i, q_i]_F$. Summing over i yields $E = k \sum_i [q_i, q_i]_F$. Dividing the real inequality $2S \le 2E$ by the positive scalar 2 proves the exact required bound

$$\sum_{i,j} [a_j q_i, a_i q_j]_F \le k \sum_i [q_i, q_i]_F.$$

No sphere normalization, analytic integration by parts, curvature identity, or external analytic premise enters this finite polynomial proof.

S21 The finite divergence contraction constant

Source: neutral angular expansion lines 2962–2966, the contraction estimate used in equation (9.18). This section is written before translation; the predecessor's finite algebra is copied without changing its definitions or statements.

Let q_i be real finite polynomials homogeneous of degree $k \geq 1$, put $s = \sum_i a_i q_i$ and $n = |\sigma|$. Then s is homogeneous degree $k-1$. Let $S = \sum_i [q_i, q_i]_F$ and $B = [s, s]_F$. The proven adjoint and finite-sum identities imply

$$\sum_i [q_i, X_i s]_F = \sum_i [a_i q_i, s]_F = [s, s]_F = B.$$

The proven product/adjoint identity gives $[X_i s, X_i s]_F = [s, s]_F + [a_i s, a_i s]_F$. Summing and applying the all-degree derivative identity to s yields

$$\sum_i [X_i s, X_i s]_F = (n + k - 1)B = LB, \qquad L = k + n - 1.$$

If σ is nonempty, $n \geq 1$ and $k \geq 1$, so $L > 0$. For any real scalar a, bilinearity and symmetry imply the exact finite identity

$$\sum_i [q_i - aX_i s, q_i - aX_i s]_F = S - 2a \sum_i [q_i, X_i s]_F + a^2 \sum_i [X_i s, X_i s]_F.$$

Take $a = L^{-1}$. The right side is $S - 2L^{-1}B + L^{-2}LB = S - L^{-1}B$. The left side is nonnegative by the actual Fischer positivity proved in the predecessor. Multiplying by positive L gives $0 \leq LS - B$, hence the exact contraction bound

$$[\operatorname{div} q, \operatorname{div} q]_F \leq (k + n - 1) \sum_i [q_i, q_i]_F.$$

The zero-polynomial case is contained in the finite identity and uses no division by B. If σ is empty, both the tuple sum and s are zero, so the conclusion is $0 \leq 0$ directly, without forming a positive denominator. The final statement can therefore include every finite σ, using this separate empty case. Harmonicity, sphere normalization, tensor connections and integration by parts are not assumptions. Interpreting the square root of this constant as an operator norm on an analytic family remains a separate bridge.

S22 Exact Euclidean shrinking-sphere geometry

This private preparation artifact addresses the exact $h = 0$ finite-vector calculation in the original manuscript, section 04-shrinking-balls, labels balls:test-vectors and balls:test-expression. The neutral preserved expansion has this calculation at lines 1522–1584. It proves no continuation, geodesic-error, or PDE conclusion, and does not replace the frozen unconditional main theorem. All inner products below are real.

Theorem S22.1 (Exact nonnegative residual)**.** *Let E be a real inner product space. Suppose $\|\omega\| = \|v\| = 1$, $\langle \omega, v\rangle = 0$, and $\|b\| \le 1/4$. Put $u = \omega + b$, $D = \|u\|^2$, $q = u/D$, and $Pz = z - \langle \omega, z\rangle \omega$. Suppose $\langle u, w\rangle = 0$ and $\|w\|^2 = 1/D$. Then*

$$R = \|P(\omega - q)\|^2 + \|P(v - w)\|^2 - \langle b, q\rangle^2 - \langle b, w\rangle^2 \ge 0.$$

In particular the requested universal cubic lower bound holds with $C = 0$.

Proof. Write $a = \langle \omega, b\rangle$, $B = \|b\|$, $m = 1 + a$, $t = \|Pb\|^2 = B^2 - a^2$, $z = \langle \omega, w\rangle$, and $k = \|Pw\|$. Cauchy–Schwarz gives $|a| \le B \le 1/4$, so $m \ge 3/4 > 0$. Also $D = m^2 + t = 1 + 2a + B^2 \ge 9/16 > 0$, and $D \le 2m$ because $B^2 \le 1$. Thus $x = m/D \ge 1/2$.

Orthogonality gives $\langle b, w\rangle = -z$ and $\langle Pb, Pw\rangle = -mz$. Cauchy–Schwarz gives $m^2 z^2 \le tk^2$. Meanwhile $k^2 + z^2 = \|w\|^2 = 1/D$. Consequently

$$\frac{m^2}{D} = m^2(k^2 + z^2) \le (m^2 + t)k^2 = Dk^2,$$

so $k^2 \ge m^2/D^2 = x^2$ and $k \ge x$.

The Lean realization uses the equivalent Cauchy–Schwarz calculation with $e = \omega - (m/D)u$. Orthogonality gives $\langle e, w\rangle = z$, while $\|e\|^2 = 1 - m^2/D$. Thus $z^2 \le (1 - m^2/D)/D$. Subtracting this from $k^2 = 1/D - z^2$ gives $k^2 \ge m^2/D^2$ directly.

The exact real projection and negative terms are

$$\|P(\omega - q)\|^2 = t/D^2, \qquad \langle b, q\rangle = (a + B^2)/D = (D - m)/D, \qquad \langle b, w\rangle = -z.$$

Since $Pv = v$ and $\|v\| = 1$, the reverse triangle inequality gives $\|P(v - w)\|^2 = \|v - Pw\|^2 \ge (1 - k)^2$. Substituting $z^2 = 1/D - k^2$ and $t = D - m^2$ yields exactly

$$\begin{aligned} R &\ge t/D^2 + (1 - k)^2 - (D - m)^2/D^2 - (1/D - k^2) \\ &= 2(k - x)(k + x - 1) \ge 0. \end{aligned}$$

Both factors are nonnegative, since $k \ge x \ge 1/2$. No choice of dimension, angle of w, or direction of v was excluded. The same proof works for $\|b\| < 1$, where $m > 0$ and $D > 0$. □

Lemma S22.2 (Positive Hessian contribution, exact Euclidean case)**.** *If a real symmetric form H satisfies $H(\xi, \xi) \le -c\beta\|\xi\|^2$ for every ξ, with $c, \beta > 0$, then for $r > 0$ the strict expression $R - r^2(H(q, q) + H(w, w))$ is at least $2cr^2\beta/D$. For $\|b\| \le 1/4$ one has $D \le 25/16$, so this is at least $(32/25)cr^2\beta > 0$.*

Proof. Both q and w have squared norm $1/D$. Apply the bound separately and add; the preceding theorem gives $R \ge 0$. □

S22.1 Explicit scalar error absorption

For comparison with the manuscript's error bookkeeping, suppose an independently justified curvature/metric perturbation estimate gives

$$S \geq Ar^2\beta - C_0 r^2 - C_1 r^3\beta^3 - C_2\varepsilon r^2\beta - C_3\varepsilon^2 r^2, \qquad A > 0, \quad C_i \geq 0.$$

This inequality is an explicit conditional algebra interface, not an adopted analytic premise. If $r\beta^2 \leq \varepsilon$, $\beta \geq B_0 \geq 1$, $\varepsilon \leq 1$, $B_0 \geq 4(C_0 + C_3)/A$, and $(C_1 + C_2)\varepsilon \leq A/4$, then

$$S \geq \tfrac{1}{2} Ar^2\beta > 0.$$

Indeed $r^3\beta^3 = r^2\beta(r\beta^2) \leq \varepsilon r^2\beta$; $C_0 + C_3\varepsilon^2 \leq C_0 + C_3 \leq A\beta/4$; the other two errors sum to at most $Ar^2\beta/4$. The vector smallness required above follows from $\|b\| = r\beta \leq \varepsilon/\beta \leq \varepsilon$ if $\beta \geq 1$, hence is ensured by additionally taking $\varepsilon \leq 1/4$. The profile condition $\inf \beta \geq c_p p$ can supply the required lower bound by choosing p first. The formal absorption lemma uses the resulting conditions $4(C_0 + C_3) \leq A\beta$ and $(C_1 + C_2)\varepsilon \leq A/4$ directly. Its separate strict-positivity corollary retains $A, r, \beta > 0$.

The exact flat calculation has no cubic defect. A cubic error is still a permissible conservative estimate, and perturbations may require it, but the present calculation alone does not establish their uniform constants or the manuscript's Jacobi, connection, continuation, or bootstrap claims.

S23 Summability in the original cross continuation

Throughout this section $\mu_j \geq 0$. This is the scalar and Banach-space convergence component of `cross:series`, expanded v001 lines 648–662. It does not construct a singular basis, prove the PDE interpolation estimate, or identify a distributional extension. Those are separate application obligations.

Lemma S23.1. *Let $\mu : \mathbb{N} \to \mathbb{R}$ be nonnegative, let $a > 0$, $C \geq 0$, and $N \in \mathbb{N}$ satisfy $Na > 1$. Suppose for all j, $\mu_j \leq C(j+1)^{-N}$. Then $\sum_j \mu_j^a$ converges.*

Proof. Monotonicity of positive real powers on nonnegative arguments gives $0 \leq \mu_j^a \leq C^a(j+1)^{-Na}$. The real p-series with exponent $-Na < -1$ is summable, including after shifting by one. Multiplication by the fixed real constant C^a preserves summability. The nonnegative comparison criterion gives the result. The proof includes $C = 0$; no lower bound on any μ_j is needed. □

Lemma S23.2. *Suppose for every $N \in \mathbb{N}$ there is $C_N \geq 0$ with $\mu_j \leq C_N(j+1)^{-N}$ for all j. For every $a > 0$, $\sum_j \mu_j^a$ converges.*

Proof. By the Archimedean property choose $N > 1/a$. Multiplication by $a > 0$ gives $Na > 1$, so apply the preceding lemma with its corresponding C_N. This uses one sufficiently high decay order, not a continuously chosen singular basis and not any positive lower singular-value bound. □

Lemma S23.3. *Let E be a complete normed additive group, and suppose $v_j \in E$, $\|v_j\| \leq M\mu_j^a$ for all j, where μ has the preceding nonnegative rapid decay and $a > 0$. Then $\sum_j v_j$ converges.*

Proof. The scalar sequence $M\mu_j^a$ is summable by the previous lemma and constant multiplication. The norm comparison theorem for a complete normed group proves summability of v. The bound itself implies its right side is nonnegative; no unnecessary sign assumption on M is needed for this abstract comparison. In the manuscript $M \geq 0$. For the cross construction, set $a = \gamma_1 + \gamma_2 - 1 > 0$. The actual summand norm bound comes from the two interpolation estimates and coefficient extension and still requires proof. □

Lemma S23.4. *If also $\mu_j \leq 1$ and $0 < p \leq a$, then $\sum_j \mu_j^a \leq \sum_j \mu_j^p$. Hence the positive margin p gives a common convergent majorant for all allowed interpolation exponents.*

Proof. For $0 \leq \mu_j \leq 1$, nonnegative real powers decrease with the exponent, so $\mu_j^a \leq \mu_j^p$. Both series are summable by rapid decay. Sum the pointwise inequality. This includes zero singular values; the actual injective restriction has strictly positive ones. □

S24 Uniform scalar assembly of the raising estimate

This component assembles the exact inequalities at `angular-expanded:after-first-ibp`, equation (9.18), and `angular-expanded:hessian-contribution`. It does not assume them as literature results or prove their analytic application. The scalar variables below stand for the integrated quantities specified there.

Let $n \geq 3$, $k \geq 1$, $d = k + n/2 - 1$, $A = 2 + 2/d$, and $\theta = 1/8$. Let $X, Y, G, W, V \geq 0$, $c_0, c_1 > 0$, $C \geq 0$, and $\beta_0 \geq 1$. Suppose

$$c_1 \leq c_0/8, \qquad c_1 \leq \beta_0, \qquad 10C \leq c_0\beta_0, \qquad \beta_0 V \leq W.$$

Let F, H, T, R_2, R_3 be real numbers satisfying

$$\begin{aligned} F &\leq k(X+T) - H + R_2, \\ F &\leq (k+n-1)X - c_1 Y + (2c_1/\beta_0)G + 8c_1k^3W, \\ T &\leq 0, \qquad H \leq -2c_0k^2W, \\ R_2 &\leq Ck^2V, \qquad R_3 \geq -Ck^3V. \end{aligned}$$

Then the exact remaining raising form obeys

$$A(dX - F - dH) + 2G + R_3 \geq \tfrac{1}{2}X + \tfrac{1}{4}c_1Y + c_0k^3W.$$

The estimate uses only the upper bound on R_2; its absolute-value bound in the manuscript is sufficient. The stronger negative trace term is discarded, making this scalar conclusion weaker and easier to apply.

Multiply the first bound for F by $1-\theta$ and the second by θ, and add. Substitute in the displayed raising form. The result is at least

$$\begin{aligned} A\{&(d - k - \theta(n-1))X + \theta c_1 Y - (1-\theta)kT - (d - 1 + \theta)H \\ &- (1-\theta)R_2 - 8\theta c_1 k^3 W\} + (2 - 2A\theta c_1/\beta_0)G + R_3. \end{aligned}$$

Here $2 \leq A \leq 4$, $d - k - \theta(n-1) = (3n-7)/8 \geq 1/4$, and $d - 1 + \theta \geq 5k/8$. Therefore the first two positive terms are at least $X/2 + c_1Y/4$. The T term is nonnegative, and the Hessian term is at least $(5/2)c_0k^3W$. The G coefficient is nonnegative because $c_1 \leq \beta_0$. The convex-bound loss is at most $4c_1k^3W \leq (c_0/2)k^3W$. As $k \geq 1$ and $\beta_0 V \leq W$, the two curvature losses sum to at most $5Ck^3V \leq (c_0/2)k^3W$. The remaining coefficient is at least $(3/2)c_0$, hence in particular at least c_0.

For the source application, $Y = \int (k/\beta)|D^+v|_F^2$, $W = \int \beta |v|_F^2$, and $V = \int |v|_F^2$. The second F inequality follows from the pointwise convex bound: $(k+n-1)\int \beta^{-1}|D^+v|_F^2 \geq Y$ and $\int \beta^{-1}|\sum_i a_i D_i^- v|_F^2 \leq G/\beta_0$. Since these are pointwise bounds, no derivative of β^{-1} is taken. Choose any $0 < c_1 \leq \min(c_0/8, 1)$ and then $\beta_0 \geq \max(1, 10C/c_0)$; all conditions are uniform in degree. This does not establish the sphere normalization, covariant integration by parts, curvature bounds, or PDE coercivity. Those remain explicit application obligations.

S25 Finite-vector robustness of the shrinking-sphere test

Private successor to shrinking-geometry-preparation. Source: original 04-shrinking-balls.tex lines 259–313, labels balls:test-vectors and balls:test-expression; neutral preserved expanded.tex lines 1526–1580. The completed predecessor is retained unchanged. Its exact flat calculation and fully proved finite-vector helpers are used, not any analytic premise. This artifact proves no Q-metric comparison, curvature estimate, continuation, Green identity, angular estimate, or PDE conclusion.

Lemma S25.1 (Minimized exact residual)**.** *Let $\|\omega\| = 1$, $\|b\| = B \le 1/4$, $u = \omega + b$, $D = \|u\|^2$, $q_0 = u/D$, and let $w_0 \perp u$ have $\|w_0\| = 1/\|u\|$. Set $Pz = z - \langle \omega, z\rangle \omega$, $k_0 = \|Pw_0\|$, $m = \langle \omega, u\rangle$, and $x = m/D$. Then*

$$\|P(\omega - q_0)\|^2 + (1 - k_0)^2 - \langle b, q_0\rangle^2 - \langle b, w_0\rangle^2 = 2(k_0 - x)(k_0 + x - 1) \ge 0.$$

Proof. The predecessor proves $D > 0$, $k_0 \ge x \ge 1/2$ by Cauchy–Schwarz on $\omega - xu$ and w_0. Its identities are $\|P(\omega - q_0)\|^2 = (D - m^2)/D^2$, $\langle b, q_0\rangle = (D - m)/D$, $\langle b, w_0\rangle = -\langle \omega, w_0\rangle$, and $k_0^2 + \langle \omega, w_0\rangle^2 = 1/D$. Substitution and expansion prove the stated factorization and positivity. □

The projection identities $\|Pz\|^2 = \|z\|^2 - \langle \omega, z\rangle^2$ and linearity imply $\|Pz\| \le \|z\|$, and therefore $|\|Px\| - \|Py\|| \le \|x - y\|$. Cauchy–Schwarz also gives $|\langle b, x\rangle - \langle b, y\rangle| \le B\|x - y\|$. For real numbers X, Y with $|X - Y| \le \eta$, $\eta \ge 0$, write $e = X - Y$ and expand $(Y + e)^2$. The inequalities $|Ye| \le |Y|\eta$ and $e^2 \le \eta^2$ give

$$Y^2 - 2|Y|\eta \le X^2 \le Y^2 + 2|Y|\eta + \eta^2.$$

These scalar comparisons will be applied before expanding norms.

Lemma S25.2 (Quantitative model size)**.** *The exact model above satisfies*

$$\|q_0\| = \|w_0\| = 1/\|u\| \le 2, \quad \|P(\omega - q_0)\| \le 2B, \quad |1 - \|Pw_0\|| \le 3B.$$

Proof. The triangle and reverse triangle inequalities give $3/4 \le s = \|u\| \le 5/4$ and $|s - 1| \le B$. The real projection equals $-Pb/D$, so its norm is at most $B/D \le (16/9)B \le 2B$. Since $u \perp w_0$, $z = \langle \omega, w_0\rangle = -\langle b, w_0\rangle$, so $|z| \le B/s$. Also $\|Pw_0 - w_0\| = |z|$. Thus

$$|1 - k_0| \le |1 - 1/s| + |1/s - k_0| \le B/s + |z| \le 2B/s \le (8/3)B \le 3B.$$

□

Theorem S25.3 (All-vector perturbation bound)**.** *For the exact model above, suppose $\|v\| = 1$, $v \perp \omega$, $\eta \ge 0$, $\|q - q_0\| \le \eta$, and $\|w - w_0\| \le \eta$. No orthogonality or norm condition is imposed on the perturbed q, w. Then*

$$R(q, w) = \|P(\omega - q)\|^2 + \|P(v - w)\|^2 - \langle b, q\rangle^2 - \langle b, w\rangle^2 \ge -16(B\eta + \eta^2).$$

This implies the requested bound with $C = 128$ as well.

Proof. Put $A = \|P(\omega - q)\|$, $A_0 = \|P(\omega - q_0)\|$, $k = \|Pw\|$. Projection contraction gives $|A - A_0|, |k - k_0| \le \eta$. The scalar square comparison and model sizes yield

$$A^2 \ge A_0^2 - 4B\eta, \qquad (1 - k)^2 \ge (1 - k_0)^2 - 6B\eta.$$

The reverse triangle inequality, $Pv = v$, and $\|v\| = 1$ give $\|P(v - w)\|^2 \ge (1 - k)^2$. For each $z = q, w$, with $z_0 = q_0, w_0$, Cauchy–Schwarz gives $|\langle b, z\rangle - \langle b, z_0\rangle| \le B\eta$ and $|\langle b, z_0\rangle| \le 2B$. Hence

$$\langle b, z\rangle^2 \le \langle b, z_0\rangle^2 + 4B^2\eta + B^2\eta^2.$$

Adding the four estimates and the minimized exact residual gives $R(q,w) \geq -10B\eta - 8B^2\eta - 2B^2\eta^2$. Since $0 \leq B \leq 1/4$, $8B^2\eta \leq 2B\eta$ and $2B^2\eta^2 \leq \eta^2/8$. Thus $R(q,w) \geq -12B\eta - \eta^2/8 \geq -16(B\eta + \eta^2)$. There is no pure $O(\eta)$ error: the imaginary term is minimized first, so its linear perturbation is proportional to $|1 - k_0| = O(B)$. □

Lemma S25.4 (Radial normalization of the tangential vector)**.** *If $u \perp w$ and $|\|w\| - 1/\|u\|| \leq \eta \leq 1/4$, then*

$$w_0 = \frac{1}{\|u\|\|w\|} w$$

is defined, is orthogonal to u, has norm $1/\|u\|$, and $\|w - w_0\| \leq \eta$.

Proof. Since $\|u\| \leq 5/4$, $1/\|u\| \geq 4/5$ and $\|w\| \geq 4/5 - 1/4 = 11/20 > 0$. Its scalar multiplier is positive. Scalar multiplication preserves orthogonality and gives $\|w_0\| = 1/\|u\|$. Finally $\|w - w_0\| = |\|w\| - 1/\|u\|| \leq \eta$ by homogeneity. □

Lemma S25.5 (Perturbed Hessian contribution)**.** *Under the all-vector perturbation hypotheses, additionally suppose $\eta \leq 1/4$ and a real form H obeys $H(\xi,\xi) \leq -c\beta\|\xi\|^2$ for every vector, with $c, \beta \geq 0$. Then for every real r,*

$$R(q,w) - r^2(H(q,q) + H(w,w)) \geq \tfrac{1}{2}cr^2\beta - 16(B\eta + \eta^2).$$

Proof. Since $s \leq 5/4$, $\|q_0\| = \|w_0\| = 1/s \geq 4/5$. The perturbation bounds and reverse triangle inequality give $\|q\|, \|w\| \geq 4/5 - 1/4 = 11/20 \geq 1/2$. Thus $\|q\|^2 + \|w\|^2 \geq 1/2$. Applying the Hessian bound twice and multiplying by $r^2 \geq 0$ gives the positive Hessian contribution; add the preceding perturbation theorem. This conditional finite-vector lemma does not claim that the manuscript profile Hessian bound is proved. □

For application bookkeeping only, if a separately proved metric-vector comparison gives $\eta \leq K|h|$ and $|h| \leq \varepsilon r$, then $16(B\eta + \eta^2) \leq 16(K\varepsilon r^2\beta + K^2\varepsilon^2 r^2)$ when $B = r\beta$. This is precisely the finite-vector portion of the manuscript error structure; the actual Q comparison and curvature estimates remain separate obligations.

This finite-vector estimate applies to a metric perturbation only after one separately proves that the actual Q-normalized vectors obey these Euclidean closeness conditions, with $\eta \leq C\|h\|$ uniformly. No such metric or curvature estimate is represented as literature here.

S26 Exact classical energy bridge

Authority: the ten frozen definitions at commit `d1205fb493b3fbe04e55f72b76c6ff0761fb5a03`, preserved exactly in `frozen-target.lean`. Tau Ceti authority is the exact source commit `d87e7f1676050bccc88913a473d5aae77156313c`, particularly `Analysis/PDE/EnergyForm/Integrated/Basic.lean` and its pointwise `EnergyForm/Basic.lean` and `Ellipticity/Basic.lean` prerequisites. This is an API correspondence proof, not adoption of an analytic premise.

Keep $E = \text{EuclideanSpace}\,\mathbb{R}(\text{Fin}\,3)$, the frozen ball $B = \{x : \|x\| < 1\}$, the exact total Fréchet derivative, and $d_i u(x) = \text{fderiv}(u, x)(e_i)$ with the frozen vectors $e_i = \text{EuclideanSpace.single}(i, 1)$. Define the vector $G_u(x)$ by those three coordinates using the canonical finite L^2 product conversion; in particular $G_u(x)_i = d_i u(x)$ by definition. Define jets $J_u(x) = (u(x), G_u(x))$ and principal coefficients

$$A_g(x) = \text{vol}(g, x)\, g(x)^{-1}.$$

No differentiability or nonsingularity is added to this correspondence: both sides retain their respective total derivative, matrix inverse, square root and Bochner integral conventions.

The pinned energy-integrand API evaluates zero drift and mass as $G_u(x)^{\mathsf{T}} A_g(x) G_u(x)$. Its exact matrix convention is `matrixBilinearForm A V.2 U.2`; on the diagonal the two vectors are identical. In coordinates this is

$$\sum_i d_i u(x) \sum_j \text{vol}(g, x) g(x)^{-1}_{ij} d_j u(x) = \left(\sum_{i,j} g(x)^{-1}_{ij} d_i u(x) d_j u(x) \right) \text{vol}(g, x).$$

The equality uses only finite distributivity and real commutativity. Integrate the everywhere equal scalar functions against volume $\restriction B$. This is exactly the frozen set integral, and proves

$$\text{energy}(g, u) = \text{energyFormIntegral}(\text{volume} \restriction B)(A_g)(0)(0)(J_u)(J_u).$$

No integrability hypothesis is needed for equality of the same total Bochner integrals. Integrability needed for estimates or minimization is separate.

The vector $G_u(x)$ also equals the total Mathlib gradient: the Riesz identity $\langle \nabla u(x), e_i \rangle = d_i u(x)$ and the coordinate inner-product rule identify every coordinate. This identification has no differentiability assumption and consequently agrees also with the total zero derivative at nondifferentiable points.

S26.1 Exact smooth competitor parameterization

For a globally smooth datum f, define

$$Z = \{w : E \to \mathbb{R} : \ w \text{ globally smooth and } \|x\| \geq 1 \Rightarrow w(x) = 0\}.$$

The frozen competitor class is exactly $\{f + w : w \in Z\}$. Indeed, if u is a frozen competitor, $w = u - f$ is globally smooth and vanishes on the prescribed exterior, and $f + w = u$. Conversely, $w \in Z$ gives the globally smooth competitor $u = f + w$ with its required exterior value. Reindexing this exact set proves, without a minimizer assumption,

$$\text{minimalEnergy}(g, f) = \inf\{\text{energy}(g, f + w) : w \in Z\}.$$

Every $w \in Z$ has topological support in the closed unit ball: nonzero values occur in the open ball, and closing the support remains in the closed ball. The closed ball is compact in this exact finite-dimensional Euclidean space, so every such w is globally compactly supported. This does *not* imply its topological support is compact inside the open ball. A smooth function vanishing to infinite order on the sphere can have topological support meeting that sphere. Embedding every such perturbation in $H^1_0(B)$ therefore needs a genuine closure/density argument.

S26.2 Concrete test-function embedding bridge

Let Ω be the exact open unit ball, bundled as an open set, and let $\phi \in \mathcal{D}(\Omega, \mathbb{R})$. Its smoothness is part of the bundle. Its support closure lies inside Ω, so $\|x\| \geq 1$ implies $\phi(x) = 0$; hence its underlying function belongs to Z. The existing pinned linear embedding

$$T\phi = \mathrm{W1p.ofTestFunction}_{\ell}(\mathrm{volume}, \Omega, 2)(\phi)$$

has value component equal to the existing test-function L^2 class and gradient component equal to the test-function gradient L^2 class. The supplied almost-everywhere representatives therefore give

$$\mathrm{jetField}(T\phi)(x) = (\phi(x), \nabla\phi(x)) = J_{\phi}(x) \quad \text{a.e. on } B.$$

Substitution into the total integrated form, followed by the proved exact classical energy identity, gives

$$\mathrm{energy}(g, \phi) = \mathrm{energyFormH1}(A_g, 0, 0)(T\phi, T\phi).$$

The existing embedding theorem puts $T\phi$ in the closed submodule $W_0^{1,2}(B)$. This proves a concrete bridge for the genuine test-function subset, not the full frozen competitor class. No coefficient bounds, ellipticity, integrability, density assertion or minimizer is required for this equality of total integrals.

S26.3 An abstract infimum lemma with explicit hypotheses

For a nonempty subset S of a topological space and a continuous real-valued functional F, suppose $F(\overline{S})$ is bounded below. Then

$$\inf F(\overline{S}) = \inf F(S).$$

One inequality follows from $S \subset \overline{S}$. For the other, let $r = \inf F(S)$. Every $x \in S$ satisfies $r \leq F(x)$, since the image is nonempty and bounded below. The closed set $F^{-1}([r, \infty))$ contains S, hence its closure. Thus r is a lower bound of $F(\overline{S})$ and $r \leq \inf F(\overline{S})$, as required. This lemma isolates only the topological infimum mechanism. Applying it to the frozen smooth competitor energy still requires the concrete Sobolev embedding of all Z, an affine functional continuous in the graph norm, nonemptiness and a genuine lower bound. Those are not granted by this lemma.

S27 Admissible coefficients, continuity and nonnegative energy

This successor uses the actual frozen `ComparatorChallenges.AnisotropicCalderon` definitions and the exact passed energy bridge, preserving all derivative and total integral conventions. Let g satisfy the frozen `IsAdmissibleMetric` and set $A(x) = \sqrt{\det g(x)}\, g(x)^{-1}$. Positive definiteness of $g(x)$ gives $\det g(x) > 0$, hence $\sqrt{\det g(x)} > 0$. The checked library inverse theorem gives positive definiteness of $g(x)^{-1}$, and its positive scalar multiplication theorem gives positive definiteness of $A(x)$. In particular $\xi^{\mathsf{T}} A(x)\xi \geq 0$ for every vector, including zero.

Each matrix entry of g is smooth, hence continuous. The finite matrix product topology therefore gives continuity of g. The determinant is continuous and everywhere nonzero. Matrix inverse is continuous at a matrix whose determinant is nonzero, so $x \mapsto g(x)^{-1}$ is continuous. The square root is globally continuous; scalar multiplication then gives continuity of A. No exterior-equality assumption is needed in these steps.

The exact energy bridge identifies the frozen energy with the zero-drift, zero-mass integrated principal form. Pointwise nonnegative quadratic density and the nonnegative-integral theorem imply $\mathrm{energy}(g, u) \geq 0$ for every function u, without assuming integrability: the total integral retains its default-zero convention. Thus every member of the set whose infimum defines `minimalEnergy` is nonnegative. The real total-infimum nonnegativity theorem gives $\mathrm{minimalEnergy}(g, f) \geq 0$ for every f, including empty competitor sets (whose total real infimum is zero). No minimizing competitor or regularity of u or f is asserted.

S27.1 Uniform ellipticity on the exact ball

Let $K = \overline{B(0,1)}$, compact in the exact three-dimensional Euclidean space. Let $S = \{\xi : \|\xi\| = 1\}$, also compact and nonempty. The continuous function $Q(x,\xi) = \xi^{\mathsf{T}} A(x)\xi$ is strictly positive on $K \times S$. Its attained minimum λ is therefore strictly positive. Rescaling a nonzero ξ to $\xi/\|\xi\|$ and using quadratic homogeneity yields $\lambda\|\xi\|^2 \leq Q(x,\xi)$; zero is immediate. The continuous matrix bilinear form has finite operator norm on K; an attained upper bound M gives $|\eta^{\mathsf{T}} A(x)\xi| \leq M\|\eta\|\|\xi\|$. Taking $\Lambda = \max(M, \lambda)$ retains that bound and ensures $0 < \lambda \leq \Lambda$. This is the exact pinned `TauCeti.PDE.UniformlyEllipticOn K A lambda Lambda` predicate. The frozen open ball B is a subset of K, so restricting the same bound via `UniformlyEllipticOn.mono_set` gives the exact bound on B. The compact minimization, rescaling and norm-bound steps give the exact quantitative predicate. They do not by themselves identify the full smooth competitor infimum with a weak Dirichlet minimum.

S28 Characteristic normalization from metric smallness

S28.1 Exact sphere normalization

At a point of $d_\gamma(x,y) = r(y) > 0$ in a normal neighborhood, parallel transport along the radial γ-geodesic identifies the endpoint tangent spaces. Write $\omega \in T_y$ for its unit direction, $b = \nabla^\gamma r$, and $u = \omega + b$. The first variation of distance and $\rho = (d_\gamma^2 - r^2)/2$ give exactly

$$d_x\rho = r\omega^\flat, \qquad d_y\rho = -ru^\flat.$$

These first-variation formulas are geometric inputs here, not Lean-proved results of this private finite-vector component. Choose a γ-orthonormal frame. The first principal metric becomes I, the second contravariant principal operator becomes a self-adjoint positive operator Q, and its covariant metric is $M = Q^{-1}$. Put $s = \|u\|$ and $t = \langle u, Qu\rangle$. The second metric gradient is $-rQu$ and has squared M-norm r^2t. The source's $a_i = \nabla_i\rho/\|\nabla_i\rho\|^2$ therefore gives

$$ra_1 = \omega, \qquad -ra_2 = q = Qu/t.$$

If $w = rv_2$ is any allowed second tangential vector, the source conditions become $\langle u, w\rangle = 0$ and $\langle w, Mw\rangle = 1/t$. Equivalently, put $\zeta = Mw$, so

$$w = Q\zeta, \quad \langle u, Q\zeta\rangle = 0, \quad \langle \zeta, Q\zeta\rangle = 1/t.$$

Conversely these conditions reconstruct precisely an allowed w. Self-adjointness gives $\langle q, Mw\rangle = \langle u, Q\zeta\rangle/t = 0$ and $\langle q, Mq\rangle = 1/t$. Thus $U_2 = q + iw$ is M-null and $\langle u, U_2\rangle = 1$. The real part q is fixed by the normalized metric normal; the two complex equations alone must not be used to infer that formula for an arbitrary real part.

For completeness, metric smallness with $\kappa < 1$ makes Q injective: if $Qz = 0$, then $\|z\| = \|Qz - z\| \le \kappa\|z\|$, hence $z = 0$. Applying this to $z = x - y$ proves injectivity. On a finite-dimensional space, the checked library's linear injectivity/surjectivity theorem makes Q bijective, so the inverse and $\zeta = Q^{-1}w$ exist without a new assumption. The same Rayleigh bound makes $\langle z, Qz\rangle > 0$ for nonzero z. The real normalized covector $\alpha = u/t$ satisfies $\langle \alpha, Q\alpha\rangle = 1/t$ and $\langle \alpha, Q\zeta\rangle = 0$; with source symmetry, the reversed mixed pairing vanishes as well. These equations expose the real and imaginary characteristic normalization explicitly.

S28.2 Operator smallness and concrete vector bounds

Assume $\|\omega\| = 1$, $\|b\| \le 1/4$, $0 \le \kappa \le 1/4$, and $\|Q - I\|_{op} \le \kappa$. The latter metric hypothesis implies, for every real vector z,

$$\|Qz - z\| \le \kappa\|z\|, \qquad |\langle z, Qz\rangle - \|z\|^2| \le \kappa\|z\|^2.$$

The second inequality is Cauchy–Schwarz on z and $(Q - I)z$. Thus $s \in [3/4, 5/4]$ and $(1 - \kappa)s^2 \le t \le (1 + \kappa)s^2$, so $t > 0$. The unperturbed model is $q_0 = u/s^2$.

Lemma S28.1 (Real characteristic-vector error)**.** *Under these hypotheses,* $\|q - q_0\| \le 4\kappa$.

Proof. The exact decomposition

$$q - q_0 = t^{-1}(Qu - u) + (t^{-1} - s^{-2})u$$

and the quadratic-form bound yield

$$\|q - q_0\| \le \kappa s/t + |t - s^2|s/(ts^2) \le 2\kappa s/t.$$

Moreover $t \ge (3/4)s^2 \ge s/2$ because $s \ge 3/4$. Therefore $2\kappa s/t \le 4\kappa$. □

Lemma S28.2 (Imaginary characteristic-vector length error)**.** *If $\langle \zeta, Q\zeta \rangle = 1/t$ and $w = Q\zeta$, then $|\|w\| - 1/s| \le 4\kappa$.*

Proof. Put $a = \langle \zeta, Q\zeta \rangle > 0$, $z = \|\zeta\|$, $d = sz$. The quadratic-form bounds at u and ζ and $ta = 1$ give

$$(1-\kappa)^2 d^2 \le 1 \le (1+\kappa)^2 d^2.$$

All factors are nonnegative, so $(1-\kappa)d \le 1 \le (1+\kappa)d$. Consequently $|d-1| \le \kappa d$ and $d \le 4/3$. The operator bound gives $|\|w\| - z| \le \kappa z$. Multiplying by s and using the triangle inequality,

$$|s\|w\| - 1| \le s|\|w\| - z| + |d-1| \le 2\kappa d \le (8/3)\kappa.$$

As $s \ge 3/4$, divide by s to obtain $|\|w\| - 1/s| \le (32/9)\kappa \le 4\kappa$. No inverse-operator or spectral closeness was assumed in this calculation. □

Theorem S28.3 (Actual characteristic vectors satisfy the perturbation interface)**.** *If additionally $\kappa \le 1/16$ and $\langle u, Q\zeta \rangle = 0$, the previous normalization lemma constructs $w_0 = w/(s\|w\|)$ with $u \perp w_0$, $\|w_0\| = 1/s$, and $\|w - w_0\| \le 4\kappa$. Together with $\|q - q_0\| \le 4\kappa$, this applies the proved constant-16 robustness theorem with $\eta = 4\kappa \le 1/4$. For every unit $v \perp \omega$, the actual vector residual obeys*

$$R(q, w) \ge -64\|b\|\kappa - 256\kappa^2.$$

If $H(\xi, \xi) \le -c\beta\|\xi\|^2$ for every vector and $c, \beta \ge 0$, then the proved perturbed-Hessian component gives

$$R(q, w) - r^2(H(q,q) + H(w,w)) \ge \tfrac{1}{2}cr^2\beta - 64\|b\|\kappa - 256\kappa^2.$$

These are finite-vector conclusions with explicit characteristic constraints, not assumptions of the desired closeness or residual.

S28.3 Inverse-metric formulation

On a finite-dimensional real inner product space, the already proved bijectivity of Q constructs a linear inverse M by the checked `LinearEquiv.ofBijective` construction. Thus $Q(Mz) = z$ and $M(Qz) = z$ for every z, without positing an external inverse operator. For a source-allowed w, take $\zeta = Mw$. The inverse identity gives $Q\zeta = w$; symmetry of the real inner product gives

$$\langle \zeta, Q\zeta \rangle = \langle Mw, w \rangle = \langle w, Mw \rangle = 1/t.$$

Also $\langle u, Q\zeta \rangle = \langle u, w \rangle = 0$. The characteristic-vector theorem therefore proves the same 4κ normalization distances and constant-$64, 256$ residual bounds directly for every w satisfying the actual inverse-metric length and tangency conditions. Here the identity $M = Q^{-1}$ is derived in finite dimension, whereas the actual source metric's identification with this inverse is ordinary metric linear algebra, not a new desired-closeness assumption. For $q = Qu/t$ and symmetric Q, $Mq = u/t$, so the same calculation yields

$$\langle u, q \rangle = 1, \quad \langle q, Mq \rangle = \langle w, Mw \rangle = 1/t, \quad \langle q, Mw \rangle = \langle w, Mq \rangle = 0.$$

These are precisely the real and imaginary parts of the null condition for $q + iw$, together with its unit defining-function normalization.

S28.4 Correspondence with h, curvature, and connections

In the manuscript $h = Q - \gamma^{-1}$ is a contravariant tensor. Writing G for the coordinate matrix of γ, its orthonormal-frame representation is $\widetilde{Q} = G^{1/2}QG^{1/2} = I + G^{1/2}hG^{1/2}$. Hence $\|\widetilde{Q} - I\|_{op} \le \|G\|_{op}\|h\|_{op}$. The finite operator portion is proved in the general form $Q - I = LHL \Longrightarrow \|Q - I\|_{op} \le \|L\|_{op}^2\|H\|_{op}$: apply submultiplicativity to $(LH)L$ and then to LH. Taking $L = G^{1/2}$ gives the displayed coordinate bound once the background-frame identification and square-root norm equality are justified. With uniformly bounded background G, take $\kappa = C_\gamma\|h\|$; no new metric-closeness premise is introduced. A coordinate tensor norm is converted to the operator norm by the fixed finite-dimensional norm comparison, whose constant must be retained. The source bootstrap $H_8 \le \varepsilon r$ implies the pointwise bound $\kappa \le C_\gamma \varepsilon r$. With $B = r\beta$ the finite-vector error just proved is bounded by $64C_\gamma\varepsilon r^2\beta + 256C_\gamma^2\varepsilon^2 r^2$.

Parallel γ-orthonormal frames make the first metric identity exact; there is no extra distance-dependent error in q or the first-variation normalization. If one instead uses γ-normal coordinates based at y, the coordinate metric at x is $I + O(Kr^2)$, not generally $I + O(r^2)$ in arbitrary coordinates. The bound depends on curvature and normal-domain constants. This coordinate metric error and the distance Hessian's Jacobi error are distinct from the pointwise h error.

For the actual product connection, write $T = \nabla^M - \nabla^\gamma$. Its exact tensor formula is

$$2M(T(X,Y),Z) = (\nabla^\gamma_X M)(Y,Z) + (\nabla^\gamma_Y M)(X,Z) - (\nabla^\gamma_Z M)(X,Y).$$

Under separately justified bounded-metric comparisons this yields $\|T\| \le C(\|h\| + \|\partial h\|)$ in fixed coordinates. The source bootstrap then gives $\|T\| \le C\varepsilon r$. The connection replacement changes the second-variable Hessian exactly by $-d_y\rho(T(w,w))$ (and the analogous real-normal term). Since $\|d_y\rho\| = r\|u\| \le (5/4)r$ and the normalized endpoint vectors are bounded, its size is at most $C\varepsilon r^2$.

Separately, the Jacobi/second-variation estimate requires a proved bound $C_K r^2$ for the squared-distance Hessian evaluated on the endpoint vectors. That curvature estimate, first variation, existence of the normal coordinates/parallel frames, the coordinate norm comparisons, and the connection estimate are explicitly not proved by the Lean finite-vector component. They must be closed before deriving source continuation. The present calculation proves the previously missing pointwise characteristic-vector normalization and error size from actual metric smallness; it does not certify the full analytic or geometric application.

S29 Restriction closure and complete cross matching

All solution spaces in this section are over $\mathbb{C}$; the real case is obtained by restriction. Inner products are linear in their first argument. Coordinate L^2 norms use Lebesgue measure. The operators are scalar second-order operators with smooth coefficients and real positive definite principal matrices. Their lower-order coefficients may be complex.

S29.1 The unique-continuation principle used below

Lemma S29.1 (Weak unique continuation)**.** *Let V be a connected coordinate domain and let L be an operator of the preceding type on V. If $u \in C^\infty(V)$ satisfies $Lu = 0$ and vanishes on a nonempty open subset of V, then $u = 0$ on V. The same assertion applies after interior elliptic regularity to an L^2_{loc} distributional solution.*

Proof. Let Z be the set of points having a neighborhood on which $u = 0$. This is nonempty and open. If $z \in \overline{Z} \cap V$, continuity of every derivative gives $\partial^\alpha u(z) = 0$ for every multi-index α. Taylor's theorem implies vanishing of infinite order at z. In coordinates centered at z, the smooth real elliptic principal coefficients satisfy the coefficient hypothesis of Hörmander's Theorem 17.2.6. The equation gives a bounded lower-order inequality $|p(x, D)u| \le C(|\nabla u| + |u|)$, which satisfies the weighted lower-order inequality in that theorem on a small ball, with any fixed parameter $0 < \sigma \le 1$. Thus $u = 0$ near z. The set Z is relatively closed, hence $Z = V$. The cited result is Hörmander [3, Theorem 17.2.6, pp. 14–17]. For distributional solutions, the local elliptic parametrix of Hörmander [3, Theorem 18.1.24, p. 87] first gives smoothness. □

S29.2 The closure of the restriction range

Lemma S29.2 (Cross expansion with restriction closure)**.** *Let $D_i \subset \mathbb{R}^n$ be bounded connected coordinate domains, $S_i \Subset D_i$ nonempty open sets, and $O_i \Subset D_i$ open sets. Let L_i be operators of the preceding type on neighborhoods of the sets where they are used. Suppose a consistent product solution F is given on*

$$(D_1 \times S_2) \cup (S_1 \times D_2), \qquad \|F\|_{D_1 \times S_2} + \|F\|_{S_1 \times D_2} \le M.$$

Suppose every L_i-solution $w \in L^2(D_i)$ satisfies

$$\|w\|_{O_i} \le C_i \|w\|_{S_i}^{\gamma_i} \|w\|_{D_i}^{1-\gamma_i}, \qquad 0 < \gamma_i < 1, \quad \gamma_1 + \gamma_2 > 1.$$

Then the singular-function series constructs a smooth product solution on $O_1 \times O_2$, agreeing with F on both actual intersections with the given cross. In the completeness argument the relevant Hilbert subspace is

$$\mathcal{K} = \overline{R\mathcal{H}_1(D_1)}^{\,L^2(S_1)}, \qquad \mathcal{H}_1(D_1) = \{w \in L^2(D_1) : L_1 w = 0 \text{ in distributions}\},$$

where R is restriction to S_1. No closedness of $R\mathcal{H}_1(D_1)$ is required.

Proof. The space $\mathcal{H}_1(D_1)$ is closed: if $w_k \to w$ in $L^2(D_1)$, then pairing with $L_1^*\varphi$ for $\varphi \in C_c^\infty(D_1)$ preserves $L_1 w = 0$. Restriction has norm at most one. Interior elliptic estimates show it is compact. Indeed, multiply by a cutoff equal to one near $\overline{S}_1$ and supported in D_1. For every integer k the resulting H^k norm is bounded by $C_k \|w\|_{D_1}$. Fourier projection on a cube containing this

support, after extension by zero, has rank $O(A^n)$ and L^2 error $O(A^{-k})$. Restricting those finite-rank approximants to S_1 proves compactness and gives

$$\mu_j \le C_k(1+j)^{-k/n} \quad \text{for every fixed } k. \tag{S29.1}$$

The cutoff estimates follow from the usual localized elliptic estimates and differentiation; see Hörmander [3, Theorem 17.1.3 and the extension by Friedrichs' lemma, pp. 6–9].

By Lemma S29.1, restriction is injective. Thus R^*R is compact, positive and self-adjoint with zero kernel. The compact self-adjoint spectral theorem supplies an orthonormal basis (w_j) of $\mathcal{H}_1(D_1)$ with

$$R^*Rw_j = \mu_j^2 w_j, \qquad 0 < \mu_j \le 1, \qquad (Rw_i, Rw_j)_{S_1} = \mu_j^2 \delta_{ij}.$$

The singular values are listed in nonincreasing order, with multiplicity. For this spectral theorem, including its kernel decomposition, see McLean [1, Theorem 2.36, pp. 47–48]. If the space is finite dimensional, all subsequent sums are finite. For completeness, if T has rank at most m, its restriction to the span of $w_1, \dots, w_{m+1}$ has a nonzero kernel vector w of norm one. Then $\mu_{m+1} \le \|Rw\| = \|(R-T)w\| \le \|R-T\|$. Applying this inequality to the Fourier approximants gives (S29.1) directly.

The vectors $e_j = Rw_j/\mu_j$ are a complete orthonormal system in $\mathcal{K}$. To verify completeness, expand any $w \in \mathcal{H}_1(D_1)$ in the w_j basis and apply the bounded map R. Thus the span of the Rw_j is dense in $R\mathcal{H}_1(D_1)$ and hence in its closure $\mathcal{K}$.

Fubini's theorem and distributional testing give $F(\cdot, y) \in \mathcal{H}_1(D_1)$ for almost every $y \in S_2$. Its first-piece expansion is

$$F(x,y) = \sum_j w_j(x) b_j(y), \qquad b_j(y) = (F(\cdot,y), w_j)_{D_1}, \quad \|b_j\|_{S_2} \le M.$$

On all of D_2, define instead

$$b_j(y) = \mu_j^{-2}(F(\cdot,y), Rw_j)_{S_1}. \tag{S29.2}$$

Restriction of the first-piece expansion proves that these definitions agree almost everywhere on S_2. Cauchy–Schwarz gives $\|b_j\|_{D_2} \le M/\mu_j$. Testing in y under the integral shows $L_2 b_j = 0$ distributionally, so each b_j is smooth in the interior. The two interpolation estimates imply

$$\|w_j\|_{O_1} \le C_1 \mu_j^{\gamma_1}, \qquad \|b_j\|_{O_2} \le C_2 M \mu_j^{\gamma_2 - 1}.$$

Put $\alpha = \gamma_1 + \gamma_2 - 1 > 0$ and choose k with $k\alpha/n > 1$. Equation (S29.1) makes

$$\sum_j \|w_j b_j\|_{O_1 \times O_2} \le C_1 C_2 M \sum_j \mu_j^{\alpha} < \infty.$$

The limit $\widetilde{F}$ solves both equations in distributions. Their sum is a scalar elliptic operator on the product, so interior regularity gives smoothness and compact-subset C^m bounds.

It remains to identify the second-piece limit. For any $z \in \mathcal{K}^{\perp}$, define the $L^2(D_2)$ function

$$q_z(y) = (F(\cdot,y), z)_{S_1}.$$

Fubini and testing show $L_2 q_z = 0$ distributionally. For almost every $y \in S_2$, the first-piece slice is a restriction of an element of $\mathcal{H}_1(D_1)$, so $q_z = 0$ on S_2. Elliptic regularity and Lemma S29.1 give $q_z = 0$ on connected D_2. Since $L^2(S_1)$ is separable, take a countable dense set in $\mathcal{K}^{\perp}$ and discard the union of its exceptional null sets. It follows that $F(\cdot, y) \in \mathcal{K}$ for almost every $y \in D_2$.

Parseval's expansion of these second-piece slices in the e_j basis has coefficients

$$(F(\cdot, y), e_j)_{S_1} = \mu_j b_j(y).$$

Consequently $\sum_j (Rw_j)b_j$ converges to F in $L^2(S_1 \times D_2)$; this also follows directly by Parseval and dominated convergence. On $S_1 \times O_2$ the same series converges absolutely, since its summand norm is at most $C_2 M \mu_j^{\gamma_2}$. Both limits must therefore agree. On $O_1 \times S_2$, the first-piece series is likewise absolutely convergent, with summand norm at most $C_1 M \mu_j^{\gamma_1}$, and agrees with the original expansion. Restrict these identities respectively to $(O_1 \cap S_1) \times O_2$ and $O_1 \times (O_2 \cap S_2)$. Uniqueness of L^2 limits proves equality with $\widetilde{F}$ there. Smoothness turns these equalities into pointwise overlap identities. The argument makes no containment assumption between S_i and O_i. □

S29.3 Outward slices in a swept family of spheres

We now specify the geometric room needed to compare a continued kernel with both parts of the original cross. Let U be a coordinate neighborhood carrying a smooth Riemannian metric γ. Let $Y \Subset U$ be the closure of a bounded domain; its boundary may have corners. Let $V \subset U$ be an open neighborhood of Y, and let $r_-, r_+ \in C^\infty(V)$ satisfy, on Y,

$$0 < r_- < r_+ \le R, \qquad |dr_-|_\gamma, |dr_+|_\gamma \le \kappa < 1. \tag{S29.3}$$

The strict inequalities persist on a smaller V.

The following normal-neighborhood hypothesis is explicit. There is a compact set $K \Subset U$ containing all points $\exp_y^\gamma(v)$ with $y \in Y$ and $|v|_\gamma \le 2R$. For every $x \in K$, $\exp_x^\gamma$ is defined and injective for $|v|_\gamma < 3R$, its image is contained in U, and these radial geodesics minimize distance, so

$$d_\gamma(x, \exp_x^\gamma(s\vartheta)) = s \quad (0 < s < 3R, |\vartheta|_\gamma = 1). \tag{S29.4}$$

The same statements hold for y in a slightly smaller neighborhood of Y and for the compact sets occurring in the collars below. In particular, all distance functions used here are smooth off the diagonal. This requires room for rays centered at x, not only for the original spheres centered at y.

Define the compact swept set and its top and side subsets by

$$\begin{aligned}
\mathcal{T} &= \{(x, y) : y \in Y,\ r_-(y) \le d_\gamma(x, y) \le r_+(y)\},\\
\mathcal{T}_+ &= \{(x, y) : y \in Y,\ d_\gamma(x, y) = r_+(y)\},\\
\mathcal{T}_\partial &= \{(x, y) \in \mathcal{T} : y \in \partial Y\}.
\end{aligned}$$

All x coordinates in these sets lie in K.

Proposition S29.3 (Agreement with the entire mixed cross)**.** *Assume $n \ge 2$ and the geometry just specified. Let $W \subset U$ be open. Let G_1, G_2 be smooth off the diagonal in $U \times U$, with*

$$L_1^x G_1 = 0, \qquad L_2^y G_2 = 0, \qquad G_1 = G_2 \text{ on } (W \times W) \setminus \{x = y\}.$$

Let F be a smooth product solution on an open neighborhood $\mathcal{N}$ of $\mathcal{T}$, avoiding the diagonal.

Suppose an initialized product solution H has already been identified with both original cross pieces on its domain:

$$H = G_1 \text{ where } y \in W, \qquad H = G_2 \text{ where } x \in W.$$

Assume there are open subsets $\mathcal{N}_+, \mathcal{N}_L, \mathcal{N}_P \subset \mathcal{N}$ such that

1. *$\mathcal{T}_+ \subset \mathcal{N}_+$, H is defined on $\mathcal{N}_+$, and $F = H$ there;*
2. *every y coordinate in $\mathcal{N}_L$ belongs to W, and $F = G_1$ on $\mathcal{N}_L$;*
3. *H is defined on $\mathcal{N}_P$, and $F = H$ there;*
4. *$\mathcal{T}_\partial \subset \mathcal{N}_L \cup \mathcal{N}_P$.*

Then, after retaining a smaller open neighborhood $\mathcal{A}$ of

$$\Sigma_- = \{(x, y) : y \in Y,\ d_\gamma(x, y) = r_-(y)\},$$

one has

$$F = G_1 \text{ on } \mathcal{A} \cap \{y \in W\}, \qquad F = G_2 \text{ on } \mathcal{A} \cap \{x \in W\}. \tag{S29.5}$$

The same identities hold throughout the swept interior $y \in Y^\circ$, $r_-(y) < d_\gamma(x, y) < r_+(y)$.

Proof. Put $m = \min_Y r_- > 0$ and $g = \min_Y(r_+ - r_-) > 0$. Compactness of $\mathcal{T}$ and openness of $\mathcal{N}$ give $\eta_0 > 0$ such that all pairs with $y \in Y$ and

$$r_-(y) - \eta_0 < d_\gamma(x, y) < r_+(y) + \eta_0 \tag{S29.6}$$

belong to $\mathcal{N}$, with $x \in K$ and the stated normal room. This follows also by contradiction: a sequence of pairs with buffer width tending to zero has a subsequence converging to $\mathcal{T}$. Choose $\eta_0 < m/4$, $\eta_0 < g/4$ and $\eta_0 < R/4$. The same compactness argument makes a top radial collar lie in $\mathcal{N}_+$, and makes the buffered side collar lie in $\mathcal{N}_L \cup \mathcal{N}_P$. Decrease the buffer width to a common $\eta > 0$ that satisfies all three requirements.

First fix $y \in Y^\circ \cap W$. The full first-variable shell

$$r_-(y) - \eta < d_\gamma(x, y) < r_+(y) + \eta$$

is connected: in normal polar coordinates it is a positive radial interval times $\mathbb{S}^{n-1}$, and $n \geq 2$. It lies in $\mathcal{N}$. The function $F(\cdot, y) - G_1(\cdot, y)$ solves L_1 on this shell and is zero on its nonempty top collar because $F = H = G_1$ there. Lemma S29.1 proves equality on the whole shell. This includes a collar on both sides of its inner sphere.

Now fix $x \in W$ and $y_0 \in Y^\circ$ satisfying the buffer condition, with distance $s_0 = d_\gamma(x, y_0)$. If (x, y_0) is already in the retained top or side seed collar, the required equality with G_2 holds there. Otherwise take the unique unit vector ϑ for which $y_0 = \exp_x^\gamma(s_0\vartheta)$, and continue its ray

$$y(s) = \exp_x^\gamma(s\vartheta), \qquad s_0 \leq s \leq 2R.$$

The normal room makes this a well-defined unit-speed minimizing geodesic in U. As long as $y(s) \in Y$, define

$$q_\pm(s) = s - r_\pm(y(s)).$$

Equation (S29.3) gives

$$q'_\pm(s) = 1 - dr_\pm(y(s))[\dot{y}(s)] \geq 1 - \kappa > 0. \tag{S29.7}$$

If initially $q_+(s_0) \geq 0$, the point is in the top collar after the buffer has been decreased, so it was already settled. Otherwise follow the ray up to the first of these two events: it meets $q_+ = 0$, or it exits Y°. One event occurs by $s = 2R$. If the ray remains in Y that long, then $q_+(2R) = 2R - r_+(y(2R)) \geq R > 0$, forcing the first event earlier.

Before the stopping event, $q_+ < 0$ and $q_- > q_-(s_0) > -\eta$ by (S29.7). Thus the complete ray segment stays in the buffer (S29.6), hence in $\mathcal{N}$. This remains true when the starting point is

slightly inside the inner sphere: q_- increases until it reaches zero and never returns to the excluded inner region. The lower bound $m - \eta > 0$ avoids the diagonal.

At an outer event, $(x, y(s))$ belongs to $\mathcal{T}_+$ and a neighborhood of it is in $\mathcal{N}_+$; there $F = H = G_2$ because $x \in W$. At a boundary event, the retained buffered side collar lies in $\mathcal{N}_L \cup \mathcal{N}_P$. On $\mathcal{N}_P$ one again has $F = H = G_2$. On $\mathcal{N}_L$, the endpoint has $y \in W$, so $F = G_1 = G_2$ because the fixed x also lies in W. Each case supplies equality on an open neighborhood in the fixed-x second-variable slice. No boundary regularity of the first exit is needed: the side seed is an open neighborhood covering every point of ∂Y, including corners.

The difference $F(x, \cdot) - G_2(x, \cdot)$ solves L_2 on the open set $\{y : (x, y) \in \mathcal{N}\}$ away from x. The ray segment and the endpoint neighborhood lie in one connected component of that set. Lemma S29.1 makes the difference zero on that component, in particular at y_0. The comparison is in the second variable with x fixed; it does not require G_2 to solve the first equation outside W.

For centers on or slightly outside ∂Y, retain only the neighborhood supplied by $\mathcal{N}_L \cup \mathcal{N}_P$. Compactness of $\Sigma_- \cap \{y \in \partial Y\}$ gives an open neighborhood V_0 of Y whose part outside Y is covered by that side seed at the inner-sphere pairs and in a smaller radial collar. Take $\mathcal{A}$ to be the union of this retained side neighborhood and a sufficiently thin inner radial collar with $y \in Y^\circ$. It is an open neighborhood of Σ_-. The preceding two slice comparisons and the seed identities prove (S29.5) on all its original-cross intersections. There is no convexity assumption on W: its relevant variable stays fixed during each comparison. The proof also applies to every point of the swept interior, without the inner buffer. □

Lemma S29.4 (Uniform normal room on a fixed patch)**.** *Let Y be compact in a coordinate patch and let γ_λ be a smooth family of smooth positive metrics on a fixed neighborhood of Y, with λ in a compact interval. There are a compact enlarged patch and a number $\rho > 0$ such that every center in a smaller enlarged patch has an injective exponential map on the tangent ball of radius ρ, all those radial segments stay in the larger patch, and their lengths equal the corresponding Riemannian distances. Consequently* (S29.4) *holds for sufficiently small R, uniformly in λ.*

Proof. Choose two nested compact coordinate neighborhoods with positive coordinate margin, containing Y in their interiors. On the larger one, the metrics have common positive lower and finite upper eigenvalue bounds, and the Christoffel symbols and their first derivatives are uniformly bounded. The geodesic equation therefore has a common existence interval for sufficiently small initial velocities, during which its solutions stay in the larger neighborhood. In coordinate velocities, uniformly over centers and parameters,

$$\exp_x^{\gamma_\lambda}(v) = x + v + O(|v|^2), \qquad D_v \exp_x^{\gamma_\lambda}(v) = I + O(|v|).$$

Decrease the velocity ball so the derivative differs from I by less than $1/2$. Integrating that derivative along a straight segment in the velocity ball proves injectivity; the inverse function theorem gives smooth normal coordinates there. Uniform metric comparison converts this common coordinate-velocity ball into a common metric-velocity ball.

Gauss' lemma gives the normal-coordinate metric in the form $ds^2 + g_s$ with g_s nonnegative on angular directions. One can see its orthogonality directly: for a radial unit tangent ∂_s and an angular variation J, torsion freeness and metric compatibility give

$$\frac{d}{ds}\langle \partial_s, J\rangle = \langle \partial_s, \nabla_J \partial_s\rangle = \tfrac{1}{2} J\langle \partial_s, \partial_s\rangle = 0,$$

and the value at the center is zero. Every curve staying in the normal ball and joining the center to radius s thus has length at least s. A competing curve that leaves the normal ball first reaches its

boundary radius and has length at least that larger radius. Hence the radial segment minimizes among all competing curves in the patch. Shrink once to retain a common boundary margin. Taking $3R < \rho$ and making the $2R$ displacement of the original centers smaller than the enlarged-patch margin supplies all the required centers and rays. □

S29.4 Application to the shrinking-sphere construction

Fix a positive $\delta < 1$ and take $r_- = r_\delta$, $r_+ = r_1$. The propagated product solution over all intermediate radii is defined on a neighborhood of $\mathcal{T}$: strict monotonicity of $s \mapsto r_s(y)$ identifies its parameter cylinder with this swept set. In the notation of `balls:profiles` and `balls:bootstrap-condition`,

$$|dr_s|_\gamma = r_s\beta_s \le \frac{\varepsilon}{\beta_s} \le \frac{\varepsilon}{cp} \qquad (\delta \le s \le 1).$$

Choose p, ε so the final quantity is less than $1/2$. This supplies (S29.3); continuity supplies its strictly smaller than one bound on a neighborhood of the fixed Y. For $\delta = 1$ there is no swept annulus: the initialized kernel already gives the desired agreement near that sphere.

The seed boundary is covered by

$$Q_L = \{y : y_n < -1/4\}, \qquad Q_P = \{y : t(y) > t_b\}.$$

For sufficiently small λ, the relevant bounded part of Q_L lies in W_λ and the prescribed kernel is $G_{1,\lambda}$. On Q_P, all intermediate radii are at least $R_* t_b^p$. The top radius has the positive minimum $m_+ = \min_Y r_1 > 0$. Choose a fixed positive-separation threshold d_0 smaller than $\min(m_+, R_* t_b^p)/4$, and retain top and Q_P collars of width smaller than that same fixed margin. On those collars use the initialized kernel H of `cross:initial`. Its already proved whole-cross agreement supplies the hypotheses for $\mathcal{N}_+$ and $\mathcal{N}_P$.

These two center regions cover the entire boundary of $Y = \{|y'| \le 1, -1 \le y_n \le 1\}$. The lower face belongs to Q_L. The upper face belongs to Q_P. On a lateral face, either $y_n < -1/4$ or $t_0 = y_n + |y'|^2 \ge 3/4$, which gives $t > t_b$ by the chosen inequality $3t_b < t(1/4)$. They give open side neighborhoods, not just values on the faces. Propagation preserves their prescriptions on these neighborhoods. When $x \in W_\lambda$, the Q_L prescription equals $G_{2,\lambda}$ because both variables lie in W_λ. When $y \in W_\lambda$, the Q_P prescription equals $G_{1,\lambda}$ by the initialization comparison.

To supply (S29.4), first fix a compact enlarged coordinate neighborhood with positive room around Y. The metrics γ_λ form a smooth uniformly positive family there for small λ, including the limiting metric at $\lambda = 0$. Lemma S29.4 gives a common small normal-neighborhood radius on this compact family of centers. The quantity $\sup_Y r_1$ is a fixed finite constant times R_*. Consequently R_* can be decreased so that all displaced x centers and their length-$3R$ rays remain in the enlarged patch and inside that common normal radius. This choice is made with open room and before the final dilation threshold. A statement only giving normal balls centered at y is insufficient for the fixed-x ray argument; the just described enlargement supplies the stronger room condition.

Uniform local comparison of d_{γ_λ} with coordinate distance converts the fixed positive threshold d_0 into a fixed positive Euclidean separation threshold for `cross:initial`. All enlarged location bounds, collars, and positive separation thresholds are fixed before its smallness bound for λ is invoked. For the selected positive δ, compactness of the swept set gives the retained buffer used in Proposition S29.3. That proposition proves agreement with $G_{1,\lambda}$ wherever $y \in W_\lambda$, and with $G_{2,\lambda}$ wherever $x \in W_\lambda$, on an open neighborhood of the entire inner sphere family. The retained buffer may depend on δ; no estimate uniform as $\delta \downarrow 0$ is used in this comparison.

S30 Uniform initialization at fixed positive separation

This supplement supplies the kernel estimate and the quantitative finite construction underlying `cross:initial`. It concerns a prescribed bounded range and a prescribed positive separation. It supplies no bound uniform as that separation tends to zero. All continuation estimates below are derived from the cross-series construction and strict scalar weights; no mixed continuation theorem is being taken as a published premise.

S30.1 The actual interior Green kernel, with its smooth correction

Fix one of the original smooth metrics on the fixed compact smooth Dirichlet domain, and write $L = -\Delta_g$. Take a coordinate compact set K around the contact point, strictly inside that domain. Let $m(y) = \sqrt{\det g(y)}$ in this chart. The ordinary Dirichlet inverse is defined by its coercive form on H^1_0, and smooth data have smooth solutions up to the boundary. These are standard inputs: McLean, *Strongly Elliptic Systems and Boundary Integral Equations*, 2000, Lemma 2.32, printed p. 43, and Theorems 4.16 and 4.18, pp. 135–139, with the trace lifting of Lemma 3.36 and Theorem 3.37, pp. 101–102. The proof of Theorem 4.18 includes the zero-order regularity step.

Extend the coefficients smoothly and elliptically across a slightly larger chart. Hörmander III, corrected second printing 1994, reprinted 2007, Theorem 18.1.24, printed p. 87, constructs a properly supported parametrix $E \in \Psi^{-2}$ with $LE - I$ smoothing. Take cutoffs $\psi = 1$ near K and $\chi = 1$ near $\operatorname{supp}\psi$, both supported in the interior chart. Set $\widetilde{E} = \chi E\psi$. For source points near K, the kernel of $L\widetilde{E} - I$ is a jointly smooth function $F(x, y)$: the parametrix error is smoothing, and the commutator $[L, \chi]E\psi$ has separated input and output supports. The latter kernel is smooth by Theorem 18.1.16, pp. 80–81. Its supports are fixed away from the original boundary.

The kernel of $\widetilde{E}$ has zero Dirichlet trace. Correct its actual error using the ordinary inverse R:

$$K_R(x, y) = K_{\widetilde{E}}(x, y) - R_x F(x, y). \tag{S30.1}$$

For a smooth source supported near K, the right side solves the correct Dirichlet equation; uniqueness identifies it with R. The correction is jointly smooth in (x, y). Indeed each y derivative of F is smooth with fixed compact x support; the fixed Sobolev inverse and boundary regularity bound every prescribed x derivative of its solution, and Sobolev embedding then bounds joint derivatives. No inverse on an unspecified negative bulk Sobolev space is used for this correction.

K_R is the kernel with respect to coordinate Lebesgue measure. The manuscript's actual Green kernel is normalized with metric volume, so

$$G(x, y) = K_R(x, y)/m(y), \qquad Rf(x) = \int G(x, y) f(y) m(y)\, dy. \tag{S30.2}$$

This conversion retains all the following estimates since m is smooth and bounded away from zero on the fixed chart.

Lemma S30.1 (Finite interior Green derivative bounds)**.** *For $n \geq 3$ and every fixed pair of multi-indices α, ζ, there are $C_{\alpha\zeta}$ and $\rho_0 > 0$ such that*

$$|\partial_x^\alpha \partial_y^\zeta G(x, y)| \leq C_{\alpha\zeta} |x - y|^{2-n-|\alpha|-|\zeta|} \quad (x, y \in K, 0 < |x - y| < \rho_0). \tag{S30.3}$$

The constants involve the original fixed inverse, interior buffers, and finitely many coefficient and symbol seminorms at the specified order.

Proof. Localize the order -2 symbol dyadically at $|\xi| \sim 2^\ell$, $\ell \geq 0$. If $j = |\alpha| + |\zeta|$, differentiation of the kernel has size at most

$$C_N 2^{\ell(n-2+j)}(1 + 2^\ell|x-y|)^{-N}.$$

The decay factor follows by repeated integration by parts in ξ; the finitely many derivatives of the localized symbol are bounded by its order -2 seminorms. For $2^\ell \leq |x-y|^{-1}$ sum the geometric series with positive exponent $n-2+j$. For the remaining indices choose $N > n-2+j$ and sum the decreasing geometric series. Both sums are bounded by $C|x-y|^{-(n-2+j)}$. The bounded low-frequency kernel, the smooth correction in (S30.1), and derivatives of m^{-1} in (S30.2) satisfy the same weaker bound after decreasing $\rho_0 \leq 1$. This proves the claim. □

For $G_\lambda(x,y) = \lambda^{n-2}G(\lambda x, \lambda y)$,

$$\partial_x^\alpha \partial_y^\zeta G_\lambda(x,y) = \lambda^{n-2+j}(\partial_X^\alpha \partial_Y^\zeta G)(\lambda x, \lambda y). \tag{S30.4}$$

Fix $B_1 < \infty$ and $d_1 > 0$. Choose λ sufficiently small that $\lambda B(0,B_1) \subset K$ and $2\lambda B_1 < \rho_0$. The powers of λ cancel in (S30.3), giving

$$\sup_{|x|,|y|\leq B_1, |x-y|\geq d_1} |\partial_x^\alpha \partial_y^\zeta G_\lambda(x,y)| \leq C_{\alpha\zeta} d_1^{2-n-j}. \tag{S30.5}$$

The case $x = y$ is excluded. The restriction $n \geq 3$ is used in the dyadic geometric sum and in the power normalization. In dimension two the zeroth derivative has a logarithmic term; that case is not part of the selected theorem or this assertion.

Applying this proof separately to the two fixed original metrics gives uniform bounds on every initial mixed-data compact set with positive separation. Its two equations retain their normalizations: under $X = \lambda x$, $L_{g_\lambda} = \lambda^2 L_g$ on pulled-back functions, and $\lambda^n \delta^g_{\lambda y}(\lambda x) = \delta^{g_\lambda}_y(x)$.

S30.2 Uniform radial interpolation on fixed templates

The weight $\theta(z) = |z - z_*|^{-1}$ has normal Hessian $2r^{-3}$, tangential Hessian $-r^{-3}$, and nonzero gradient on a fixed annulus. Its strict-weight sum is $r^{-3} > 0$. The scalar commutator proof in `cross:carleman` makes its constants uniform under small smooth coefficient perturbations. For completeness it applies on the whole fixed annulus without requiring a single constant between all normal and tangential Hessian values: use $m(z) = 3r^{-3}/2$ in the auxiliary pairing $4\tau\,\mathrm{Re}(mA_\tau v, v)$. Then $\mathrm{Hess}\,\theta + mg$ is positive definite and $\mathrm{Hess}\,\theta(e,e) - m > 0$ at the Euclidean coefficients. The variable multiplier adds only the bounded zero-order term $-2\tau\Delta_g m\,|v|^2$, by integrating $\mathrm{Re}(m(-\Delta_g)v, v)$. Cauchy–Schwarz bounds the auxiliary pairing by $\|A_\tau v\|^2/2 + C\tau^2\|v\|^2$. The positive τ^3 term absorbs it and the extra multiplier term for large τ. Strict margins persist under small perturbations. This derives the needed estimate with fixed radial cutoffs. Hörmander IV, Chapter XXVIII, Proposition 28.3.3 and its proof, pp. 240–241, give the corresponding strict-symbol positivity framework; the particular radial cutoff inequality here is a direct scalar calculation, not a quoted quantitative three-ball theorem.

Let $r_A < r_B < R < R_C < R_D$ be fixed radii inside a solution domain, and assume an observed ball contains the inner cutoff support $r_A \leq r \leq r_B$ with room for an interior gradient estimate. Use a cutoff zero before r_A, one from r_B through R_C, and zero beyond R_D. Interior estimates bound its commutator terms by the observed norm and the large-domain norm. Since θ decreases, the weighted estimate on the target ball of radius R, adding its already observed inner part, gives

$$\|u\|_{B_R} \leq C\left(e^{\tau(\theta(r_A)-\theta(R))}\|u\|_S + e^{-\tau(\theta(R)-\theta(R_C))}\|u\|_D\right) \quad (\tau \geq \tau_0). \tag{S30.6}$$

Any harmless inverse polynomial Carleman prefactors are bounded for $\tau \geq \tau_0 \geq 1$. Normalize the large norm to one and balance the exponentials. The result is

$$\|u\|_{B_R} \leq C\|u\|_S^{\gamma}\|u\|_D^{1-\gamma}, \qquad \gamma = \frac{\theta(R) - \theta(R_C)}{\theta(r_A) - \theta(R_C)} \in (0,1). \tag{S30.7}$$

If the balancing parameter is below τ_0, increasing C covers the estimate by the trivial large-norm bound. The zero observed norm case follows from ordinary unique continuation. All constants are uniform for fixed radii, cutoffs, ellipticity and finite coefficient bounds. They need not be uniform when two of the fixed radii coalesce.

Normalize the first plane-crossing template by its scale s. Write ϑ_0 for its small geometric constant (the source calls this τ_0); it is distinct from the Carleman lower threshold. In `cross:initial`, put $p = x_0 - \vartheta_0 s e_n$, choose $\vartheta_0 < 1/16$, and keep the outer cutoff in a fixed concentric ball of radius less than $3\vartheta_0 s$. Take $r_A = (1-2\alpha)\vartheta_0 s$, an inner cutoff ending before the observed radius $(1-\alpha)\vartheta_0 s$, and target radius $R = (\vartheta_0 + c)s$. Keep R_C a fixed amount beyond $\vartheta_0 s$. Formula (S30.7) tends to one as $\alpha, c \downarrow 0$. The cutoff constants may grow in this limit; choose positive values once and freeze them before choosing λ. The target contains a ball about x_0 with radius cs.

The second template is the fixed annulus $1/2 < |z| < 2$, its observed ball about $-e_n$, and a target compact annulus slightly larger than $4/5 \leq |z| \leq 6/5$. Join finitely many small target balls to the observed ball by finite chains whose enlarged balls remain inside this connected annulus. Applying (S30.7) along each chain and taking a finite cover gives a fixed $\gamma_2 > 0$. Each chain contributes the product of its positive exponents; the minimum of finitely many such products is positive. This construction precedes the choice of α, c, so γ_2 is independent of those choices. Choose the latter so that $\gamma_1 + \gamma_2 \geq 1 + a_0$ for some fixed $a_0 > 0$. Fixed reference shapes and radii determine these exponents, independently of the eventual bounded range of centers and scales. On any specified compact range of those parameters, the same templates work for sufficiently small λ because the rescaled coefficients converge smoothly and uniformly to the Euclidean ones.

S30.3 Uniform size of the actual cross-series extension

For the solution restriction $R : \mathcal{H}(D_1) \to L^2(S_1)$, take a fixed interior cutoff and a fixed containing cube. At any prescribed integer order k, interior estimates give $\|\chi u\|_{H^k} \leq C_k \|u\|_{L^2(D_1)}$. Fourier truncation at frequency A has rank at most CA^n and error at most $C_k A^{-k}$. The finite-rank characterization of singular values therefore gives $\mu_j \leq C_N (j+1)^{-N}$ for every fixed N. The constants are uniform under the preceding fixed coefficient and geometric bounds. McLean, Theorem 2.36, printed p. 47, supplies the compact self-adjoint spectral decomposition of R^*R. Its zero eigenspace is absent by unique continuation. The finite-rank singular value bound also follows directly from that decomposition: a rank r map has a kernel vector in the first $r+1$ singular directions, so its error norm is at least μ_{r+1}. No lower bound on μ_j, continuous choice of basis, or invertibility of the restriction on its entire target is needed.

For compatible cross data with the sum of the two L^2 norms at most M, the source's coefficient formula and Cauchy–Schwarz give $\|b_j\|_{S_2} \leq M$, $\|b_j\|_{D_2} \leq M/\mu_j$. The two interpolation estimates consequently imply

$$\|w_j b_j\|_{L^2(O_1 \times O_2)} \leq C M \mu_j^{\gamma_1 + \gamma_2 - 1}.$$

Since $0 < \mu_j \leq 1$ and $\gamma_1 + \gamma_2 - 1 \geq a_0$, choose one integer N with $N a_0 > 1$. Then

$$\|F_{\mathrm{ext}}\|_{L^2(O_1 \times O_2)} \leq C M C_N^{a_0} \sum_{j \geq 0} (j+1)^{-N a_0}. \tag{S30.8}$$

The bound is uniform. The distributional limit solves both equations. Interior estimates for their elliptic sum give each fixed C^m bound on a smaller product with fixed buffer. Agreement with both cross pieces uses orthogonality to the closure of the restriction range, as derived in S29.2. This does not require the restriction range itself to be closed and does not assert that arbitrary local solutions extend to D_1. This derives the norm estimate for the construction in `cross:series`, rather than postulating a bounded continuation operator. The construction is linear once a singular system is chosen, and its bound is independent of that choice.

Every initial cross compact has positive separation at these templates: the first large ball lies at distance at most $4\vartheta_0 s$ from x_0, whereas the second annulus starts at $s/2$. Hence the separation is at least $(1/2 - 4\vartheta_0)s > 0$. On a fixed range $s \geq s_{\min} > 0$, (S30.5) supplies a uniform M.

S30.4 A reference seed width selected before the final tube

Retain smaller first balls and narrower second annuli inside the output products just obtained. Choose their relative widths using the reference template margins. There is a fixed positive retained first-ball width c_{ret}; choose it small compared with the unused first-ball and annulus buffers. This choice is made on the unit reference shapes. It is independent of the final location bound and the eventual positive separation range. Compact ranges determine only how small λ must be for those templates to work uniformly.

Overlapping retained products have scales comparable to their common $|x - y|$, with fixed comparison factors. In equal orientations, lower the near-plane variable; in opposite orientations both variables are in retained near-plane balls. On a fixed positive-scale range choose $0 < \eta_0 \ll \min(c_{\mathrm{ret}}, \alpha\vartheta_0)s_{\min}$. The simultaneous downward translation by $h = \max(x_n, y_n, 0) + 2\eta_0$ in the opposite-orientation case is smaller than both unused output buffers, after the retained width is chosen sufficiently small. Its endpoint has both variables below $-\eta_0$. The equal-orientation path is even simpler and uses only its near-plane variable. Both comparison paths have open room in the larger output products. Source agreement on the observed slices, single-variable unique continuation in the connected annuli, and then unique continuation along these paths identify the germs with the actual lower cross and with one another.

Choose a fixed $c_0 > 0$ small compared with these retained relative widths. If the smaller nonnegative height is x_n, put $x_0 = (x', 0)$ and $s = |y - x_0|$. For $x_n < c_0|x - y|$, triangle inequalities make s comparable with $|x - y|$, put x in the retained first ball, and put y in the retained annulus. Negative heights are covered by the same near-plane templates or the originally given cross farther below. Thus this construction gives the coarse seed of `cross:coarse-region`, with uniform bounds on each fixed compact separated range. The reference constant c_0 has been chosen before the final transverse tube is compactified. After that tube is fixed one may enlarge the bounded location and scale range and reduce the dilation threshold; one does not reselect c_0 from the enlarged range.

The observed balls and comparison endpoints have fixed strictly negative heights. Since $q(0) = dq(0) = 0$, $q(\lambda z')/\lambda = O(\lambda|z'|^2)$ on every bounded range. Therefore all the required truncated lower sets lie in W_λ for sufficiently small λ. This check includes the largest bounded range eventually used, not just the original target range.

S30.5 Quantitative finite propagation through the frozen cylinder

Use the exact tube geometry and choice order in `cross:tubes`–`cross:tube-error`: choose the already fixed c_0, then $M > B + 3$, excluded-neighborhood sizes, $\sigma > 0$, the positive a interval, T_0, and finally $e_0 > 0$. The base $y_n \geq -B - 1$, $t = M + y_n + e_0|y'|^2 \leq T_0$ is compact. The positive separation lower bound is $a_{\min}(M - B - 1)^{1+\sigma} > 0$. Outside the upward-closed seed, $r/t \leq 1/c_0$,

so $b = (1+\sigma)(r/t)\nabla t$ stays in a bounded range. Its norm also has a positive lower bound on the compact cylinder. The projected distance-Hessian square is bounded below by $c|b|^2$ away from the two excluded neighborhoods; those neighborhoods lie in the seed. The error is at most $C(\sigma + T_0 e_0)|b|^2$ and is absorbed by the ordered choices. Thus both partial gradients and the strict Hessian margin have uniform positive minima on the closed non-seed part. This is a check of the specific original geometry, not an assumed continuation estimate.

Here are the finite constants behind `cross:propagation`. At a reference non-seed point the strict product Hessian criterion constructs two strict scalar weights and a quadratic gap. Its finite-dimensional splitting proof can be retained: normalized dual positive forms D with $D\,d\rho = 0$ have equal opposite-normal variance and tangential block traces. Their tangential extreme points have rank one, because a rank $r \geq 2$ admits a nonzero symmetric perturbation preserving the two traces. The strict Hessian test is positive on all those extreme points. Compactness first supplies a positive gap there and then a finite K with $\operatorname{Hess}\rho + K(d\rho)^{\otimes 2} > P_1 \oplus P_2$. Prescribing the two factor gradients and Hessians gives the strict weights and the quadratic gap. The local cutoff construction in `cross:criterion`, with its fixed R, l, e, produces two compact data products at parameter values strictly above the central level, interpolation exponents with positive sum margin, and a larger output neighborhood with a smaller buffered output box.

These constructions may be chosen smoothly for centers in a small reference neighborhood. Indeed the splitting forms at its central point retain their strict inequalities nearby; prescribe the varying gradients using quadratic coordinate polynomials with those fixed forms. The coordinate inverses for the strict weights, cutoffs, quadratic gap, and data products then vary smoothly. Compactness gives a finite collection of such parameterized local templates. Their minimum strictness, data-level offsets, output widths, exponent margins, and input/output buffers are positive. The scalar Carleman calculation and (S30.8) give a uniform local L^2-to-C^m construction bound, say A_m. No continuous choice of singular bases is needed. A single small coefficient neighborhood preserves all these finite choices.

Let $d_* > 0$ be a common vertical half-width of the smaller outputs, and take a fixed step $h < d_*/4$. Start at a level contained in the given top neighborhood and move downward by steps of size h. At a reached level a_k, the compact data of each local construction lie above $a_k + \eta_*$ for a common $\eta_* > 0$, or in the given seed. They are already available. A finite collection of output boxes covers that level; the number has a uniform bound from the fixed base atlas and the retained horizontal output radii. Their smaller outputs cover the next slab with fixed room. On every overlap component a vertical segment reaches a common height above a_k, where both germs agree with the preceding extension. Unique continuation glues them. Upward closure gives agreement with the seed by the same argument. This slab argument supplies agreement before advancing further; an arbitrary unordered graph of local boxes would not by itself prove it.

Define M_k to bound the extension on the finitely many buffered compacts of the reached region needed as data for subsequent steps, and let M_{seed} bound the corresponding prescribed-region compacts. The local L^2 norm is bounded by their fixed volumes times $M_k + M_{\mathrm{seed}}$. Buffered interior estimates and a finite cover give, after increasing one fixed constant $A \geq 1$,

$$M_{k+1} \leq A(M_k + M_{\mathrm{seed}}), \qquad M_k \leq A^k M_0 + M_{\mathrm{seed}} \sum_{j=1}^{k} A^j. \tag{S30.9}$$

There are at most $N = \lceil (a_{\max} - a_{\min})/h \rceil + 2$ stages. Every constant and this finite stage bound was selected at the reference geometry before λ. Increasing the output regularity changes only the finite coefficient order and the constants, not the argument. Initial and seed bounds come from

(S30.5) and the first cross-series construction. Hence the full fixed-separation initialization has a uniform finite C^m bound.

After choosing the entire bounded cylinder, all reference templates, all comparison paths and buffers, and all finite coefficient orders, choose λ_0 so the rescaled coefficients stay in the preceding small neighborhood on every used bounded set, the Green estimates apply there, and the truncated lower sets lie in W_λ. No constant in (S30.9) depends on $0 < \lambda \le \lambda_0$. Dependence on the fixed target separation, including very large constants or a very small λ_0, is allowed.

S30.6 Exact conclusion and retained limits

This derivation establishes the claimed type of fixed-positive-separation bound in `cross:initial` for its actual rescaled Green data and its specified construction, at the mathematical authoring level. To obtain an open neighborhood and compact-subset bounds, perform the construction with slightly larger location bounds and a slightly smaller positive separation before restricting to the prescribed set. The source's final connected punctured-domain comparison identifies the result with both full lower pieces; ordinary unique continuation identifies constructed solutions and is not used to manufacture them.

The estimate supplies exactly the upper cutoff-region input in `balls:functional`: once p, R_*, t_b and the reference geometry are fixed, its spheres have $r \ge R_* t_b^p > 0$ and lie in one fixed bounded range. Choose the fixed-separation range with room first, then reduce λ_0. Smooth Dirichlet boundary constructions at these positive radii preserve every specified finite angular norm and first center derivative. No uniform derivative bound for the mixed Green kernel on the collapsing spheres is inferred.

This is a new authoring supplement, not a fresh independent review. The standard inverse, parametrix, scalar calculus, spectral theory and unique-continuation interfaces still require their exact reviewed source translations and formal proofs. The preceding project-specific splitting, strict tube test, cross-series range agreement and gluing are derived steps, not literature premises. The frozen main theorem, its definitions, and its trust policy are unchanged. No axiom is added.

S31 Inverse derivatives and the connection replacement error

This successor derivation supplies explicitly scoped finite and actual-field components. The previously reviewed manuscript is retained as history. The source application is the second-metric connection replacement in the shrinking-sphere Hessian test. The Levi-Civita identity used to identify that connection is Baird–Wood, *Harmonic morphisms between Riemannian manifolds*, 2003, Section2.1, equation(2.1.5), printed p.27. Its tensor subtraction against a background metric gives the Koszul identity below. Its geometric identification is not assumed to be Lean-proved by the finite calculation.

Let E be a real inner product space. Let $Q, M : E \to E$ be bounded linear operators with $QM = I$, and assume

$$\|Qz - z\| \le \kappa\|z\| \quad (z \in E), \qquad 0 \le \kappa < 1.$$

Then $\|Qz\| \le (1+\kappa)\|z\|$. Also, with $z = Mx$,

$$\|Mx\| \le \|QMx\| + \|QMx - Mx\| \le \|x\| + \kappa\|Mx\|, \qquad \|Mx\| \le (1-\kappa)^{-1}\|x\|.$$

These statements do not need a spectral theorem or a symmetry assumption.

Lemma S31.1 (Derivative of a genuine inverse field)**.** *Let $Q(y), M(y)$ have Fréchet derivatives DQ, DM at y_0 and be two-sided inverses on a neighborhood of y_0. For every direction X and vector Z,*

$$DM[X]Z = -M(y_0)(DQ[X](M(y_0)Z)).$$

Proof. Differentiate $Q(y)M(y) = I$ using the bounded bilinear composition rule: $Q(y_0)DM[X] + DQ[X]M(y_0) = 0$. Apply the left inverse $M(y_0)$ and evaluate at Z. The derivative of the constant identity is zero; uniqueness of the Fréchet derivative gives the displayed equality. Neighborhood equality suffices; no global inverse-field identity is needed. □

If $\|DQ[X]Z\| \le d\|X\|\|Z\|$, with $d \ge 0$, the preceding inverse norm bound gives

$$\|DM[X]Z\| \le \frac{d}{(1-\kappa)^2}\|X\|\|Z\|. \tag{S31.1}$$

All inverses in these formulas have positive denominators.

Theorem S31.2 (Quantitative Koszul consequence)**.** *Suppose $T : E \times E \to E$ and DM satisfy, for every X, Y, Z,*

$$2\langle M(T(X,Y)), Z\rangle = \langle DM[X]Y, Z\rangle + \langle DM[Y]X, Z\rangle - \langle DM[Z]X, Y\rangle. \tag{S31.2}$$

With the preceding actual inverse identities and derivative estimate,

$$\|T(X,Y)\| \le \frac{3(1+\kappa)}{2(1-\kappa)^2} d\|X\|\|Y\|.$$

In particular, $0 \le \kappa \le 1/4$ gives

$$\|T(X,Y)\| \le \frac{10}{3} d\|X\|\|Y\|. \tag{S31.3}$$

Proof. Cauchy–Schwarz and (S31.1) bound each term on the right of (S31.2) by $d(1-\kappa)^{-2}\|X\|\|Y\|\|Z\|$. Take $Z = M(T(X,Y))$. If this vector is zero, $QM = I$ gives $T(X,Y) = 0$. Otherwise divide its positive norm out of the resulting squared-norm inequality, obtaining $\|M(T(X,Y))\| \le (3/2)d(1-\kappa)^{-2}\|X\|\|Y\|$. Apply Q and its norm bound. For $\kappa \le 1/4$, use $1+\kappa \le 5/4$ and $(1-\kappa)^{-1} \le 4/3$; the resulting constant is $(3/2)(5/4)(16/9) = 10/3$. □

Lemma S31.3 (Hessian replacement at the shrinking-sphere scale)**.** *Let $r, \varepsilon, d \geq 0$, $d \leq \varepsilon r$, let $\ell : E \to \mathbb{R}$ be a bounded linear covector with $\|\ell\| \leq (5/4)r$, and suppose $\|q\|, \|w\| \leq 2$. Under* (S31.3)*,*

$$|\ell(T(q,q)) + \ell(T(w,w))| \leq \frac{100}{3}\varepsilon r^2.$$

Proof. The left side is at most $\|\ell\|(\|T(q,q)\| + \|T(w,w)\|)$, hence at most $(5r/4)(10d/3)(\|q\|^2 + \|w\|^2)$. The last sum is at most8, so the product is at most $(100/3)rd$. Substitute $d \leq \varepsilon r$. No lower bound on r is needed. □

For the intended application $\ell = d_y\rho$ and $T = \nabla^M - \nabla^\gamma$. The Hessian difference is exactly $-d_y\rho(T(X,Y))$, by subtracting the covariant derivatives of the one-form $d\rho$. In a background parallel orthonormal frame, ℓ has norm $r\|\omega + b\| \leq 5r/4$. The previously derived characteristic comparisons give $\|q\|, \|w\| \leq 4/3 + 1/4 < 2$. For the actual covariant derivative of the contravariant metric, differentiating the identity $QM = I$ uses the background connection product rule. Its size is controlled by $\|\nabla^\gamma h\|$, which fixed coordinate bounds control by $C(\|h\| + \|\partial h\|)$. Thus the bootstrap $H_8 \leq \varepsilon r$ gives the required $d \leq C\varepsilon r$ with a retained background constant. The finite calculation supplies the replacement error from that actual derivative size, rather than assuming the desired connection bound.

Remaining obligations are explicit: the geometric identification of the Koszul tensor, covariant/coordinate derivative comparisons and bounded frame constants, the geodesic first-variation formulas, the profile Hessian, normal-domain existence and Jacobi curvature estimate. This calculation does not prove the full source continuation criterion or main theorem. No standard or novel mathematical axiom is installed.

S31.1 An explicit coordinate tensor without a Koszul premise

For a complete real inner product space, let $J : E \to E^*$ be the Fréchet–Riesz isometric isomorphism, $J(v)(z) = \langle v, z\rangle$. Let $D : E \to \mathcal{L}(E,E)$ be bounded linear. For fixed X, Y the bounded functional

$$F_{X,Y} = J(D[X]Y) + J(D[Y]X) - J(Y) \circ (Z \mapsto D[Z]X)$$

satisfies $F_{X,Y}(Z) = \langle D[X]Y, Z\rangle + \langle D[Y]X, Z\rangle - \langle D[Z]X, Y\rangle$ by symmetry of the real inner product. Define the coordinate tensor

$$\Gamma_{Q,D}(X,Y) = \tfrac{1}{2}Q(J^{-1}F_{X,Y}).$$

If $MQ = I$, linearity and the Riesz identity give $2\langle M\Gamma_{Q,D}(X,Y), Z\rangle = F_{X,Y}(Z)$ exactly. Thus the quantitative estimates above apply to this explicit tensor without assuming their desired norm bound or a Koszul identity as an extra premise. To identify it with the coordinate Levi–Civita tensor one uses the derivative $D = DM$ of the actual covariant metric, the usual commuting coordinate fields, and the standard Koszul formula. That geometric identification remains a separate application obligation.

The symmetry of the metric derivative can also be obtained from actual field calculus: if $M(y)$ is self-adjoint in a neighborhood of x and has derivative DM there, differentiate $\langle M(y)X, Z\rangle = \langle M(y)Z, X\rangle$ for fixed X, Z. Uniqueness of scalar Fréchet derivatives gives $\langle DM[U]X, Z\rangle = \langle DM[U]Z, X\rangle$ for every direction U. Consequently $F_{X,Y} = F_{Y,X}$ and $\Gamma_{Q,DM}(X,Y) = \Gamma_{Q,DM}(Y,X)$. These derivative and algebraic identities are distinct from the unproved full Riemannian/geodesic application. Combining the genuine inverse-field derivative identity with this construction gives a theorem directly for differentiable fields: if $Q(y)M(y) = I$ near x, $M(x)Q(x) = I$, $\|Q(x)z - z\| \leq \kappa\|z\|$, $\kappa \leq 1/4$, and $\|DQ(x)[X]Z\| \leq d\|X\|\|Z\|$ with $d \geq 0$, then $\|\Gamma_{Q(x),DM(x)}(X,Y)\| \leq (10/3)d\|X\|\|Y\|$.

The pointwise right inverse follows by evaluating the neighborhood equality at x; the derivative relation follows by differentiating that equality. There is no assumed connection-size conclusion in this field theorem.

S31.2 Combining the characteristic and connection calculations

Let $\|\omega\| = 1$, $\|b\| \le 1/4$, $0 \le \kappa \le 1/16$, $\|Q - I\| \le \kappa$, $QM = MQ = I$, and put $u = \omega + b$, $t = \langle u, Qu\rangle$, $q = t^{-1}Qu$. Suppose $\langle u, w\rangle = 0$ and $\langle w, Mw\rangle = t^{-1}$. The checked characteristic comparison supplies $\|q - u/\|u\|^2\| \le 4\kappa$ and a vector w_0 with $\|w_0\| = \|u\|^{-1}$, $\|w - w_0\| \le 4\kappa$. The triangle inequality gives $\|u\| \ge 3/4$, hence $\|u/\|u\|^2\| = \|u\|^{-1} \le 4/3$ and $\|q\|, \|w\| \le 4/3 + 4\kappa \le 19/12 < 2$.

For a profile Hessian H satisfying $H(X, X) \le -c\beta\|X\|^2$ with $c, \beta \ge 0$, the already checked metric characteristic theorem gives

$$\begin{aligned} S_0 &\ge \tfrac{1}{2}cr^2\beta - 64\|b\|\kappa - 256\kappa^2, \\ S_0 &= \|\Pi_\omega(\omega - q)\|^2 + \|\Pi_\omega(v - w)\|^2 \\ &\qquad - \langle b, q\rangle^2 - \langle b, w\rangle^2 - r^2\big(H(q, q) + H(w, w)\big). \end{aligned}$$

where $\|v\| = 1$, $\langle \omega, v\rangle = 0$. Take the explicit coordinate tensor constructed from differentiable inverse fields and assume their derivative bound $d \le \varepsilon r$, $r, \varepsilon, d \ge 0$, $\|\ell\| \le 5r/4$. The connection contribution $e = \ell(\Gamma(q, q)) + \ell(\Gamma(w, w))$ obeys $|e| \le (100/3)\varepsilon r^2$ by the vector bounds just derived. Consequently

$$S_0 - e \ge \tfrac{1}{2}cr^2\beta - 64\|b\|\kappa - 256\kappa^2 - \tfrac{100}{3}\varepsilon r^2.$$

This is a derived lower bound for the explicit finite expression. Geodesic first variation, curvature/Jacobi errors, profile Hessian identification and the eventual strict positivity after parameter choices remain additional application obligations; they are not hidden in this statement.

S32 Concrete regularized radius calculus

This successor addresses the actual scalar radius profile in the selected proof route. The first softplus derivative formula holds for every nonzero delta; its positivity and geometric use require positive delta. The total-function algebraic sigmoid and deltaDerivative positivity identities also have the larger real domains recorded in the catalog. The previous exact review package is retained as history. Let $\delta > 0$, $t \in \mathbb{R}$, and define

$$e = e^{t/\delta}, \quad L = \log(1+e), \quad \ell_\delta(t) = \delta L, \quad a_\delta(t) = \frac{e}{\delta(1+e)L}.$$

Both e and L are strictly positive, so every denominator below is nonzero. Ordinary differentiation gives

$$\ell_\delta'(t) = \frac{e}{1+e} \in (0,1), \qquad (\log \ell_\delta)'(t) = a_\delta(t) > 0,$$

and, by the quotient rule,

$$a_\delta'(t) = \frac{e(L-e)}{\delta^2(1+e)^2L^2} < 0.$$

For the sign, apply the checked real logarithm inequality $\log x < x - 1$ for $x > 0$, $x \neq 1$ to $x = 1 + e$. Also $a_\delta \ell_\delta = e/(1+e) < 1$, and hence $a_\delta \leq \ell_\delta^{-1}$.

For $t \geq 0$, monotonicity of the logarithm applied to $e < 1 + e$ gives $t \leq \ell_\delta(t)$. For $t \leq 0$, $0 < e \leq 1$, so $0 < \ell_\delta(t) \leq \delta \log 2$. Therefore on the nonpositive side the profile collapses as $\delta \downarrow 0$; on the positive side it stays bounded below by t. For fixed t, differentiation in δ gives

$$\partial_\delta \ell_\delta(t) = L - \frac{(t/\delta)e}{1+e} > 0.$$

If $s = t/\delta \leq 0$, the subtracted term is nonpositive and $L > 0$. If $s \geq 0$, then $L > s$ and $se/(1+e) \leq s$, so the difference is positive. Consequently $0 < \delta \leq 1$ gives $\ell_\delta(t) \leq \ell_1(t)$; on a fixed interval $t \leq T$, increasingness in t then gives $\ell_\delta(t) \leq \log(1 + e^T)$.

These scalar derivatives supply the sign in the actual profile Hessian: for a depth function $t(y)$ with $\mathrm{Hess}_\gamma t \leq -c_t \gamma$, and $f = \log R_* + p \log \ell_\delta(t)$,

$$\mathrm{Hess}_\gamma f = p a_\delta \,\mathrm{Hess}_\gamma t + p a_\delta' \, dt \otimes dt \leq -p c_t a_\delta \gamma.$$

If $|dt|_\gamma \leq C_t$ and $\beta = p a_\delta |dt|_\gamma$, the last bound is at most $-(c_t/C_t)\beta\gamma$. No profile concavity conclusion is assumed in the scalar derivative calculation.

For $p \geq 2$, $R_* \geq 0$, and $0 \leq b \leq p C_t a_\delta$, the radius $r = R_* \ell_\delta^p$ satisfies

$$r b^2 \leq R_* p^2 C_t^2 \ell_\delta^{p-2}.$$

Indeed $a_\delta \ell_\delta \leq 1$, hence $\ell_\delta^p a_\delta^2 \leq \ell_\delta^{p-2}$ by factoring $\ell_\delta^p = \ell_\delta^{p-2}\ell_\delta^2$ and using positivity. On a bounded t interval, the preceding upper bound on ℓ_δ makes the right side uniformly $O(R_*)$ for each fixed p. In particular, for $0 < \delta \leq 1$, $t \leq T$ and $B_T = \log(1 + e^T) > 0$, monotonicity of nonnegative integer powers gives $r b^2 \leq R_* p^2 C_t^2 B_T^{p-2}$ with exactly the same hypotheses.

For clarity the finite Hessian consequence can be stated independently of its geometric identification. In a real normed space, let a covector η satisfy $\|\eta\| \leq C_t$ with $C_t > 0$, let $c_t \geq 0$, and let $H(X,X) \leq -c_t \|X\|^2$. Define

$$\mathcal{H}(X,Y) = p a_\delta H(X,Y) + p a_\delta' \eta(X)\eta(Y), \qquad \beta = p a_\delta \|\eta\|.$$

The actual scalar derivative signs proved above imply $\mathcal{H}(X,X) \le -pa_\delta c_t \|X\|^2$. The covector bound implies $\beta \le pa_\delta C_t$; multiplying by $c_t/C_t \ge 0$ and reversing signs therefore gives

$$\mathcal{H}(X,X) \le -(c_t/C_t)\beta\|X\|^2.$$

Thus this form has the required concavity bound derived from the depth Hessian and the actual radius coefficients. Its identification with the covariant Hessian of the composed depth profile still requires the stated chain rule and geometric depth construction.

The complete higher derivative bounds, covariant chain-rule identification, depth-function conformal Hessian, cutoff support geometry and application to the global continuation argument remain separate obligations. All formal claims will be bounded to precisely the proved scalar calculations.

S33 Actual composed profile: first and second coordinate derivatives

Let E be a real normed space, $\delta > 0$, $p \in \mathbb{N}$, $c \in \mathbb{R}$, and $t : E \to \mathbb{R}$. Put $F(y) = c + p \log \ell_\delta(t(y))$, with the actual positive softplus radius $\ell_\delta(s) = \delta \log(1 + e^{s/\delta})$. Write $a = a_\delta$ and $a' = a'_\delta$ for the already derived first and second logarithmic radius derivatives. At a point x, suppose t has Frechet derivative $T \in \mathcal{L}(E, \mathbb{R})$ and the derivative field $y \mapsto Dt(y)$ has derivative $H \in \mathcal{L}(E, \mathcal{L}(E, \mathbb{R}))$. Assume also that t is differentiable on a neighborhood of x. This is an ordinary regularity hypothesis satisfied by a smooth depth function, not a Hessian or continuation conclusion. The scalar chain rule and zero derivative of c give

$$DF(x) = pa(t(x))T.$$

The same identity holds in the neighborhood. Differentiate the actual differential field with the bounded bilinear scalar multiplication rule. The chain rule applied to $a \circ t$ yields

$$D(DF)(x)[X][Y] = pa(t(x))H[X][Y] + pa'(t(x))T(X)T(Y).$$

The formal second derivative is taken of $y \mapsto \operatorname{fderiv} F(y)$ itself. Its equality with the displayed product field near x is established pointwise using the previous first derivative theorem, and then the derivative congruence rule applies. No candidate second derivative is assumed to be the derivative of F.

For any prescribed coordinate connection tensor $\Gamma : E \times E \to E$, define

$$\mathcal{H}_\Gamma u(X,Y) = D(Du)(x)[X][Y] - Du(x)(\Gamma(X,Y)).$$

Substitution and scalar linearity prove the actual coordinate identity

$$\mathcal{H}_\Gamma F(X,Y) = pa(t(x))\mathcal{H}_\Gamma t(X,Y) + pa'(t(x))T(X)T(Y).$$

This identity holds for every tensor Γ, and can be instantiated with the already constructed coordinate metric tensor. Identifying this coordinate formula with the manifold covariant Hessian remains a separate geometric obligation. Since the checked sign is $a' < 0$, on the diagonal the second term is nonpositive. If $\mathcal{H}_\Gamma t(X,X) \le -c_t\|X\|^2$ with $c_t \ge 0$ and $\|T\| \le C_t$ with $C_t > 0$, the actual chain formula and the previously checked profileHessianForm bound give

$$\mathcal{H}_\Gamma F(X,X) \le -(c_t/C_t)(pa(t(x))\|T\|)\|X\|^2.$$

This is conditional on the depth Hessian estimate, which must still be constructed from the geometric setup. It does not assume the profile estimate itself, a Jacobi bound, or any original continuation conclusion. The frozen unconditional main is unchanged.

S33.1 Exact second coefficient and actual second derivative norm

With $e = e^{s/\delta}$ and $L_0 = \log(1+e)$, the explicit scalar formulas give

$$a'_\delta(s) + a_\delta(s)^2 = \frac{eL_0}{\delta^2(1+e)^2L_0^2} > 0.$$

Thus $|a'_\delta| \le a_\delta^2$, uniformly in the actual positive delta and real depth variable. This proves the second-order scalar bound, rather than the full higher-order family. The norm of a rank-one bounded map $U \mapsto S(U)V$ is at most $\|S\|\|V\|$. Applying this checked operator-norm inequality, the triangle inequality and the scalar bound to the actual profile second differential yields

$$\|D(DF)(x)\| \le pa_\delta(t(x))\|H\| + pa_\delta(t(x))^2\|T\|^2.$$

The identity with the actual second derivative field was proved above. Uniform depth derivative bounds and the lower profile-gradient bound needed to express all higher derivatives in powers of beta remain separate obligations.

S34 Coordinate conformal connection and actual exponential depth

This component follows the original shrinking-balls depth construction, not a replacement profile. Work in a real Hilbert space and at a coordinate point let M be the covariant metric operator, $Q = M^{-1}$, $D = DM$, and $T = Dt_0$. Assume the two inverse identities and symmetry of M; define $g(X,Y) = \langle MX, Y\rangle$ and $G = QJ^{-1}T$, where J is the real Riesz isometry. Then $g(G,X) = T(X)$. For $a > 0$ and the conformal derivative data

$$M_\gamma = aM, \quad Q_\gamma = a^{-1}Q, \quad D_\gamma[X]Y = aD[X]Y - 2LaT(X)MY,$$

the explicit Riesz-built coordinate connection obeys

$$\Gamma_\gamma(X,Y) = \Gamma_g(X,Y) - L\{T(X)Y + T(Y)X - g(X,Y)G\}.$$

To prove the identity, insert the conformal inverse/derivative data into the already proved coordinate Koszul identity. Expand inner products by scalar linearity, use the symmetry of M and $g(G,Z) = T(Z)$, and compare with the base Koszul identity. This gives equality of the M-pairing against every Z; the real inner-product separation and $QM = I$ prove vector equality. Thus the connection formula is derived from the actual explicit tensor construction, rather than supplied as an additional mathematical premise. For a differentiable field $M(y)$ and differentiable $t_0(y)$, the product and exponential chain rules prove that $M_\gamma(y) = e^{-2Lt_0(y)}M(y)$ has derivative D_γ at the point. The contravariant scaling gives the inverse identities. Its identification with a manifold Levi–Civita connection remains a geometric interpretation obligation.

For $L \neq 0$ and any reference level s_0 define the actual depth primitive

$$d_L(s) = \frac{e^{-2Ls_0} - e^{-2Ls}}{2L}.$$

Ordinary calculus gives $d_L' = e^{-2Ls}$ and $d_L'' = -2Le^{-2Ls}$. With $s_0 = 1/8$ this is precisely the integral $\int_{1/8}^{s} e^{-2Lu}\,du$ from the manuscript. The checked fundamental theorem of calculus proves this exact interval-integral identification; the primitive therefore has precisely the manuscript normalization and derivative. At $t_0 = 0$ and $L > 0$, this depth is strictly negative, and its derivative is positive everywhere. Let $t = d_L \circ t_0$, $a = e^{-2Lt_0(x)}$, $H = D(Dt_0)(x)$. Assuming differentiability of t_0 on a neighborhood and differentiability of its actual derivative field at x, the scalar and bounded product chain rules give

$$Dt = aT, \qquad D(Dt)[X][Y] = aH[X][Y] - 2LaT(X)T(Y).$$

Use the actual conformal connection from above in the coordinate Hessian definition. The two rank-one terms cancel exactly:

$$\mathcal{H}_{\Gamma_\gamma} t(X,Y) = a\{\mathcal{H}_{\Gamma_g} t_0(X,Y) - LT(G)g(X,Y)\}.$$

This is the original depth Hessian formula in coordinate form. No strict concavity conclusion is assumed in deriving it.

Suppose $g(X,X) \geq 0$, $\mathcal{H}_{\Gamma_g} t_0(X,X) \leq Cg(X,X)$, $T(G) \geq m > 0$, $C \geq 0$, and $c > 0$. Choose $L = (C + c)/m > 0$, before any later shrinking-radius choice. The displayed identity gives

$$\mathcal{H}_{\Gamma_\gamma} t(X,X) \leq (C - Lm)ag(X,X) = -c\, g_\gamma(X,X).$$

Thus a quantitative positive depth concavity constant follows from explicit ordinary metric/Hessian and gradient bounds. Obtaining uniform bounds for the actual scaled $g_{1,\lambda}$ on the cylinder, interpreting the coordinate Hessian geometrically, geodesic/Jacobi errors, and the original analytic continuation application remain separate formal obligations. No external premise, novel conclusion or change to the frozen unconditional target is installed.

S35 Relative coordinate connections and covariant inverse derivatives

The manuscript's small connection difference is measured relative to the conformal reference γ. Ordinary coordinate DQ can be large even when $Q = \gamma^{-1}$, so a direct global-coordinate use of the earlier field derivative bound would be invalid. This component derives the relative tensor estimate and keeps the coordinate-to-manifold interpretation explicit. At a point represented in a real Hilbert space, let M, Q be two-sided inverse operators, M symmetric, and let B_X denote an arbitrary torsion-free background coordinate connection: $B_X Y = B_Y X$. Write D_Q, D_M for ordinary derivatives, with their actual inverse-field identity. Let $A_X = (B_X)^*$ be the checked bounded adjoint. The covariant derivative data obey their defining coordinate identities

$$C_Q[X]Z = D_Q[X]Z + B_X(QZ) + Q(A_X Z), \quad C_M[X]Z = D_M[X]Z - A_X(MZ) - M(B_X Z).$$

These are tensor derivative definitions, not connection-size or continuation conclusions. Apply M to the first identity at MZ, use both inverse identities and the ordinary inverse derivative formula, and cancel the background terms. This proves

$$C_M[X]Z = -M(C_Q[X](MZ)).$$

The actual ordinary inverse derivative formula was already derived by differentiating the neighborhood field inverse identity; the relative identity therefore has the same genuine derivative foundation.

Let Γ_M be the explicit Riesz/Koszul coordinate connection from the ordinary covariant derivative D_M, and put $T(X,Y) = \Gamma_M(X,Y) - B_X Y$. The base Koszul identity, metric symmetry, adjoint pairing, and background torsion identity give

$$2\langle MT(X,Y), Z\rangle = \langle C_M[X]Y, Z\rangle + \langle C_M[Y]X, Z\rangle - \langle C_M[Z]X, Y\rangle.$$

In detail the six background correction terms reduce to $-2\langle MB_X Y, Z\rangle$: the $B_X Z = B_Z X$ pair cancels by metric symmetry, the $B_Y Z = B_Z Y$ pair cancels directly, and $B_X Y = B_Y X$ combines the final two terms. No bound for B itself is required. If $\|Q - I\| \le \kappa \le 1/4$ and $\|C_Q[X]Z\| \le d\|X\|\|Z\|$, $d \ge 0$, the already proved inverse/Koszul norm calculation now applies to C_Q, C_M, T, giving

$$\|\Gamma_M(X,Y) - B_X Y\| \le (10/3)d\|X\|\|Y\|.$$

For $d \le \varepsilon r$, $\|\ell\| \le 5r/4$, and $\|q\|, \|w\| \le 2$, its two diagonal Hessian replacement terms obey the same $(100/3)\varepsilon r^2$ bound, with the relative tensor. The coordinate Hessian definitions also give exactly

$$\mathcal{H}_{\Gamma_M} u(X,Y) - \mathcal{H}_B u(X,Y) = -Du(x)(\Gamma_M(X,Y) - B_X Y).$$

Taking B to be the reference γ connection and identifying these explicit covariant derivative formulas with ∇^γ still requires the actual geometric coordinate/frame bridge. The finite proof does not claim that raw global-coordinate DQ satisfies the manuscript's smallness hypothesis. No novel Hessian/continuation conclusion, external premise, or change to the frozen main target is made.

S36 Uniform lower slope of the actual radius profile

This successor component addresses the original large-beta choice through the actual profile derivative. The PDE application remains open. Let $\delta > 0$, $s \in \mathbb{R}$, $e = e^{s/\delta}$, $L_0 = \log(1+e)$, $\ell = \delta L_0$, and $a = e/[\delta(1+e)L_0]$. The checked inequality $L_0 < e$ yields

$$a(\delta + \ell) = \frac{e(1+L_0)}{(1+e)L_0} > 1.$$

If $0 < \delta \le 1$ and $s \le T$, the already proved softplus upper bound gives $\delta + \ell \le 1 + \log(1+e^T) =: U$, with $U > 0$. Since $a > 0$, it follows that $a \ge U^{-1}$. This is uniform even when s/δ is arbitrarily negative; no positive lower bound on the radius itself is asserted.

Let $t : E \to \mathbb{R}$ be a genuinely differentiable depth function with $t(x) \le T$ and $\|Dt(x)\| \ge m \ge 0$. For the actual profile $F = c + p \log \ell_\delta(t)$, the proved first derivative identity and the norm of scalar multiplication give

$$\|DF(x)\| = pa\|Dt(x)\| \ge \frac{pm}{U}.$$

For fixed $m > 0, U > 0$ and any prescribed threshold B, the Archimedean property supplies a natural $p \ge 2$ with $pm/U > B$. Thus the same p works at every point where the stated depth and gradient bounds hold, uniformly in positive delta at most one. Uniform metric/depth bounds on the actual manifold, higher derivative estimates and analytic application remain separate obligations; no novel conclusion or external premise is assumed.

S37 Actual interval endpoint representation and Jacobi error

This successor derivation expands the interval part of the original moving-endpoint squared-distance Hessian calculation. It does not assume that a candidate integral expression solves an ODE, or install a geometric Jacobi estimate as a premise. Let E be a real Banach space and $j, j', f : \mathbb{R} \to E$ have genuine derivatives $Dj = j'$ and $Dj' = f$ at every point of $[0,1]$, with f continuous there. Write $B = j(0)$, $A = j(1)$. For $0 \le t \le 1$ put

$$I_1(t) = \int_0^t sf(s)\,ds, \qquad I_2(t) = \int_t^1 (1-s)f(s)\,ds.$$

The derivative of $sj'(s) - j(s)$ is $sf(s)$, and the derivative of $(1-s)j'(s) + j(s)$ is $(1-s)f(s)$. The vector-valued fundamental theorem of calculus, including the endpoints, therefore gives

$$I_1 = tj'(t) - j(t) + B, \qquad I_2 = -(1-t)j'(t) + A - j(t).$$

Multiplying these equations by $1-t$ and t and adding gives the exact Dirichlet Green representation

$$j(t) = (1-t)B + tA - (1-t)I_1 - tI_2.$$

Subtracting gives the genuine derivative identity

$$j'(t) - (A - B) = I_1 - I_2.$$

If $\|f(s)\| \le C$ on the interval, $C \ge 0$, Bochner norm comparison with the scalar functions Cs and $C(1-s)$ yields

$$\|I_1(t)\| \le Ct^2/2, \qquad \|I_2(t)\| \le C(1-t)^2/2.$$

Thus the actual displacement from the affine endpoint interpolation and the actual derivative displacement obey

$$\|j(t) - ((1-t)B + tA)\| \le Ct(1-t)/2 \le C/8, \qquad \|j'(t) - (A-B)\| \le C/2.$$

For a genuinely solved Jacobi equation in a parallel frame, $f(s) = -K(s)j(s)$, suppose $\|K(s)z\| \le k\|z\|$ and $0 \le k \le 1$. Let $J = \max_{[0,1]} \|j\|$ and $M = \max(\|A\|, \|B\|)$; continuity on the compact interval gives this actual maximum. Taking $C = kJ$ above yields $J \le M + kJ/8$, hence $J \le 8M/7$ (in particular $J \le 2M$). The displacement and derivative errors then have the concrete bounds $kM/4$ and kM, respectively. If the Jacobi coefficient arises from curvature along a speed-r geodesic, $k = K_0 r^2$.

In a Hilbert space the affine derivative is constant, and the integral of the error derivative is zero by its zero endpoints. Consequently the cross term in the index-form derivative energy vanishes exactly. The remaining error is bounded by $(k^2 + 4k)M^2 \le 5kM^2$, giving a sufficient $O(r^2)(\|A\|^2 + \|B\|^2)$ moving-endpoint Hessian error. Actual manifold parallel transport, first/second variation, curvature identification and uniform geometric bounds remain separate formal obligations. The exact interval estimates above have privately compiled proofs from genuine derivative data; acceptance still requires the full trusted check and isolated review. The subsequent Hilbert index-form energy cancellation and geometric obligations remain unproved, as does the frozen main.

S38 Daytime successor scope and evidence boundary

This successor retains the exact frozen v002 theorem and all ten definitions. It appends complete upfront constructions for the actual geometric, angular coefficient, radial PDE, differential commutator and Sobolev components below. Development compilation is distinct from trusted acceptance. The unconditional main remains unproved. Published interfaces require three independent source audits, and any adopted interface is a visible proof parameter; human ratification remains pending. Earlier supplements are historical authoring records. Their then-open descriptions and precursor hashes do not assert the current status or identify current inputs. The source-bound ledger and final exact check/review receipts determine the current accepted supporting scopes.

Exact published-source quantifier. The proposed standard parameter quantifies over every globally C^∞ real $f : \mathbb{R}^3 \to \mathbb{R}$ with $f(x) = 0$ whenever $\|x\| = 1$, and concludes that its actual smooth Sobolev jet belongs to the existing Tau Ceti H^1_0 test-function graph closure. Exterior-zero smooth perturbations are only its downstream specialization. No compact ambient support, exterior vanishing, derivative trace, metric condition or PDE assumption occurs in this parameter. McLean Theorems 3.40(ii), 3.37 and the recorded graph-topology/test-class representation justify this exact sphere-zero type; three independent source audits and supervisor reconciliation are required for provisional adoption, with human ratification pending.

S38.1 Bundling the actual metric connection field

Let E be a complete real Hilbert space, $Q : E \to E$ a bounded operator and $D : E \times E \to E$ a bounded bilinear map. Define

$$K_D(X,Y) = D(X,Y) + D(Y,X) - (D(\cdot,X))^*Y.$$

The real adjoint is a linear isometry on bounded operators; flip is a linear isometry on bounded bilinear maps. Consequently $D \mapsto K_D$ is itself a bounded linear map on the Banach space of bounded bilinear maps. Construct

$$\Gamma_{Q,D}(X,Y) = \tfrac{1}{2}QK_D(X,Y)$$

as an actual curried bounded bilinear operator by postcomposition with Q. The adjoint identity gives $\langle K_D(X,Y),Z\rangle = \langle D(X,Y),Z\rangle + \langle D(Y,X),Z\rangle - \langle D(Z,X),Y\rangle$. The Riesz inverse of the previously constructed exact Koszul functional is therefore precisely $K_D(X,Y)$. This proves that the new bundled operator is pointwise equal to the already checked `coordinateConnection`, without changing its definition or requiring a desired connection identity as a hypothesis. When $Q = M^{-1}$ and $D = DM$, it is exactly the constructed coordinate metric connection used in the original calculation. Inverse and metric/geometric identifications remain explicit application obligations. For actual fields $Q(x), M(x)$ set $\Gamma(x) = \Gamma_{Q(x),DM_x}$, using the actual Fréchet derivative of M. If Q is C^n and M is C^m with $n+1 \le m$, then DM is C^n. The bounded linear map $D \mapsto K_D$ preserves C^n regularity, and bounded operator postcomposition is a continuous bilinear map. Multiplication by $1/2$ preserves regularity. Hence the actual constructed field Γ is C^n. In particular globally smooth Q, M yield globally smooth Γ. This conclusion is derived from actual input-field regularity, rather than assuming regularity of an abstract connection field. For every X, Y, the two direct terms of K_D are bounded by $\|D\|\|X\|\|Y\|$. The adjoint term has the same bound because its operator norm equals that of $D(\cdot, X)$. Thus $\|K_D(X,Y)\| \le 3\|D\|\|X\|\|Y\|$ and $\|\Gamma_{Q,D}\| \le \frac{3}{2}\|Q\|\|D\|$. These are larger total algebraic scopes; genuine metric derivative and Levi-Civita interpretations require the actual derivative/inverse hypotheses. No new external premise, main proof, uniform family normal-neighborhood or geodesic distance-Hessian conclusion is asserted.

Genuine derivative and compact-domain consequences. For smooth Q, M the constructed field Γ is smooth, so it is actually differentiable at every x, with derivative equal to its Fréchet derivative, and the derivative field $D\Gamma$ is smooth. For every compact $K \subset E$, continuity gives finite scalar bounds C_0, D_0 on $\|\Gamma\|$ and $\|D\Gamma\|$ on K. Replacing them by $C = \max(C_0, 0)$ and $D = \max(D_0, 0)$ makes both nonnegative without losing either bound. These constants are derived from the actual constructed smooth fields. A family uniform in additional parameters still requires a common compact parameter domain and actual field smoothness there; no such metric-family realization is silently assumed.

S39 Genuine initial-value consequences of the endpoint estimate

Let E be a complete real normed space and let $j, j', f : \mathbb{R} \to E$ have actual first and second derivatives $Dj(t) = j'(t)$ and $Dj'(t) = f(t)$ at every $t \in [0,1]$, including ambient two-sided derivatives at both endpoints. All interval statements below restrict these real-domain functions; no weaker one-sided endpoint interface is claimed. Assume f is continuous and $\|f(t)\| \le k\|j(t)\|$ for $0 \le k \le 1/2$. Put $M = \max(\|j(0)\|, \|j(1)\|)$. The already proved endpoint derivative estimate gives $\|j'(0) - (j(1) - j(0))\| \le kM$.

Zero initial derivative. If $j'(0) = 0$, the last estimate reads $\|j(1) - j(0)\| \le kM$. The triangle inequality yields $\|j(1)\| \le \|j(0)\| + kM$; the same upper bound holds for $\|j(0)\|$. Thus $M \le \|j(0)\| + kM$, whence $M \le 2\|j(0)\|$ and $\|j(1) - j(0)\| \le 2k\|j(0)\|$. This is a consequence of the genuine ODE data, not an assumed Jacobi endpoint estimate.

Zero initial position. If $j(0) = 0$ and $j'(0) = V$, then $M = \|j(1)\|$ and $\|j(1) - V\| \le k\|j(1)\|$. The triangle inequality gives $\|j(1)\| \le \|V\| + k\|j(1)\|$, hence $\|j(1)\| \le 2\|V\|$ and $\|j(1) - V\| \le 2k\|V\|$.

Endpoint operator and injectivity. Suppose a bounded real-linear map $T : E \to E$ is supplied together with, for every V, actual such functions having $j_V(0) = 0$, $j_V'(0) = V$ and $j_V(1) = TV$. Applying the previous bound to each V proves $\|(T - I)V\| \le 2k\|V\|$, so $\|T - I\| \le 2k$. If $k < 1/2$, then $\|TV\| \ge (1 - 2k)\|V\|$, and $TV = 0$ implies $V = 0$. Therefore T is injective. The complete-space Neumann construction below supplies the stronger inverse consequence. No exponential map, geodesic, parallel frame or parameter differentiability is constructed here. The actual geometric application must still identify the endpoint operator and prove uniform coefficient smallness and the required parameter bounds.

Checked Neumann inverse. For a complete normed space, bounded endomorphisms form a complete normed ring. If $k < 1/2$, the proved $\|I - T\| \le 2k < 1$ permits the convergent geometric series $\sum_{n \ge 0} (I - T)^n$, a two-sided bounded inverse of T. The checked mathlib construction `Units.oneSub` supplies this unit and `ContinuousLinearMap.isHomeomorph_of_isUnit` supplies the homeomorphism consequence. This paragraph adds precisely this inverse consequence to the preceding scoped preparation; it still does not identify T with an exponential differential or provide a parameter derivative bound.

S40 The actual smooth inverse of a positive metric operator field

Let E be a finite-dimensional complete real Hilbert space and let $M : E \to \mathcal{L}(E, E)$ be smooth, with $\langle M(x)X, X\rangle > 0$ for every x and every $X \neq 0$. For fixed x, positivity implies injectivity: if $M(x)X = M(x)Y$ and $X \neq Y$, then $M(x)(X - Y) = 0$ while its inner product with $X - Y$ is strictly positive, a contradiction. Finite dimensionality makes injectivity equivalent to surjectivity. The checked bounded-operator unit/bijection theorem therefore makes every $M(x)$ a unit.

Define the actual field $Q(x) = \text{Ring. inverse}(M(x))$. Two-sided unit inversion gives $Q(x)M(x) = I = M(x)Q(x)$, and hence $Q(x)(M(x)X) = X$ and $M(x)(Q(x)X) = X$ for every X. The checked analytic inversion theorem for complete normed algebras is C^∞ at every unit. Composing it with the actual smooth M at each x proves that Q is smooth. This constructs the inverse field, rather than assuming a second arbitrary smooth field.

Consequently the actual coordinate connection formed from (Q, M) is smooth, has genuine derivatives, and admits derived compact-domain connection and derivative bounds. The existing coordinate field construction and ODE flow lemmas can now be applied with this actual inverse. Identifying the operator field with the frozen matrix-valued metric, proving its self-adjointness and the full Levi–Civita property, and deriving common parameter-domain bounds remain separate obligations. Finite dimensionality is explicitly used only for positivity-to-surjectivity; the selected dimension-three target satisfies it.

S41 Actual local coordinate geodesic flow from the constructed connection

Let E be a complete real Hilbert space and let $Q, M : E \to \mathcal{L}(E, E)$ be smooth. Use the actual constructed bounded bilinear field $\Gamma(x) = \frac{1}{2}Q(x)K_{DM(x)}$. Define its autonomous phase vector field on $E \times E$ by $F(x, v) = (v, -\Gamma(x)(v, v))$. Composition with the coordinate projections, the continuous bilinear operator evaluation, negation and pairing prove that F is smooth. In particular it is C^1 at each initial phase point.

The checked mathlib Picard–Lindelof theorem for a C^1 vector field gives, for each initial phase (x_0, v_0) and time t_0, a positive phase-ball radius a and time radius ϵ, such that for every phase point $z \in \overline{B}((x_0, v_0), a)$ there is an actual differentiable curve α with $\alpha(t_0) = z$ and $\alpha'(t) = F(\alpha(t))$ on $(t_0 - \epsilon, t_0 + \epsilon)$. Writing $q = \pi_1\alpha$ and $v = \pi_2\alpha$, differentiation of the continuous linear projections gives actual $q' = v$ and $v' = -\Gamma(q)(v, v)$. The proof therefore constructs coordinate solutions of the geodesic equation from smooth fields; it does not assume their existence or their derivative identities.

The field Q must still be identified with M^{-1} and M with the selected metric before these curves are metric Levi–Civita geodesics. This local solution theorem does not yet prove a minimizing geodesic, a convex normal neighborhood, smooth parameter dependence of the flow, parallel frames, an exponential differential or a distance Hessian. The phase-ball/time constants are local to a fixed smooth field, not uniform over the shrinking/dilation metric family.

S42 Exact frozen Cartesian metric to operator correspondence

For the selected $E = \mathbb{R}^3$ with its fixed Euclidean norm, the checked matrix-to-linear-endomorphism algebra equivalence, followed by the finite-dimensional linear-to-continuous-linear equivalence, gives an algebra equivalence $A \mapsto \mathcal{A}(A)$ from real 3×3 matrices to bounded endomorphisms of E. Its underlying linear map is continuous because its domain is finite dimensional. Thus it is a smooth linear map. For an exact frozen admissible metric g, define $M_g(x) = \mathcal{A}(g(x))$. The entrywise smoothness in the frozen definition gives smoothness of the whole matrix field by the finite product rule, hence actual smoothness of M_g by composition. Positive definiteness of $g(x)$ makes it a unit; the algebra equivalence carries that unit to $M_g(x)$.

The actual inverse field $Q_g = \mathrm{Ring.inverse} \circ M_g$ is therefore smooth. It is exactly $\mathcal{A}(g(x)^{-1})$: matrix nonsingular inversion equals ring inversion and the algebra equivalence carries unit inverses to unit inverses. No separately supplied inverse field or smoothness assumption is needed. Consequently the constructed coordinate connection from (Q_g, M_g) is smooth, with genuine derivative and nonnegative derived connection/derivative bounds on every compact domain. These are actual consequences of the exact frozen metric hypotheses. The full coordinate Levi–Civita interpretation, geodesic normal domains, parallel frames, minimizing distance, shrinking-family uniformity and novel analytic continuation remain separate obligations.

Actual coordinate torsion and metric compatibility. Hermitian symmetry of the positive Cartesian matrix gives $\langle M_g X, Z\rangle = \langle M_g Z, X\rangle$. Differentiating this actual identity, using actual smoothness, gives $\langle DM_g[U]X, Z\rangle = \langle DM_g[U]Z, X\rangle$. The previously proved Koszul construction with the actual two-sided inverse now makes $\Gamma_g(X,Y) = \Gamma_g(Y,X)$. Its actual Koszul identity is $2\langle M_g\Gamma_g(X,Y), Z\rangle = \langle DM_g[X]Y, Z\rangle + \langle DM_g[Y]X, Z\rangle - \langle DM_g[Z]X, Y\rangle$. Adding the identity with Y, Z exchanged, the cross terms cancel and metric/derivative symmetry gives $\langle M_g\Gamma_g(X,Y), Z\rangle + \langle M_g Y, \Gamma_g(X,Z)\rangle = \langle DM_g[X]Y, Z\rangle$. Thus torsion symmetry and coordinate metric compatibility are derived for the actual frozen metric field, rather than assumed. Registration with an intrinsic manifold connection library and all normal-domain, variation and distance identifications remain separate tasks.

Exact metric local flow application. Substitute the actual smooth fields (Q_g, M_g) into the previously constructed Picard–Lindelof phase field. For every initial phase and initial time there is a common positive phase-ball and time interval with actual position/velocity curves satisfying $q' = v$ and $v' = -\Gamma_g(q)(v,v)$ for all initial phases in that ball. This removes both arbitrary-field smoothness assumptions from the local coordinate ODE conclusion and uses only the frozen admissible metric hypotheses. It supplies no minimizing/convex-normal or smooth-parameter conclusion.

Actual metric geodesic energy first integral. For the exact admissible frozen metric and actual differentiable curves q, v at a time t, assume the genuine equations $q'(t) = v(t)$ and $v'(t) = -\Gamma_g(q(t))(v(t), v(t))$. Set $e(t) = \langle M_g(q(t))v(t), v(t)\rangle$. The actual chain and bilinear derivative rules give

$$e' = \langle DM_g(q)[v]v, v\rangle + \langle M_g(q)v', v\rangle + \langle M_g(q)v, v'\rangle.$$

The already derived metric compatibility identity at $X = Y = Z = v$ identifies the first term with $\langle M_g\Gamma_g(v,v), v\rangle + \langle M_g v, \Gamma_g(v,v)\rangle$. Substitute the actual velocity equation; the two acceleration terms cancel these two terms exactly. Hence the genuine derivative of the actual metric energy is zero. No constancy, speed bound, geodesic variation or minimizing property is assumed to obtain this

conclusion. Constancy on a connected existence interval and a uniform velocity bound require their subsequent explicit mean-value/ellipticity bridge.

Auxiliary matrix norm and coercion convention. For the continuity/derivative proof of the finite matrix-to-operator map, use the entrywise supremum norm on Cartesian matrices. Its topology is the finite product topology already used by the frozen entrywise smoothness hypotheses. No frozen metric, energy, vector norm or isometry convention changes. Operator inverse identities use explicit values of mapped units, avoiding ambiguous coercions at the type of units.

S42.1 Direct endpoint-pair error

The genuine derivative estimate also gives a shorter route to a sufficient error bound without the energy cross-term calculation. In a real Hilbert space put $d = A - B$, $e_0 = j'(0) - d$, $e_1 = j'(1) - d$. Then

$$\langle j'(1), A\rangle - \langle j'(0), B\rangle - \|A - B\|^2 = \langle e_1, A\rangle - \langle e_0, B\rangle.$$

Cauchy–Schwarz, $\|e_i\| \le kM$, and $\|A\|, \|B\| \le M$ bound its absolute value by $2kM^2$. This is sufficient for the same original $O(K_0 r^2)(\|A\|^2 + \|B\|^2)$ error after $k = K_0 r^2$. The endpoint-pair expression must still be identified with the actual moving-endpoint squared-distance Hessian by geometric variation. The interval calculus does not assume that identification or conclude the main theorem.

S42.2 Actual interval index identity and endpoint error

Let E be a complete real Hilbert space, and let $j, j', f : \mathbb{R} \to E$ satisfy $Dj(s) = j'(s)$ and $Dj'(s) = f(s)$ at every $s \in [0,1]$, with f continuous on this interval. Derivative existence supplies continuity of j and j' on $[0,1]$. Define the actual interval index quantity by

$$I(j, j', f) = \int_0^1 (\|j'(s)\|^2 + \langle f(s), j(s)\rangle)\, ds.$$

The real inner-product derivative rule, applied to $F(s) = \langle j'(s), j(s)\rangle$, gives $F'(s) = \langle f(s), j(s)\rangle + \langle j'(s), j'(s)\rangle$. The second term is exactly $\|j'(s)\|^2$. Both terms are continuous, hence integrable. The interval fundamental theorem therefore proves

$$I(j, j', f) = \langle j'(1), j(1)\rangle - \langle j'(0), j(0)\rangle.$$

This endpoint equality is derived from genuine differential-equation data; it is not supplied as a desired identity hypothesis. If additionally $0 \le k \le 1$ and $\|f(s)\| \le k\|j(s)\|$ on $[0,1]$, put $M = \max(\|j(0)\|, \|j(1)\|)$ and $d = j(1) - j(0)$. The previously proved actual interval compact-maximum argument and Green bounds imply $\|j'(s) - d\| \le kM$ for every $s \in [0,1]$. Expanding both endpoint inner products gives the exact difference

$$I(j, j', f) - \|d\|^2 = \langle j'(1) - d, j(1)\rangle - \langle j'(0) - d, j(0)\rangle.$$

The triangle inequality and Cauchy–Schwarz bound its absolute value by $kM(\|j(1)\| + \|j(0)\|) \le 2kM^2$. For the intended geodesic application, a parallel frame would give $f = -r^2\mathcal{R}j$ and hence $k = Kr^2$ from an actual curvature-operator bound. Parallel-frame construction, the actual curvature bound, and the identification of this index quantity with the two-endpoint distance Hessian remain separate unproved geometric obligations. This calculation asserts neither geodesic existence nor a distance-Hessian conclusion.

S42.3 A genuine bounded operator Jacobi equation

Now let $\mathcal{R}(s)$ be an actual bounded linear operator on E, with $\|\mathcal{R}(s)\| \le K$ on $[0,1]$, where $K \ge 0$ and $Kr^2 \le 1$. Assume the genuine derivatives $Dj = j'$ and $Dj' = -r^2\mathcal{R}(s)j(s)$, and continuity of the displayed force. Bounded-operator evaluation gives $\| - r^2\mathcal{R}(s)j(s)\| = r^2\|\mathcal{R}(s)j(s)\| \le r^2K\|j(s)\|$. Therefore the preceding interval identity and estimate apply with $k = Kr^2$ and the actual force $f(s) = -r^2\mathcal{R}(s)j(s)$. They prove an index error at most $2Kr^2M^2$, for every real r. This is a conditional result about an actual bounded-operator ODE, with its force instantiated explicitly; it does not assume a Jacobi-norm or index-error conclusion. It still makes no claim that a supplied operator field is the curvature of the metrics in the frozen target. If the actual operator field $\mathcal{R}$ is continuous on $[0,1]$, even force continuity follows rather than being supplied separately: $Dj = j'$ implies continuity of j, operator evaluation $(\mathcal{R}, j) \mapsto \mathcal{R}j$ is continuous, and multiplication by the fixed scalar $-r^2$ is continuous. The operator-equation error estimate therefore applies directly to continuous bounded operator fields with genuine first and second derivatives of j.

S42.4 Actual coordinate curvature operator and Jacobi index bound

Let E be a real Hilbert space and let $\Gamma(x)$ be an actual continuously bilinear map $E \times E \to E$. Regard Γ as a field into the Banach space of curried bounded bilinear maps. At a point x where Γ is Fréchet differentiable, write $D\Gamma_x$ for its actual derivative. For fixed v define the bounded linear operator

$$\mathcal{R}_x(v)X = D\Gamma_x[X](v,v) - D\Gamma_x[v](X,v) + \Gamma_x(X,\Gamma_x(v,v)) - \Gamma_x(v,\Gamma_x(X,v)).$$

Each term is linear and continuous in X, and the operator is constructed explicitly using curried-map flips and composition. It is the displayed coordinate curvature expression $R(X,v)v$; identifying a supplied connection field with the Levi-Civita connection of the frozen metric remains separate. For total Lean functions the operator body uses actual `fderiv`; algebraic bounds remain valid on the total domain, while a `HasFDerivAt` hypothesis identifies it with a genuine supplied derivative.

Bounded-map evaluation gives $\|D\Gamma_x[X](v,v)\| \le \|D\Gamma_x\|\|X\|\|v\|^2$; the permuted derivative term has the same bound. The two composition terms are each at most $\|\Gamma_x\|^2\|X\|\|v\|^2$. The triangle inequality therefore proves

$$\|\mathcal{R}_x(v)X\| \le 2(\|D\Gamma_x\| + \|\Gamma_x\|^2)\|v\|^2\|X\|, \qquad \|\mathcal{R}_x(v)\| \le 2(\|D\Gamma_x\| + \|\Gamma_x\|^2)\|v\|^2.$$

If actual field and derivative bounds are $\|\Gamma_x\| \le C$ and $\|D\Gamma_x\| \le D$, with $C, D \ge 0$, this yields $\|\mathcal{R}_x(v)\| \le 2(D + C^2)\|v\|^2$. No desired curvature-operator norm conclusion is supplied as a hypothesis. Along an actual position $q(s)$ and velocity $v(s)$ on $[0,1]$, suppose $\|\Gamma_{q(s)}\| \le C$, $\|D\Gamma_{q(s)}\| \le D$, and $\|v(s)\| \le V$, where $C, D, V \ge 0$. Then the actual operator $\mathcal{R}_{q(s)}(v(s))$ is bounded by $K = 2(D + C^2)V^2$. If this operator field is continuous and the genuine interval derivatives satisfy $Dj = j'$, $Dj' = -r^2\mathcal{R}_{q(s)}(v(s))j(s)$, with $Kr^2 \le 1$, the previously proved actual-operator Jacobi index estimate applies and yields an error at most $2Kr^2 \max(\|j(0)\|, \|j(1)\|)^2$. Here both the curvature bound and the Jacobi index estimate are derived from the actual supplied fields. The field bounds, differentiability/continuity, parallel-frame realization, actual geodesic variation and distance-Hessian identification remain explicit application obligations. This result supplies neither a new literature premise nor a proof of the main theorem. Operator continuity can also be derived. Suppose the actual composed fields $\Gamma(q(s))$, $D\Gamma_{q(s)}$ and $v(s)$ are continuous on $[0,1]$. The flip of a bounded bilinear map is a linear isometry, so it preserves continuity. Evaluation and composition of bounded linear maps are continuous bilinear operations. These facts give continuity of each of the four displayed operator terms and hence of $\mathcal{R}_{q(s)}(v(s))$. Thus the Jacobi index estimate needs only these concrete field/derivative/velocity continuity hypotheses, rather than assuming continuity of the desired curvature operator. Their derivation from a smooth metric and a genuine geodesic remains a separate application obligation. Coordinate curvature and parallel-frame curvature need an explicit change of frame. Given actual bounded linear maps $T : E \to E$ from frame vectors to coordinate vectors and $S : E \to E$ in the reverse direction, construct $\widetilde{\mathcal{R}} = S\mathcal{R}T$. Its norm is at most $\|S\|\|\mathcal{R}\|\|T\|$. Thus actual bounds $\|S\| \le A$, $\|T\| \le B$ and the preceding coordinate estimate give $\|\widetilde{\mathcal{R}}\| \le A\,2(D + C^2)V^2B$. Continuity follows from continuity of S, T and the derived coordinate operator field via bounded-map composition. The same actual Jacobi index argument then applies to the force $-r^2\widetilde{\mathcal{R}}(s)j(s)$, with the small coefficient $K = A\,2(D + C^2)V^2B$. No inverse or isometry assumption is needed for this larger algebraic norm statement. For its geometric interpretation, the supplied maps still must be identified with inverse parallel-frame maps, and their actual uniform norm bounds must be proved. This prevents treating coordinate curvature as already represented in a parallel orthonormal frame.

S43 Actual metric speed constancy and global velocity control

This is a supporting construction for the coordinate geodesics used in the original shrinking-ball argument. It uses exactly a frozen admissible Cartesian metric g on $\mathbb{R}^3$, not an arbitrary independently supplied inverse or metric-compatible connection. Put $M_g(x)X = g(x)X$ in the frozen Euclidean vector norm, $Q_g = M_g^{-1}$, and let Γ_g be the actual smooth Koszul connection constructed earlier. Define the squared metric speed

$$e_g(q,v)(t) = \langle M_g(q(t))v(t), v(t)\rangle.$$

It is a curve quantity, distinct from the frozen scalar-function Dirichlet functional. Ordinary matrix multiplication and the real Euclidean inner product give $\langle M_g(x)X, X\rangle = X^T g(x)X$ exactly.

Constancy on an actual open interval. Let $q, v : \mathbb{R} \to \mathbb{R}^3$ satisfy genuine ambient derivatives $q'(t) = v(t)$ and $v'(t) = -\Gamma_g(q(t))(v(t), v(t))$ at every point of $I = (a, b)$. Metric compatibility and the already derived chain rule imply $e_g'(t) = 0$ at every such point. Hence e_g is differentiable on I. The real mean-value theorem on the convex, open, preconnected interval gives $e_g(s) = e_g(t)$ for all $s, t \in I$. No constancy or minimizing property is supplied as a hypothesis; no endpoint derivatives are required outside I.

An actual global quadratic lower bound. The scalar function $F(x, X) = \langle M_g(x)X, X\rangle$ is continuous. On the compact set $\overline{B}(0,1) \times S(0,1)$ it is strictly positive, because a unit vector is nonzero and the frozen $g(x)$ is positive definite. Compactness gives $c > 0$ with $c \le F(x, X)$ on this set. For arbitrary $X \neq 0$ normalize by $\|X\|^{-1}$; bilinearity gives $F(x, \|X\|^{-1}X) = \|X\|^{-2}F(x, X)$, hence $c\|X\|^2 \le F(x, X)$. At $X = 0$ both sides vanish. Put $\lambda = \min(c, 1) > 0$. For $\|x\| \le 1$ the lower bound follows from $\lambda \le c$. For $\|x\| \ge 1$ the exact frozen exterior identity $g(x) = I$ gives $F(x, X) = \|X\|^2$, and $\lambda \le 1$ gives the same bound. Thus $\lambda\|X\|^2 \le F(x, X)$ for every x, X.

Velocity bound along the actual coordinate solution. Apply this global inequality at $x = q(t)$, $X = v(t)$, and substitute the proved interval constancy at any initial time $s \in I$. Then

$$\lambda\|v(t)\|^2 \le e_g(q,v)(s), \qquad \|v(t)\|^2 \le e_g(q,v)(s)/\lambda \quad (t \in I).$$

Positivity of λ justifies the division. This supplies a bound for a fixed actual admissible metric and supplied genuine local solutions. It does not assert global flow existence, smooth initial-data dependence, normal or minimizing neighborhoods, parallel frames, Jacobi realization, a distance Hessian, or constants uniform over the manuscript's shrinking metric family. Those applications remain explicit obligations. No published result enters as an unproved premise, and the frozen main type and definitions are unchanged.

S44 Angular functional preparation — upfront expansion

Written before candidate Lean. Mathematical source: original preserved manuscript build/sections/05-angular-estimate.tex, lines 263–415 (weighted raising proof), especially actual polynomial differential contraction in the convex bound; current expanded v009 lines 2888–3040. Existing checked algebra: canonical Calderon/FischerPolynomial.lean, HarmonicRaising.lean, FischerEstimates.lean, FischerContraction.lean. Selected target v002 remains exact and is not replaced by any conditional theorem here.

S44.1 Concrete directional differential and weighted derivative operators

For a finite coordinate type sigma, let F(p,q)=sum_alpha alpha! p_alpha q_alpha, exactly the canonical Fischer inner product. Define W_f(q)=sum_i f_i q_i using scalar polynomial multiplication C(f_i), and a_f(v)=W_f(partial_i v). These are actual polynomial operators, not norm variables. All real scalars f_i are permitted; B=sum_i f_iˆ2 >=0.

Prove F(W_f q,W_f q) <= B sum_i F(q_i,q_i). If B>0, expand the nonnegative expression sum_i F(q_i-C(f_i/B)W_f q,q_i-C(f_i/B)W_f q). Its cross term is F(W_f q,W_f q), by proven Fischer bilinearity, and its scalar-square term is B times that same square. The expression is Q-Bˆ{-1}S >=0; multiplying by B proves the assertion. If B=0, each f_i=0 and W_f q=0, so no inverse issue is hidden.

For actual homogeneous v of degree k, the canonical Euler/Fischer identity gives sum_i F(partial_i v,partial_i v)=k F(v,v). Substitution proves F(a_f v,a_f v)<=k B F(v,v). Under actual gradient smallness sum_i f_iˆ2 <= betaˆ2, this yields the source's bound k betaˆ2 F(v,v). This hypothesis is a source profile condition; coercivity is not hypothesized.

For a supplied vector polynomial w_i (interpreted as base covariant derivative at a point, interpretation still open), define D_iˆ+ v=w_i+C(k f_i)v and D_iˆ-v=-w_i+C(k f_i)v. Then their divergence sum obeys div(Dˆ+v)+div(Dˆ-v)=C(2k)a_f v. Expand div(Dˆ+v)=-div(Dˆ-v)+C(2k)a_f v. Positivity of the squared norm of the difference yields ||p+q||_Fˆ2<=2||p||_Fˆ2+2||q||_Fˆ2. Together with the directional contraction this proves F(div Dˆ+,div Dˆ+)<=2 F(div Dˆ-,div Dˆ-)+8 kˆ3 betaˆ2 F(v,v). If each w_i is homogeneous of degree k, the established divergence contraction supplies the other bound (k+n-1) sum_i F(D_iˆ+,D_iˆ+). Multiply the two derived inequalities by 1-zeta and zeta, for 0<=zeta<=1, and add to prove exactly the pointwise convex bound. This is a genuine actual-operator instance of the source pointwise step, independent of unproved base integration identities.

S44.2 Infinite degree coefficient families

Define N_0=1 and N_{k+1}=(n+2k)N_k as the source's normalization (finite product). With n>0 each N_k>0. Define normalized square E_k(p)=F(p,p)/N_k; normalized vector square is its coordinate sum. Families remain explicit functions from N to polynomials; no unproved identification with sphere L2 is claimed. On q(k) homogeneous degree k+1, actual lowering is div q(k) in degree k, with E_k(div q(k)) <= (k+n)(n+2k) sum_i E_{k+1}(q(k)*i). For harmonic q(k), actual sphere-normalized raising is C(n+2(k+1)) harmonicRaising(k) q(k), with E*{k+2}(raised q(k)) <= (n+2(k+1))(k+2) sum_i E_{k+1}(q(k)_i). The upper bound uses the exact harmonic-raising norm identity and finite mixed contraction bound, discarding only its nonnegative correction term.

For any nonnegative degree weight omega_k and a summable right-hand weighted energy, summability of the actual output squares follows from the nonnegative comparison theorem; monotonicity of tsum proves the infinite-family energy inequality. The operator square energy is an

actual infinite sum of factorial-weighted coefficient expressions, so this removes a finite-only algebra restriction. This does not establish a completed Hilbert domain or smooth sphere convergence.

For harmonic h of degree k+2, F(harmonicRaising(k)q,h)=sum_i F(q_i,partial_i h), since its radius-square correction pairs to zero. The normalization recursion cancels the raising scalar, proving E-pairing adjoint identity exactly between degrees k+1 and k+2; corresponding termwise infinite series equality follows by tsum congruence (a Hilbert adjoint assertion additionally needs domain/density control).

S44.3 Remaining application obligations and evidence boundary

No new axiom or unproved novel conclusion is introduced. Still unproved here: probability-sphere/Fischer normalization; identification of smooth functions with coefficient families and finite Sobolev convergence; covariant polynomial/tensor differentiation and curvature slot norms; compactly supported base integration by parts; Hessian profile applicability; actual geodesic moving-trace/PDE differential operators, their symbol bounds, cutoff estimates and the global uniqueness connection. This preparation cannot prove main or the complete weighted raising/coercivity estimate. Compilation is development evidence only; source correspondence, independent review and trusted checking belong to supervisor integration. Actual serving model/effort/tokens unknown.

S44.4 Additional exact model norm identity (expanded before authoring)

For q+ harmonic and homogeneous degree k+1 and any q-, set c=n+2(k+1), b=n+2k. The actual normalized raising/lowering square difference is E_{k+2}(C(c)HR_k(q+))-E_k(div(q-)) = [c(Q+cross-4/(4k+2n)F(div(q+),div(q+)))-b F(div(q-),div(q-))]/N_{k+1}. Here Q and cross are the actual coefficient expressions from the canonical harmonic raising identity. Derive the two exact normalized square recurrences by expanding Fischer scalar multiplication and N_{j+1}=(n+2j)N_j, then substitute the proven harmonic raising identity and combine the two fractions. For q±=D±v this is precisely the source Fock-difference calculation at degree k+1, before base integration. No integration-by-parts term is assumed or disguised as a parameter.

S44.5 Actual pairing domains (expanded before authoring)

Define the scalar-family gradient by partial_i h_k. If h_k is homogeneous of degree k+2, its degree k+1 normalized vector square equals (k+2)(n+2(k+1)) E_{k+2}(h_k), by Euler's Fischer identity and the normalization recursion. Thus a summable scalar spectral budget yields an actual summable gradient square family.

To make the adjoint series convergence observable rather than only a termwise identity, prove |F(p,q)| <= (F(p,p)+F(q,q))/2 using positivity at p-q and p+q. Division by positive N gives the corresponding normalized inequality. Summing the finite coordinate pairings and applying the triangle inequality bounds the absolute pairing by half the q vector square plus half the gradient vector square. For nonnegative weights, summability of these two square budgets implies absolute summability of the actual coefficient pairing series. The exact normalized raising adjoint identity transfers that convergence to the raising pairing. This is a concrete coefficient domain statement; completion/density and identification with smooth sphere functions remain unproved application obligations.

S44.6 First radial trace correction (expanded before authoring)

The preserved original angular section lines 105–140 requires B1 of order zero in the normalized radial family. The concrete drift derivative identified by the Green author is B1 v=-(b dot partial_z v)/(2m+n-2) on harmonic degree m>=1, B1 constants=0. Source: preserved normalized principal operator and rescaling discussion, with the explicit derivation in work/green-transfer-preparation/EXPANSION.md (authoring interface, independent correspondence not yet accepted). In this coefficient unit define this polynomial boundary correction exactly, using degree m and card sigma=n.

For m=k+1, c=2m+n-2=n+2k>0. The directional derivative is homogeneous degree k and its Fischer square is <=m $|b|^2$ F(v,v). Since N_m=c N_{m-1}, division by c^2 N_{m-1} gives normalized square <=(m/c)$|b|^2$ E_m(v). If n>=2 then m/c<=1/2 uniformly; in n3 the exact factor is m/(2m+1). For degree zero the directional derivative vanishes by the proved gradient energy identity and positive definiteness, so the separately defined correction is zero. Summing with any nonnegative degree weight, a summable input weighted normalized energy and fixed b gives summability and the uniform output coefficient-energy bound. This is the genuine coefficient-space order-zero first correction, retaining the separate Green/PDE trace realization obligation and the degree shift in any Sobolev comparison.

S45 Actual radial Dirichlet variation: upfront expansion v001

Authorship by green-transfer-author, 2026-10-09. This is not an independent audit. All work remains private. Frozen v002 target and ten definitions are unchanged. Actual serving model, effort and tokens are unknown; requested Sol/high is not observed metadata.

S45.1 Source and exact conventions

Original sections 03/04/05/06 have been read, including actual cross existence, Green flux and local harmonicity, moving trace expansion, mixed base-angular coercivity and bootstrap parameter order. The current full expansion is expanded/calderon-ball-expanded-v009.tex, especially angular-expanded:trace and fixed-separation-supp:limits. No Green continuation, density bound, metric uniqueness or complete pseudodifferential estimate is assumed here.

The first equation is Delta_g u=0 (positive Cartesian principal part); its Green inverse is -Delta_g. The reference gamma=exp(-2 L t0)g has the same solution equation after multiplication by exp(2 L t0):

```
L_gamma = Delta_gamma + (n-2)L <dt0,du>_gamma.
```

Thus in gamma-normal coordinates at y the principal first radius derivative vanishes and the drift at the center is b(y)=(n-2)L grad_gamma t0(y). For n=3 it is L grad_gamma t0(y). After x=exp_y^gamma(r z) and multiplication by r^2, the family is

```
L_(y,r) = Delta_z + r b(y) dot grad_z + r² R_(y,r).
```

The smooth remainder includes both second-order metric terms and first-order drift variation. There is no zero-order potential. The radial trace is the outward radial derivative on the unit sphere, not the conormal operator on the physical first-metric ball.

S45.2 Complete first-variation proof

Fix a harmonic homogeneous polynomial v of degree k >=1, and set q=b dot grad v. Commutation of Cartesian derivatives gives Delta q=0; Euler homogeneity gives degree(q)=k-1. Set d=2k+n-2>0 and

```
w(z)=(1-|z|²) q(z)/(2 d).
```

The radius-square product rule gives Delta((1-$|z|^2$)q)=-2d q, so Delta w=-q. On |z|=1, w=0. These are the inhomogeneous equation and boundary condition of the first radius derivative of the genuine Dirichlet solution: differentiating L_r v_r=0 and the fixed boundary data gives Delta v'_0=-b dot grad v, v'_0|sphere=0. Standard Dirichlet uniqueness identifies v'_0=w; identification with a true family requires the explicit standard inverse/differentiability interface, and will not be asserted as a formal consequence of polynomial identities alone.

Its radial trace is z dot grad w=-q/d on the unit sphere. Consequently

```
B1 v= - b dot A v / (2k+n-2).
```

For k=0, q=0 and w=B1v=0. For k=1, q=b dot a is constant for v=a dot z; w=(1-$|z|^2$)(b dot a)/(2n), B1v=-(b dot a)/n. In dimension three the denominator is 2k+1, and the degree-one trace is -b dot a/3. This fixes sign, dimension and normalization.

S45.3 Explicit operator identity and complete order-zero first correction

There is a second construction that identifies the entire first correction, rather than only its Sobolev norm. For the constant center drift model L_r=Delta+r b dot grad, put u=exp(-r b dot z/2)v. Direct differentiation gives

```
L_r u=exp(-r b dot z/2)(Delta-r²|b|²/4)v.
```

Its radial Dirichlet trace is therefore exactly

```
Lambda_r=M_exp(-r b.omega/2) Lambda_Helmholtz(r²|b|²/4)
        M_exp(r b.omega/2) - r M_(b.omega)/2.
```

At r=0 the Helmholtz operator is B; the potential depends on r^2, so its first radius derivative is zero whenever the usual differentiable Dirichlet inverse interface is supplied. Thus

```
B1=(1/2)[B,M_(b.omega)]-(1/2)M_(b.omega).
```

This agrees with the forcing formula. Indeed the boundary multiplication of a degree-k harmonic v has raising part (b.z)v-$|z|^2$(b.Av)/d and lowering part (b.Av)/d, with d=2k+n-2. The former is harmonic of degree k+1 by the radius-square Laplacian identity, and the latter has degree k-1. Applying B to each and subtracting the multiplication gives precisely -b.Av/d. This formula handles k=0 as well because [B,M]1=M1.

Given the standard Euclidean B in Psi^1 and the standard scalar angular commutator rule, this expression is in the complete Psi^0 topology, with smooth kernels included. The center dependence is only through smooth b and orthonormal trivialization. This identifies the required first correction using an actual PDE gauge transformation; it does not prove the second Taylor remainder in complete Psi^1 topology for the variable metric family. That remaining remainder is exactly the standard parameter-dependent Dirichlet-calculus obligation, not a novel transfer bound.

S45.4 Degree-uniform strength and actual derivative budget

The Fischer-to-probability-sphere factor is N_k=n(n+2)...(n+2k-2). Euler and Fischer adjunction give sum_i F(A_i v,A_i v)=k F(v,v); N_k/N_(k-1)=d. Hence sum_i $||A_i v||^2$_sphere=k d $||v||^2$_sphere. Cauchy-Schwarz gives

```
||B1 v||²_sphere <= |b|² k/d ||v||²_sphere <= (|b|²/2)||v||²_sphere  (n>=2).
```

For an H^s fiber norm the output degree k-1 has weight k^(2s) and the input weight (k+1)^(2s). The ratio is bounded by 2^(2|s|), proving B1:H^s->H^s for each real s, uniformly in degree. Finite sums extend by density. Proving the exact sphere/Fischer integration correspondence remains separate; the separate angular-functional-author packet proves the explicit Fischer normalization and summable all-degree coefficient-energy bound; its sphere integration correspondence remains a separate obligation. This radial packet does not silently call Fischer energy the sphere norm.

B1 is linear in b. Every base derivative at fixed orthonormal trivialization is the same multiplier with the corresponding derivative of b, plus bounded frame connection terms. Through six base derivatives it therefore uses only the corresponding fixed first-metric/conformal background derivatives, with no h derivative and no radius derivative. The first correction is order zero by this explicit spectral computation. This proves Sobolev mapping control; it does not by itself prove the complete classical symbol and smoothing-kernel topology required by the source commutator calculus.

For r=r_delta, the contribution f_i r B1 lies in the bounded order-zero remainder since r beta <= epsilon/beta. Its jth base derivatives require derivatives of f through j+1 and radius through j: |partialˆa r|<=C_a r betaˆa and |partialˆa f_i|<=C_a betaˆ(a+1), so |partialˆj(r f_i)|<=C_j r betaˆ(j+1)<=C_j epsilon betaˆ(j-1). Thus j<=6 consumes f through7, within source8. Angular differentiation acts on A and the degree multiplier, not on h. Products with normalized second-metric coefficients use h through at most8 as already stated in the source, but that full coefficient/error-form argument is not proved by this packet.

S45.5 Cutoff and remaining original obligations

On lower cutoff region transfer is evaluation and these actual rescaled Dirichlet operators apply. On upper cutoff region r>=R_* t_bˆp>0, the source finite-separation continuation provides the distinct mixed-kernel input; the lower estimate cannot replace it. The uniform shrinking density bound still requires exact trace-operator decomposition in complete Psi topology, genuine raising estimate, retained mixed perturbation error forms and Green-local harmonicity. No outside-hypothesis counterexample is claimed.

Planned Lean correspondence: polynomial evaluated Frechet derivative -> true Cartesian Laplacian -> drift forcing zero-boundary solution -> evaluated outward radial trace -> k=0/1 dimension-three checks -> normalized all-degree coefficient bound. Previously compiled Angular-Polynomial already proves the generic formal radius forcing; this packet adds its actual derivative interpretation and drift specialization.

S45.6 Coordinate-space limit of the Lean bridge

The evaluated Fréchet and Cartesian Laplacian statements use the finite coordinate function space (Fin 3 -> Real), with its ordinary finite Pi norm. Its equivalence to the frozen EuclideanSpace R3 is a separate representation bridge; no frozen definition is replaced. The differential statements are valid on this genuine finite-dimensional normed coordinate space.

S46 Uniform mixed base/angular error form: complete coefficient derivation v001

Authoring by green-transfer-author, 2026-10-09. Original source: section05-angular-estimate.tex from angular:norm-comparison onward; exact current expansion: angular-expanded:P-decomposition, U-bounds, real-principal-part, error-types. This is an actual operator derivation, not an independent verdict or a Lean acceptance. The final estimate below is derived from the actual trace coefficient classes and standard scalar angular calculus. The complete parameter-dependent trace-calculus interface remains open. No novel coercivity, transfer density bound or uniqueness is taken as a premise.

S46.1 Coordinate model and source coefficient classes

Use a fixed finite orthonormal trivialization of the tangent sphere bundle. The sphere probability measure is fixed there; base measure is mu(y)dy with mu>0 smooth, bounded above and below, and fixed smooth background bounds. Horizontal differentiation is partial_i plus a fixed angular first-order connection operator. Replacing it by partial_i contributes only the lower base orders included below.

Write the actual normalized trace decomposition P=2T+U+J as

```
T^dagger = K^l partial_l + M,
U = a^{ij} partial_i partial_j + B^i partial_i + C,
Z = T^flat - T^dagger,
a^{ij}=r Q^{ij}=a^{ji} real scalar, angular independent.
```

The reference raising coefficient has K^l in Psi^1 with fixed smooth base bounds. M is in beta Psi^2. Z is in beta Psi^1. With j base derivatives, these have allowances K: fixed background (or the weaker beta^j), M: beta^(j+1), Z: beta^(j+1). The U coefficients obey

```
partial^j a in epsilon beta^(j-2) Psi^0,
partial^j B in epsilon beta^(j-1) Psi^1,
partial^j C in epsilon beta^j Psi^2.
```

Only the finitely many derivatives counted below are needed. Psi means scalar classical angular operators with the complete symbol and smoothing seminorms sufficient for the indicated Sobolev mappings. For scalar angular A in Psi^m and D in Psi^n, [A,D] is in Psi^(m+n-1). In particular, this rule applies to K,B,M,C: they are scalar operators on sphere functions, despite carrying a coordinate tensor index. It would not hold for an arbitrary matrix-valued angular principal symbol.

These classes are derived from the actual C_i^*=r^-1 A_i^*+f_i B+E_i+F_i, E_i in r Psi^1, F_i in Psi^0, with six allowed base derivatives. The normalized second equation has no ordinary coordinate drift (harmonic coordinates); in the gamma connection its drift is b_2^k=Q^{ij}Gamma_gamma^k_ij. Consequently its derivatives do not cost an extra derivative of h.

The coefficient a is exactly rQ. The first-base coefficient consists of 2rQ f B, 2h A^*, 2rQ(E+F), r b_2, and the fixed horizontal connection terms; its size is epsilon beta^-1 Psi^1. The zero-base term has (h/r)A^*A^*, rQ(f_i f_j B^2+f_{j;i}B), and products with E/F and drift. Cross products A^*F and b_2 A^* have uniform order one and are assigned to J. The others have size epsilon Psi^2. The derivative of r^-1 cancels the degree shift [B,A^*]=A^*; this cancellation is exact and is required to identify 2T.

For each j used here, Leibniz and the source profile yield

```
|partial^j r| <= C_j r beta^j,
|partial^j(r^-1)| <= C_j r^-1 beta^j,
|partial^j(h/r)| <= C_j epsilon beta^j,
r beta^2 <= epsilon, beta>=1.
```

Angular differentiation does not differentiate h(y). Fixed angular symbol regularity is supplied by the first metric, not by infinitely many small h derivatives.

S46.2 Exact commutator expansion with no third base derivative

In the following formulas products and commutators are genuine ordered angular operator products. Coefficient derivatives are ordinary base derivatives. Summation is understood, and a is symmetric.

For H=K^l partial_l+M, direct Leibniz expansion gives [H,U] as the following seven groups:

```
[K partial, a partial^2]
  = K^l(partial_l a^{ij}) partial_i partial_j
    -2a^{ij}(partial_i K^l) partial_j partial_l
    -a^{ij}(partial_i partial_j K^l) partial_l.

[K partial, B partial]
  = [K^l,B^i] partial_l partial_i
    +K^l(partial_l B^i) partial_i
    -B^i(partial_i K^l) partial_l.

[K partial, C]
  = [K^l,C] partial_l + K^l(partial_l C).

[M,a partial^2]
  = -2a^{ij}(partial_i M) partial_j
    -a^{ij}(partial_i partial_j M).

[M,B partial]
  = [M,B^i] partial_i - B^i(partial_i M).

[M,C] = [M,C].
```

There is no unlisted [M,a]partial² term because a is angular independent. The cubic base term in the first line is (K^l a^{ij}-a^{ij}K^l)partial_l partial_i partial_j=0. This is exact cancellation, not a bound that loses a base derivative.

The second-base coefficients are in epsilon beta^-1 Psi^1: K partial a has that size, a partial K is smaller, and [K,B] loses one angular order. The first-base coefficients are in epsilon Psi^2: K partial B, B partial K, [K,C], a partial M and [M,B] all satisfy this. The zero-base coefficients are in epsilon beta Psi^3: K partial C, a partial²M, B partial M and [M,C] satisfy it, with lower angular orders included by Sobolev embedding. The first base derivative of a second-base coefficient is in epsilon Psi^1, which is exactly what the one integration by parts requires.

S46.3 The adjoint difference and quadratic perturbation

Sphere adjoints are stars. Since a is real, angular independent and symmetric, integration by parts gives exactly

```
V := U^dagger-U = D^i partial_i + E,
D^i = 2 mu^-1 partial_j(mu a^{ij}) - B^i - (B^i)^*,
E = mu^-1 partial_i partial_j(mu a^{ij})
    -mu^-1 partial_i(mu (B^i)^*) + C^* - C.
```

Hence D is in epsilon beta^-1 Psi^1 and E is in epsilon Psi^2. There is no second-base part in V. This is the precise use of reality of rQ; angular independence alone would not cancel the leading adjoint part of a complex coefficient.

Now [U^dagger,U]=[V,U]. Substituting K=D,M=E into the previous exact expansion proves no third base derivative and the stronger classes

```
base2: epsilon^2 beta^-2 Psi^1,
base1: epsilon^2 beta^-1 Psi^2,
base0: epsilon^2 Psi^3.
```

They lie in the required epsilon^2 beta^-1 Psi^1, epsilon^2 Psi^2 and epsilon^2 beta Psi^3 since beta>=1. A derivative of its base2 coefficient remains controlled by epsilon^2 beta^-1 Psi^1. There is no concealed fourth base derivative.

For UZ the exact expansion is

```
UZ = a^{ij}Z partial_i partial_j
     +(2a^{ij}partial_i Z+B^j Z) partial_j
     +a^{ij}partial_i partial_j Z+B^i partial_i Z+CZ.
```

Its classes are respectively epsilon beta^-1 Psi^1, epsilon Psi^2, and epsilon beta Psi^3. Its adjoint can be treated in the quadratic pairing without expanding additional derivatives.

The source error operator is

```
E_err = 2[H,U] + 2[H,U]^dagger + [V,U]
        -2(UZ+(UZ)^dagger).
```

The equality [U^dagger,T]=[T^dagger,U]^dagger is exact. In its real quadratic form, the adjoint partners have the same real part. Thus it suffices to estimate the displayed operators themselves; no additional base derivatives are consumed by writing out those partners.

S46.4 Actual mixed Sobolev form bound

Let phi be smooth, compactly supported in the base, with arbitrary angular degree; the zero-mean condition is needed only for the subsequent raising estimate. Define

```
X=||partial phi||_(H^(1/2)),  Y=||phi||_(H^(3/2)).
```

For a second-base term S^{ij}partial_i partial_j with S in epsilon beta^-1 Psi^1, one integration by parts against mu dy gives

```
-<S^{ij}partial_j phi, partial_i phi>
-<mu^-1 partial_i(mu S^{ij})partial_j phi,phi>.
```

The first is bounded by C epsilon beta^-1 X^2 pointwise in y, using Psi^1:H^(1/2)->H^(-1/2). The differentiated coefficient is epsilon Psi^1, so the second is bounded by C epsilon X Y (a stronger angular norm on phi is allowed). First-base epsilon Psi^2 terms map H^(1/2)->H^(-3/2), and are bounded by the same C epsilon X Y. Zero-base epsilon beta Psi^3 terms map H^(3/2)->H^(-3/2), and are bounded by C epsilon beta Y^2.

Pointwise weighted Young gives 2XY<=beta^-1 X^2+beta Y^2. Integrating in y therefore proves

```
|Re<E_err phi,phi>| <= C epsilon integral(beta^-1 X^2+beta Y^2)
                     <= C epsilon N(phi)^2,
```

for epsilon<=1. Constants are independent of delta, even when beta is unbounded above. No base Fourier cutoff, bound on partial^2 phi, estimate on its third derivatives, or comparison of base and angular frequencies has been inserted. These are exactly the mixed strengths retained by the source raising estimate.

In coordinates partial differs from the horizontal nabla by an angular Psi^1 operator with fixed bounded coefficients. Then

```
beta^-1||partial phi||_(1/2)^2
  <=2 beta^-1||nabla phi||_(1/2)^2+C beta^-1||phi||_(3/2)^2,
```

and the latter term is controlled by beta||phi||_(3/2)^2. A fixed finite base partition produces first and zero order errors with the same classes. Thus the coordinate estimate is the claimed intrinsic form bound.

S46.5 Explicit finite derivative budget

The longest path through the quadratic perturbation is a coefficient a differentiated twice in E=V_0, then once in [D partial,U] or twice in [E,a partial^2], and finally once when a second-base coefficient is integrated by parts. Inspecting actual terms sharpens the count: [E,a partial^2] has at most first base order, so it is NOT integrated by parts. It uses at most four base derivatives of a. The second-base coefficients from [D partial,U] are D partial a, a partial D and [D,B]; their final derivative uses a through order3 and B through order2. The other first/zero coefficients use a through4, B through3 and C through2 at most. For [H,U], the analogous maxima are a through2, B through2, C through1; M and K are fixed first-metric/profile quantities. UZ requires Z through2 and U through1 if its second-base term is differentiated. All maxima are within the source allowance of five U coefficient derivatives.

Constructing C includes one derivative of a trace remainder S_i. Taking C through order2 needs S_i through3; the conservative five-coefficient-derivative envelope would need S_i through6. Thus the source's six trace derivatives suffice. Those trace coefficients contain f_i, so their sixth base derivative uses f through7. h enters only as h/r, h A^*, Q and b_2=Q Gamma; five coefficient derivatives need h through5, while a deliberately conservative budget of four error derivatives plus one construction derivative plus two trace/profile allowances stays below8. No angular symbol derivative hits h. Fixed first-metric derivatives can be higher and remain background bounds; they are not asserted to obey epsilon r smallness.

S46.6 Result and precise remaining obligations

At the mathematical authoring level, the mixed error form follows from the actual scalar trace coefficient classes, the reality/angular independence of rQ, and standard angular scalar calculus. The explicit formulas show why the high base frequencies do not cause a loss and why the eight small derivatives of h suffice. No admissible quasimode or failure was found in this comparison. This is not a claim that the entire section05 or main theorem has been proved.

Remaining: formal realization of the complete parameter-dependent scalar sphere Psi algebra and Sobolev mappings; exact trace Taylor remainder including smoothing topology; genuine weighted raising estimate on the curved sphere bundle; full source Green continuation/local harmonicity and cutoff trace bounds; reviewed published interfaces and kernel checking. The explicit PDE trace packet is independently useful and formally compiled separately.

S47 Exact smooth energy to Tau Ceti H1 bridge, upfront expansion v001

Authority: the ten frozen definitions at d1205fb493b3fbe04e55f72b76c6ff0761fb5a03. Actual Tau W1p is the closed subspace of L2(value,gradient) satisfying distributional first derivative identities with Lebesgue measure, with Euclidean product graph norm. Actual W1p0 is closure of ofTestFunction_l in this space. These definitions are not changed.

1. Any continuous ambient vector-valued function restricted to B has finite L2 norm: its norm is bounded on compact closedBall(0,1), B is contained in that closed ball, Lebesgue(B) is finite, and MemLp.of_bound applies. In particular a globally smooth real f and its Euclidean gradient have L2 representatives. Classical differentiation plus local integrability of f and its Frechet derivative gives Tau HasWeakFDerivOn. Almost-everywhere replacement by these L2 representatives preserves the weak derivative identity. W1p.mk therefore gives an actual element J(f) with value f and gradient grad f almost everywhere on B. J is linear on the vector space of globally smooth real functions, shown by uniqueness of W1p value and the L2 operations. Energy_eq_energyFormIntegral and the jet identities give energy g f = actual energyFormH1(a_g,0,0)(J(f),J(f)), a_g=sqrt(det g)*gˆ-1.

2. External candidate, not yet adopted: for a globally smooth real w vanishing on norm x >=1, J(w) belongs to actual w1p0Submodule. McLean Cambridge 2000 Theorem 3.40(ii), printed105-106 (PDF120-121), s=1, k=2, gives H1_0(B)=ker trace. The ball is C1,1 and w has ordinary zero trace; Theorem3.37 printed102 identifies trace with boundary restriction for smooth functions. Source H1 is Fourier restriction norm; Theorem3.30(ii), printed92, identifies it with W1 based on weak L2 derivatives with equivalent norms, and the H1_0 definition printed77 is closure of compactly supported smooth functions. For real w, real parts of complex approximants preserve compact support and convergence. Tau graph norm equals L2 value squared plus L2 gradient squared, so its topology is the source W1 topology. This is the representation justification for the narrow candidate, not a premise of full data equivalence.

3. Every ball test function lies in smoothZeroExterior (already checked), and its J equals Tau ofTestFunction_l by value uniqueness. Thus the image under J of the frozen smooth zero-exterior functions contains the test-function range; item2 puts it inside the closed H1_0 subspace. Consequently its closure equals exactly the existing H1_0. For smooth datum f, the frozen competitors are exactly f+w. The energy is a continuous quadratic functional F_f(v)=a(J(f)+v,J(f)+v) on H1 whenever the actual Tau coefficient-integrand is essentially bounded. If the quadratic form is nonnegative on all H1 elements, its image is bounded below by0. The existing sInf_image_closure_eq then equates the frozen real infimum with the actual affine H1_0 infimum. No smooth minimizer or exterior extension of a minimizer is required.

4. For bounded symmetric coefficients and actual coercivity of the restricted Tau form, Lax-Milgram on H1_0 applied to the continuous functional v -> -a(J(f),v) produces w_f. Set U_f=J(f)+w_f; then a(U_f,v)=0 for all H1_0 v. Bilinearity and symmetry give a(U_f+v,U_f+v)=a(U_f,U_f)+a(v,v). Nonnegativity gives the minimizer, strict coercivity uniqueness. Existing metric coefficient preparation, not an external assumption of novel coercivity, is intended to discharge boundedness and uniform ellipticity. This standard elliptic energy coercivity is distinct from the project's original angular continuation/coercivity estimates.

5. The minimizer map in f is linear by Lax-Milgram uniqueness. Its mixed energy form is symmetric bilinear and has diagonal minimalEnergy(g,f). Equality of frozen diagonal data gives equality of mixed smooth-data forms by real polarization using f+h and f-h. These are actual variational boundary pairings on smooth data modulo H1_0. Completion to abstract trace-space DN equality still needs the source trace-space identification and density; it is not asserted from a placeholder DN operator.

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6. Coefficient hypotheses in4 are discharged from exact IsAdmissibleMetric, using copied private MetricCoefficient/MetricUniform proofs. Smooth entries give continuous g; positive determinant gives continuous inverse and positive sqrt determinant. A=sqrt(det g)gˆ-1 is continuous and positive definite, hence symmetric. Its bilinear norm has a finite compact-ball upper bound. Its quadratic value on compact closedBall times unitSphere has a positive minimum. Homogeneity gives a positive global lower ellipticity constant on the ball, with no smallness assumption. Tau memLp_energyIntegrand_of_bounds then gives the actual bounded form. Tau Poincare on a ball-contained domain with R1 gives the graph-norm diagonal lower bound lambdaˆ2/(2lambda*(4+1)), strictly positive. Tau isCoercive_energyFormH1L0 packages this actual checked inequality; no analytic coercivity premise is assumed. These copied proof sources are reused from the earlier metric preparation, with an explicit Integrated.Symmetry import needed by their energy-nonnegativity lemma. The new metric application therefore has only frozen admissibility plus the narrow source density parameter, and produces an actual affine W1p minimizer and weak equation. Its existence does not assert a smooth minimizer.

S48 Radial forcing and trace in the exact frozen Euclidean space

The preceding evaluated polynomial proof uses finite Cartesian functions with their Pi norm. Here we transfer it to the exact frozen $\mathbb{R}^3 = \text{EuclideanSpace}\,\mathbb{R}(\text{Fin}\,3)$; we do not replace that space or its norm. Let $C : \mathbb{R}^3 \simeq_L \mathbb{R}^{\{0,1,2\}}$ be the checked `PiLp.continuousLinearEquiv` with exponent two. Its underlying map is the coordinate function and $C(e_i)$ is the ordinary single-coordinate vector. For a finite polynomial p define $P_p(x) = \text{eval}(Cx, p)$.

The genuine Fréchet chain rule gives $DP_p(x) = D\,\text{eval}_p(Cx) \circ C$. Thus $DP_p(x)[v] = \sum_i v_i\,\text{eval}(Cx, \partial_i p)$. Applying this to the actual frozen coordinate derivative $d_i P_p(x) = DP_p(x)[e_i]$ gives exactly $d_i P_p(x) = P_{\partial_i p}(x)$. Applying the same identity once more and summing gives the true Cartesian Laplacian $\sum_i d_i(d_i P_p)(x) = P_{\Delta_{\text{pol}} p}(x)$. Every derivative here is an actual total Fréchet derivative in the unchanged frozen norm.

The Euclidean square-norm identity is $\|x\|^2 = \sum_i x_i^2$. Hence $\|x\| = 1$ implies the sphere equation used by the evaluated polynomial proof. For harmonic homogeneous p of degree $k \geq 1$ and $b \in \mathbb{R}^3$, its explicit drift response $W = P_{\text{driftFirstVariation}(k,Cb,p)}$ has

$$\sum_i d_i(d_i W)(x) = -DP_p(x)[b], \qquad W(x) = 0 \quad (\|x\| = 1),$$

by substitution into the already proved formal Laplacian and forcing identities. On that same actual Euclidean sphere the chain rule and the proved radial formula give

$$DW(x)[x] = -(2k+1)^{-1} DP_p(x)[b].$$

For the coordinate polynomial $p = z_j$, the right side is $-b_j/3$; constant polynomials have zero response. These are genuine frozen-coordinate PDE and outward radial derivative identities. They do not yet identify the explicit response with the derivative of a variable-metric Dirichlet solution family, a conormal trace, or a full uniform Green-transfer estimate. Those standard inverse/parameter and original analytic applications remain open.

S49 Exact smooth energy to Tau Ceti H1 bridge, upfront expansion v001

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4. For bounded symmetric coefficients and actual coercivity of the restricted Tau form, Lax-Milgram on H1_0 applied to the continuous functional v -> -a(J(f),v) produces w_f. Set U_f=J(f)+w_f; then a(U_f,v)=0 for all H1_0 v. Bilinearity and symmetry give a(U_f+v,U_f+v)=a(U_f,U_f)+a(v,v). Nonnegativity gives the minimizer, strict coercivity uniqueness. Existing metric coefficient preparation, not an external assumption of novel coercivity, is intended to discharge boundedness and uniform ellipticity. This standard elliptic energy coercivity is distinct from the project's original angular continuation/coercivity estimates.

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S49.1 7. Abstract bundled bilinear route (before Lean, 2026-10-09 00:48 UTC)

Let V be a real normed vector space, W a real Hilbert space, i:W→V a bounded linear inclusion, B:V×V→R a bounded bilinear form, and B0:W×W→R a bounded bilinear form with explicit representation B0(w,v)=B(iw,iv). Assume B0 coercive. The affine forcing for datum d∈V is ℓ(v)=−B(d,iv), a continuous linear functional because it is the negative of composition of B(d, ·) with i. Let w0=B0ˆ{-1}ℓ via the existing checked IsCoercive.solutionOfFunctional API, and U=d+iw0. That API gives B0(w0,v)=ℓ(v). Thus B(iw0,iv)=−B(d,iv), and bilinearity yields B(U,iv)=0 for every v. This orthogonality statement needs neither symmetry nor ellipticity beyond the already supplied coercivity of B0.

Assume separately B symmetric. For any v∈W, bilinear expansion gives

$$B(U+iv,U+iv) = B(U,U) + B(U,iv) + B(iv,U) + B(iv,iv).$$

Orthogonality and symmetry set the two cross terms to0. The representation turns the last term into B0(v,v). Coercivity implies B0(v,v)≥0. Given arbitrary affine competitor d+iv, use z=v−w0 and linearity of i to identify d+iv=U+iz; consequently B(U,U)≤B(d+iv,d+iv). This is the actual attained affine minimum, with no smooth density premise. Smooth-density is used only later in the separately passed exact frozen infimum correspondence.

Specialization: V is the actual Tau W1p volume ballOpens2, W its actual W1p0, i the subtype continuous linear map, B the bundled energyFormH1L, B0 the bundled energyFormH1L0. Prove

the representation through the published checked equality lemmas energyFormH1L0_apply and energyFormH1L_apply, with explicit arguments rather than rewriting integrated energy definitions. Convert final equalities through those same evaluation lemmas only once. The actual drift/mass functions are zero functions; normalize them explicitly and never conflate scalar0 with a function at an unresolved type. Tau has global shortcut seminormed/normed-space instances plus local normed-group/inner-product-space W1p0 instances: specialization must register the exact local W1p0 instances expected by IsCoercive.solutionOfFunctional; abstract lemmas quantify the structures once, avoiding repeated reconstruction of nested Sobolev graph spaces.

This route changes proof organization only. It adds no mathematical premise, changes no frozen definition or target, and remains private development until actual compilation and root's canonical current-input checks.

S50 Exact smooth energy to Tau Ceti H1 bridge, upfront expansion v001

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3. Every ball test function lies in smoothZeroExterior (already checked), and its J equals Tau ofTestFunction_l by value uniqueness. Thus the image under J of the frozen smooth zero-exterior functions contains the test-function range; item2 puts it inside the closed H1_0 subspace. Consequently its closure equals exactly the existing H1_0. For smooth datum f, the frozen competitors are exactly f+w. The energy is a continuous quadratic functional F_f(v)=a(J(f)+v,J(f)+v) on H1 whenever the actual Tau coefficient-integrand is essentially bounded. If the quadratic form is nonnegative on all H1 elements, its image is bounded below by0. The existing sInf_image_closure_eq then equates the frozen real infimum with the actual affine H1_0 infimum. No smooth minimizer or exterior extension of a minimizer is required.

4. For bounded symmetric coefficients and actual coercivity of the restricted Tau form, Lax-Milgram on H1_0 applied to the continuous functional v -> -a(J(f),v) produces w_f. Set U_f=J(f)+w_f; then a(U_f,v)=0 for all H1_0 v. Bilinearity and symmetry give a(U_f+v,U_f+v)=a(U_f,U_f)+a(v,v). Nonnegativity gives the minimizer, strict coercivity uniqueness. Existing metric coefficient preparation, not an external assumption of novel coercivity, is intended to discharge boundedness and uniform ellipticity. This standard elliptic energy coercivity is distinct from the project's original angular continuation/coercivity estimates.

5. The minimizer map in f is linear by Lax-Milgram uniqueness. Its mixed energy form is symmetric bilinear and has diagonal minimalEnergy(g,f). Equality of frozen diagonal data gives equality of mixed smooth-data forms by real polarization using f+h and f-h. These are actual variational boundary pairings on smooth data modulo H1_0. Completion to abstract trace-space DN equality still needs the source trace-space identification and density; it is not asserted from a placeholder DN operator.

Private compilation uses an exact ten-definition prefix of the frozen source, with the admitted main omitted. Integration must import the actual unchanged target module. No target theorem is proved or altered here, no premise is installed, and human ratification remains pending. Actual serving model/effort/tokens unknown.

6. Coefficient hypotheses in4 are discharged from exact IsAdmissibleMetric, using copied private MetricCoefficient/MetricUniform proofs. Smooth entries give continuous g; positive determinant gives continuous inverse and positive sqrt determinant. A=sqrt(det g)gˆ-1 is continuous and positive definite, hence symmetric. Its bilinear norm has a finite compact-ball upper bound. Its quadratic value on compact closedBall times unitSphere has a positive minimum. Homogeneity gives a positive global lower ellipticity constant on the ball, with no smallness assumption. Tau memLp_energyIntegrand_of_bounds then gives the actual bounded form. Tau Poincare on a ball-contained domain with R1 gives the graph-norm diagonal lower bound lambdaˆ2/(2lambda*(4+1)), strictly positive. Tau isCoercive_energyFormH1L0 packages this actual checked inequality; no analytic coercivity premise is assumed. These copied proof sources are reused from the earlier metric preparation, with an explicit Integrated.Symmetry import needed by their energy-nonnegativity lemma. The new metric application therefore has only frozen admissibility plus the narrow source density parameter, and produces an actual affine W1p minimizer and weak equation. Its existence does not assert a smooth minimizer.

S50.1 7. Abstract bundled bilinear route (before Lean, 2026-10-09 00:48 UTC)

Let V be a real normed vector space, W a real Hilbert space, i:W→V a bounded linear inclusion, B:V×V→R a bounded bilinear form, and B0:W×W→R a bounded bilinear form with explicit representation B0(w,v)=B(iw,iv). Assume B0 coercive. The affine forcing for datum d∈V is ℓ(v)=−B(d,iv), a continuous linear functional because it is the negative of composition of B(d, ·) with i. Let w0=B0ˆ{-1}ℓ via the existing checked IsCoercive.solutionOfFunctional API, and U=d+iw0. That API gives B0(w0,v)=ℓ(v). Thus B(iw0,iv)=−B(d,iv), and bilinearity yields B(U,iv)=0 for every v. This orthogonality statement needs neither symmetry nor ellipticity beyond the already supplied coercivity of B0.

Assume separately B symmetric. For any v∈W, bilinear expansion gives

$$B(U+iv,U+iv) = B(U,U) + B(U,iv) + B(iv,U) + B(iv,iv).$$

Orthogonality and symmetry set the two cross terms to0. The representation turns the last term into B0(v,v). Coercivity implies B0(v,v)≥0. Given arbitrary affine competitor d+iv, use z=v−w0 and linearity of i to identify d+iv=U+iz; consequently B(U,U)≤B(d+iv,d+iv). This is the actual attained affine minimum, with no smooth density premise. Smooth-density is used only later in the separately passed exact frozen infimum correspondence.

Specialization: V is the actual Tau W1p volume ballOpens2, W its actual W1p0, i the subtype continuous linear map, B the bundled energyFormH1L, B0 the bundled energyFormH1L0. Prove

the representation through the published checked equality lemmas energyFormH1L0_apply and energyFormH1L_apply, with explicit arguments rather than rewriting integrated energy definitions. Convert final equalities through those same evaluation lemmas only once. The actual drift/mass functions are zero functions; normalize them explicitly and never conflate scalar0 with a function at an unresolved type. Tau has global shortcut seminormed/normed-space instances plus local normed-group/inner-product-space W1p0 instances: specialization must register the exact local W1p0 instances expected by IsCoercive.solutionOfFunctional; abstract lemmas quantify the structures once, avoiding repeated reconstruction of nested Sobolev graph spaces.

This route changes proof organization only. It adds no mathematical premise, changes no frozen definition or target, and remains private development until actual compilation and root's canonical current-input checks.

S50.1.1 7b. Abstract uniqueness and datum additivity (before Lean)

If d+iw is orthogonal to iW, then bilinearity gives B(iw,iv)=−B(d,iv)=ℓ(v), hence B0(w,v)=ℓ(v). The existing checked uniqueness theorem IsCoercive.eq_solutionOfFunctional gives w=w0. Thus the harmonic member of this affine class is unique. For d1,d2, the sum of their corrections solves the forcing for d1+d2 because B and i are linear and the forcing is additive in d; uniqueness therefore gives correction(d1+d2)=correction(d1)+correction(d2), and consequently U(d1+d2)=U(d1)+U(d2). This proof uses no symmetry. With symmetry, expand B(U1+U2,U1+U2): the two cross terms are equal and their sum is2B(U1,U2). Thus2B(U1,U2)=B(U(d1+d2),U(d1+d2))−B(U1,U1)−B(U2,U2). This generic polarization is later converted to exact minimalEnergy only after the conditional density/minimum correspondence.

S51 Exact successor component correspondence

S51.1 JacobiEndpoint

For actual ambient C2 interval functions, expand the endpoint inner-product difference from the squared endpoint displacement into the two endpoint derivative errors. Under continuous forcing bounded by k times the solution,0<=k<=1, its absolute value is at most2k endpointSize squared. The geometric distance-Hessian interpretation remains open.

Exact public declaration inventory is preserved in the registered claim record and source manifest; the full construction and proof above supplies every supporting helper in this scope.

S51.2 JacobiIndex

Define the actual Bochner index integral of norm(j-prime) squared plus inner(f,j). Genuine first/second derivatives on Icc0,1 and continuous f imply its exact endpoint pairing by differentiating inner(j-prime,j) and FTOC; consequently its error from the squared endpoint displacement is at most2k endpointSize squared under the derived small-force assumptions.

Exact public declaration inventory is preserved in the registered claim record and source manifest; the full construction and proof above supplies every supporting helper in this scope.

S51.3 JacobiCurvature

For an actual bounded operator family R and genuine j-prime-prime=minus r squared Rj, derive forcing comparison from normR<=K>=0. If Kr squared<=1, the actual index error is at most2Kr squared endpointSize squared. A continuous R gives actual forcing continuity from genuine solution continuity. No manifold Jacobi realization is assumed.

Exact public declaration inventory is preserved in the registered claim record and source manifest; the full construction and proof above supplies every supporting helper in this scope.

S51.4 CoordinateCurvature

Construct the actual coordinate curvature contraction from a bounded bilinear connection and its actual Frechet derivative, with the selected Baird-Wood sign convention. Derive bound2(D+C squared)V squared from explicit connection/derivative/normalized-velocity bounds, continuity for smooth actual fields, and the corresponding actual index error. Explicit transported composition includes both supplied frame norm factors; normalized q-prime=r v and actual parallel-frame/Jacobi/distance-Hessian identifications remain open.

Exact public declaration inventory is preserved in the registered claim record and source manifest; the full construction and proof above supplies every supporting helper in this scope.

S51.5 ConnectionField

Construct bounded linear adjoint/Koszul and curried connection maps, prove equality to the previously fixed coordinateConnection using the Riesz pairing, and apply them to actual smooth Q,M and actual fderivM. Prove finite/infinite regularity, actual derivative and derivative-field regularity, derived compact-domain bounds, and3/2 normQ normDM operator bound. No family-uniform normal-neighborhood conclusion is included.

Exact public declaration inventory is preserved in the registered claim record and source manifest; the full construction and proof above supplies every supporting helper in this scope.

S51.6 JacobiInitial

For genuine ambient two-sided C2 interval data and small forcing0<=k<=1/2, derive endpoint bounds from zero initial derivative or position. A supplied bounded endpoint map realized by these solutions differs from identity by norm<=2k; for k<1/2 the checked Neumann unit construction gives a bounded inverse and homeomorphism. The map is not assumed to be an exponential differential; that realization remains open.

Exact public declaration inventory is preserved in the registered claim record and source manifest; the full construction and proof above supplies every supporting helper in this scope.

S51.7 MetricInverse

In finite-dimensional real Hilbert space, strict quadratic positivity gives injectivity and surjectivity of each actual metric operator, hence it is a unit. Define its actual ring inverse, prove both inverse evaluations, smoothness by analytic inversion, and actual constructed connection smoothness. This derives inverse fields rather than supplying arbitrary inverses.

Exact public declaration inventory is preserved in the registered claim record and source manifest; the full construction and proof above supplies every supporting helper in this scope.

S51.8 GeodesicFlow

Define actual phase field(q,v)=(v,minus Gamma(q)(v,v)) from the constructed smooth connection and prove it smooth. Checked Picard-Lindelof yields genuine local phase solutions on an open time interval for nearby initial values, and projection derivatives yield actual q-prime=v,v-prime=minusGamma(q)(v,v). Minimization, smooth parameter dependence, global existence and shrinking-family-uniform domains remain open.

Exact public declaration inventory is preserved in the registered claim record and source manifest; the full construction and proof above supplies every supporting helper in this scope.

S51.9 MetricOperator

Construct the exact continuous algebra equivalence from frozen Fin3 matrices to frozen Euclidean bounded operators. For exact admissible g, prove smoothness, unit positivity, symmetry and derivative symmetry, exact matrix inverse correspondence, actual connection smoothness, torsion symmetry, Koszul identity and metric compatibility. Derive compact connection bounds, genuine local metric coordinate-geodesic solutions and actual zero derivative of squared metric speed. Complete manifold/normal/distance-Hessian realization remains open.

Exact public declaration inventory is preserved in the registered claim record and source manifest; the full construction and proof above supplies every supporting helper in this scope.

S51.10 MetricGeodesicEnergy

Prove exact operator quadratic-form equality, actual interval constancy of metric speed by the mean-value theorem, global positive quadratic lower bound derived from compact closed ball times unit sphere and exact Euclidean exterior, and the consequent fixed-metric geodesic velocity bound. No extra ellipticity assumption, global flow or family-uniform bound is asserted.

Exact public declaration inventory is preserved in the registered claim record and source manifest; the full construction and proof above supplies every supporting helper in this scope.

S51.11 AngularFunctional

Define actual finite-polynomial contraction/directional derivative and supplied-point D-plus/minus=w plus/minus kf v. Derive directional Fischer square<=k sumf squared F(v,v) by positivity/Euler, exact divergence relation, the2divDminus squared+8k cubed beta squared F(v,v) inequality, and the source pointwise convex bound from actual homogeneity and gradient/profile bounds. Base derivatives, curvature and integration identities are not assumed as conclusions.

Exact public declaration inventory is preserved in the registered claim record and source manifest; the full construction and proof above supplies every supporting helper in this scope.

S51.12 AngularCoefficient

Define actual Nk normalization, normalized Fischer pairing/squares and concrete graded lowering/raising/gradient families. Derive positive recurrence/ratio, degree-wise bounds and exact normalized Fock difference; genuine nonnegative input square summability gives actual output summability and tsum bounds. Prove adjoint pairing identity, gradient spectral energy, and absolute pairing convergence from actual energy budgets. Define the actual first drift correction, prove its degree-uniform normalized square factor m/(n+2m-2)<=1/2 for n>=2, constantszero and genuine weighted infinite bound. These are coefficient-family statements, with sphere identification, completed Hilbert domains, curved IBP and full coercivity still open.

Exact public declaration inventory is preserved in the registered claim record and source manifest; the full construction and proof above supplies every supporting helper in this scope.

S51.13 GreenRadialTrace

Construct actual polynomial evaluation Frechet differential and true twice-Frechet Cartesian Laplacian in the finite Pi coordinate norm. For harmonic homogeneous degree k>=1 and fixed driftb, the explicit polynomial w=(1-|z| squared)D_bv/[2(2k+n-2)] solves genuine Delta w=minusD_bv, has exactly zero sphere boundary value, and actual outward radial trace minusD_bv/(2k+1) in dimension3. Constantsgivezero and degreeone gives minusb_j/3. Identification with the true variable-metric Dirichlet-family radius derivative and the frozen Euclidean coordinate equivalence remain explicit separate obligations.

Exact public declaration inventory is preserved in the registered claim record and source manifest; the full construction and proof above supplies every supporting helper in this scope.

S51.14 PrincipalCommutator

On genuine globally smooth functions valued in a real normed algebra, define firstBase=K times actualderiv and secondBase=real a scalar times actualsecondderiv. Derive actual first and second product derivatives and prove their commutator equals a-prime K u-doubleprime minus2a K-prime u-doubleprime minus a K-doubleprime u-prime, with no third derivative even for noncommuting K. This is a one-direction cancellation, not the full sphere pseudodifferential estimate.

Exact public declaration inventory is preserved in the registered claim record and source manifest; the full construction and proof above supplies every supporting helper in this scope.

S51.15 SmoothSobolevBridge

Construct genuine Tau W1p jets of every frozen globally smooth real datum from actual value/gradient finite-ball L2 and classical weak differentiation, and prove actual value/gradient

representatives, source graph norm, exact energyFormH1 equality, affine additivity and true test-function embedding. Define narrow SmoothBoundaryZeroClosure as an explicit standard source parameter, not an axiom. From that parameter prove the closure of ALL frozen smooth zero-exterior competitors is the existing Tau H1_0 closure and the exact image and real infimum correspondence for continuous bounded energy, with bounded-below hypothesis still explicit. Admissible-metric Lax-Milgram attainment and removal of boundedness hypotheses are separate unless supplied in a later module.

Exact public declaration inventory is preserved in the registered claim record and source manifest; the full construction and proof above supplies every supporting helper in this scope.

S51.16 GreenEuclidean

Use the actual exponent-two PiLp continuous linear equivalence to evaluate finite polynomials on the exact frozen EuclideanSpace R3. Prove genuine Frechet chain derivatives and unchanged frozen Cartesian d, actual iterated-d Laplacian equals polynomial Laplacian, and actual Euclidean unit-sphere coordinate square identity. Consequently derive genuine frozen-R3 drift forcing, zero sphere boundary and outward radial derivative minusD_bv/(2k+1), including degreeone minusb_j/3. No coordinate norm is replaced. True variable-metric Dirichlet-family derivative, conormal trace and complete uniform Green transfer remain separate.

Exact public declaration inventory is preserved in the registered claim record and source manifest; the full construction and proof above supplies every supporting helper in this scope.

S51.17 AffineForcing

Construct the actual continuous StrongDual functional on the existing Tau H1_0 space by composing the bounded energy form applied to the actual smooth datum jet with the genuine subtype inclusion, then negating. Prove its exact value minus a(Jf,v) from the checked Tau representation identity. This is actual nonzero-boundary Lax-Milgram forcing, conditional only on explicit coefficient-integrand boundedness, without assuming a minimizer or any novel analytic estimate.

Exact public declaration inventory is preserved in the registered claim record and source manifest; the full construction and proof above supplies every supporting helper in this scope.

S51.18 AffineBilinear

For genuine bounded bilinear forms B on V and B0 on the complete real Hilbert space W, actual bounded inclusion i and explicit B0(u,v)=B(iu,iv), use checked Lax-Milgram coercivity to construct the actual affine correction and solution. Derive genuine orthogonality, exact symmetric quadratic splitting and attained minimum over every actual affine competitor. These are bundled algebraic variational statements; the concrete Tau coefficient specialization, frozen energy infimum equality and unconditional main remain separate until actually proved.

Exact public declaration inventory is preserved in the registered claim record and source manifest; the full construction and proof above supplies every supporting helper in this scope.

S51.19 AffineBilinearData

Use checked Lax-Milgram uniqueness to prove actual correction equality from affine orthogonality, correction and solution additivity in the real datum, and exact symmetric real polarization of the genuine affine solution energy. This remains abstract bounded-bilinear algebra, with concrete Tau specialization and equality to frozen minimalEnergy separate.

Exact public declaration inventory is preserved in the registered claim record and source manifest; the full construction and proof above supplies every supporting helper in this scope.